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MIL-HDBK-510-1(USAF)
13 November 2008

SUPERSEDING
MIL-HDBK-510(USAF)
1 October 2007

DEPARTMENT OF DEFENSE
HANDBOOK

AEROSPACE FUELS CERTIFICATION

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AMSC N/A FSG 91GP

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FOREWORD

1. This handbook is approved for use by the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

2. This handbook provides guidance on the certification of alternative fuels, fuel alternates and fuel additives. This military handbook provides guidance for evaluating and certifying aviation fuels and aviation fuel additives and is consistent with MIL-HDBK-516, Airworthiness Certification Criteria; and MIL-HDBK-514, Operational Safety, Suitability, & Effectiveness for the Aeronautical Enterprise. The criteria are applicable to the four "value streams" of aircraft systems, base, infrastructure, support equipment & vehicles.

3. Section 2 provides a listing of applicable documents. Section 3 provides pertinent definitions and acronyms. Sections 4 and 5 provide guidance and approach for planning and conduct of a fuel certification program using a disciplined systems engineering process.

4. Appendix A provides fuel requirements decomposition and traceability. Appendix B provides recommendations for fuel properties/characteristic testing. Appendix C provides baseline aerospace fuel property information or reference thereto. Appendix D provides a material compatibility test protocol and baseline information. Appendix E provides a toxicity test protocol and baseline information. Appendix F provides the fire protection and survivability/vulnerability. Appendix G provides aircraft propulsion fuels qualification/certification process. Appendix H provides aerospace fuels infrastructure requirements. Appendix I provides information obtained from the various risk assessments conducted for each candidate fuel. Appendix J is reserved. Appendix K provides an environmental assessment. Appendix L provides a property traceability index.

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CONTENTS

Paragraph		Page
1.	SCOPE	1
1.1	Scope	1
1.1.1	Secretary of the Air Force direction	1
1.1.2	The certification process	1
1.1.3	Candidate fuel characteristics	2
1.2	Baseline fuels	2
1.2.1	Fundamental fuel characteristics	2
1.2.2	Jet fuel evolution	3
1.2.3	Russian fuel parallel evolution	4
1.3	Alternative fuels	5
1.3.1	Fuel source specification	5
1.3.2	Potential coal and natural gas fuel sources	5
1.3.3	Applicable alternative fuels	6
1.4	Fuel Additives	6
1.4.1	JP-8 Required Additives	6
1.4.1.1	Fuel System Icing Inhibitor (FSII)	6
1.4.1.2	Static Dissipater Additives (SDA)	7
1.4.1.3	Corrosion Inhibitor/Lubricity Improver (CI/LI)	7
1.4.1.4	Metal Deactivator Additive (MDA)	7
1.4.1.5	Antioxidants	8
1.4.1.6	Thermal Oxidative Stability Improver (+100) Additive	8
1.4.1.6.1	Extensive testing performed	8
1.4.1.6.2	Equivalent or reduced maintenance	8
2.	APPLICABLE DOCUMENTS	9
2.1	General	9
2.2	Government documents	9
2.2.1	Specifications, standards, and handbooks	9
2.2.2	Other Government documents, drawings, and publications	10
2.3	Non-Government publications	10
3.	DEFINITIONS	11
3.1	Alphabetical listing of acronyms and abbreviations	15
4.	GENERAL GUIDANCE	19
4.1	Aerospace fuel certification process goals	19

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4.2 Primary tasks 19
4.3 Expeditionary fuels 19
4.4 Improved fuels and additives 19
4.5 Fuels from alternative or unconventional resources 19
4.6 Segregation of candidate fuels and additives 20
4.6.1 Certification fuel 20
4.6.2 Fleet implementation 20
5. DETAILED REQUIREMENTS 21
5.1 Aerospace fuel certification process 21
5.1.1 Entrance criteria 21
5.1.1.1 Fuel property/characteristic testing 21
5.1.1.2 Material compatibility testing 21
5.1.1.3 Toxicity 21
5.1.1.4 Fire protection and survivability/vulnerability 21
5.1.1.5 Aircraft propulsion 21
5.1.1.6 Aerospace fuels infrastructure 21
5.1.2 Conduct Subset 1 testing 22
5.1.2.1 Fuel property/characteristic testing 22
5.1.2.2 Material compatibility testing 22
5.1.2.3 Toxicity 22
5.1.2.4 Fire protection and survivability/vulnerability 22
5.1.2.5 Aircraft propulsion 22
5.1.2.6 Aerospace fuels infrastructure 23
5.1.2.7 Conduct Subset 2 testing 23
5.1.2.8 Fuel property/characteristic testing 23
5.1.2.9 Material compatibility testing 23
5.1.2.10 Toxicity 23
5.1.2.11 Fire protection and survivability/vulnerability 23
5.1.2.12 Aircraft propulsion 24
5.1.2.13 Aerospace fuels infrastructure 24
5.1.3 Conduct Subset 3 testing 24
5.1.3.1 Fuel property/characteristic testing 24
5.1.3.2 Material compatibility testing 24
5.1.3.3 Toxicity testing 24
5.1.3.4 Fire protection and survivability/vulnerability 24
5.1.3.5 Aircraft propulsion 25

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Check the source to verify that this is the current version before use.

5.1.3.6 Aerospace fuels infrastructure 25
5.2 Develop initial certification plan 25
5.2.1 Evaluate initial certification plan 25
5.2.2 Finalize certification plan 25
5.2.3 Execute certification plan 25
5.2.4 Share verification data 25
5.2.5 Certify or reject fuel 25
5.3 Risk assessment process 26
5.3.1 The AFCO risk assessment process 27
5.3.1.1 AFCO risk assessment considerations 29
5.3.1.1.1 Business case 29
5.3.1.1.2 Benefits case 29
5.3.1.1.3 Risks and negative impacts 29
5.3.1.1.3.1 Definitions of qualitative impact descriptors 30
5.3.1.1.3.1.1 Minor or low 30
5.3.1.1.3.1.2 Medium 30
5.3.1.1.3.1.3 Major or high 30
5.3.2 The single manager risk assessment process 31
5.3.2.1 Single manager risk assessment considerations 33
5.3.2.1.1 Business case 33
5.3.2.1.2 Benefits case 33
5.3.2.1.3 Risks and negative impacts 33
5.3.3 The high-level decision authority risk assessment process 34
5.3.3.1 High-level decision authority risk assessment considerations 36
5.3.3.1.1 Business case 36
5.3.3.1.2 Benefits case 36
5.3.3.1.3 Risks and negative impacts 36
5.4 Field service evaluation 36
5.4.1 Data collection and evaluation 37
5.4.2 Pacer programs 37
5.4.3 Teardown inspections 37
5.5 Certification criteria elevation potential 38
6. NOTES 38
6.1 Intended use 38
6.2 Subject term (key word) listing 38

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Check the source to verify that this is the current version before use.

APPENDICES

APPENDIX A REQUIREMENTS DECOMPOSITION AND TRACEABILITY 39

APPENDIX B FUEL PROPERTIES/CHARACTERISTIC TESTING.....44

APPENDIX C BASELINE AEROSPACE FUEL PROPERTY INFORMATION 56

APPENDIX D EVALUATING THE COMPATIBILITY OF ADDITIVES OR ALTERNATIVE FUELS WITH FUEL SYSTEM MATERIALS.....178

APPENDIX E TOXICITY TEST PROTOCOL AND ESOH BASELINE INFORMATION 208

APPENDIX F FIRE PROTECTION AND SURVIVABILITY/VULNERABILITY 234

APPENDIX G AIRCRAFT PROPULSION FUELS CERTIFICATON PROCESS.....248

APPENDIX H AEROSPACE FUELS INFRASTRUCTURE REQUIREMENTS.....257

APPENDIX I RISK ASSESSMENT RESULTS 262

APPENDIX J CERTIFICATION LESSONS LEARNED 264

APPENDIX K ENVIRONMENTAL, SAFETY AND OCCUPATIONAL HEALTH....265

APPENDIX L FUEL PROPERTY TRACEABILITY INDEX 277

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FIGURES

Number		Page
FIGURE 1.	Aerospace fuels qualification/certification process	26
FIGURE 2.	AFCO risk assessment process	28
FIGURE 3.	Single manager risk assessment process	32
FIGURE 4.	High-level decision authority risk assessment process	35
FIGURE A-1.	Functional System Decomposition	39
FIGURE A-2.	Requirements decomposition and traceability	43
FIGURE C-1.	Total acid histogram JP-8 (2005) *Courtesy of PQIS*	66
FIGURE C-2.	Total acid average JP-8 (year trend 2001-2005) *Courtesy PQIS 2005*	66
FIGURE C-3.	Aromatic content histogram JP-8 (2005) *Courtesy of PQIS 2005*	70
FIGURE C-4.	Aromatic content average JP-8 (year trend 2001-2005) *Courtesy PQIS 2005*	70
FIGURE C-5.	Bulk modulus as a function of fuel temperature for JP-5/JP-8/Jet A/Jet A-1 *[CRC 2004]*	75
FIGURE C-6.	Bulk modulus as a function of fuel pressure for JP-5/JP-8/Jet A/Jet A-1	76
FIGURE C-7.	Cetane index histogram JP-8 (2005) *Courtesy of PQIS*	78
FIGURE C-8.	Cetane index average JP-8 (year trend 2001-2005) *Courtesy PQIS 2005*	78
FIGURE C-9.	Density histogram JP-8 (2005) *Courtesy PQIS 2005*	83
FIGURE C-10.	Density average JP-8 (year trend 2001-2005) *Courtesy PQIS 2005*	84
FIGURE C-11.	Typical density as a function of temperature *[CRC, 1983]*	85
FIGURE C-12.	Density versus temperature limitations	86
FIGURE C-13.	Thermal expansion [CRC 2004]	87
FIGURE C-14.	Dielectric constant as a function of temperature *[CRC, 2004]*	89
FIGURE C-15.	Dielectric constant as a function of density *[CRC World Fuel Survey, 2006]*	90
FIGURE C-16.	Initial boiling point distillation histogram JP-8 (2005) *Courtesy of PQIS*	92
FIGURE C-17.	Recovered distillation histogram JP-8 (2005) *Courtesy of PQIS*	93
FIGURE C-18.	10% Recovered distillation average JP-8 (year trend 2001-2005) *Courtesy PQIS 2005*	93
FIGURE C-19.	20% Recovered distillation histogram JP-8 (2005) *Courtesy of PQIS*	94
FIGURE C-20.	50% Recovered distillation histogram JP-8 (2005) *Courtesy of PQIS*	94
FIGURE C-21.	90% Recovered distillation histogram JP-8 (2005) *Courtesy of PQIS*	95
FIGURE C-22.	Final boiling point distillation histogram JP-8 (2005) *Courtesy of PQIS*	96

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FIGURE C-23. Final boiling point distillation average JP-8 (year trend 2001-2005) *Courtesy PQIS 2005* 96
FIGURE C-24. Fuel conductivity as a function of temperature *[CRC, 2004]* 99
FIGURE C-25. JP-5 enthalpy data *(Courtesy CRC)* 101
FIGURE C-26. Existent gum histogram JP-8 (2005) *Courtesy of PQIS* 103
FIGURE C-27. Filtration time histogram JP-8 (2005) *Courtesy of PQIS* 105
FIGURE C-28. Filtration time average JP-8 (year trend 2001-2005) *Courtesy of PQIS 2005* 106
FIGURE C-29. Fuels flammability limits versus altitude (CRC, 2004) 109
FIGURE C-30. Fuels flammability limits versus altitude [DOT, 1998] 110
FIGURE C-31. Flash point histogram JP-8 (2005) *Courtesy of PQIS* 112
FIGURE C-32. Flash point average JP-8 (year trend 2001-2005) *Courtesy PQIS 2005* 112
FIGURE C-33. Freezing point histogram JP-8 (2005) *Courtesy of PQIS* 115
FIGURE C-34. Freezing point average JP-8 (year trend 2001-2005) *Courtesy PQIS 2005* 115
FIGURE C-35. FSII Histogram JP-8 (2005) *Courtesy of PQIS* 117
FIGURE C-36. Gas chromatograph of FT synthetic fuel (Courtesy AFRL, USAF) 119
FIGURE C-37. Gas chromatograph of JP-8 119
FIGURE C-38. Net heat of combustion histogram JP-8 (2005) *Courtesy of PQIS* 121
FIGURE C-39. Net heat of combustion average JP-8 (year trend 2001-2005) *Courtesy PQIS 2005* 122
FIGURE C-40. Reserved 122
FIGURE C-41. Hydrogen content histogram JP-8 (2005) *Courtesy of PQIS* 125
FIGURE C-42. Hydrogen content average JP-8 (year trend 2001-2005) *Courtesy PQIS 2005* 125
FIGURE C-43. Minimum spark ignition energy *[CRC, 2004]* 129
FIGURE C-44. Fuels flammability limits versus altitude [DOT, 1998] 129
FIGURE C-45. Naphthalene histogram JP-8 (2005) *Courtesy of PQIS* 131
FIGURE C-46. Naphthalene average JP-8 (year trend 2001-2005) *Courtesy PQIS 2005* 132
FIGURE C-47. Ostwald coefficient as a function of temperature *[CRC, 2004]* 134
FIGURE C-48. Particulate matter histogram JP-8 (2005) *Courtesy of PQIS* 136
FIGURE C-49. Particulate matter average JP-8 (year trend 2001-2005) *Courtesy PQIS 2005* 136
FIGURE C-50. Smoke point histogram JP-8 (2005) *Courtesy of PQIS* 140
FIGURE C-51. Smoke point average JP-8 (year trend 2001-2005) *Courtesy PQIS 2005* 141
FIGURE C-52. Specific heat as a function of temperature *[CRC,2004]* 143

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Check the source to verify that this is the current version before use.

FIGURE C-53. Mercaptan sulfur histogram JP-8 (2005) *Courtesy of PQIS*..... 145
FIGURE C-54. Mercaptan sulfur average JP-8 (year trend 2001-2005) *Courtesy PQIS 2005*..... 146
FIGURE C-55. Total sulfur histogram JP-8 (2005) *Courtesy of PQIS*..... 147
FIGURE C-56. Total sulfur average JP-8 (year trend 2001-2005) *Courtesy PQIS 2005*... 148
FIGURE C-57. World average jet fuel surface tension. *Courtesy CRC World Fuel Survey*..... 150
FIGURE C-58. Surface tension as a function of temperature *[CRC, 2004]*..... 151
FIGURE C-59. Thermal conductivity as a function of temperature *[CRC, 2004]*..... 153
FIGURE C-60. Reserved..... 154
FIGURE C-61. Thermal stability (JFTOT ΔP) histogram JP-8 (2005) *Courtesy of PQIS*..... 156
FIGURE C-62. Jet fuel break point distribution *[CRC World Fuel Survey, 2006]*..... 156
FIGURE C-63. Vapor pressure as a function of temperature [CRC, 2004]..... 163
FIGURE C-64. TS-1 Vapor pressure as a function of temperature *[courtesy Boeing]*..... 164
FIGURE C-65. Heat of vaporization as a function of temperature *[CRC, 2004]*..... 165
FIGURE C-66. Velocity of sound versus temperature *[CRC World Fuel Survey, 2006]*.... 167
FIGURE C-67. Viscosity (@-20°C) histogram JP-8 (2005) *Courtesy of PQIS*..... 169
FIGURE C-68. Viscosity (@-20°C) average JP-8 (year trend 2001-2005) *Courtesy PQIS 2005*..... 169
FIGURE C-69. Viscosity as a function of temperature *[CRC, 2004]*..... 170
FIGURE C-70. Water separation index histogram JP-8 (2005) *Courtesy of PQIS*..... 173
FIGURE C-71. Water solubility versus temperature for aircraft fuels *(CRC, 2004)*..... 176
FIGURE C-72. Water solubility histogram JP-8 (2005) *Courtesy of PQIS*..... 177
FIGURE K-1. Environmental impact analysis process.....267
FIGURE K-2. Request for environmental impact analysis..... 268
FIGURE K-3. Change in particle number emissions of TF33 engines burning JP-8 and FT blend..... 273
FIGURE K-4. Change in particle mass EI for TF33 engines using FT blend relative to JP-8..... 274
FIGURE L-1. Index spreadsheet.....283

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Check the source to verify that this is the current version before use.

TABLES

Number		Page
TABLE I.	Summary of major jet fuel characteristics (additives explained in 1.4)	4
TABLE II.	Fuel compositional analysis via ASTM D2425	6
TABLE B-I.	System safety related fuel property/characteristic	46
TABLE B-II.	System safety and performance related fuel properties	47
TABLE B-III.	System performance related fuel properties/characteristics	52
TABLE B-IV.	System durability, and supportability related fuel properties/characteristics	54
TABLE C-I.	Acid number, total fuel property rating value	67
TABLE C-II.	Reserved	67
TABLE C-III.	Additive compatibility fuel property rating value	68
TABLE C-IV.	Reserved	68
TABLE C-V.	Aromatics fuel property rating value	71
TABLE C-VI.	Reserved	71
TABLE C-VII.	Autoignition temperature fuel property rating value	73
TABLE C-VIII.	Reserved	73
TABLE C-IX.	Bulk modulus fuel property rating value	76
TABLE C-X.	Reserved	76
TABLE C-XI.	Calculated cetane index fuel property rating value	79
TABLE C-XII.	Reserved	79
TABLE C-XIII.	Cetane number fuel property rating value	80
TABLE C-XIV.	Reserved	80
TABLE C-XV.	Copper strip corrosion fuel property rating value	81
TABLE C-XVI.	Reserved	81
TABLE C-XVII.	Density fuel property rating value at 15°C	85
TABLE C-XVIII.	Density fuel property rating value at temperature, *T*	86
TABLE C-XIX.	Thermal expansion fuel property rating	87
TABLE C-XX.	Dielectric constant (as a function of T) fuel property rating value	90
TABLE C-XXI.	Reserved	90
TABLE C-XXII.	Suggested distillation specification requirement	91
TABLE C-XXIII.	Reserved	92
TABLE C-XXIV.	Distillation curve fuel property rating value	92
TABLE C-XXV.	10% Distillation curve fuel property rating value	94
TABLE C-XXVI.	20% Distillation curve fuel property rating value	94

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:45Z
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TABLE C-XXVII.	50% Distillation curve fuel property rating value	95
TABLE C-XXVIII.	90% Distillation curve fuel property rating value	95
TABLE C-XXIX.	Distillation curve fuel property rating value	97
TABLE C-XXX.	Distillation curve fuel property rating value for residue	97
TABLE C-XXXI.	Distillation curve fuel property rating value for loss	97
TABLE C XXXII.	Distillation curve fuel property rating value for slope	97
TABLE C-XXXIII.	Distillation curve fuel property rating value for slope	97
TABLE C-XXXIV.	Electrical conductivity at standard T fuel property rating value	99
TABLE C-XXXV.	Electrical conductivity versus temperature	100
TABLE C-XXXVI.	Enthalpy versus temperature fuel property rating value	102
TABLE C-XXXVII.	Reserved	102
TABLE C-XXXVIII.	Existent gum fuel property rating value	104
TABLE C-XXXIX.	Reserved	104
TABLE C-XL.	Filtration time fuel property rating value	106
TABLE C-XLI.	Reserved	106
TABLE C-XLII.	Flame speed test fuel property rating value	107
TABLE C-XLIII.	Reserved	107
TABLE C-XLIV.	Flammability limits @ 25°C fuel property rating value	110
TABLE C-XLV.	Reserved	110
TABLE C-XLVI.	Flash point fuel property rating value	113
TABLE C-XLVII.	Reserved	113
TABLE C-XLVIII.	Freezing point fuel property rating value	116
TABLE C-XLIX.	Reserved	116
TABLE C-L.	FSII fuel property rating value	117
TABLE C-LI.	Reserved	117
TABLE C-LII.	Gas chromatograph fuel property rating value	120
TABLE C-LIII.	Reserved	120
TABLE C-LIV.	Net heat of combustion fuel property rating value	122
TABLE C-LV.	Reserved	122
TABLE C-LVI.	Hot surface ignition fuel property rating value	123
TABLE C-LVII.	Reserved	123
TABLE C-LVIII.	Hydrogen content fuel property rating value	126
TABLE C-LIX.	Reserved	126
TABLE C-LX.	Lubricity fuel property rating value	127
TABLE C-LXI.	Reserved	127

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:45Z
Check the source to verify that this is the current version before use.

TABLE C-LXII.	Minimum spark ignition energy fuel property rating value	130
TABLE C-LXIII.	Reserved	130
TABLE C-LXIV.	Naphthalenes fuel property rating value	132
TABLE C-LXV.	Reserved	132
TABLE C-LXVI.	Ostwald coefficient / gas solubility fuel property rating value	134
TABLE C-LXVII.	Reserved	134
TABLE C-LXVIII.	Particulate matter fuel property rating value	137
TABLE C-LXIX.	Reserved	137
TABLE C-LXX.	Pour point fuel property rating value	138
TABLE C-LXXI.	Reserved	138
TABLE C-LXXII.	Saybolt color fuel property rating value	139
TABLE C-LXXIII.	Reserved	139
TABLE C-LXXIV.	Smoke point fuel property rating value	141
TABLE C-LXXV.	Reserved	141
TABLE C-LXXVI.	Specific heat as a function of temperature fuel property rating value	143
TABLE C-LXXVII.	Reserved	143
TABLE C-LXXVIII.	Storage Stability fuel property rating value	144
TABLE C-LXXIX.	Reserved	144
TABLE C-LXXX.	Mercaptan sulfur fuel property rating value	146
TABLE C-LXXXI.	Reserved	146
TABLE C-LXXXII.	Total sulfur fuel property rating value	148
TABLE C-LXXXIII.	Reserved	148
TABLE C-LXXXIV.	Surface tension fuel property rating value at 25° C	151
TABLE C-LXXXV.	Reserved	151
TABLE C-LXXXVI.	Thermal conductivity fuel property rating value	153
TABLE C-LXXXVII.	Reserved	153
TABLE C-LXXXVIII.	Reserved	154
TABLE C-LXXXIX.	Reserved	154
TABLE C-XC.	Thermal stability fuel property rating value	157
TABLE C-XCI.	Thermal stability fuel property rating value	157
TABLE C-XCII.	Reserved	157
TABLE C-XCIII.	Trace metals property rating value	159
TABLE C-XCIV.	Trace organics property rating value	160
TABLE C-XCV.	Vapor pressure versus temperature fuel property rating value	166
TABLE C-XCVI.	Reserved	166

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TABLE C-XCVII.	**Viscosity fuel property rating value: aircraft**	**171**
TABLE C-XCVIII.	**Viscosity fuel property rating value: support equipment & vehicles**	**171**
TABLE C-XCIX.	**Reserved**	**171**
TABLE C-C.	**Water reaction interface rating fuel property rating value**	**172**
TABLE C-CI.	**Reserved**	**172**
TABLE C-CII.	**Water separation index fuel property rating value**	**173**
TABLE C-CIII.	**Note 9 of Table 2 of MIL-DTL-83133F: "The minimum microseparometer rating using a Micro-Separometer (MSEP) shall be as follows:"**	**174**
TABLE C-CIV.	**Water solubility fuel property rating value**	**177**
TABLE C-CV.	**Reserved**	**177**
TABLE D-I.	**Jet reference fuel**	**187**
TABLE D-II.	**Nonmetallic materials, tests, and test temperatures**	**191**
TABLE D-III.	**Metallic materials, tests, and test temperatures**	**195**
TABLE D-IV.	**Complete list of materials**	**196**
TABLE E-1	**Toxicity test overview index**	**210**
TABLE G-I.	**Propulsion system entrance criteria**	**251**
TABLE G-II.	**Propulsion System Subset 1 Properties**	**252**
TABLE G-III.	**Propulsion System Subset 2 Properties**	**253**
TABLE G-IV.	**Propulsion System Subset 3 Properties**	**253**
TABLE L-I.	**Fuel property/test method application**	**277**

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1. SCOPE

1.1 Scope.

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This military handbook documents a lean, knowledge based process to evaluate, approve, and certify fuels and fuel additives for use in all Air Force equipment. This document was developed to fill the knowledge and experience gaps that currently exist when considering all aspects of the Air Force enterprise related to fuels in a single integrated and cost effective manner instead of a system by system evaluation. This document defines an approach and process to assure that a kerosene type fuel is suitable for aviation, support equipment & vehicles, interchangeable in the logistics infrastructure, and meets Air Force standards related to the environment, safety and health. Any new fuel or fuel additive will be compared to JP-8 in terms of safety of operation, performance, durability, survivability, material compatibility, environmental impact, safety and health. If the new fuel or fuel additive is found to be suitable, it will be included in an update of the MIL-DTL-83133F procurement specification.

1.1.1 Secretary of the Air Force direction.

In 2006 the Secretary of the Air Force, the Honorable Michael Wynne, recognized the need to develop fuels from domestic alternative and unconventional resources as a way to reduce the US dependence on foreign oil. This dependence is a national security issue and could potentially impact military operations in the future. Promising information was evolving on Fischer-Tropsch fuels that could be produced from domestic resources such as coal, natural gas, biomass, petroleum coke and blends of biomass with fossil sources. Fischer Tropsch fuels produced from coal were being used by military and commercial aircraft in South Africa and demonstrated improved characteristics compared to petroleum derived fuels in studies being conducted by the US Air Force, Army and Navy. To accelerate the development and pave the way for use by the US Air Force, Mr. Wynne directed an US Air Force Material Command (AFMC) team to demonstrate the use of this fuel in a manned aircraft, the B-52, in 2006. In addition, he directed an AFMC team to develop a lean, knowledge based process for the approval and certification of this fuel as well as future generations of fuel and fuel additives.

1.1.2 The certification process.

The current process for certifying fuel requires each weapon system manager to independently determine if the fuel or fuel additive is fit for purpose, meets operational, performance, durability, safety and other weapon system considerations, and then document the suitability for use as a primary, alternative or emergency fuel. Lessons learned from the conversion from JP-4 to JP-8, implementation of the thermal stability improving “+100” additive and the use of Russian TS-1 provided the basis to develop a lean knowledge based process to replace the current process. In addition, the current process emphasis for fuels has primarily focused on aviation with minimal focus on other key elements of the Air Force enterprise such as logistics, support equipment & vehicles, environment, health and safety. A new process that includes all Air Force equipment would allow a more cost effective and streamlined approach to field a fuel or a fuel additive. The process that is documented in this handbook does not replace or supplant the role of the weapon system single manager. Within this handbook, the single manager will be used to reference the person (s) that has (have) ultimate responsibility for

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approving fuel or fuel additives for Air Force process(es), weapon system, piece(s) of equipment or logistics infrastructure for which they are responsible. The term single manager (SM) will include not only weapon system or equipment managers, but persons responsible for environmental, operational, safety, health or logistics processes related to fuels. Ultimately each SM will need to make the final determination of suitability for use; however this standardized process provided the key technical data needed for single manager evaluations. The intent is to continuously improve the fidelity of this handbook with lessons learned during the qualification / certification of fuels or fuel additives, using Fisher-Tropsch fuel blends as the initial validation.

1.1.3 Candidate fuel characteristics.

This handbook is for fuel candidates or fuel additives having the general characteristics of a kerosene fuel that meets safety of flight, performance, durability and operational characteristics comparable to the baseline fuel. The intent is to compare the new fuel or fuel additive to the baseline fuel and assess the potential to incorporate the new fuel into the MIL-DTL-83133F specification and to document the suitability for use in a weapon system as either a primary fuel, alternative fuel, or emergency fuel. To warrant consideration, the candidate fuel or fuel additive will meet the following requirements for consideration: 1) logistical availability in an expeditionary operation or 2) improve weapon system performance, durability, operational characteristics, or logistics or 3) be produced from an alternative or unconventional domestic resource. Candidate fuels or fuel additives will demonstrate technical maturity in the form of test reports, technical data, property lists, documentation on the resource and processing used to make the fuel or additive, and information on the environmental, safety and health aspects. In addition, information on the economic viability of the fuel or additive will be provided. This information should include cost estimates, volume of fuel or additive that can be produced, any requirements for special handling or logistics, environmental impacts, and general production information. If the entrance criteria to initiate an evaluation that will lead to the approval and certification of the fuel or fuel additive are met, then the handbook process will be used to provide the data set that includes requirements from all aspects of the Air Force enterprise, which will be used by the SMs for the approval/certification process. As lessons are learned during the execution of the process, they will be used to update this handbook.

1.2 Baseline fuels.

The early pioneers in gas turbine development, Whittle in England and Von Ohain in Germany, faced a wide variety of options in choosing a fuel for gas turbines. Whittle had considered diesel fuel, but ended up choosing illuminating kerosene because of an expected requirement for a lower freeze point than that available with diesel. In contrast, Von Ohain originally demonstrated his turbine engine with hydrogen, but vehicle considerations led to a switch to liquid fuel. The world's first turbojet-powered flight was made on 27 August 1939 in a Heinkel 178 aircraft burning aviation gasoline. However, most of the jet engines developed before the end of World War II utilized conventional kerosene as a fuel. The first jet fuel specification was the Directorate of Engine Research and Development 2482 (DERD 2482), published in England in 1947.

1.2.1 Fundamental fuel characteristics.

As engines and specifications developed, it became apparent that several fuel properties were fundamental to bounding the envelope of jet fuel characteristics. High-altitude operation

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meant fuel freeze point required attention. However, the lower the freeze point, the lower the fraction of crude oil that was suitable, therefore, freeze point had to be balanced against availability. Higher fuel volatility/vapor pressure aided vaporization-controlled engine performance requirements such as altitude relight, which had to be traded against boil-off and entrainment losses from fuel tanks at altitude (as well as safety concerns from explosive mixtures in tank vapor spaces.) In the United States, JP-1, JP-2, and JP-3 were (ultimately unsuccessful) attempts to balance the conflicting requirements of volatility, freeze point, and availability/cost. Two fuels emerged in the late 1940s and early 1950s from this chaotic situation: a wide-cut naphtha/ kerosene mixture called JP-4 in the United States (MIL-F-5624 in 1950) and a kerosene fuel with a -50°C (-58°F) freeze point (DERD-2494 in England and Jet A-1 in ASTM D-1655 in the United States). This freeze point was arrived at through a significant research effort. ASTM D-1655 also specified Jet A with a –40°C (-40°F) freeze point. The Jet A-1 freeze point was changed from -50°C to -47°C (-58°F to -53°F) in the late 1970s to increase the number of petroleum feedstocks that could be used to manufacture the fuel and thus increase availability and reduce cost. The differences between the major aviation gas turbine fuels are summarized in Table I. Civil aviation currently uses Jet A-1 (or its equivalent) throughout the world outside of the FSU (former Soviet Union), except for domestic carriers in the United States, who use Jet A to reduce cost. The freeze point value in the specification is often the most difficult (and costly) for the refiner to meet. Military aircraft used JP-4 until converting to JP-8 in the 1980s. JP-8 (currently MIL-DTL-83133F, see A.3.1) is essentially Jet A-1 with three military-specified additives. The conversion to JP-8 occurred primarily to improve the safety of aircraft, although the "single fuel for the battlefield" concept (and the similarity of jet fuel to diesel fuel) is centered on the use of aviation kerosene in all U.S. Air Force and U.S. Army aircraft, support equipment & vehicles. A similar process is occurring in the U.S. Navy, where the large variety of liquid fuels has shrunk down to just two, JP-5 (for aircraft) and F-76 diesel for all other liquid fuel requirements. The Navy also uses JP-8 in many Naval Air Stations on land. Logistical considerations may drive commercial jet fuel to be the single battlefield fuel in the future. Currently, the military consumes about 10% of the jet fuel produced in the United States, so the commercial market is dominant.

1.2.2 Jet fuel evolution.

The history of the evolution of conventional, widely available jet fuels from the late 1950s to the present is mainly the story of the evolution of test methods and fuel additives to maintain the integrity of the jet fuel supply and to improve safety and correct operational problems. Specialty fuels were developed for various applications throughout the second half of the 20th century. In the early 1950s, JP-5 (included in MIL-F-5624) was developed. JP-5 is a high-flash-point (60°C/140°F) aviation kerosene used onboard U.S. Navy ships to enhance safety. The development of higher Mach aircraft led to several specialty fuels. As flight velocity increases, aerodynamic heating leads to larger amounts of heat being rejected to the fuel, both in the tanks and in the engine, leading to vapor pressure and thermal stability concerns. The cutoff point between the use of conventional Jet A-1/JP-8 fuels and specially produced fuels is between Mach 2.2 and 3. Thus, the Mach 2.2 Concorde uses Jet A-1, whereas the Mach 3 XB-70 and the Mach 3 SR-71 used specialty fuels. JP-6 (MIL-F-25656) was a low-volatility kerosene developed for the Mach 2 XB-70. The Mach 3 SR-71 required JP-7 (MIL-T-38219), a low-volatility/ high thermal stability, highly processed (low sulfur and aromatics) kerosene. The U-2 high-altitude reconnaissance aircraft required both improved thermal stability and lower freeze

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point in its fuel (JP-TS, MIL-T-25524) because of its high-altitude, long-duration cruise. These specialty fuels gave higher performance than conventional aviation kerosenes, at the expense of higher fuel and logistical costs (JP-7 and JP-TS are roughly three times the cost of JP-8 and Jet A-1). The accepted operational temperature limits of these various fuels are approximately 163°C (325°F) for Jet A/Jet A-1/JP-8/JP-5, 219°C (425°F) for JP-TS, and 288°C (550°F) for JP-7. These temperatures are the maximum temperature achieved by the fuel, typically in the engine nozzles. The accepted limit for gas turbine operation is approximately Mach 4.

1.2.3 Russian fuel parallel evolution.

Russian jet fuels underwent a parallel evolution throughout this period. In most areas, current Russian fuels TS-1, RT, and Russian specifications (GOST 10227) are interchangeable with Jet A-1/JP-8, with the exception of the US additive package listed in the MIL-DTL-83133F. The main difference between fuels TS-1 and RT is in the area of thermal stability: TS-1 is a straight-run fuel, whereas RT is hydrotreated. By comparison with Jet A-1/JP-8, TS-1 and RT are lighter (have a lower initial boiling point and 10% recovery point in distillation) and have a correspondingly lower flash point and freeze point. Thus, worldwide there are three major specifications in civil use: ASTM D1655, British Defence Standard (Def Stan) 91-91, (successor to DERD 2494), (See A.3.3) and Russian GOST 10227-86. International oil companies have created the "Joint Check List" to standardize jet fuel deliveries worldwide under Jet A-1/Def Stan 91-91. The International Air Transport Association has also issued guidance material for its members codifying the Jet A/Jet A-1/TS-1 specifications. Two specialty Russian fuels are specified in GOST 12308: T-8V, a higher density/higher flash-point kerosene and T-6, a high-density kerosene (specific gravity 0.84 versus 0.8 for Jet A-1/JP-8), which has no commercial or military counterpart in Europe or the United States. U.S. Air Force programs in the 1980s demonstrated the production of fuels similar to T-6, but no specification was published in the absence of user requirements. Historically, flight characteristics have driven changes in aviation fuels. Now logistics are also driving the introduction of alternatives. This establishes the need to document fuel property requirements for the future developer to understand why all properties are important.

TABLE I. Summary of major jet fuel characteristics (additives explained in 1.4).

	JP-8	Jet A	Jet A-1	TS-1
Flash point (min), °C (°F)	38 (100)	38 (100)	38 (100)	28 (82)
Freeze Point (max), °C (°F)	-47 (-53)	-40 (-40)	-47 (-53)	n/a*
Additives	CI/LI, FSII, SDA	None	SDA	n/a**

* Russian freeze point test method differs from other fuels, but has been found to meet the -47 °C (-53°F) spec in all cases.

** Russian additives are chemically different than their Western counterparts, although they fill the same roles (icing inhibitor, static dissipater, etc.)

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1.3 Alternative fuels.

The term "alternative" fuels, as defined in this handbook, is used to differentiate between kerosene-type jet fuels produced from crude oil and synthetic fuels produced from non-crude oil sources such as coal, natural gas, biomass, etc. The meaning of the phrase “alternative fuel” can vary depending on who is using it. In some scientific circles, alternative fuels refer to the use of hydrogen or ethanol as a fuel. The alternative fuels investigated in this handbook are intended to be used as the baseline fuels as used in military equipment. Therefore, an alternative fuel should emulate the baseline fuel’s properties to increase its fungibility within military assets.

1.3.1 Fuel source specification.

As of 2008, the MIL-DTL-83133F specification limits the source of the fuel used in Section 3.1: "The fuel supplied under this specification shall be refined hydrocarbon distillate fuel oils containing additives in accordance with 3.3. The feed stock from which the fuel is refined shall be crude oils derived from petroleum, tar sands, oil shale, or mixtures thereof." Def Stan 91-91 has similar language for Jet A-1: "The fuel shall consist wholly of hydrocarbon compounds derived from conventional sources including crude oil, natural gas liquid condensates, heavy oil, oil shale and oil sands, and qualified additives as listed in Annex A." Thus current jet fuel (as of 2008) can (and does) contain molecules from conventional crude oil (petroleum) as well as components derived from Canadian tar sands. The properties of jet fuels derived from petroleum, oil/tar sands, and shale are somewhat different, but are similar enough to be lumped together in the specification. In the U.S., oil shale is not currently (2008) a source of jet fuel components, but that could change.

1.3.2 Potential coal and natural gas fuel sources.

Presently, coal and natural gas (CH_4) are two fuel sources not included by the MIL-DTL-83133F specification which have potential to enter the jet fuel market. Techniques such as "indirect" liquefaction, the Fischer-Tropsch process, or "direct" liquefaction can yield hydrocarbon fuel components similar to current petroleum based jet fuels. The Fischer-Tropsch (F-T) process converts synthesis gas, a mixture of carbon monoxide and hydrogen, into a "wax" - a collection of long chain paraffinic hydrocarbon molecules. With some further changes by such processes as hydrocracking and isomerization, hydrocarbons suitable for use in jet fuel are produced. Coal and natural gas are converted into the F-T feedstock synthesis gas through "gasification". The Fischer-Tropsch process was used by the Germans during World War II to produce liquid fuels from coal, and has been used since the 1970’s in South Africa to produce liquid fuels also from coal. In several countries which have a surplus of natural gas, F-T plants are producing liquid fuels and chemicals from the available natural gas. Fischer-Tropsch fuels are composed almost entirely of n- and iso-paraffins, and are absent the cyclo-paraffins and aromatics found in petroleum-derived fuels. Fuels produced by direct liquefaction of coal preserve the ring structure of coal, but require extensive hydrotreatment to meet jet fuel specifications. Thus, a jet fuel derived from direct liquefaction is composed primarily of cyclo-paraffins. Table II compares the chemical composition among average petroleum-based, F-T, and coal liquefaction jet fuels.

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TABLE II. Fuel compositional analysis via ASTM D2425.

	World survey average jet fuel, vol %	F-T Jet A-1	Jet fuel from coal liquefaction
n- + iso-paraffins	58.78	99.7	0.6
monocycloparaffins	10.89	<0.2	46.4
dicycloparaffins	9.25	<0.2	47.0
tricycloparaffins	1.08	<0.2	4.6
alkyl benzenes	13.36	<0.2	0.3
indans+tetralins	4.9	<0.2	1.1
naphthalene	0.13	<0.2	<0.2
substituted naphthalenes	1.55	<0.2	<0.2

1.3.3 Applicable alternative fuels.

The alternative fuels applicable to this handbook may not be limited to sources derived from a fossil fuel. New research is investigating biofuels created from items such as biomass, animal fat, vegetable oil, switch grass, algae, etc. as potential liquid fuel sources. Biofuels may be gasified into carbon monoxide and hydrogen. These gas streams would then follow a Fischer-Tropsch process leading to a jet fuel. A different approach starts with bio-oils which are hydrotreated thereby creating long, unbranched hydrocarbons. These hydrocarbon chains are fed into an isomerization reactor which reforms long, unbranched hydrocarbons into the various combinations of branched, unbranched, and ringed hydrocarbons comprising jet fuels. These two processes for biofuels represent a small fraction of the ideas currently under investigation. As with coal liquefaction and F-T processing of natural gas, alternative fuels from biological sources are different than petroleum derived jet fuels. These differences come in the form of chemical composition and contamination which effect the fuels physical properties and thereby its performance in engines. Due to the environmental benefits and domestic location of biofuels, the military cannot ignore their potential as an alternative fuel.

1.4 Fuel Additives.

1.4.1 JP-8 Required Additives.

1.4.1.1 Fuel System Icing Inhibitor (FSII).

Ethylene Glycol Monomethyl Ether (EGME) was the original FSII, introduced to inhibit free water from freezing after a 1958 aircraft incident. The first FSII (MIL-I-27686) formulation consisted of 87.3% EGME and 12.7% glycerol. The initial use of glycerol was to protect the sealants and coatings used in fuel tanks from attack by EGME. Field experience with MIL-I-27686 indicated it was causing problems due to glycerol not completely dissolving in the fuel. The specification was changed several times reducing the concentration of glycerol before finally removing it completely. EGME was later replaced with Diethylene Glycol Monomethyl Ether (DiEGME) due to carcinogenic concerns with EGME and is currently (2008) the only FSII (MIL-I-85470) listed in the MIL-DTL-83133F specification. Additional work is underway to

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determine the “lower effective concentration” of DiEGME and, in a separate effort, to identify a FSII less aggressive toward fuel tank sealants and coatings. Interestingly, due to DiEGME being widely accepted as a biostat, any changes to the concentration or type of icing inhibitor will need to address the effects of same on micro-organisms.

1.4.1.2 Static Dissipater Additives (SDA).

Static electrical charge can be generated when unlike surfaces move across one another, e.g., a glass rod and a wool cloth. Similarly, when jet fuel (a relatively poor conductor) moves through a hose, valve, or filter, a static charge can be created. If the charge does not dissipate more rapidly than it is produced, a discharge in the form of a spark could be experienced. The energy of this spark may induce an explosion if the fuel vapor and air mixture is within flammability limits. The rate of charge dissipation is proportional to the electrical conductivity of the fuel. Military jet fuels require the use of a conductivity improver additive - Static Dissipator Additive (SDA). The SDA does not prevent charge generation but, enhances the rate of charge dissipation by increasing fuel conductivity. Currently, the only static dissipator available for use in aviation fuel is Octel’s Stadis 450 additive.

1.4.1.3 Corrosion Inhibitor/Lubricity Improver (CI/LI).

CI/LI additives are primarily composed of fatty acids and/or their derivatives. They were originally introduced to provide pipelines with corrosion protection for direct shipment of fuel to government facilities. In 1954 the USAF issued the Inhibitor, Corrosion, Fuel Soluble specification (MIL-I-25017) and has maintained a Qualified Product List of approved additive manufacturers/formulations. The original MIL-I-25017 requirements were that “the inhibitor (a) must protect ferrous metals when added to fuels in low concentrations, (b) must not be extracted from the fuel by water that might be present with the fuel, (c) must be soluble in all Avgas and jet fuels of interest, (d) must be low in ash, (e) must be compatible with all other approved fuel additives, (f) must not adversely affect the fuel performance in the aircraft.” The first US specification mandating use of MIL-I-25017 was MIL-F-5624 Revision C in May 1955. In 1965 the mandatory use of MIL-I-25017 corrosion inhibitor (CI) was deleted but, was soon reinstated after occurrences of aircraft engine fuel control malfunctions were experienced and attributed to the CI removal. It was determined that MIL-I-25017 additives increased the lubricating properties of the fuel itself. When MIL-I-25017 was later revised, the title changed to Inhibitor, Corrosion/Lubricity Improver (CI/LI), Fuel Soluble to reflect that the material was both a corrosion inhibitor and a lubricity improver.

1.4.1.4 Metal Deactivator Additive (MDA).

Even in trace quantities, active metals such as copper or zinc may degrade thermal oxidative stability of aviation fuel by catalytic action. The function of metal deactivators is to form stable complexes with such metals so as to inhibit their ability to promote the formation of deposits (fuel oxidation). Currently, the only metal deactivator additive approved for use in MIL-DTL-83133F is N, N’-disalicylidene-1, 2-propane diamine. The allowed concentration of active material used on initial batching of fuel at the refinery is limited to 2.0 mg/L. Cumulative addition of metal deactivator is capped at 5.7 mg/L if fuel is re-treated and should only be permitted if the fuel is determined to contain active metals. MDA cannot be used in military jet fuel unless the supplier has obtained consent from the procuring activity and end user.

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1.4.1.5 Antioxidants.

Reactive compounds in aviation turbine fuel respond to oxygen from the small amount of dissolved air in fuel setting off a chain of oxidation reactions that result in the formation of peroxides, soluble gums, and insoluble particulates. Antioxidants interrupt this chain of reactions when present before the fuel is exposed to the atmosphere. When fuel is hydrogen treated, the natural antioxidants found in straight run fuels are removed resulting in a less stable fuel. For this reason, an approved antioxidant will be added (during rundown into feed/batch tankage) to all USAF jet fuel containing blending stocks that have been hydrogen treated.

1.4.1.6 Thermal Oxidative Stability Improver (+100) Additive.

In the late 1980's the Air Force Research Laboratory (AFRL) initiated an effort to improve the thermal stability of USAF used jet fuel through the use of additives. After several years of evaluating hundreds of additives, one single additive was selected. The additive selected is currently manufactured by GE Betz (formerly BetzDearborn) and is designated, for DoD applications, as Spec-Aid® 8Q462. It is also available commercially in the U.S. as Turboline FS100 and overseas as Aeroshell Performance Additive 101 (APA101). This additive improved thermal stability of jet fuel by up to 100°F (38°C) – increasing fuel thermal heat sink by 50% when used at a dosage rate of 256 mg/L. Currently (2008), JP-8 is the only USAF jet fuel to utilize this additive and JP-8 fuel additized with this additive is designated JP-8+100.

1.4.1.6.1 Extensive testing performed.

Since its introduction to the field in 1994, hundreds of thousands of flight hours have been logged using this additive in JP-8 with no flight safety or performance issues. Extensive testing has been done on the additive and it has been found to be fully compatible with materials used in aircraft systems—both airframe and engine—and it has been shown to be of no impact to flight operations—including altitude relights and fuel gauging systems. Toxicologically, the additive is no more hazardous than the baseline fuel.

1.4.1.6.2 Equivalent or reduced maintenance.

Many studies have been conducted to determine if the additive has any impact on aircraft engine maintenance. In most cases, the impact has been that maintenance due to coking and fouling where the additive is in use has been reduced. The amount of the reduction is dependent upon the engine, local maintenance procedures and policies and the flight mission. There have also been documented instances when use of the additive was terminated for one reason or another and after additive use was discontinued, engine maintenance due to coking and fouling increased. When use of the additive was resumed, maintenance requirements were reduced. There have been no instances where the additive had a negative impact on flight operability or maintenance. When initially fielded, there was concern that the additive's detergent/dispersant elements might cause filter/coalescer problems but recent tests conducted at Southwest Research Institute (SwRI®) have shown that Spec-Aid® 8Q462 is no worse than the baseline fuel for filter coalescers meeting API/IP 1581 5th Edition specifications. Current AF policy regarding handling of JP-8+100 is that it can be returned to bulk fuel storage in a 1:1 ratio.

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2. APPLICABLE DOCUMENTS.

2.1 General.

The documents listed below are not necessarily all of the documents referenced herein, but are those needed to understand the information provided by this handbook.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks.

The following specifications, standards, and handbooks form a part of this document to the extent specified herein.

DEPARTMENT OF DEFENSE SPECIFICATIONS

JSSG-2000	Joint Service Specification Guide, Air System
JSSG-2001	Joint Service Specification Guide, Air Vehicle
JSSG-2006	Joint Service Specification Guide, Aircraft Structures
JSSG-2007	Joint Service Specification Guide, Engines, Aircraft, Turbine
JSSG-2009	Joint Service Specification Guide, Air Vehicle Subsystems
MIL-DTL-83133	Turbine Fuel, Aviation, Kerosene Type, JP-8 (NATO F-34), NATO F-35, and JP-8+100 (NATO F-37)

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-882	System Safety
MIL-STD-3004	Quality Assurance/Surveillance for Fuels, Lubricants and Related Products

DEPARTMENT OF DEFENSE HANDBOOKS

MIL-HDBK-514	Operational Safety, Suitability, and Effectiveness for the Aeronautical Enterprise
MIL-HDBK-515	Weapon Systems Integrity Guide (WSIG)
MIL-HDBK-516	Airworthiness Certification Criteria

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

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2.2.2 Other Government documents, drawings, and publications.

The following other Government documents, drawings, and publications form a part of this document to the extent specified herein.

UNITED STATES AIR FORCE POLICY DIRECTIVES

AFPD 62-6	USAF Aircraft Airworthiness Criteria
AFPD 63-12	Assurance of Operational Safety, Suitability & Effectiveness

(Copies of these directives are available at http://www.e-publishing.af.mil.)

UNITED STATES AIR FORCE TECHNICAL ORDERS (TOs)

TO 42B1-1-14	Fuels for USAF Aircraft

(Copies of this Technical Order may be obtained by calling (937) 255-8050, DSN 785-8050. It is also available through The Defense Technical Information Center Website and from DTIC Headquarters, 8725 John J. Kingman Rd., Ft. Belvoir VA 22060-6218; telephone (800) 225-3842. Approved users may access Technical Orders at https://www.toindex-s.wpafb.af.mil/ and qualified recipients may obtain copies of this Technical Order at https://afkm.wpafb.af.mil.)

2.3 Non-Government publications.

The following document forms a part of this document to the extent specified herein.

UNITED SOVIET SOCIALIST REPUBLIC STATE STANDARD

GOST 10227-86	Jet Fuels

(Copies of this document may be obtained at http://store.ihs.com/specsstore.)

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3. DEFINITIONS.

A

Acceptable Products: An acceptable product is one which can be used in place of another product for extended periods without technical advice.

Additives**:** Compounds used to impart new properties to a product or to improve a property which it already possesses - for example, mixed tertiary butylphenols when added to a fuel to improve its resistance to oxidation.

Airworthiness: The property of a particular air system configuration to safely attain, sustain, and terminate flight in accordance with the approved usage and limits.

Alternative Fuels: An alternative fuel is any fuel determined to be substantially not petroleum, (e.g., non crude oil sources for liquid hydrocarbons), that yields energy security benefits and environmental benefits. The term "alternative" fuels, as defined in this handbook, is used to differentiate between kerosene-type jet fuels produced from crude oil and similar fuels produced from alternative sources such as coal, natural gas or biomass.

Alternative Fuels Certification Office (AFCO): The government team responsible for the implementation and coordination of the fuel certification process

Alternate Fuel: An alternate fuel is one on which the air vehicle can be flown without operational restrictions but which can have long-term durability or maintainability impact if used for continuous operation (multiple flights). Alternate fuels are used only on an occasional or intermittent basis. Use of an alternate fuel should cause no adverse effect on the air vehicle mission(s).

Approved Fuel: A fuel(s) approved for use in an aircraft with restrictions or limitations defined, if any.

B

Baseline Fuel: Baseline fuel is defined as a kerosene type turbine fuel that established this handbook's pass/fail criteria to which all candidate fuels will be compared. JP-8 in accordance with MIL-DTL-83133F was chosen as the baseline fuel for the handbook as of April 2007.

Blending: Blending refers to the procedures by which predetermined quantities of two or more similar products are homogenously mixed to upgrade one of the products or to produce an intermediate grade or quality. The term is also used to define the injection of additives, such as corrosion or icing inhibitors, into fuels.

C

Certified Fuel: Fuel(s) first approved (to a standard) for use in an original system certification for flight.

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Clean (Clear) and Bright: Clean is the absence of visible solids, a cloud, a haze, an emulsion, or free water in the product. Bright is the sparkle of clean, dry product in transmitted light.

Commingling: Commingling is the mixing of two or more products of different ownership or grade.

D

Decomposition of Requirements: For this handbook, requirements are broken down into parts from requirements documents until the relevant fuel properties/characteristics are identified.

Derived Requirements: Derived requirements trace back to a driving requirement. For this handbook, the derived requirements are the relevant fuel properties/characteristics, and/or interfaces with other systems and other elements.

E

Emergency Substitute: A product which may be used in an emergency only, in place of another product, after local exploration of adequacy of the product, using locally available technical manuals and personnel. However, technical advice will be sought from the nation's service, as soon as possible after the event as to whether or how long; the emergency substitute could safely be retained in use.

Emergency Fuel: A fuel which may cause significant damage to the engine or other systems; therefore, its use is to be limited to one flight. The applicable aircraft flight manual or system manager should be consulted regarding operating restrictions and post flight maintenance actions necessary when using an emergency fuel. Examples of conditions that might warrant use of emergency fuels

- Accomplishing an Important Military Mission
- Countering Enemy Actions
- Emergency Evacuation Flights
- Emergency Aerial Refueling

Entrance Criteria: Key information required to make a determination whether to initiate the fuel certification process. All fuel candidates will meet the general characteristics of a kerosene fuel that meets safety, performance, durability and operational characteristics comparable to the baseline fuel.

F

Finish System: Any organic coating (eg. Corrosion inhibiting) or other surface preparation (eg. Acid anodizing) that could exist on equipment.

Fit for Purpose: A classification of property types which refers to properties inherent of a fuel that are not controlled by specification.

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Fuel Holdup: A layer of frozen fuel attached to the inner wall of a fuel tank

Fungibility: Fully interchangeable

G

Gap Analysis: Refers to the study and comparison performed to identify those missing pieces of information between the fuels information outlined in the Mil Handbook and documented for the candidate fuel/fuel additive and the systems level analysis of the weapon system, piece of equipment, environmental, safety, occupational health or logistics process required to qualify or certify the fuel for use.

H

High Flash Point Kerosene Type: High flash point kerosene fuel has essentially the same characteristics as kerosene type fuels, but with a minimum flash point of 60°C (140°F). This higher flash point fuel is required by the Navy for fire safety purposes aboard aircraft carriers.

I

J

K

Kerosene Fuels: Petroleum distillates with an approximate boiling range of 165° − 290°C (330° − 550°F).

Kerosene Type: Hydrocarbon liquid that has similar chemical and physical properties / characteristics as kerosene fuel.

L

M

Micron: One micron (micrometer, 10^{-6} meter) is a thousandth part of one millimeter.

N

O

Operational Safety: The condition of having acceptable risk to life, health, property, and environment caused by a system or end-item when employing that system or end-item in an operational environment.

Operational Suitability: The degree to which a system or end-item can be placed satisfactorily in field use, with consideration given to availability, compatibility, transportability, interoperability, reliability, wartime use rates, maintainability, full-dimension protection, operational safety, human factors, architectural and infrastructure

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compliance, manpower supportability, logistics supportability, natural environmental effects and impacts, and documentation and training requirements.

P

Primary Fuel: The fuel(s) on which the air vehicle is designed to operate continuously without restrictions and is (are) also used to demonstrate contract compliance for complete steady state and transient operating conditions.

Q

Qualified Fuel: Fuel(s) that is (are) certified for use in a system with no restrictions.

R

Requirements Decomposition: For this handbook, requirements are broken down into parts from source requirements documents, (JSSG, etc.) until the relevant fuel properties/characteristics are identified.

S

Single Manager: The single individual specifically designated, under the integrated weapon system management architecture, to be responsible for the life cycle management of a system or end-item. The Single Manager is the program manager vested with full authority, responsibility, and resources to execute and support an approved Air Force program. The term, “Single Manager” will include not only weapon system or equipment managers, but persons responsible for environmental, operational, safety, health or logistics processes related to fuels.

Subset 1 Testing: Fuel properties/characteristics critical to personnel safety, system safety and/or system performance.

Subset 2 Testing: Fuel properties/characteristics critical to system performance and/or durability. This second subset also contains component level tests that do not directly correlate to a fuel property.

Subset 3 Testing: Fuel properties/characteristics critical to the system durability, and supportability requirements.

Suitable for Use: May be a certified, qualified, approved, or other fuel which may or may not have flight restrictions.

Standardized Product: A product is deemed to be standardized when it conforms to specifications which either have the same technical requirements, or which, in the opinion of the responsible working party, have equivalent technical requirements.

Synthetic Fuel: Any fuel produced from coal, natural gas or biomass.

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System: A specific grouping of subsystems, components, or elements designed and integrated to perform a military function.

T

Technology Readiness Level (TRL) 6: System/subsystem model or prototype demonstration in a relevant environment (ground or space).

U

Unified Product: A standardized product which is used by all Allied nations for (a) given use(s).

V

Value Stream: All the steps (both value added and non-value added) in a process that the customer is willing to pay for in order to bring a product or service through the main flows essential to producing that product or service.

W

Wide-Cut Type: Mixtures of gasoline and kerosene distillate fractions with an approximate boiling range of 35° − 315°C (95° − 600°F).

X

Y

Z

3.1 Alphabetical listing of acronyms and abbreviations.

ACGIH	American Conference of Governmental Industrial Hygienists
AFCO	Alternative Fuels Certification Office
AFGS	Air Force Guide Specification
AFIOH	Air Force Institute for Occupational Health
AFMC	Air Force Material Command
AFOSR	Air Force Office of Scientific Research
AFPD	Air Force Prime Directive
AFPET / AFPA	Air Force Petroleum Agency
AFPFMP	Alternate Fuel Production Facility Measurement Process
AFRL	Air Force Research Laboratory
AFRL/MLRX	Air Force Research Laboratory, Materials and Manufacturing Directorate

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AMS	Aerospace Materials Specifications
AMT	Accelerated Mission Test
APA101	Aeroshell Performance Additive 101
API	American Petroleum Institute
ARP	Aerospace Recommended Practice
ASI	Advertising Specialty Institute
ASTM	American Society for Testing and Materials
ATSDR	Agency for Toxic Substances and Disease Registry
BEE	Bioenvironmental Engineering
BMS	Boeing Materials Specifications
CDC-ATSDR	Centers for Disease Control and Prevention, Agency for Toxic Substances and Disease Registry
CHPPM	Center for Health Promotion and Preventive Medicine
CI/LI	Corrosion Inhibitor/Lubricity Improver
CRC	Coordinated Research Council
cSt	Centistokes
DESC	Defense Energy Support Center
DiEGME	Diethylene Glycol Monomethyl Ether
DoD	Department of Defense
E^3	Electromagnetic Environmental Effects
EGME	Ethylene Glycol Monomethyl Ether
EPA	Environmental Protection Agency
ESOH	Environmental Safety and Occupational Health
FSE	Field Service Evaluation
FSII	Fuel System Icing Inhibitor
F-T	Fischer-Tropsch
Gal	Gallon
HDBK	Handbook
HEL	Higher Explosive Limit
HHA	Health Hazard Assessment
HRI	Hazard Risk Index
HSMCT	Hot Section Materials Compatibility Test
IH	Industrial Hygiene

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IPT	Integrated product Team
JRF	Jet Reference Fuel
JFTOT	Jet Fuel Thermal Oxidative Test
JSSG	Joint Service Specification Guide
LEL	Lower Explosive Limit
MDA	Metal Deactivator Additive
MIE	Minimum Ignition Energy
MIL	Military
MLSA	Medical Laboratory Scientists Association
MSDS	Material Safety Data Sheet
NAC-AEGL	National Advisory Committee for Acute Exposure Guidelines for Hazardous Substances
NATO	North Atlantic Treaty Organization
NEP	Neutral Endopeptidase
NHRC	Naval Health Research Center
NIC	Notice of Intended Changes
NRC	National Research Council
OEM	Original Equipment Manufacturer
OSS&E	Operational Safety, Suitability, & Effectiveness
OPPTS	Office of Prevention, Pesticides and Toxic Substances
PDF	Portable Document Format (Adobe Acrobat)
PPE	Personal Protective Equipment
ppm	Parts per million
PRTG	Propulsion Directorate, Fuels Branch
PQIS	Petroleum Quality Information System
RAP	Risk Assessment Process
REL	Recommended Exposure Level
SAE	Society of Automotive Engineers
SDA	Static Dissipater Additives
SE&V	Support Equipment & Vehicles
SM	Single Manager
SME	Subject Matter Expert
SOF	Safety of Flight

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STEL	Short-Term Exposure Limit
SVP	Shared Verification Products
TLV	Threshold Limit Value
TPHCWG	Total Petroleum Hydrocarbon Criteria Working Group
TRL	Technology Readiness Level
TWA	Time-Weighted Average
UEL	Upper Explosive Limit
UFC	Unified Facilities
UFGS	Unified Facility Guide Specification
USAF	United State Air Force

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4. GENERAL GUIDANCE.

4.1 Aerospace fuel certification process goals.

The aerospace fuel certification process applies the tenets of systems engineering which involves decomposing requirements to the lowest level and developing the verification activity required to demonstrate that the requirements are achieved. The main goal of the aerospace fuel certification process is to ensure the desired level of safety, performance, durability, supportability, interoperability, etc. with the least possible economic burden to Air Force systems, equipment and infrastructure. Additionally, the process is to certify the fuel in the most cost-effective manner by minimizing duplication of effort and maximizing sharing of data between all systems.

4.2 Primary tasks.

The primary tasks involved in certifying an aerospace fuel for use in Air Force systems involves executing the certification process, evaluating the risk using the risk assessment process and performing field service evaluations as appropriate.

4.3 Expeditionary fuels.

The Alternative Fuel Certification Office (AFCO) will evaluate any kerosene type fuel that is available in a location of expeditionary military activities, which has been used for aviation but has not been certified by the single managers or their equivalents as suitable for use in US Air Force equipment. In general, the Defense Energy Support Center (DESC), combatant commander, and/or NATO commander will identify the fuel and request an evaluation of the expeditionary fuel to ease logistics burden, reduce cost, and / or more effectively conduct operations. Since the fuel has previously been used in aviation, fuel use information related to the aircraft type, engine system, and any other equipment will be required along with the fuel properties and characteristics defined later in this section.

4.4 Improved fuels and additives.

The AFCO will evaluate advanced fuels and fuel additives that have been developed in military research and development programs or by the commercial sector that have been demonstrated to a Technology Readiness Level (TRL) of 6 or higher. These new fuels / additives will demonstrate an equal or improved performance, operability, durability, maintenance, cost or reduced logistics footprint when compared to JP-8 and all relevant test data, test reports, and related documentation to determine technical maturity and economic viability will be provided the AFCO.

4.5 Fuels from alternative or unconventional resources.

The AFCO will evaluate fuels produced from alternative and unconventional resources that are kerosene type fuels that can be used for aviation, support equipment & vehicles and can be interchanged freely with petroleum derived fuels. In general, these fuels will be evaluated by military research and development organizations or the commercial sector and be at a TRL of 6 or higher and demonstrated to be equal to or better than the baseline fuel. The AFCO will be provided with information on the resource used to produce the fuel, general information on the processes used to manufacture the fuel, all relevant test data, test reports and related documentation to determine technical maturity and economic viability. In addition, information related to the environmental, safety and health aspects of the fuel are required.

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4.6 Segregation of candidate fuels and additives.

During the certification process two important issues will be considered regarding the segregation of a candidate fuel and/or additive from the normally supplied grades of fuel used for operations. The first issue applies to the fuel or additive used for the certification process and associated testing. The second issue applies after the decision to implement use of the fuel or additive.

4.6.1 Certification fuel.

During the certification process, the certification product (whether a fuel, a fuel blend, or a fuel-additive mixture) used for testing will be kept separate from fuel in the operational fuel supply. Use and special cleaning procedures of separate tanks and fuel transfer systems will be implemented. The purpose is to avoid contamination of the normal fuel supply and to avoid compromising the certification product by uncontrolled blending with other fuel stocks. When the candidate fuel is a blend, blending should be done using defined procedures provided by the applicable Service Control Point (e.g., AFPET for USAF) to ensure the proper mix. Under no circumstances should the certification product be mixed with the standard fuel supply beyond limits specified by the applicable Service Control Point. At completion of the certification, blend back ratios will be provided in writing by the applicable Service Control Point. Blend-back will not begin until these procedures are received by the test location.

4.6.2 Fleet implementation.

The simplest fleet implementation is when a fuel can be mixed with, or fully replace, existing fuel stocks. This involves only limited transition planning, with the transition being transparent to the user. When a fuel or additive cannot be mixed with existing stocks, the sustainment and infrastructure impacts will be fully identified. A plan for transition will be included as a part of the decision to implement use of the new fuel or additive. Similarly when the new fuel or additive drives more restrictive operating limitations or performance impacts or can be implemented to only a limited fleet, it will be necessary to keep it permanently segregated and specifically identified by its own grade designation. The sustainment and infrastructure implications of such segregation may be significant and will be given proper weight in the implementation decision. Again, a transition plan coordinated through the applicable Service Control Point will be part of that implementation decision process.

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5. DETAILED REQUIREMENTS.

5.1 Aerospace fuel certification process.

The aerospace fuel certification process is applicable to any candidate fuel for utilization in aerospace applications. The certification process herein is consistent with Air Force Policy Directive (AFPD) 63-12 which states that the SM has OSS&E responsibility and AFPD 62-6 which states that the SM also has Airworthiness Certification responsibility. The role of the AFCO in this certification process is to minimize duplication of effort to the maximum extent possible by working with each SM to identify opportunities for sharing verification test and analysis results herein referred to as shared verification products (SVP). The fuel certification process is depicted on Figure 1 and each step of the process is described below.

5.1.1 Entrance criteria.

The AFCO will ensure that all entrance criteria testing requirements identified in Appendix B, D, E, F, G, and H is accomplished prior to execution of the certification process. The AFCO will evaluate the entrance criteria test results for the candidate fuel and compare the data to the baseline results described in Appendix C, D, E, F, G and H. Based on this comparison and considering performance, cost and risk, the AFCO will determine whether or not to proceed with the certification process. If the AFCO determines the candidate fuel cannot be certified for use, the AFCO will document this conclusion with the supporting rationale in Appendix J.

5.1.1.1 Fuel property/characteristic testing.

The entrance criteria fuel property/ characteristic tests described in Appendix B will be conducted and the results compared to the baseline results in Appendix B.

5.1.1.2 Material compatibility testing.

The entrance criteria material compatibility tests described in Appendix D will be conducted and the results compared to the baseline results contained in Appendix D.

5.1.1.3 Toxicity.

The entrance criteria toxicity tests described in Appendix E will be conducted and the results compared to the baseline results contained in Appendix E.

5.1.1.4 Fire protection and survivability/vulnerability.

The entrance criteria fire protection tests described in Appendix F will be conducted and the results compared to the baseline results contained in Appendix F.

5.1.1.5 Aircraft propulsion.

The entrance criteria aircraft propulsion tests described in Appendix G will be conducted and the results compared to the baseline results contained in Appendix G.

5.1.1.6 Aerospace fuels infrastructure.

The entrance criteria logistics tests described in Appendix H will be conducted and the results compared to the baseline results contained in Appendix H.

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5.1.2 Conduct Subset 1 testing.

The AFCO will ensure that the Subset 1 testing requirements identified in Appendix B, D, E, F, G, and H are conducted to evaluate the primary set of fuel properties/characteristics, to ensure compatibility with the materials commonly used in Air Force equipment that comes into contact with fuel and fuel vapors, and to ensure personnel safety. This initial subset of tests is critical to personnel safety, system safety, and/or system performance. For each test result that did not meet the baseline requirements, the AFCO will initiate the risk assessment process described in 5.3 to determine the impact to Air Force systems and personnel, additionally involving the affected SM as appropriate. The AFCO will evaluate the results of Subset 1 testing, the entrance criteria testing, and the risk assessments and determine if the candidate fuel:

(a) Cannot be certified for use.

(b) Is the same as the baseline fuel.

(c) Can be certified for utilization without further testing.

(d) Requires additional testing.

If the AFCO determines the candidate fuel cannot be certified for use, the fuel will be rejected, the process will be terminated, and the AFCO will document this conclusion with the supplemental test data in Appendix J. If the AFCO determines that the candidate fuel is the same as the baseline fuel, the AFCO will recommend that the fuel be accepted and that the JP-8 specification be updated accordingly. This determination will then be documented in Appendix J and the process will be terminated. If the AFCO determines the candidate fuel can be certified without additional testing, this recommendation will be provided to each SM with supporting rationale and the process will proceed as described in 5.2. If the AFCO determines additional testing is required on the fuel candidate, the process will be continued as described in 5.1.3.

5.1.2.1 Fuel property/characteristic testing.

The Subset 1 fuel property/ characteristic tests described in Appendix B will be conducted and the results compared to the baseline results in Appendix B.

5.1.2.2 Material compatibility testing.

The Subset 1 material compatibility tests described in Appendix D will be conducted and the results compared to the baseline results also contained in Appendix D.

5.1.2.3 Toxicity.

The Subset 1 toxicity tests described in Appendix E will be conducted and the results compared to the baseline results also contained in Appendix E.

5.1.2.4 Fire protection and survivability/vulnerability.

The Subset 1 fire protection tests described in Appendix F will be conducted and the results compared to the baseline results contained in Appendix F.

5.1.2.5 Aircraft propulsion.

The Subset 1 aircraft propulsion tests described in Appendix G will be conducted and the results compared to the baseline results also contained in Appendix G.

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5.1.2.6 Aerospace fuels infrastructure.

The Subset 1 logistics tests described in Appendix H will be conducted and the results compared to the baseline results contained in Appendix H.

5.1.2.7 Conduct Subset 2 testing.

The AFCO will ensure that the Subset 2 testing requirements identified in Appendix B, D, F, G, and H are conducted (if required) to evaluate the secondary set of fuel properties/ characteristics, to ensure compatibility with the materials commonly used in Air Force equipment that comes into contact with fuel and fuel vapors and to ensure personnel safety. Additionally, the component/system level tests described in Appendix B not directly correlated to a fuel property/characteristic will be performed and evaluated. This second subset of tests is critical to system performance and/or durability and a failure to meet these requirements will be thoroughly evaluated. For each test result that does not meet the baseline requirements, the AFCO will initiate the risk assessment process described in 5.3 to determine the impact to Air Force systems and personnel, additionally involving the affected SM as appropriate. The AFCO will evaluate the results of Subset 2 testing, all previous testing, and the risk assessments to determine if the candidate fuel:

(a) Cannot be certified for use.

(b) Is the same as the baseline fuel.

(c) Can be certified for utilization without further testing.

(d) Requires additional testing.

If the AFCO determines the candidate fuel cannot be certified for use or the candidate fuel is the same as the baseline fuel, the AFCO will document this conclusion with supporting rationale in Appendix L and the process will be terminated. If the AFCO determines the candidate fuel can be certified without additional testing, this recommendation will be provided to each SM with supporting rationale and the process will proceed as described in 5.2. If the AFCO determines additional testing is required, the process will be continued as described in 5.1.3.

5.1.2.8 Fuel property/characteristic testing.

The Subset 2 fuel property/characteristic and component/system tests described in Appendix B will be conducted and the results compared to the baseline results in Appendix B.

5.1.2.9 Material compatibility testing.

The Subset 2 material compatibility tests described in Appendix D will be conducted and the results compared to the baseline results contained in Appendix D.

5.1.2.10 Toxicity.

The Subset 2 toxicity tests described in Appendix E will be conducted and the results compared to the baseline results contained in Appendix E.

5.1.2.11 Fire protection and survivability/vulnerability.

The Subset 2 fire protection tests described in Appendix F will be conducted and the results compared to the baseline results contained in Appendix F.

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5.1.2.12 Aircraft propulsion.

The Subset 2 aircraft propulsion tests described in Appendix G will be conducted and the results compared to the baseline results contained in Appendix G.

5.1.2.13 Aerospace fuels infrastructure.

The Subset 2 logistics tests described in Appendix H will be conducted and the results compared to the baseline results contained in Appendix H.

5.1.3 Conduct Subset 3 testing.

The AFCO will ensure that the Subset 3 testing requirements identified in Appendix B, D, F, G, and H are conducted (if required) to evaluate the final set of fuel / additive properties/characteristics, to ensure compatibility with the materials commonly used in Air Force equipment that comes into contact with fuel, and to ensure personnel safety. In addition, the component/system level tests described in Appendix B not correlated to a fuel property results will be conducted and evaluated. This third subset of tests is generally related to durability, supportability, interoperability, etc. and a failure to meet these requirements will be evaluated. For each test result that did not meet the baseline requirements, the AFCO will initiate the risk assessment process described in 5.3 to determine the impact to Air Force systems and personnel, additionally involving the affected SM as appropriate. The AFCO will evaluate the results of Subset 3 testing, all previous testing, and the risk assessments to determine if the candidate fuel:

(a) Cannot be certified for use.

(b) Can be certified for utilization

(c) Can be certified with limitations and appropriate documentation

If the AFCO determines the candidate fuel cannot be certified for use, the AFCO will document this conclusion with supporting rationale in Appendix J. If the AFCO determines the candidate fuel can be certified, or can be certified with limitations and appropriate documentation, this recommendation will be provided to each SM with supporting rationale and the process will proceed as described in 5.2.

5.1.3.1 Fuel property/characteristic testing.

The Subset 3 fuel property/characteristic and component/system tests described in Appendix B will be conducted and the results compared to the baseline results in Appendix B.

5.1.3.2 Material compatibility testing.

The Subset 3 material compatibility tests described in Appendix D will be conducted and the results compared to the baseline results contained in Appendix D.

5.1.3.3 Toxicity testing.

The Subset 3 toxicity tests described in Appendix E will be conducted and the results compared to the baseline results contained in Appendix E.

5.1.3.4 Fire protection and survivability/vulnerability.

The Subset 3 fire protection tests described in Appendix F will be conducted and the results compared to the baseline results contained in Appendix F.

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5.1.3.5 Aircraft propulsion.

The Subset 3 aircraft propulsion tests described in Appendix G will be conducted and the results compared to the baseline results contained in Appendix G.

5.1.3.6 Aerospace fuels infrastructure.

The Subset 3 logistics tests described in Appendix H will be conducted and the results compared to the baseline results contained in Appendix H.

5.2 Develop initial certification plan.

Each SM will develop an initial certification plan and supporting rationale considering the recommendation from the AFCO. If the SM agrees with the AFCO recommendation, the certification plan should simply state this conclusion and no additional effort is required. If the SM determines additional testing or gap analysis is required to certify the candidate fuel in his/her respective system, the SM will provide a certification plan including the requirements impacted and the proposed verification analysis and/or testing activities identified using their gap analysis and the described risk assessment process described in 5.3. Each SM will submit their initial certification plan to the AFCO for evaluation. For passenger-carrying commercial derivative aircraft, the basic aircraft will comply with AFPD 62-4. Any fuels that are not currently FAA approved for use on the aircraft will undergo FAA approval for that aircraft. Approvals may be by amended type certificate or supplemental type certificate. For commercial derivative hybrid aircraft complying with AFPD 62-5, the preferred approval process is by FAA certification.

5.2.1 Evaluate initial certification plan.

The AFCO will evaluate each SM's initial certification plan. For initial certification plans that involve additional testing or analysis, the AFCO will identify opportunities to eliminate duplication of effort and to maximize the sharing of verification products. The AFCO will recommend to each SM changes to his/her initial certification plan that will minimize the cost to the Air Force while still providing sufficient data for the SM to certify (or reject) the fuel for utilization.

5.2.2 Finalize certification plan.

The SM and AFCO will finalize the certification plan and the SVP list. The AFCO will maintain the SVP list and ensure this information is accessible to all impacted SM.

5.2.3 Execute certification plan.

The SM will execute the certification plan and provide all SVP to the AFCO.

5.2.4 Share verification data.

The AFCO will provide the SVP generated by the SMs and provide the SVPs to the appropriate SM for his / her evaluation.

5.2.5 Certify or reject fuel.

The SM will evaluate the SVPs, resulting from executing the certification plan described in 5.2.3 and 5.2.4 and determine if the fuel can be certified for use in his / her system considering safety, performance, durability, supportability, interoperability, etc. If the fuel can be certified for use, the SM will update the appropriate Technical Orders authorizing use of the fuel /

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additive to include any applicable limitations or restrictions. If the fuel cannot be certified for use, this result will be provided to the AFCO with supporting rationale and the AFCO will incorporate these results and the supporting rationale in Appendix J.

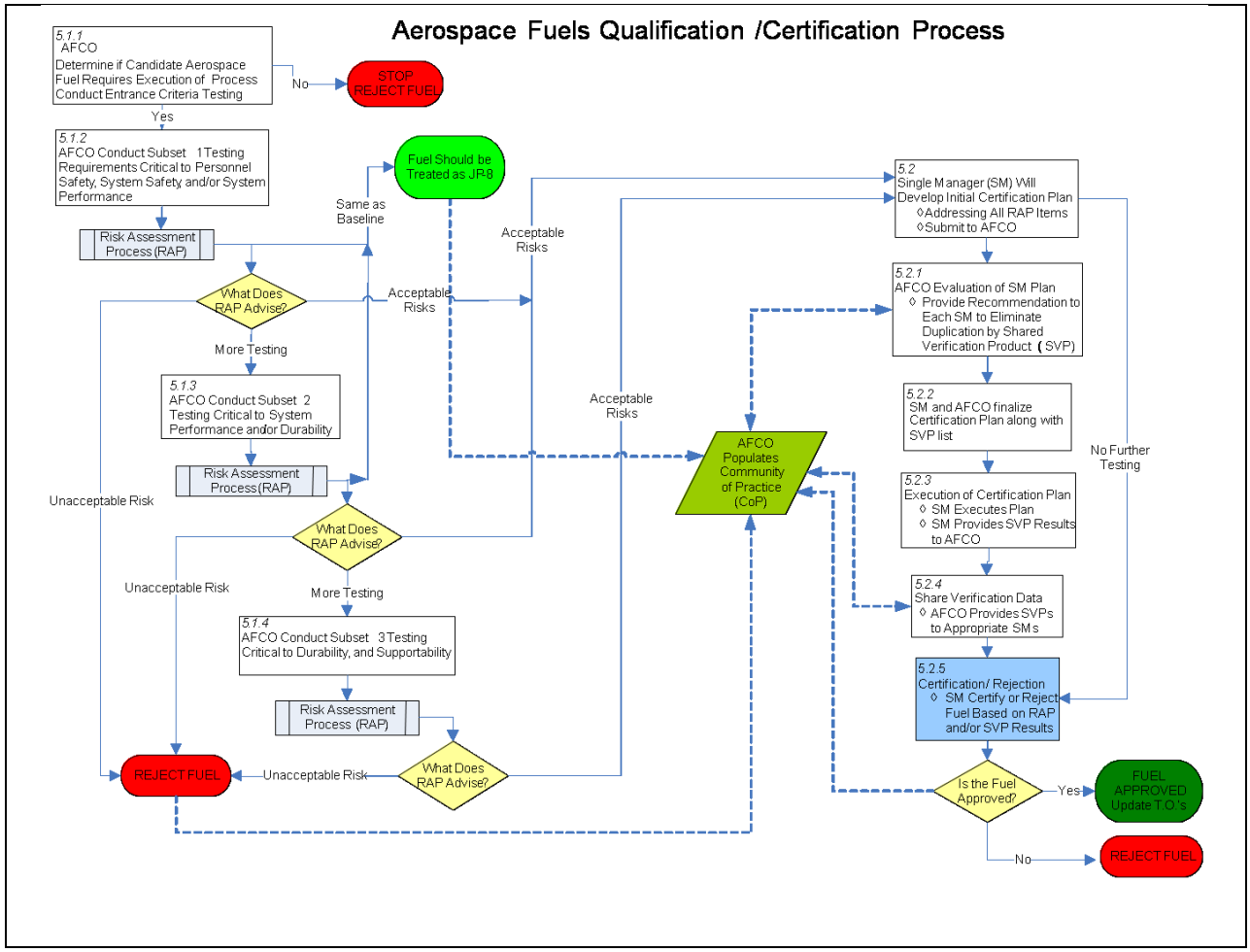


FIGURE 1. Aerospace fuels qualification/certification process.

5.3 Risk assessment process.

The risk assessment process is applied when candidate fuel or additive properties fall outside the baseline requirements described in the appendices below. The objective is to address potential risks and impacts driven by those deviations from the common experience with the baseline fuel. The risk assessments take on three distinct sequential styles depending on the timing and the responsibility level of the decision authority. The first style is used by the AFCO when a candidate fuel or additive has been proposed for certification. This assessment has a general whole-fleet focus and is based on whether foreseeable benefits, risks, and impacts balance in favor of the potential benefits. If they do, the candidate passes and is handed to the Single Managers, who apply the second style of risk assessment. This focuses on the specific system and tries to quantify, where possible, the benefits, risks, and impacts with the aim of determining whether or not the system should be certified to use the candidate fuel or additive. All Single Manager assessments are consolidated and along with an AFCO recommendation are transmitted to the high level decision authority. The final style is then applied by the high level

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decision authority responsible for the ultimate decision on whole-fleet certification. This again focuses on the whole fleet, and has several potential outcomes, depending on the aggregate of the certification decisions of the Single Managers.

5.3.1 The AFCO risk assessment process.

Properties that are outside the military and commercial operational experience with the baseline fuel (and its additives) are the key drivers of AFCO decisions about whether or not to proceed with the certification process for a candidate fuel or additive. Such a situation is especially likely for new additives. Since the AFCO does not have detailed insight into the specifics of all systems, its decisions are necessarily based on limited information. Because of this, the risk assessment takes on a qualitative characteristic in determining if risks or impacts associated with the out-of-bounds properties are sufficiently significant to overcome the potential benefit of the candidate (e.g., lower cost, greater availability, better functionality, improved life, etc.). The systems engineering process applies as part of the assessment, but other considerations, like those listed in B.4.1, are involved as well. If the chemistry of the candidate deviates little from the existing military and commercial operational experience, the AFCO could decide to proceed with Single Manager certification without knowing or considering the impacts of Subset 2 and Subset 3 properties, opting instead to rely on the expertise of the Single Managers' organizations to assess the applicability and influence of these properties on their individual certification decisions. The AFCO risk assessment process is shown in the diagram on Figure 2.

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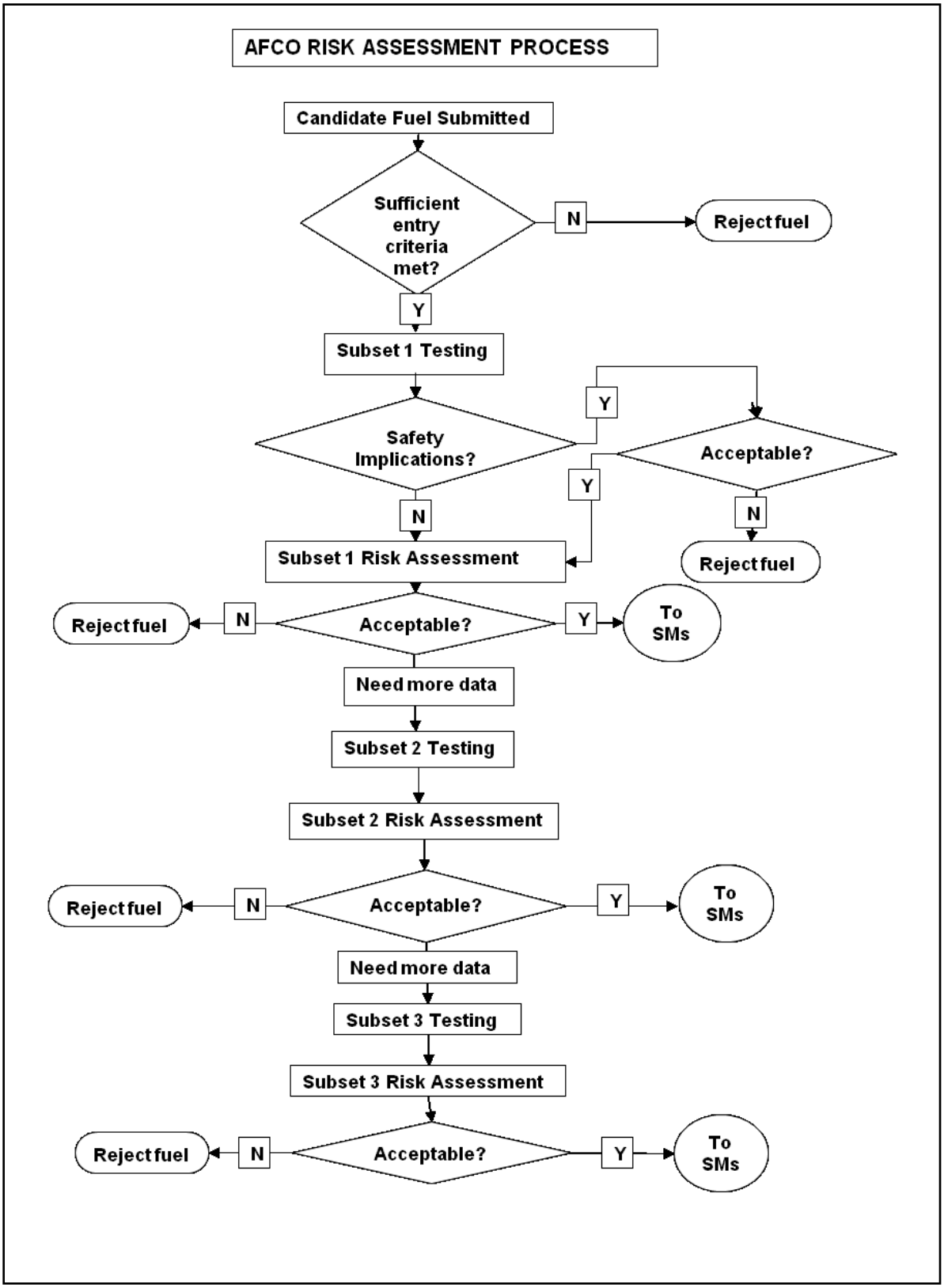


FIGURE 2. AFCO risk assessment process.

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5.3.1.1 AFCO risk assessment considerations.

5.3.1.1.1 Business case.

Potentials in terms of economic viability, produceability, technical maturity, logistics availability, etc., addressed in B.4.1 are part of the decision process. Though difficult to quantify in terms of their overall fleet impact, these potentials can be the first filter of whether or not to proceed with the certification process. If the candidate shows insufficient potential in these areas, it likely will not warrant the time, effort, and expense of certifying it for use in its intended fleet.

5.3.1.1.2 Benefits case.

Advantageous impacts of using the candidate fuel or additive form a key element of the balance between risks and the other impacts of using the candidate fuel. The advantageous impacts have to balance all of the negative impacts, including the cost of certifying the candidate for its intended fleet.

5.3.1.1.3 Risks and negative impacts.

The following outline shows the risks and impacts which may need to be considered and a qualitative ranking method for each:

Safety impacts (none, minor, medium, major)

Mitigating actions (none, easy, moderate, hard)

Change procedures (minor, medium, major)

Change operating limits (minor, medium, major)

Change hardware (minor, medium, major)

Potential costs (low, medium, high)

Potential costs beyond mitigating actions (e.g., system loss/damage) (low, medium, high)

Operational/performance impacts (none, minor, medium, major)

Change limitations (minor, medium, major)

Change procedures (minor, medium, major)

Potential costs beyond mitigating actions (e.g., reduced system efficiency requiring more fuel, reduced range, reduced payloads) (low, medium, high)

Mitigating actions (none, minor, medium, major)

Change procedures (minor, medium, major)

Change hardware (minor, medium, major)

Potential costs (low, medium, high)

Supportability impacts (none, minor, medium, major)

Change procedures (minor, medium, major)

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Change maintenance interval (minor, medium, major)

Change spare parts requirement (minor, medium, major)

Potential costs (low, medium, high)

Mitigating actions (none, minor, medium, major)

Change procedures (minor, medium, major)

Change hardware (minor, medium, major)

Potential costs (low, medium, high)

Durability impacts (none, minor, medium, major)

Change procedures (minor, medium, major)

Change maintenance (minor, medium, major)

Change parts replacement interval (minor, medium, major)

Potential costs (low, medium, high)

Mitigating actions (none, minor, medium, major)

Change procedures (minor, medium, major)

Change hardware (minor, medium, major)

Potential costs (low, medium, high)

Both safety and operational considerations have the potential to have impacts in addition to any impacts caused by the mitigating actions that may be applied because of the use of the candidate fuel or additive. For example, such impacts could include an increase (temporary or permanent) in the aircraft mishap rate due to increased frequency of engine flameouts (safety) or reduced range, payload, or loiter time due to lower energy content of the fuel (operations). Note that the Hazard Risk Index (HRI) is not included in the AFCO assessment. While the HRI concepts of frequency and severity can be used, the AFCO has insufficient insight into risks associated with individual systems to be able to aggregate them into a meaningful HRI for the entire intended fleet for the candidate fuel or additive.

5.3.1.1.3.1 Definitions of qualitative impact descriptors.

5.3.1.1.3.1.1 Minor or low.

Change can be made locally and/or minor change to system Technical Orders. One time training session required no long-term detrimental impact.

5.3.1.1.3.1.2 Medium.

Change(s) require modification to system Technical Orders, formal training required for crew and maintainers, with long-term impact to personnel and equipment. It may require increased purchases of additional equipment. Component redesign may be required to retain full mission capability.

5.3.1.1.3.1.3 Major or high.

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Extensive alterations to Technical Orders, mission planning, maintenance procedures/personnel, formal training required, and long-term impact to entire fleet. This will require additional funds for maintenance, component redesign, or increased spare parts purchase. Component redesign will be required or full mission capability may never be attained.

5.3.2 The single manager risk assessment process.

Once the candidate fuel or additive has been approved by the AFCO for continuation in the certification process and submitted to the affected Single Managers, the Single Manager's systems engineering, risk assessment, and decision-making processes are applied to determine if it is appropriate to certify the system for the use of the candidate fuel or additive. The Single Manager risk assessment and associated systems engineering processes look at the same things as the AFCO processes, but focused specifically on the Single Manager's system, or systems, and at a much more detailed level, at a greater depth with the objective of providing some quantitative definition to some or all of the relevant risks and impacts. The Single Manager risk assessment process is shown in the diagram on Figure 3.

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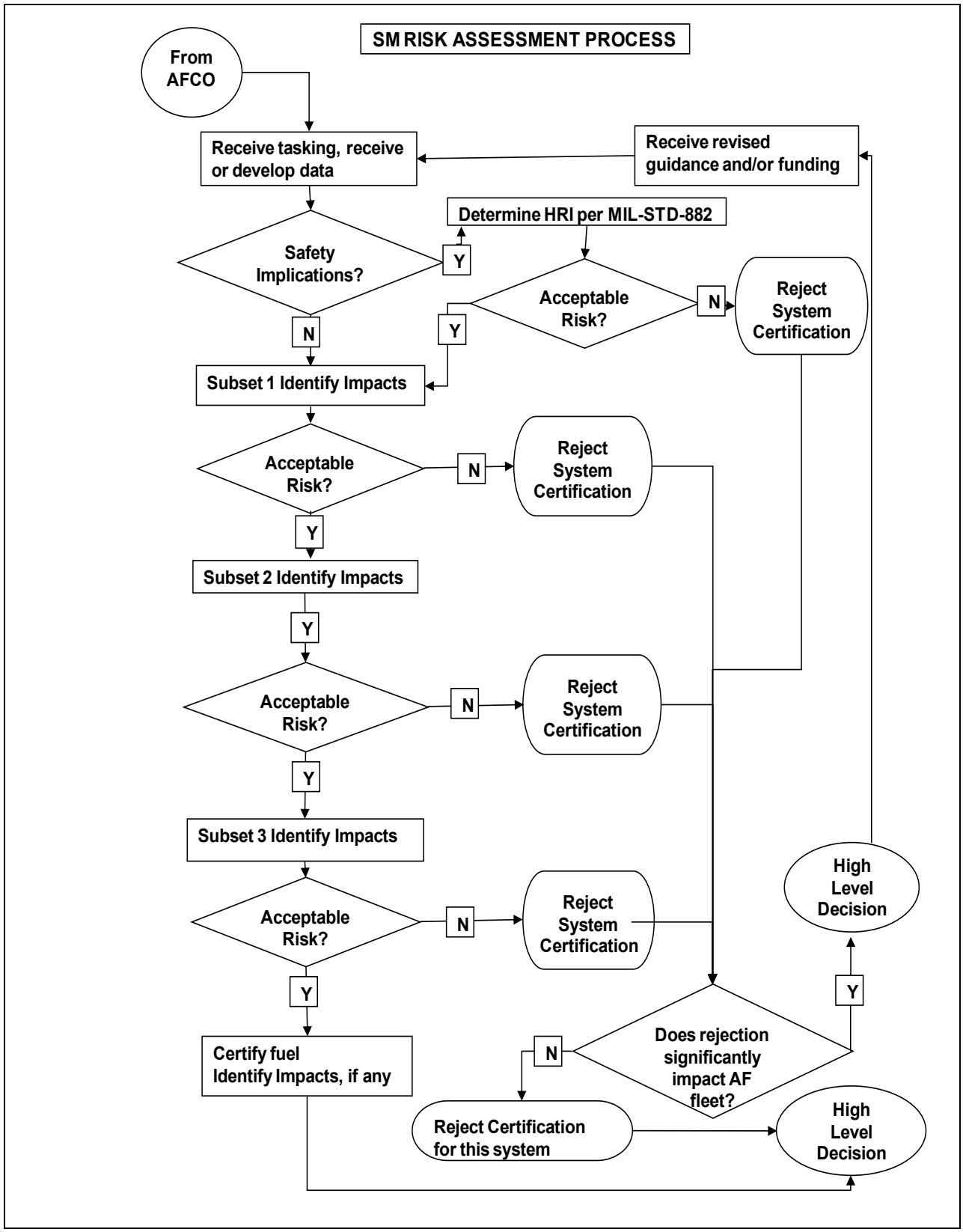


FIGURE 3. Single manager risk assessment process.

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5.3.2.1 Single manager risk assessment considerations.

5.3.2.1.1 Business case.

The business considerations from the Single Managers' viewpoint focus on their systems' life-cycle costs associated with the use of the candidate fuel or additive. There should be no need for the Single Managers to assess the viability or producibility of a candidate. Logistics availability, however, is a significant element of life cycle cost and direction received or assumptions made about this need to be clearly identified with the assessment results.

5.3.2.1.2 Benefits case.

Benefits of using the candidate fuel or additive in the Single Manager's system(s) should be identified and quantified, where possible. While it is desirable for benefits to outweigh risks and negative impacts for the Single Manager's system(s), a neutral result should normally be sufficient grounds for certifying the candidate. Further, it is possible that negative results for one Single Manager's system(s) are offset by positive results for other Single Managers' systems. In such a case, input from the High-Level decision process addressed in 5.3.3 could still drive certification for all effected systems, including the one(s) with negative results.

5.3.2.1.3 Risks and negative impacts.

The desired quantitative results involve a fair degree of uncertainty, given that they are dependent on the accuracy (or lack of accuracy) of predictive models and their underlying assumptions. These uncertainties should be identified along with the results. The following outline shows the risks and impacts which may need to be considered:

Safety impacts

- Hazard Risk Index (HRI)
  - Frequency and Severity of Impact
- Mitigating actions (none, easy, moderate, hard)
  - Change procedures -- identify
  - Change operating limits -- identify
  - Change hardware -- identify
  - Potential costs -- quantify
- Potential costs beyond mitigating actions (e.g., system loss/damage) -- quantify
  - Operational/performance impacts (none, minor, medium, major)
  - Change limitations -- identify
  - Change procedures -- identify
  - Potential costs beyond mitigating actions -- quantify
  - Mitigating actions
  - Change procedures -- identify
  - Change hardware -- identify

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Potential costs -- quantify

Supportability impacts (none, minor, medium, major)

Change procedures -- identify

Change maintenance interval -- identify

Change spare parts requirement -- identify

Potential costs -- quantify

Mitigating actions

Change procedures -- identify

Change hardware -- identify

Potential costs -- quantify

Durability impacts (none, minor, medium, major)

Change procedures -- identify

Change maintenance -- identify

Change parts replacement interval -- identify

Potential costs -- quantify

Mitigating actions

Change procedures -- identify

Change hardware -- identify

Potential costs -- quantify

5.3.3 The high-level decision authority risk assessment process.

Because most candidate fuels or additives are applicable to multiple systems rather than to a single system, a decision authority at a higher level than a Single Managers (e.g., Headquarters USAF, DoD) needs to accept the risks and impacts, both negative and positive, associated with the use of a candidate fuel or additive and authorize it for broader operational use for the intended fleet and for implementation into the logistics system supporting that operational use. While the output of this process is an authorization, it does not necessarily mean that candidate fuel or additive will actually be available for operational use at the time of the decision, nor does the authorization require that availability. There are situations, however, in which the requirement to use the candidate fuel or additive is mandated as a part of the high-level decision in order to ensure timely receipt of the benefits it provides. This would be especially relevant if the candidate provides a significant safety benefit. Figure 4 shows the High-Level decision process. One of the unique aspects of this process is a feedback loop that provides for adjustment of direction or funding from the High Level to potentially change some or all affected Single Managers' negative certification decisions. For example, additional funding could allow modification of the system to mitigate the negative impacts of the candidate fuel or additive or revised direction could allow acceptance of system performance losses or life cycle cost increases for the sake of achieving other benefits to the whole fleet or its logistics system. The AFCO's role in this process is twofold. First, the AFCO collects and consolidates

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all of the Single Managers' certification decisions, associated rationale, and supporting data and makes it available to the decision authority in the form desired by the decision authority. Second the AFCO recommends a course of action, whether it is to reject the fuel, to use the feedback loop to force reconsideration of negative certification decisions, or to proceed with certifying the fleet. If certifying the fleet has negative consequences, the AFCO will identify those consequences to the decision authority along with the AFCO's recommendation.

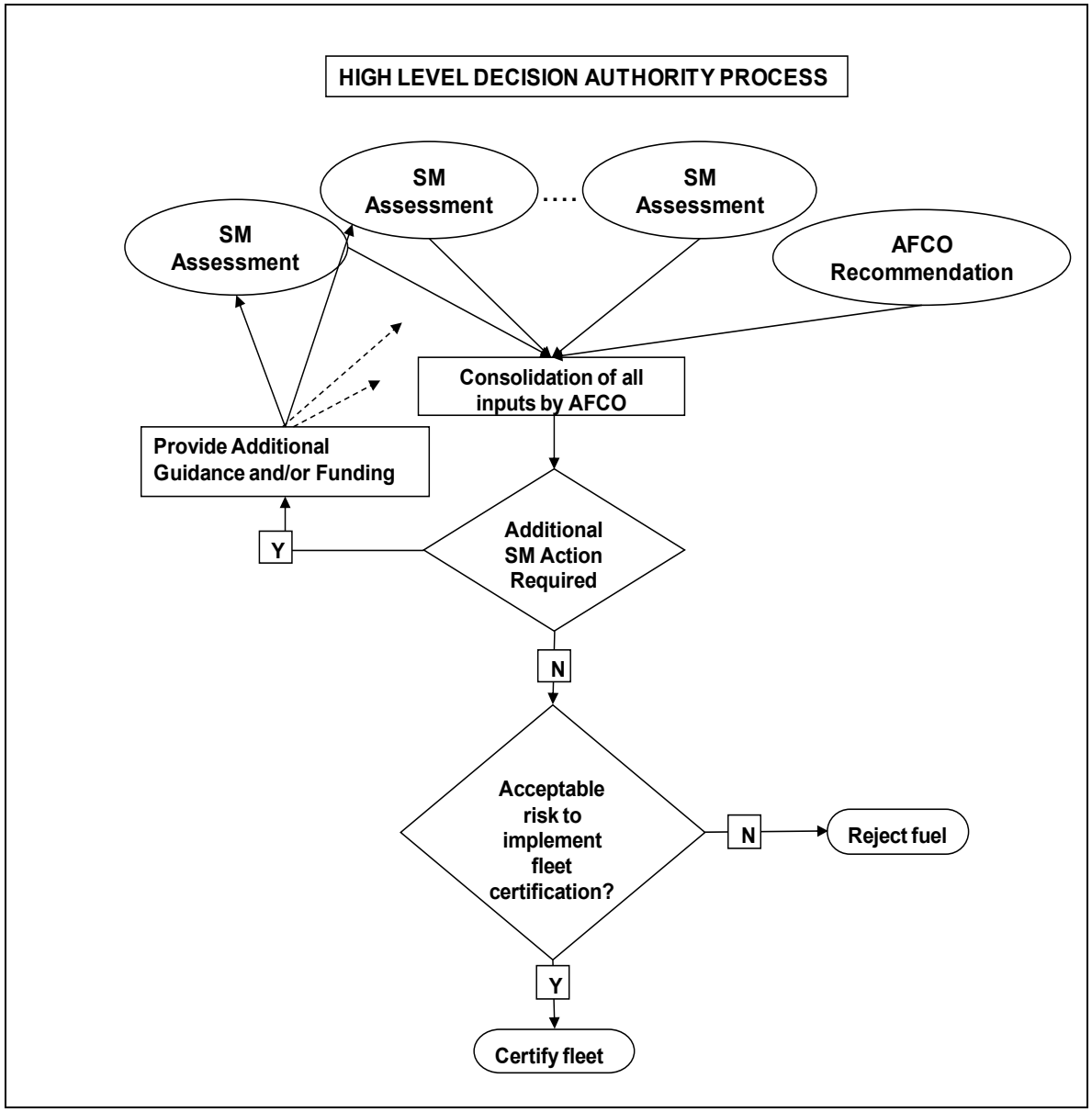


FIGURE 4. High-level decision authority risk assessment process.

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5.3.3.1 High-level decision authority risk assessment considerations.

5.3.3.1.1 Business case.

At the high-level, the business considerations revolve around the impacts of meeting mission requirements for the entire affected fleet, the impacts on its life cycle cost, and the impacts on the logistics system that have to be able to effectively and efficiently handle the candidate fuel or additive.

5.3.3.1.2 Benefits case.

The balance between benefits and risks and negative impacts have to favor the benefits clearly, but from the perspective of the entire affected fleet. This means that some elements may sum to be negative, but overall, the benefits have to override all of those negatives.

5.3.3.1.3 Risks and negative impacts.

The following outline shows the risks and impacts which may need to be considered if issues remain when the fleet is certified:

Accepting changes in system capability or availability for some systems

Accepting additional sustainment costs/impacts for some systems

Dealing with the uncertified systems

Unique fuel supply requirements

Providing funding for system modifications which then enable certification

Transitioning to the new fuel or additive

The most desirable logistics implementation would be for the new fuel or additive to be capable of being mixed with the old with no impact to system performance or operability.

If the new fuel or additive must be kept separate from the old significant logistics issues arise.

A potential need for dual fuel capability at transitioning bases.

Significant additional workload in defueling and handling the removed fuel.

5.4 Field service evaluation.

The SM will consider the need to conduct field service evaluations (FSEs) to monitor durability and service life. This task is consistent with the force management activities described in the Aircraft Structural Integrity Program, Engine Structural Integrity Program, Mechanical System Integrity Program and others to ensure continued airworthiness and is essential for certification of new fuels as well. In particular, when the relationship between durability of fuel pumps, sealants, finish systems, o-rings, etc. cannot be adequately correlated to fuel properties/characteristics or component/system tests, a field service evaluation may be warranted. However, unless degradations in durability or supportability are considered likely for a candidate fuel or additive, the FSE does not need to be completed before certification of the system with the candidate fuel or additive. Rather, the system can be certified (allowing but not mandating use of the candidate fuel or additive) and the FSE can be performed at a later time,

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more consistent with the availability of fuel, supporting infrastructure, and/or evaluation assets. The FSE can take two approaches:

(1) Broadening the experience with the candidate fuel or additive over a larger sample of fleet elements (e.g., aircraft) and

(2) Gathering quantitative data on changes in durability and supportability.

Depending on how the FSE is structured, it can focus on either or both of these approaches. The first can be addressed in a relatively short period of time using multiple elements of a fleet (e.g., multiple aircraft). The second is likely to be much longer in duration to ensure that the impacts on durability and supportability clearly show themselves before the FSE ends. In either case, the more systems (e.g., aircraft) that are involved, the more statistically relevant the data will be, but also the more expensive in terms of impacts on cost, logistics, and operations (e.g., the cost of FSE fuel itself, segregating the FSE fuel supply, limiting where aircraft can operate so that they can almost always get the FSE fuel, etc.) The complexity, expense, and operational impacts associated with doing an FSE have to be balanced against the potential benefits that can be obtained from it. The results have to favor the benefits clearly to justify doing an FSE. One of the primary considerations needs to be whether an unadulterated comparison can be made between the results of the FSE with the new fuel or additive and the previous operational experience without it (i.e., the baseline against which the FSE results are compared). If multiple changes relative to the baseline are being evaluated in a single FSE (e.g., revised combustor fuel nozzles in addition to the change in fuel or additive) it may not be possible to identify which FSE results are associated with which specific change, thereby losing a significant benefit of doing the FSE. The SM will consider options contained in 5.4.1, 5.4.2, and 5.4.3 to obtain the engineering data required to adjust the maintenance intervals as appropriate.

5.4.1 Data collection and evaluation.

The SM will determine what data should be collected and evaluated to monitor durability. These data could include part replacement history, part overhaul history, inspection results, and maintenance trends by subsystem such as fuel, propulsion, etc.

5.4.2 Pacer programs.

The SM will consider the utilization of pacer programs to monitor durability. These programs could include exchanging of components between systems to maximize exposure (lead-the-fleet) that would be subjected to thorough evaluation such as a teardown inspection as described in 5.4.3.

5.4.3 Teardown inspections.

The SM will consider conducting periodic teardown inspections to monitor durability. Teardown inspections involve disassembly and inspection of a component, typically to a greater extent than what occurs during routine or special maintenance. The SM will determine what data should be collected and evaluated to monitor durability. This data could include part replacement history, part overhaul history, inspection results, and maintenance trends by subsystem such as fuel, propulsion, etc.

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5.5 Certification criteria elevation potential.

During the certification process, the potential exists that certain fuel properties once deemed at the lower subset levels (or perhaps were not previously considered at all) may be elevated to higher subset levels. As a result, already certified platforms may need to be reinvestigated. The AFCO will work with the SMs on a case-by-case basis to determine whether or not additional testing/investigation is necessary to ensure that certification is complete for these systems.

6. NOTES.

6.1 Intended use.

This handbook provides guidance on the certification of alternative fuels, fuel alternates and fuel additives.

6.2 Subject term (key word) listing.

Alternative fuels

JP-8

Synthetic Paraffinic Kerosene (SPK)

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APPENDIX A

REQUIREMENTS DECOMPOSITION AND TRACEABILITY

A.1 SCOPE.

A.1.1 General.

This Appendix contains a discussion of the process of requirements decomposition of airworthiness criteria as defined in MIL-HDBK-516 traced to requirements in the JSSG as well as other military specifications and handbooks for support equipment, ground refueling equipment, and logistical infrastructure. The concept of using a "Systems Engineering V", as described on Figure A-1, should be used as the approach for decomposing requirements. This concept identifies requirements at the highest level then traces them to lower level requirements. Additionally, properties should be identified for each requirement. Finally, based on the results of the certification process identified in Appendix B, risks are identified and fuel applicability is assessed.

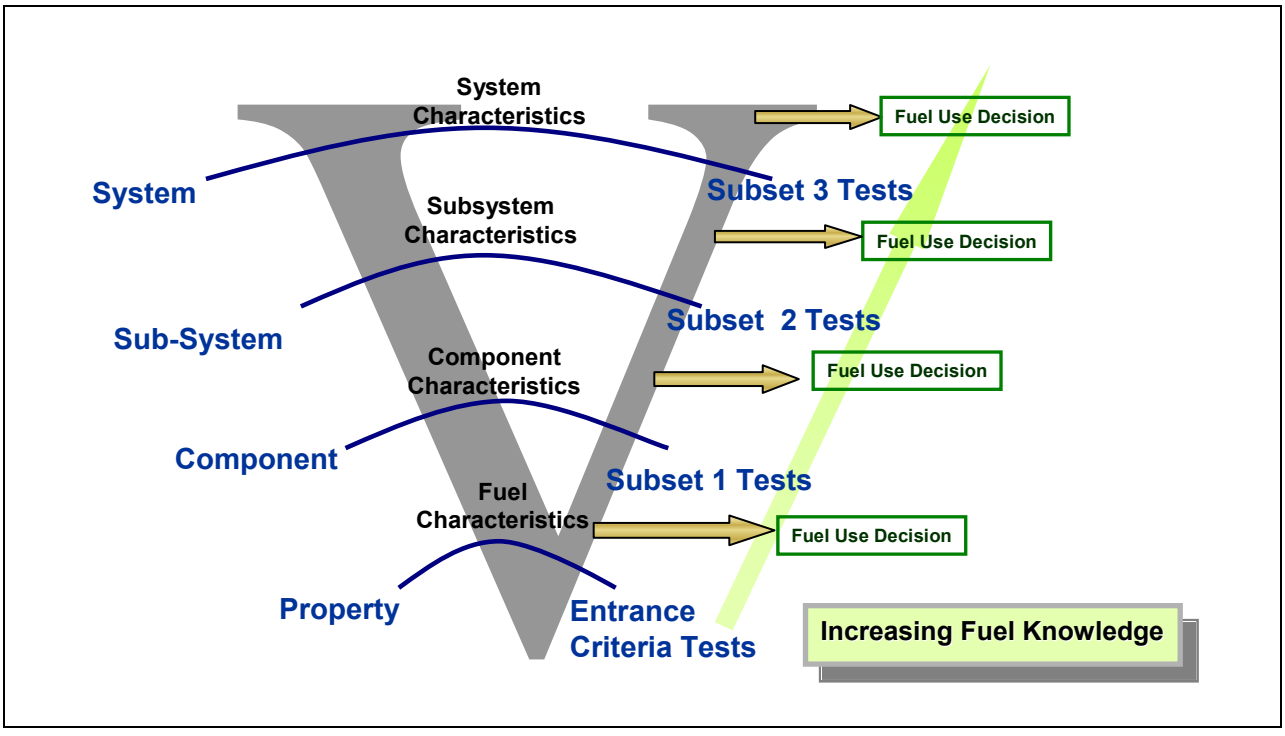


FIGURE A-1. Functional System Decomposition.

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A.2 APPLICABLE DOCUMENTS.

A.2.1 General. Applicable documents that identify requirements for OSS&E should be used to identify requirements and verification methods that may be affected by a change in fuel type or fuel configuration. The JSSG can be used to provide requirements and detailed guidance for requirements and verification methods, which are described in each JSSG appendix. A list of recommended documents for various systems is identified.

A.2.2 Government documents.

A.2.2.1 Specifications, standards, and handbooks.

The following specifications, standards and handbooks form a part of this document to the extent specified herein.

DEPARTMENT OF DEFENSE SPECIFICATIONS

AFGS-87139	Landing Gear Systems
AFGS-87233	Support Systems and Equipment

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-464	Electromagnetic Environmental Effects, Requirements for Systems
MIL-STD-1787	Aircraft Display Symbology

DEPARTMENT OF DEFENSE HANDBOOKS

MIL-HDBK-514	Operational Safety, Suitability, and Effectiveness for the Aeronautical Enterprise
MIL-HDBK-516	Airworthiness Certification Criteria

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

JOINT SERVICES SPECIFICATION GUIDES (JSSG) CD

(This product is available from ASC/ENRS, 2530 Loop Road West, WPAFB OH 45433-7101; Engineering.Standards@wpafb.af.mil.)

A.2.2.2 Other government documents, drawings, and publications. The following other Government publications provide additional information related to the fuels certification process and are included in this document for information.

DoD 4140-25M	Procedures for Management of Petroleum Products
TR-89-D-22E	Vol.V, USAAVSCOM, Aircraft Crash Survival Design Guide

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UFC 3-460-03	Operation and Maintenance: Maintenance of Petroleum Systems
UFGS-09 97 13.17	Unified Facilities Guide Specification
EO13149	"Greening the Government Through Federal Fleet and Transportation Efficiency", Presidential Documents, April 2000
EO13123	"Greening the Government Through Efficient Energy Management" Presidential Documents, June 1999

(Copies of these document are available online at The Defense Technical Information Center Website and are available from DTIC Headquarters, 8725 John J. Kingman Rd., Ft. Belvoir VA 22060-6218; telephone (800) 225-3842.)

AIR FORCE TECHNICAL ORDERS (TO)

Technical Order 42B1-1-1, Fuels for USAF Piston and Turbine Support Equipment and Administrative Vehicles, USAF, December 2006

(Copies of this Technical Order may be obtained by calling (937) 255-8050, DSN 785-8050. It may be also be obtained from The Defense Technical Information Center Website and from DTIC Headquarters, 8725 John J. Kingman Rd., Ft. Belvoir VA 22060-6218; telephone (800) 225-3842. Approved users may access Technical Orders at https://www.toindex-s.wpafb.af.mil/.)

A.2.3 Non-Government publications.

SAE INTERNATIONAL

SAE ARP 1256	Procedure for the Continuous Sampling and Measurement of Gaseous Emissions from Aircraft Turbine Engines, Society of Automotive Engineers, January 1990
SAE ARP 1258	Qualification of Hydraulic Tube Joints to Specified Flexure Fatigue Requirements, Society of Automotive Engineers, September 1993

(Copies of these documents are available from SAE International at www.sae.org or SAE World Headquarters, 400 Commonwealth Drive, Warrendale PA 15096-0001. Approved users may access the documents online at www.ihs.com.)

Aviation Fuel Systems, Buna N, Viton, API 78-24-28

(Copies of this document are available online at The Defense Technical Information Center Website and are available from DTIC Headquarters, 8725 John J. Kingman Rd., Ft. Belvoir VA 22060-6218; telephone (800) 225-3842.)

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A.3 DEFINITIONS.

A.3.1 Alphabetical listing of terms and definitions.

Derived Requirements: Characteristics needed to complete the requirements set for item design that are dependent on the nature of the item solution for their initial identification and have a functional relationship to each other. Derived requirements show the traceability from higher general system level requirements to detailed requirements.

Requirements Decomposition: For this handbook, requirements are broken down into parts from source requirements documents, (e.g., JSSG, etc.) until the relevant fuel properties/characteristics are identified.

A.4 DECOMPOSITION PROCESS.

The process for identifying fuel properties with all applicable requirements can be developed using the following process. The flow chart describing this process is depicted on Figure A-2.

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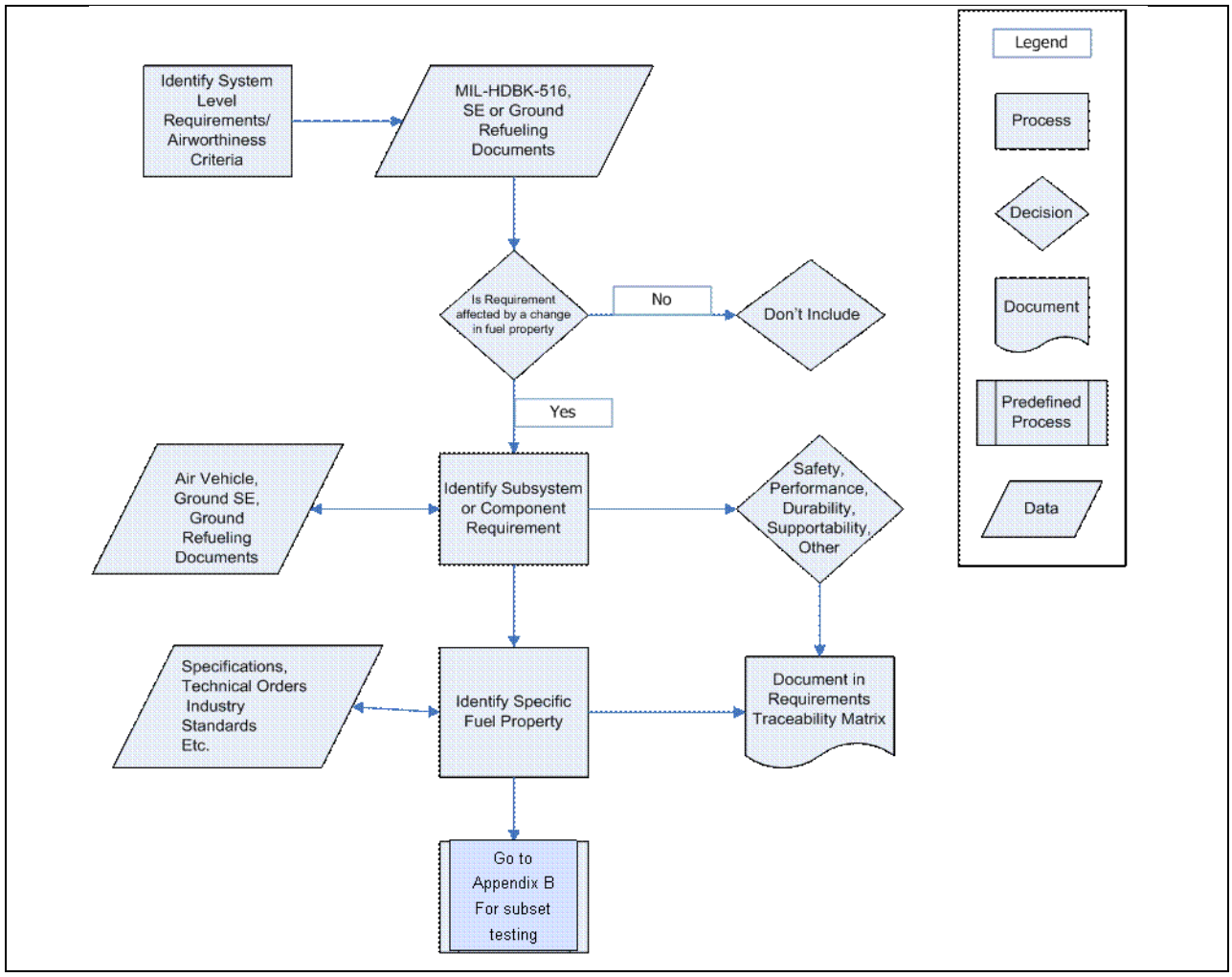


FIGURE A-2. Requirements decomposition and traceability.

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APPENDIX B

FUEL PROPERTIES/CHARACTERISTIC TESTING

B.1 SCOPE.

B.1.1 General.

This Appendix defines where each fuel property/characteristic belongs and the significance of its placement in the fuels certification process. This Appendix also includes component tests that should be conducted to further evaluate the candidate fuel for those attributes that cannot be directly related to a fuel property. The Air Force Subject Matter Experts (SMEs) evaluated all fuel properties and characteristics based on the requirements decomposition process that correlated requirements to safety, performance, durability and supportability, as described in Appendix A. The properties and characteristics were further categorized according to the number of times a property or characteristic affected a requirement and the experiences of the handbook authors.

B.2 APPLICABLE DOCUMENTS.

B.2.1 General.

The documents listed below are not necessarily all of the documents referenced herein but are those needed to fully understand the information provided by this handbook.

B.2.2 Government documents.

B.2.2.1 Specifications, standards, and handbooks.

The following specifications, standards and handbooks form a part of this document to the extent specified herein.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-83133	Turbine Fuels, Aviation, Kerosene Types, JP-8, (NATO F-34), NATO F-35, and JP-8+100 (NATO F-37)
AFGS-87233	Department of Defense Air Force Guide Specification Support Systems and Equipment

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

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B.2.3 Non-Government publications.

The following documents form a part of this document to the extent specified herein.

ASTM INTERNATIONAL

ASTM D975	Standard Specification for Diesel Fuel Oils

SAE INTERNATIONAL

SAE-ARP1247	Support Equipment, Aerospace Ground, Motorized and Non-motorized, General Requirements For

(Approved users may access these documents online at www.ihs.com. They are also available at www.astm.org and www.sae.org.)

B.3 DEFINITIONS.

B.3.1 Alphabetical listing of terms and definitions.

Entrance Criteria: Key information required to make a determination whether to initiate the fuel certification process. All fuel candidates have to meet the general characteristics of a kerosene fuel that meets safety, performance, durability and operational characteristics comparable to the baseline fuel.

SE&V: Support Equipment and Vehicles, all deployable aircraft support equipment on the ground including vehicles, powered equipment such as air conditioners, fuel handling equipment, including flight line and deployable assets.

Subset 1 Testing: Fuel properties/characteristics critical to personnel safety, system safety and/or system performance.

Subset 2 Testing: Fuel properties/characteristics critical to system performance and/or durability. This second subset also contains component level tests that do not directly correlate to a fuel property.

Subset 3 Testing: Fuel properties/characteristics critical to the system durability, and supportability requirements.

B.4 ENTRANCE CRITERIA AND FUEL PROPERTIES/CHARACTERISTIC SUBSET TESTING.

B.4.1 Entrance Criteria.

Key information is required for the Alternative Fuel Certification Office AFCO to make a determination whether to initiate the certification process. All fuel candidates have to meet the general characteristics of a kerosene fuel that meets safety, performance, durability and

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operational characteristics comparable to the baseline fuel. The intent is to compare the new fuel or fuel additive to the baseline fuel and assess the potential to incorporate the new fuel into the MIL-DTL-83133F specification. To warrant consideration, the candidate fuel or fuel additive has to meet the following requirements for consideration: 1) logistical availability in an expeditionary operation or 2) meet or exceed weapon system performance, durability, operational characteristics, or logistics or 3) be produced from a domestic alternative or unconventional domestic resource. Data will be provided to the AFCO to prove technical maturity of the fuel or fuel additive including test reports, technical data, property lists, information on the resource and processing used to make the fuel or additive, and information on the environmental, safety and health aspects. In addition, information on the economic viability of the fuel or additive has to be provided to the AFCO. This information should include cost and time estimates for production, volume of fuel or additive that can be produced, any requirements for special handling or logistics, environmental impacts, and general production information.

B.4.1.1 The Entrance Criteria definition. The entrance criteria are determined by the following, and failure to meet the provided pass criteria could constitute rejection of the fuel. As a minimum the following fuel property or fuel characteristics data have to be provided.

1. A chemical description of the fuel, Gas Chromatograph (GC) to include molecular composition. Test Method and Pass Criteria are defined in Appendix C.

2. Material Safety Data Sheet (MSDS) issued by the supplier.

3. Environment, Safety and Occupational Health (ESOH) review.

4. All property tests as required per MIL-DTL-83133F.

5. The fuel properties defined in Table B-I are related to system safety. Failure to meet the provided pass criteria should constitute rejection of the fuel.

TABLE B-I. System-safety related fuel property/characteristic.

Fuel Property and Significance	Test Method	Pass Criteria
Flash Point, affects combustibility. It is also a leading factor determining fire safety in fuel handling.	Appendix C	Temperature Range: 28°C to 68°C
Freezing Point, affects low temperature fuel behavior. It can cause issues with pumps and nozzle operations.	Appendix C	Max -40°C
Viscosity @ -20 °C, affects pumpability over the operating temperature range. It also relates to droplet size in sprays produced by burner nozzles.	Appendix C	Max 8.0 cSt

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B.4.2 Subset 1 Fuel property/characteristics. Subset 1 fuel properties/characteristics are critical to personnel safety, system safety and/or system performance.

B.4.2.1 Subset 1 fuel property/characteristic tests. The following Fuel Property/Characteristic tests as described in Table B-II should be conducted to characterize the fuel:

TABLE B-II. System safety and performance related fuel properties.

Fuel Property/Characteristic and Significance	Test Method and Evaluation Criteria
VOLATILITY	
Autoignition Temperature, a factor determining fire safety.	Appendix C
Vapor Pressure True, versus Temperature indicates venting loss of light ends at altitude and in hot climates. Also relates to cold starting. Affects vapor losses, vapor lock effects on pumping, and engine starting characteristics.	Appendix C
Hot Surface Ignition, lowest temperature required for spontaneous ignition of a substance by a hot surface. Affects fire protection design.	Appendix C
Flame Speed affects personnel safety during fire emergency and fire protection design.	Appendix C
COMBUSTION	
Flammability Limits, affects propulsion system, fuel systems design, fuel tank safety design, and fire protection design.	Appendix C
FLUIDITY	
Viscosity versus Temperature (-40 ^{0}C to 60 ^{0}C), affects pumping fuel over the operating temperature range. Relates to droplet size in sprays produced by burner nozzles.	Appendix C
Density versus Temperature (-40 ^{0}C to 60 ^{0}C), affects aircraft weight and balance calculations. Also relates to specific energy, thermal expansion, flow calculations, fuel gauging, and metering devices.	Appendix C
Bulk Modulus versus Pressure, measure of the compressibility of a fluid. Important factor in servomechanisms, where natural frequency of the system is directly proportional to the square root of the bulk modulus.	Appendix C

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TABLE B-II. System safety and performance related fuel properties - Continued.

Fuel Property/Characteristic and Significance	Test Method and Evaluation Criteria
CONTAMINANTS	
Water Solubility, affects to preclude free water.	Appendix C
Trace Elements, affects propulsion system hot section and thermal stability.	Appendix C
ELECTRICAL CHARACTERISTICS	
Dielectric Constant versus Density versus Temperature (-40 ^{0}C to 90 ^{0}C), direct effect on accuracy of dielectric compensated gauging systems.	Appendix C
OTHERS	
Lubricity refers to the effectiveness of lubricating moving parts such as pumps and control units.	Appendix C
Additives Compatibility, important in the compatibility and performance of approved DoD Fuel Mandatory Additives.	Appendix C
Storage Stability, affects shelf life and determination of peroxide formation.	Appendix C
Specific Heat versus Temperature (-40 ^{0}C to 150 ^{0}C), the amount of heat-energy that a fuel can absorb at a specific temperature, recorded in kJ/kg K., direct impact to the amount of heat that the fuel can absorb, critical to heat exchanger design.	Appendix C
Surface Tension versus Temperature (-10 ^{0}C to 60 ^{0}C), important in gas evolution and solubility. It also affects atomization characteristics of the fuel and hence combustion.	Appendix C
Thermal Conductivity versus Temperature (30 ^{0}C to 200 ^{0}C), refers to effectiveness of fuel as primary heat sink	Appendix C

B.4.2.2 Component level evaluation Subset 1.

B.4.2.2.1 Auxiliary and Emergency Power Units (APU/EPU) evaluation. APUs and EPUs that perform an in-flight emergency function should be evaluated for impacts that may affect their ability to start and operate at anticipated pressure altitudes and thermally soaked conditions. Differences in key fuel property characteristics that influence fuel supply pressure, atomization, vaporization and ignition would be indicators of the need for additional component and or unit level evaluations. Key properties for assessment would include but not be limited to flash point, flammability, freeze point, viscosity, and vapor pressure.

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B.4.2.2.2 Support Equipment and Vehicles (SE&V).

B.4.2.2.2.1 SE&V certification process. This section provides information to aid in the determination of a fuel's suitability for use in the compression ignition (CI) engine fleet of Air Force support equipment and vehicle fleet. It is intended to provide the AFCO and SM a methodology to identify, evaluate and mitigate safety, performance, and durability and supportability risks associated with a candidate fuel for the SE&V.

Currently there are 133 engine models, 247 vehicles and 50 types of support equipment. Clearly a new fuel cannot be tested in each application without incurring high costs or limited availability of the test fuel. An approach to this problem is to select a test engine or engine component which is known to be sensitive to fuel properties. A data base of Air Force SE&V could be built and maintained which captures the technical description of the engine and fuel system. The data base would then be used to identify engines or engine subsystems which are known to be sensitive to a particular fuel property and the selection would be the test article in the subset testing.

B.4.2.2.2.1.1 Fuel functions. SE&V rely on fuel for three purposes: 1) provide heat of combustion for power generation in diesel type engines and spray combustion heaters, 2) lubricate fuel pumps and injection equipment, and 3) provide cooling for the fuel injection system.

B.4.2.2.2.1.2 Power generation. There are several steps as the fuel makes it way from the tank to the combustion chamber. Safely stored fuel in the tank has to be picked up by the transfer pump and conveyed to the injection pump, intermittently pressurized to high pressures, sprayed into the engine cylinder at the correct time and with the correct droplet size and spray pattern, vaporized, mixed with air and burnt within the time available. To perform these functions it has to have certain properties within acceptable ranges such as, vapor pressure, viscosity, lubricity, cetane, density, and bulk modulus.

The fuel is also used in personal heaters which utilize a combustion can, spray nozzle, and spark ignition. The output of the combustor heats cabin air via a heat exchanger. The requirements imposed on the fuel for this application are similar to the turbine engine though not as critical.

B.4.2.2.2.1.3. Lubrication. The diesel engine fuel injection equipment is comprised of precision hardened and ground steel elements with extremely close tolerances. The fuel provides both hydrodynamic and boundary lubrication of these critical areas. The demands on the fuel to perform as a lubricant are higher for the diesel engine compared to the gas turbine application.

In hydrodynamic lubrication a journal rotates in a bearing shell, separated by an extremely thin film of oil. The journal is made of a harden steel which is carefully ground and polished. The bearing is made of a soft metal which, when it is installed, is designed to deform and conform to the shape of the journal. Thus a very small clearance is formed between the journal and the bearing. When the journal begins to turn it lifts slightly, creating a wedge shaped clearance. The oil is pumped, or dragged, into this wedge shaped space between the journal and

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bearing by virtual of its viscosity. Too low a viscosity will not lift the journal sufficiently, resulting in damage as the rotating surfaces will contact.

Not all moving parts can be designed to utilize hydrodynamic lubrication where except for startup, never contact each other. Boundary lubrication occurs when two surfaces contact each other without the benefit of the hydrodynamic layer maintaining separation. The property of a lubricant which applies in boundary lubrication is called lubricity. A lubricious fluid contains compounds which plate out on the surfaces and provide an interface between them. In effect the compound takes the wear, not the surfaces. And the compound prevents the fusing, or galling, of the surfaces. Very low concentrations of the compounds can be effective as the layers can be only molecules thick. Additives are usually used to improve fuel lubricity to an acceptable level.

B.4.2.2.2.1.4 Heat removal. The fuel carries away heat generated by friction of the fuel injection system by returning the unused heated fuel to the fuel tank. The ability of the fuel to remove excessive heat relies on fuel properties such as specific heat, density, and thermal stability in extreme temperature conditions. The demands for the fuel to serve as a coolant in the diesel engine are not as severe as in the aircraft.

B.4.2.2.2.2 SE&V properties. In addition to the fuel properties for air vehicles the following are of significance to SE&V:

Flash Point, same as for air vehicles systems.

Cetane Index, a calculated measure of the of the diesel ignition quality of a fuel. Cetane Index is based on a correlation between cetane number and distillation data for hydrocarbon fuels. Operation on a fuel with too low a value will immediately damage the engine. A Cetane Index minimum of 37 provides protection from over stressing the diesel engine structure.

Lubricity, for compression ignition engines, BOCLE scar diameter less than 0.65 mm is to protect the injection system.

Viscosity @ 40 °C, affects hydrodynamic lubrication of the fuel injection system.

Freezing Point, same as for air vehicles systems.

Viscosity @ -20 °C, same as for air vehicles systems.

Vapor Pressure, True versus Temperature, same as for air vehicles systems.

Flammability Limits, same as for air vehicles systems.

Heat of Vaporization, same as for air vehicles systems.

Heat of Combustion, impacts Brake Specific Fuel Consumption (BSFC) of engines and heat output of heaters.

Density, same as for air vehicles systems.

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B.4.2.2.2.2.1 Subset 1, safety and performance testing. The lubricity test method correlation and MIL-PRF-25017G CI/LI response is given by the following. The BOCLE (Ball on Cylinder Lubricity Evaluator) lubricity test, ASTM 5001, although not required in the JP-8 specification, is routinely reported for JP-8 as the BOCLE test method was originally devised for turbine engine applications. A different lubricity test, the HFRR (High Frequency Reciprocating Rig) at 60°C, ASTM 6079, is the specification requirement given in ASTM 975 for diesel fuel. Generally commercial diesel engines are approved by their manufacturer for use on 1-D. Manufacturers warranties for JP-8 generally require the fuel to meet the 1-D specifications for cetane, viscosity and lubricity.

B.4.2.2.3 Additional equipment evaluations. The SM will initiate the risk assessment process and determine if additional component related evaluation/testing should be accomplished.

B.4.3 Subset 2 Fuel properties/characteristics. Subset 2 fuel properties/characteristics are critical to system performance and/or durability. This second subset also contains component level tests that do not directly correlate to a fuel property.

B.4.3.1 Subset 2 Fuel property/characteristic tests. The following Fuel Property/Characteristic Tests as described in Table B-III should be conducted to further characterize the proposed fuel:

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TABLE B-III. System performance related fuel properties/characteristics.

Fuel Property/Characteristic and Significance	Test Method and Evaluation Criteria
Hot Surface Ignition, under turbulent airflow lowest temperature required for spontaneous ignition of a substance by a hot surface. Affects fire protection design.	Appendix C
Thermal Expansion, another approach in showing the effect of temperature on the density of a fluid. Important to fuel tank ullage.	Appendix C
Ignition Energy, Minimum the least amount of energy required for a spark discharge to ignite an optimum stoichiometric mixture. Critical to fire safety. (Can be tested simultaneously with flammability limits.)	Appendix C
Ostwald Coefficient, determines solubility of gases (Oxygen, Nitrogen) in fuel. Affects design of fuel pumps.	Appendix C
Cetane Number, affects diesel engine performance.	Appendix C
Electrical Conductivity versus Temperature (-40 ^{0}C to 90 ^{0}C), particularly important with regards to logistical issues when the fuel needs to dissipate static electricity which has built up during transportation.	Appendix C
Pour Point, affects low temperature fuel pumping and is critical to support equipment design.	Appendix C
Velocity of Sound, used by some fuel quantity gauging systems and as a method for determining Bulk Modulus when used with Density.	Appendix C

B.4.3.2 Critical component level evaluation Subset 2.

B.4.3.2.1 APU and Emergency Power Unit (EPU) evaluation. Fuel characteristics that may impact durability, such as thermal stability's influence on nozzle coking, should also be considered for their potential impact to the system's in-flight operability reliability. Recommended evaluation methods may include combustor rig testing, unit level cold soaked altitude tests and or field service evaluations. Units that perform ground only functions may be suited to analytical assessments along with field service evaluations.

B.4.3.2.2 SE&V. In addition to the fuel properties for air vehicles the following are of significance to SE&V:

Cetane Number, a measure of the ignition quality of a fuel in a diesel engine as determined in an actual engine. Value can differ from cetane index if the fuel is a non-hydrocarbon, and/or has a discontinuous distillation curve.

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Bulk Modulus versus Temperature (-40 °C to +90°C), affects the compressibility of the fuel which impacts the fuel injection system ability to accurately deliver fuel to the engine cylinder with the correct timing.

B.4.3.2.2.1 Subset 2, performance verification.

Impact of Fuel on Pump Performance
Pump component testing of a representative pump(s)
Full capacity (flow rate, pressure) versus time

Impact of Fuel on Engine Performance
Engine dynamometer testing of a representative engine(s) to compare the effect of the test fuel on:
Full power
Fuel consumption
Exhaust emissions
Noise
Idle stability

Impact of fuel on Engine Cold Start Performance
Evaluation of representative engine(s) installed in a cold test chamber or at a cold weather location to determine the impact of the fuel on cold start.

Impact of Fuel on Air Force Portable Heater Cold Start Performance
Test cold start of portable heater installed in cold chamber or at a cold weather location.

B.4.3.2.3 Additional equipment evaluations/testing. The SM will continue to assess the system risk and determine if additional component and or subsystem level evaluation/testing should be accomplished.

B.4.4 Subset 3 Fuel properties/characteristics. Subset 3 is composed of fuel properties/characteristics that are critical to the system durability, and supportability requirements.

B.4.4.1 Subset 3 fuel property/characteristic tests. The following Fuel Property/Characteristic tests as described in Table B-IV should be conducted to further characterize the fuel:

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TABLE B-IV. System durability and supportability related fuel properties/characteristics.

Fuel Property/Characteristic and Significance	Test Method
Enthalpy versus Temperature (0 ^{0}C to 250 ^{0}C), affects the ability to cool fuel wetted components	Appendix C

B.4.4.2 Critical component level evaluation Subset 3.

B.4.4.2.1 Fuel System Icing Inhibitor (FSII) rig test. Conduct FSII additive concentration performance rig test with candidate fuel or blends per Appendix C.

B.4.4.2.2 SE&V Evaluation. In addition to fuel properties for air vehicles the following are of significance to SE&V:

Thermal Stability, same as for air vehicle systems.

Thermal Conductivity, same as for air vehicle systems.

Specific Heat versus Temperature (-40 °C to +150 °C), same as for air vehicle systems.

B.4.4.2.2.1 Subset 3 performance/durability/supportability verification. The third subset tests for diesel engines are long-term engine durability testing which could be conducted on engine dynamometer and vehicle fleet studies. Long-term tests, such as a captive fleet test, Engine Manufactures Association 400 Hour Endurance Test or the Army's 210 Hour Wheeled Vehicle Test Cycle, will reveal issues of fuel, lubricant, and engine interactions.

Impact of fuel on injection pump durability

Evaluating Diesel Fuel Lubricity by an Injection Pump Rig 500 hour duration. Poor lubricity and or low viscosity will result in fuel delivery decreasing below established minimum or complete failure of pump before 500 hour.

High Temperature Engine Durability Test

Operation of an engine at high temperature condition to determine engine/fuel/lubricant interactions

Field Testing of SE&V Fleet

Field test SE&V at an Air Force base location which typically experiences wide seasonal weather variations or extremes and mission requires a number of different SE&V. Supply base with test fuel to be used in a selected number of SE&V. Monitor fuel used, maintain maintenance records, all fuel related discrepancies are to be fully documented. Consider testing at location where an aircraft is also being tested as the fuel will be readily available. Operate for a sufficiently long period (e.g., 1 full year) to capture seasonal weather impacts.

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B.4.4.2.3 Additional equipment evaluations. The SM will continue to assess the system risk and determine if additional system level evaluation/testing should be accomplished. The SM should consider field service evaluations to monitor durability and service life—in particular, the relationship between durability of pumps, sealants, finish systems, o-rings, etc.

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APPENDIX C

BASELINE AEROSPACE FUEL PROPERTY INFORMATION

C.1 SCOPE.

The scope of this Appendix is to educate the reader on basic definitions, test methods and established values for each of the properties discussed in the Handbook. Listed within each property section below, in addition to the aforementioned items, are noteworthy comments and historical data, specific to the property or its known behavior in certain systems. Additionally, all of the properties in this section are presented in a table. This table is a rating value, consisting of a green, and in some cases, a yellow and a red weighting. These colors and their corresponding values have been established by AFMC SME to reflect “safe for use” (green values), “unknown if safe for use, more testing required” (yellow values), and “unsafe for use” (red values). These values have been created as a guide for the reader and are based on past experiences and prior fuel behavior knowledge.

Unless otherwise noted, the summaries of the ASTM Standards included in this Appendix have been written by the USAF. ASTM has neither approved nor endorsed the summaries as written by USAF. The complete ASTM Standards may be purchased direct from ASTM International, 100 Barr Harbor Drive, West Conshohocken, PA 19428, phone: 610-832-9585, fax: 610-832-9555, e-mail: service@astm.org, website: www.astm.org.

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C.2 APPLICABLE DOCUMENTS.

C.2.1 General. The documents listed below are not necessarily all of the documents reference herein, but are those needed to understand the information provided by this Appendix.

C.2.2 Government documents.

C.2.2.1 Specifications, standards, and handbooks.

The following specifications, standards, and handbooks form a part of this document to the extent specified herein.

Joint Services Specification Guides (JSSG) CD

(This product is available from ASC/ENRS, 2530 Loop Road West, WPAFB OH 45433-7101; Engineering.Standards@wpafb.af.mil.)

FEDERAL SPECIFICATIONS

FTS 791C, Method 6053: Lubricants, Liquid Fuels, and Related Products; Methods of Testing, Federal Test Standards, September 1986

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-5624	Turbine Fuel, Aviation, Grades JP-4 and JP-5, USAF, January 2004
MIL-F-16884	Fuel, Naval Distillate, USN, May 1995
MIL-PRF-25017	Inhibitor, Corrosion / Lubricity Improver, Fuel Soluble, USAF, March 2006
MIL-DTL-38219	Turbine Fuel, Low Volatility, JP-7, USAF, August 1998
MIL-DTL-83133	Turbine Fuel, Aviation, Kerosene Type, JP-8, (NATO F-34), NATO F-35, and JP-8+100 (NATO F-37), USAF, April 2008

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-3004	Quality Assurance/Surveillance for Fuels, Lubricants, and Related Products, DLA, April 2004

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

C.2.2.2 Other Government documents, drawings, and publications.

The following other Government documents, drawings, and publications form a part of this document to the extent specified herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:45Z
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"Fuels and Lubricants Influence on Turbine Engine Design and Performance," I.E. Sumey, General Electric, AFRL-TR-73-54, November 1974

Petroleum Quality Information System (PQIS) 2005, Defense Energy Support Center, DLA, December 2006

TO 42B-1-1-1 Fuels for USAF Piston and Turbine Support Equipment and Administrative Vehicles, USAF, July 2006

TO 42B-1-1-14 Fuels for USAF Aircraft, USAF, May 2006

(Copies of these documents may be obtained from The Defense Technical Information Center Website and are available from DTIC Headquarters, 8725 John J. Kingman Rd., Ft. Belvoir VA 22060-6218; telephone (800) 225-3842. Approved users may access Technical Orders at https://www.toindex-s.wpafb.af.mil/.)

C.2.3 Non-Government publications.

The following documents form a part of this document to the extent specified herein.

ASTM INTERNATIONAL

ASTM D56 Standard Test Method for Flash Point by Tag Closed Cup Tester, American Society for Testing and Materials International, May 2005

ASTM D86 Standard Test Method for Distillation of Petroleum Products at Atmospheric Pressure, American Society for Testing and Materials International, January 2007

ASTM D93 Standard Test Methods for Flash Point by Pensky-Martens Closed Cup Tester, American Society for Testing and Materials International, November 2006

ASTM D97 Standard Test Method for Pour Point of Petroleum Products, American Society for Testing and Materials International, November 2006

ASTM D129 Standard Test Method for Sulfur in Petroleum Products (General Bomb Method)-British Standard 4454, American Society for Testing and Materials International, January 2005

ASTM D130 Standard Test Method for Corrosiveness to Copper from Petroleum Products by Copper Strip Test, American Society for Testing and Materials International, May 2005

ASTM D156 Standard Test Method for Saybolt Color of Petroleum Products (Saybolt Chromometer Method), American Society for Testing and Materials International, November 2005

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ASTM D323 Standard Test Method for Vapor Pressure of Petroleum Products (Reid Method), American Society for Testing and Materials International, August 2006

ASTM D381 Standard Test Method for Gum Content in Fuels by Jet Evaporation, American Society for Testing and Materials International, November 2004

ASTM D445 Standard Test Method for Kinematic Viscosity of Transparent and Opaque Liquids (and the Calculation of Dynamic Viscosity)-Designation: D 445–06; British Standard 2000, American Society for Testing and Materials International, May 2006

ASTM D613 Standard Test Method for Cetane Number of Diesel Fuel Oil, American Society for Testing and Materials International, May 2005

ASTM E659 Standard Test Method for Autoignition Temperature of Liquid Chemicals, American Society for Testing and Materials International, August 2005

ASTM E681 Standard Test Method for Concentration Limits of Flammability of Chemicals (Vapors and Gases), American Society for Testing and Materials International, June 2004

ASTM D924 Standard Test Method for Dissipation Factor (or Power Factor) and Relative Permittivity (Dielectric Constant) of Electrical Insulating Liquids October 2004

ASTM D971 Standard Test Method for Interfacial Tension of Oil Against Water by the Ring Method, October 2004

ASTM D976 Standard Test Method for Calculated Cetane Index of Distillate Fuels, American Society for Testing and Materials International, December 2006

ASTM D1094 Standard Test Method for Water Reaction of Aviation Fuels, American Society for Testing and Materials International, December 2005

ASTM D1266 Standard Test Method for Sulfur in Petroleum Products (Lamp Method), American Society for Testing and Materials International, February 2003.

ASTM D1298 Standard Test Method for Density, Relative Density (Specific Gravity), or API Gravity of Crude Petroleum and Liquid Petroleum Products by Hydrometer Method, American Society for Testing and Materials International, June 2005.

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ASTM D1319 Standard Test Method for Hydrocarbon Types in Liquid Petroleum Products by Fluorescent Indicator Adsorption, American Society for Testing and Materials International, November 2006.

ASTM D1322 Standard Test Method for Smoke Point of Kerosine and Aviation Turbine Fuel, American Society for Testing and Materials International, June 2002.

ASTM D1655 Standard Specification for Aviation Turbine Fuels, American Society for Testing and Materials International, December 2006

ASTM D1840 Standard Test Method for Naphthalene Hydrocarbons in Aviation Turbine Fuels by Ultraviolet Spectrophotometry, American Society for Testing and Materials International, November 2003

ASTM E2071 Standard Practice for Calculating Heat of Vaporization or Sublimation from Vapor Pressure Data, American Society for Testing and Materials International, March 2005

ASTM D2276 Standard Test Method for Particulate Contaminant in Aviation Fuel by Line Sampling, American Society for Testing and Materials International, December 2006

ASTM D2386 Standard Test Method for Freezing Point of Aviation Fuels, American Society for Testing and Materials International, January 2006

ASTM D2622 Standard Test Method for Sulfur in Petroleum Products by Wavelength Dispersive X-ray Fluorescence Spectrometry, American Society for Testing and Materials International, November 2005

ASTM D2624 Standard Test Methods for Electrical Conductivity of Aviation and Distillate Fuels, American Society for Testing and Materials International, December 2006

ASTM D2779 Standard Test Method for Estimation of Solubility of Gases in Petroleum Liquids, American Society for Testing and Materials International, October 2002

ASTM D2887 Standard Test Method for Boiling Range Distribution of Petroleum Fractions by Gas Chromatography, American Society for Testing and Materials International, December 2006

ASTM D3120 Standard Test Method for Trace Quantities of Sulfur in Light Liquid Petroleum Hydrocarbons by Oxidative Microcoulometry, American Society for Testing and Materials International, December 2006

ASTM D3227 Standard Test Method for (Thiol Mercaptan) Sulfur in Gasoline, Kerosine, Aviation Turbine, and Distillate Fuels (Potentiometric

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Method), American Society for Testing and Materials International, November 2004

ASTM D3240 Standard Test Method for Undissolved Water in Aviation Turbine Fuels, American Society for Testing and Materials International, November 2005

ASTM D3241 Standard Test Method for Thermal Oxidation Stability of Aviation Turbine Fuels (JFTOT Procedure), American Society for Testing and Materials International, December 2006

ASTM D3242 Standard Test Method for Acidity in Aviation Turbine Fuel, American Society for Testing and Materials International, November 2003

ASTM D3338 Standard Test Method for Estimation of Net Heat of Combustion of Aviation Fuels, American Society for Testing and Materials International, November 2005

ASTM D3343 Standard Test Method for Estimation of Hydrogen Content of Aviation Fuels, American Society for Testing and Materials International, November 2005

ASTM D3701 Standard Test Method for Hydrogen Content of Aviation Turbine Fuels by Low Resolution Nuclear Magnetic Resonance Spectrometry, American Society for Testing and Materials International, November 2006

ASTM D3948 Standard Test Method for Determining Water Separation Characteristics of Aviation Turbine Fuels by Portable Separometer, American Society for Testing and Materials International, December 2005

ASTM D4052 Standard Test Method for Density and Relative Density of Liquids by Digital Density Meter-X´A6´hNX´A6´hM, American Society for Testing and Materials International, April 2002

ASTM D4054 Standard Practice for Evaluating the Compatibility of Additives with Aviation-Turbine Fuels and Aircraft Fuel System Materials, American Society for Testing and Materials International, March 1993

ASTM D4294 Standard Test Method for Sulfur in Petroleum and Petroleum Products by Energy-Dispersive X-Ray Fluorescence Spectrometry, American Society for Testing and Materials International, November 2003

ASTM D4529 Standard Test Method for Estimation of Net Heat of Combustion of Aviation Fuels, American Society for Testing and Materials International, January 2006

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ASTM D4809 Standard Test Method for Heat of Combustion of Liquid Hydrocarbon Fuels by Bomb Calorimeter (Precision Method), American Society for Testing and Materials International, December 2006

ASTM D4952 Standard Test Method for Qualitative Analysis for Active Sulfur Species in Fuels and Solvents (Doctor Test), American Society for Testing and Materials International, November 2002

ASTM D5001 Standard Test Method for Measurement of Lubricity of Aviation Turbine Fuels by the Ball-On-Cylinder Lubricity Evaluator (BOCLE), American Society for Testing and Materials International, December 2006

ASTM D5006 Standard Test Method for Measurement of Fuel System Icing Inhibitors (Ether Type) in Aviation Fuels, American Society for Testing and Materials International, December 2003

ASTM D5191 Standard Test Method for Vapor Pressure of Petroleum Products (Mini Method), American Society for Testing and Materials International, May 2006

ASTM D5304 Standard Test Method for Assessing Middle Distillate Fuel Storage Stability by Oxygen Overpressure, American Society for Testing and Materials International, May 2006

ASTM D5452 Standard Test Method for Particulate Contamination in Aviation Fuels by Laboratory Filtration, American Society for Testing and Materials International, December 2006

ASTM D5453 Standard Test Method for Determination of Total Sulfur in Light Hydrocarbons, Spark Ignition Engine Fuel, Diesel Engine Fuel, and Engine Oil by Ultraviolet Fluorescence, American Society for Testing and Materials International, May 2006

ASTM D5972 Standard Test Method for Freezing Point of Aviation Fuels (Automatic Phase Transition Method), American Society for Testing and Materials International, November 2005

ASTM D6793 Standard Test Method for Determination of Isothermal Secant and Tangent Bulk Modulus, American Society for Testing and Materials International, April 2002

ASTM D7111 Standard Test Method for Determination of Trace Elements in Middle Distillate Fuels by Inductively Coupled Plasma Atomic Emission Spectrometry (ICP-AES), American Society for Testing and Materials International, April 2004

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(Copies of these documents may be ordered online at www.astm.org; and from ASTM International, 100 Barr Harbor Drive, PO Box C700, West Conshohocken PA 19428-2959; (610) 832-9585. Approved users may access the documents online at www.ihs.com.)

"*Data Book on Hydrocarbons, Application to Process Engineering*," J. B. Maxwell, University of Michigan, 1950

"Effects of JP-8+100 Fuel on Fuel Gauging System Performance," BF Goodrich Aerospace, Military Fuels and Integrated Systems, Cage code: 89305, 1995

"Handbook of Aviation Fuel Properties, 2004, Third Edition," Report No. 635, Coordinating Research Council, Inc. (CRC)

"Military Jet Fuels," C.R. Martel, AFWAL-TR-87-2062, USAF, November 1987

"Properties of Aircraft Fuels," Henry C. Barnett and Robert R. Hibbard, National Advisory Committee for Aeronautics (NACA), NACA 3276, August 1956.

"World Fuel Sampling Program," Hadaller, O., Johnson, J., Coordinating Research Council Report 647, June 2006.

Defence Standard 91-91: Turbine Fuels, Aviation Kerosine Type, Jet A-1, NATO Code: F-35, Joint Service Designation: AVTUR, Ministry of Defence, March 2006

ISO-20823 Petroleum and related products -- Determination of the flammability characteristics of fluids in contact with hot surfaces Manifold ignition test-First Edition, International Organization for Standardization, August 2003

(Copies of these document are available online at The Defense Technical Information Center Website and are available from DTIC Headquarters, 8725 John J. Kingman Rd., Ft. Belvoir VA 22060-6218; telephone (800) 225-3842.)

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C.3 BACKGROUND.

There is a long list of fuel properties that are relevant to the operation of Air Force systems. These properties are discussed in detail in this Appendix. The MIL-DTL-83133F specification controls a relatively small number of properties and describes permitted additives. A number of these properties directly affect Air Force mission performance, such as density and heat of combustion. Some properties address operability (freeze point, low temperature, viscosity), safety (flash point, conductivity) and combustion behavior (hydrogen content, aromatics). Many properties in the specification are included to ensure that the fuel has not become contaminated (particulates, existent gum, etc.). In general, the properties included in the specification are assessed by fairly simple, standardized "quality control" type of tests. Ensuring that a fuel is "fit for purpose" might require other, more complex tests. These types of tests are also included in this Appendix as well as other Appendices.

The fuel specification typically allows a range of property values, or specifies a maximum or minimum property value. Thus the specification provides the boundaries of a "box" that bounds the fuel composition. Kerosene fuels contain hundreds, if not thousands, of hydrocarbons - with the mixture varying with crude source and processing conditions. Thus, describing fuel properties becomes a statistical exercise. Fuel specification test results are compiled by the DESC in the PQIS database. Summary reports are available on-line at http://www.desc.dla.mil). Available statistical data on properties are discussed in this Appendix and referenced as "PQIS database". An excellent source of fuel property data is the CRC Report 635, Handbook of Aviation Fuel Properties. This report is typically referenced in this Appendix as "CRC, 2004". A third source of data referenced in this Appendix is CRC Report 647, "World Fuel Sampling Program, June 2006".

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C.4 ACID NUMBER, TOTAL -- ENTRANCE CRITERIA.

C.4.1 Definition.

Acid number is a measure of the acidity of a fuel. It is the amount of potassium hydroxide (KOH) in milligrams required to neutralize (pH = 7) one gram of a substance. High acid numbers are not desirable in fuels. The higher the acid number, the greater the acidic property of the fuel which would correspond to a greater affinity of the fuel to negatively react with components.

C.4.2 Standard Acid Number Test Methods.

C.4.2.1 Acidity by ASTM D3242.

The fuel sample is dissolved in a mixture of toluene and isopropyl alcohol containing a small amount of water. The resulting single phase solution is blanketed by a stream of nitrogen bubbling through it and is titrated with alcoholic potassium hydroxide to an end point indicated by the color change of p-naphtholbenzene. Results are reported in mg KOH/g.*

C.4.3 Comments.

Acid level is related to how aggressively the fuel degrades components, hoses and nozzles in aircraft and logistical supply chain. Therefore it is best suited for a Subset 2 test which addresses durability. However, more research should be carried out to investigate possible safety links which would move it into a Subset 1 test. In the JP-8+100LT program in AFRL/RZTG, it was found that a JP-8 additive that improved low-temperature fuel flowability caused the fuel to fail the acid number test. However, the fuel still passes the Jet Fuel Thermal Oxidative Test (JFTOT) thermal stability test and showed no problems during combustion testing, so it was concluded that the acid number result was an artifact in this case. This test measures both strong and weak acids, so it has been suggested that the test might be modified to differentiate between the two, on the assumption that the "bad actors" would be the more-strongly-acidic species. Note that the Jet A specification requirement (<0.1 mg KOH/g) is different from the MIL-DTL-83133F specification (<0.015 mg KOH/g), although it is rare to find a Jet A fuel that exceeds 0.015 mg KOH/g. In a recent test program evaluating material response to a fuel containing the maximum level permitted in Jet A, an existing low-acid fuel was spiked with 133 mg/L m-cresol and 267 mg/L cyclohexylbutyric acid. (400 mg/L total) to create a 0.1 mg KOH/g fuel. The testing found that the metallic and non-metallic materials on the short list (Appendix F) were unaffected by the 0.1 acid number fuel.

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C.4.4 Data / Property Occurrence.

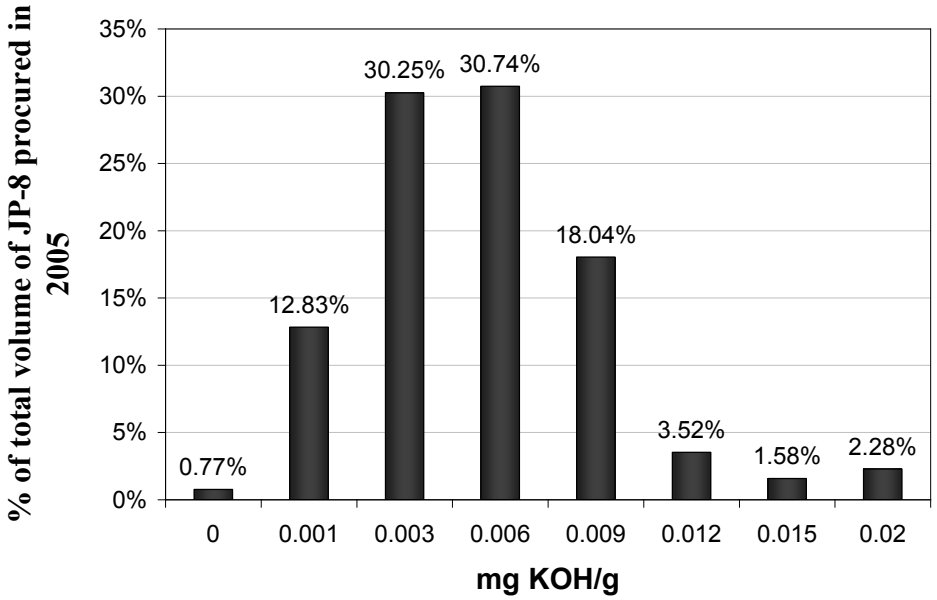


FIGURE C-1. Total acid histogram JP-8 (2005) ***Courtesy of PQIS.***

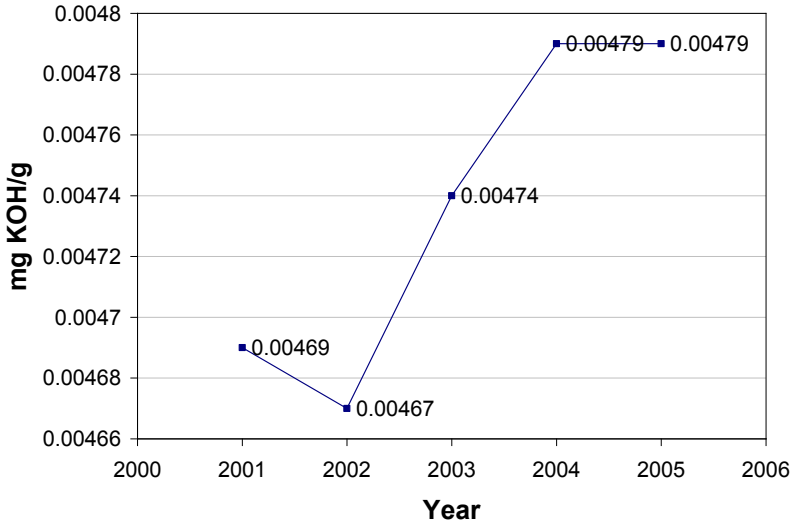


FIGURE C-2. Total acid average JP-8 (year trend 2001-2005) ***Courtesy PQIS 2005.***

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TABLE C-I. Acid number, total fuel property rating value.

	New Fuel
R	>0.10
Y	0.015 to 0.10
G	0 to 0.015

TABLE C-II. Reserved.

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C.5 ADDITIVE COMPATIBILITY – SUBSET 1.

C.5.1 Definition.

To increase certain performance characteristics of the fuel, or to mitigate certain risks possible with military missions using typical jet fuel, the USAF has certified use of various additives for its fuel. Any new fuel or additive that would be introduced into a USAF system needs to have the compatibility of that fuel / additive candidate tested with the currently used additives to ensure both physical compatibility (miscibility) as well as equal or better performance than baseline within the fuel.

C.5.2 Standard Additive Compatibility Test Methods.

C.5.2.1 Additive Compatibility by ASTM D4054: Tests are conducted with Jet A/A-1 fuel at four times the maximum additive concentration as recommended by the additive supplier. The samples are then split into two equal parts and both samples are placed into dark cold storage (0°C) for 24 hours and then inspected for indications of precipitation, cloudiness, darkening, or other visual evidence of incompatibility. The samples are then warmed, remixed, and held at 38°C for 24 hours and then inspected again for the same evidence of incompatibility as the cooled samples. Similar tests are preformed for additive to additive comparisons as well.

C.5.3 Comments.

C.5.4 Data / Property Occurrence.

TABLE C-III. Additive compatibility fuel property rating value.

	New Fuel
G	Not miscible
R	Miscible

TABLE C-IV. Reserved.

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C.6 AROMATICS – ENTRANCE CRITERIA.

C.6.1 Definition. Aromatics are a class of hydrocarbons that are typically visualized as containing a six-carbon ring with alternating single and double bonds between the carbon atoms. Since aromatics are such a broad group of hydrocarbons, there exists a large variety of aromatics contained within traditional petroleum based jet fuels.

C.6.2 Standard Aromatics Test Methods.

C.6.2.1 Aromatic Content by ASTM D1319. Fuel is percolated through a column of silica gel containing special fluorescent dyes. When desorbed by alcohol, the fuel separates into three layers (olefins, aromatics and saturates (paraffins)) of differing hydrocarbon types which become visible under ultraviolet light. The relative length of each band is translated into the volume percent of each hydrocarbon type.

C.6.3 Comments. Aromatics subset placement needs some further research. If the functional decomposition indicates a large number of safety related items then aromatics would be placed in Subset 1 testing otherwise Subset 2. The minimum value needs to be evaluated. Ground support equipment placed an upper limit at 35%, consistent with operation with diesel fuel. The jet fuel aromatic specification AN-F-58a (JP-3) lowered the aromatic level to 25 vol% in March 1949, where it has remained. This was done to "control carbon-forming tendency" (presumably soot) [Barnett and Hibbard, 1956]. The lower limit is still unclear, although the Jet A-1 specification cites 8% in an appendix dealing with synthetic fuels (Def-Stan 91-91, Annex D). It has been suggested that continuing research on levels with <8% aromatics would be advisable, since millions of gallons of JP-8 are burned each year with <8% aromatics, according to the PQIS database.

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C.6.4 Data /Property Occurrence.

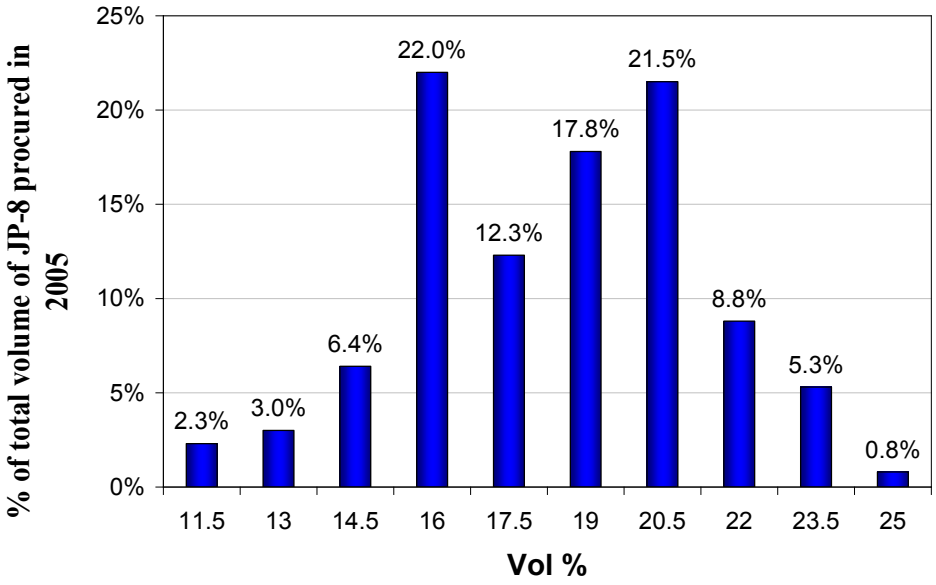


FIGURE C-3. Aromatic content histogram JP-8 (2005) ***Courtesy of PQIS 2005.***

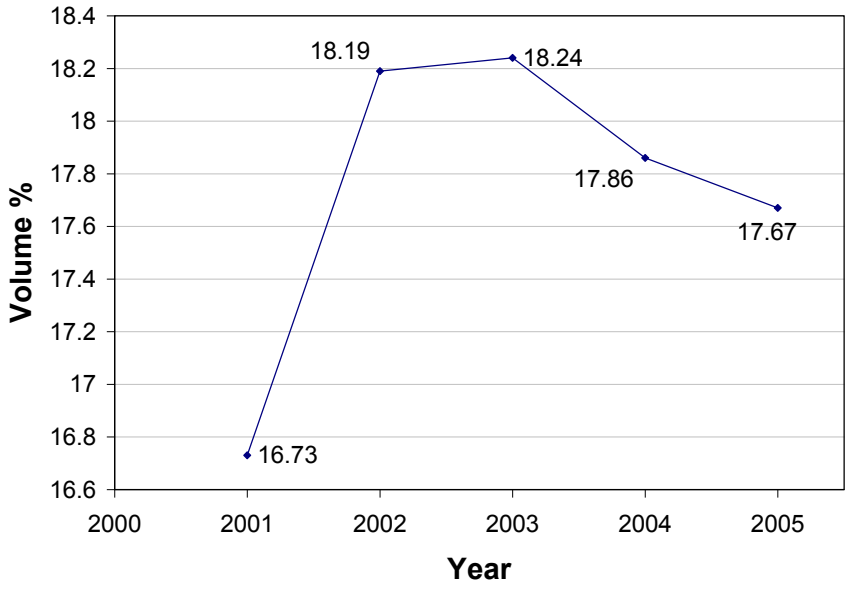


FIGURE C-4. Aromatic content average JP-8 (year trend 2001-2005) ***Courtesy PQIS 2005.***

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TABLE C-V. Aromatics fuel property rating value.

	New Fuel
R	>25%
G	8% to 25%
Y	0% to 8%

TABLE C-VI. Reserved.

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C.7 AUTOIGNITION TEMPERATURE – SUBSET 1.

C.7.1 Definition. Autoignition temperature is also referred to as spontaneous ignition temperature, self-ignition temperature, and by the acronyms AIT and SIT. AIT is the lowest temperature to which a combustible mixture has to be raised, so that the rate of heat evolved by the exothermic oxidation reaction will overbalance the rate at which heat is lost to the surroundings and cause ignition. The autoignition temperature of a jet fuel is not well understood or firmly fixed because of the number of factors that affect it. For example:

"The auto-ignition temperature of fuels will vary because of a variety of factors (ambient pressure, dwell time, fuel type, etc.), but the value generally accepted without further substantiation for kerosene type fuels, such as Jet A, under static sea level conditions, is 450°F." [FAA Advisory Circular 25.981]

"Comparisons of autoignition temperatures for hydrocarbon fuels … indicate that minimum AIT temperatures range between 400-500°F ... It should be noted there is no unique threshold temperature for hot-surface ignition since it is influenced by numerous factors, such as geometry of surface—whether concave or convex, whether in a closed environment or open environment such as the hot-manifold test, local air velocities, and residence time of the fluid." [DOT/FAA/AR-98/26].

The CRC Handbook shows values for autoignition temperature for JP-4,5,7,8 that range from 238-246°C (460-475°F) [also Zabetakis, 1965]. This indicates that the autoignition temperature is a fairly weak function of fuel composition, so the lower limit 200°C (392°F) is adopted as a conservative limit.

C.7.2 Standard Autoignition Temperature Test Methods.

C.7.2.1 ASTM E659. A small, metered sample of the product to be tested is inserted into a uniformly heated 500-ml glass flask containing air at a predetermined temperature. The contents of the flask are observed in a dark room for 10 min following insertion of the sample, or until autoignition occurs. Autoignition is evidenced by the sudden appearance of a flame inside the flask and by a sharp rise in the temperature of the gas mixture. The lowest internal flask temperature (T) at which hot-flame ignition occurs for a series of prescribed sample volumes is taken to be the hot-flame AIT of the chemical in air at atmospheric pressure. Ignition delay times (ignition time lags) are measured in order to determine the ignition delay-ignition temperature relationship.*

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C.7.3 Data / Property Occurrence.

TABLE C-VII. Autoignition temperature fuel property rating value.

	New Fuel
Y	>260°C
G	200 to 260°C
Y	<200°C

TABLE C-VIII. Reserved.

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C.8 BULK MODULUS – SUBSET 1.

C.8.1 Definition. The bulk modulus of a substance measures that substance's resistance to uniform compression. It is defined as the pressure increase needed to affect a given relative decrease in volume. A fluid with a high bulk modulus shows a small change in volume for a given change in pressure and is, therefore, difficult to compress. Applying this principle, a fuel used to activate hydraulic equipment should have a high bulk modulus of the fuel to make a responsive system.

C.8.2 Standard Bulk Modulus Test Methods.

C.8.2.1 Bulk Modulus by ASTM D6793. Bulk moduli can be calculated from P–V–T measurements or determined directly from ultrasonic velocity measurements. The P–V–T measurements method is used primarily in isothermal conditions. Since it uses actual measured quantities rather than their derivatives, the ultrasonic velocity method is more accurate and yields the adiabatic bulk modulus directly. Figures C-5 and C-6 (JP-5, Jet A, Jet A-1, and JP-8) (CRC, 2004) show the adiabatic bulk moduli versus pressure and temperature. The bulk modulus of JP-4 is typically slightly less (~50 MPa) than that of JP-5/JP-8.

C.8.3 Comments. Bulk modulus impacts the velocity that pressure changes propagate through the fuel. A deviation from the baseline fuel for which the injection system was calibrated will significantly change the fuel injection timing and shape. Acceptable limits for a test fuel and the different types of injection equipment are not known at this time and will be the subject of further research.

The relationship between Bulk Modulus and Velocity of Sound is given by the following equations:

Velocity of Sound = Square Root of (Bulk Modulus / Density),

or

Bulk Modulus = Density x (Velocity of Sound Squared).

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C.8.3 Data / Property Occurrence.

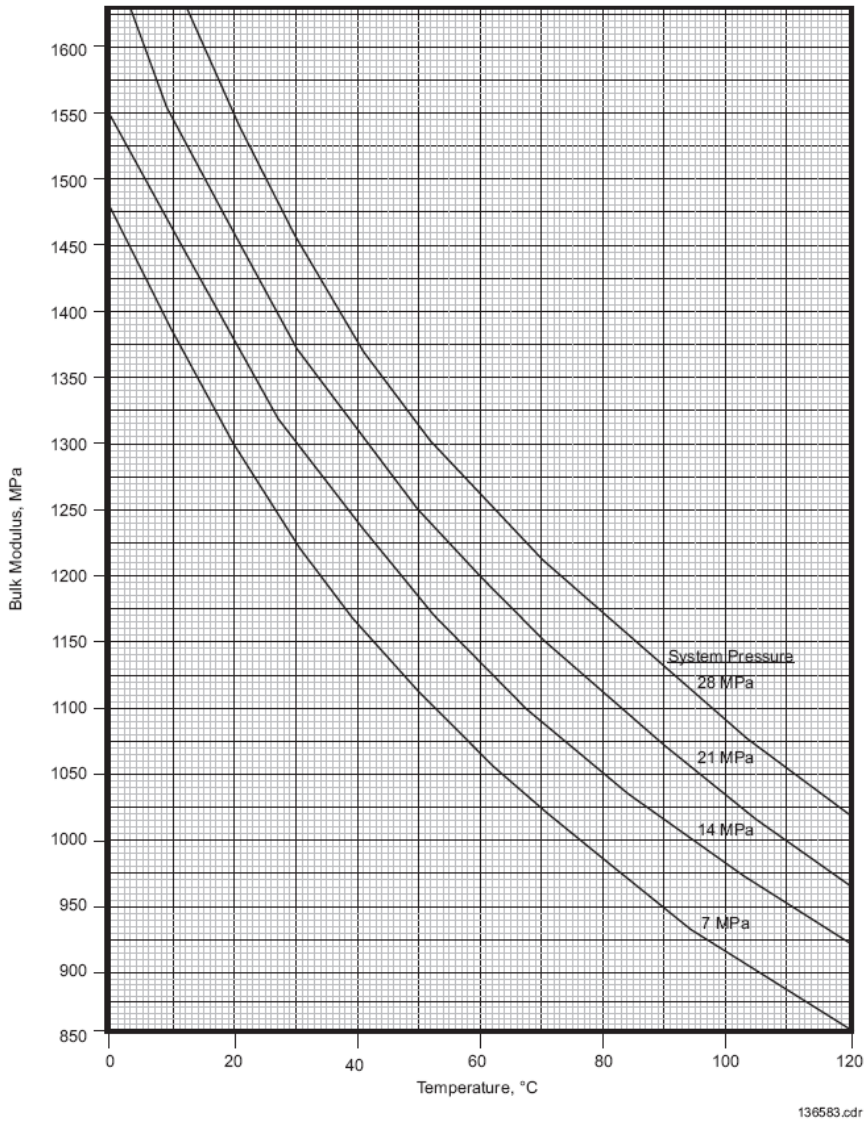


FIGURE C-5. Bulk modulus as a function of fuel temperature for JP-5/JP-8/Jet A/Jet A-1 ***[CRC 2004].***

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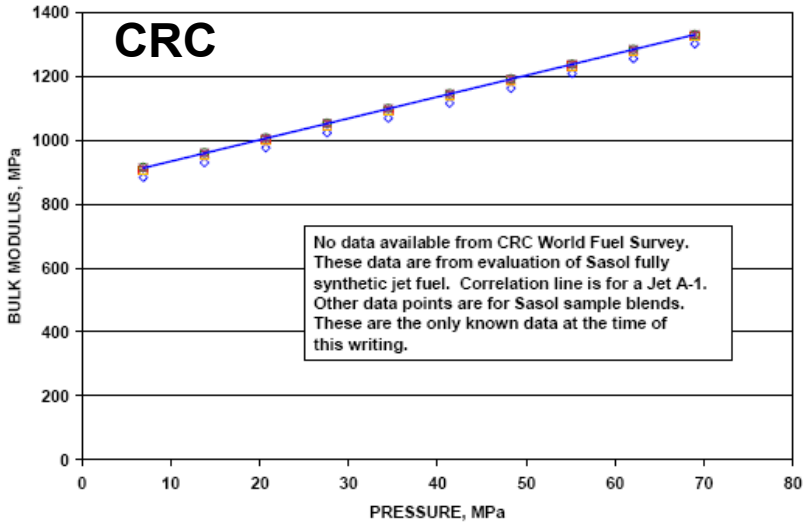


FIGURE C-6. Bulk modulus as a function of fuel pressure for JP-5/JP-8/Jet A/Jet A-1.

TABLE C-IX. Bulk modulus fuel property rating value.

	New Fuel
Y	>JP-8/JP-5 values
G	JP-8/JP-5 values
Y	<JP-8/JP-5 values

TABLE C-X. Reserved.

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C.9 CALCULATED CETANE INDEX – ENTRANCE CRITERIA.

C.9.1 Definition. The Calculated cetane index is a mathematical estimation of cetane number. Cetane number is the measure of a fuel's ignition delay; the time period between the start of injection and start of combustion (ignition) of the fuel.

C.9.2 Cetane Index Test Methods.

C.9.2.1 Calculated Cetane Index ASTM D976. The Calculated cetane index is determined from the following equations:*

$$\text{Calculated cetane index} = -420.34 + 0.016G^2 + 0.192G \log M + 65.01(\log M)^2 - 0.0001809 M^2$$

or

$$\text{Calculated cetane index} = 454.74 - 1641.416D + 774.74D^2 - 0.554B + 97.803(\log B)^2$$

Where: G = API gravity

M = mid-boiling temperature, °F

D = density at 15°C, g/mL

B = mid-boiling temperature, °C

C.9.3 Comments. Cetane number is an experimental measurement relevant to the operation of diesel engines. The cetane index is a calculation that provides an estimate of the cetane number. For fuels outside of the validation range (i.e., petroleum based) of the cetane index correlation, the index may not provide a realistic estimate of the cetane number. Baseline cetane values established in this section are based on historical performance of fuels. Low cetane values can cause operational problems in diesel engines. It has also been seen that as the cetane number begins to increase past 65, performance impacts are observed due to combustion timing effects in diesel engines.

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C.9.4 Data / Property Occurrence.

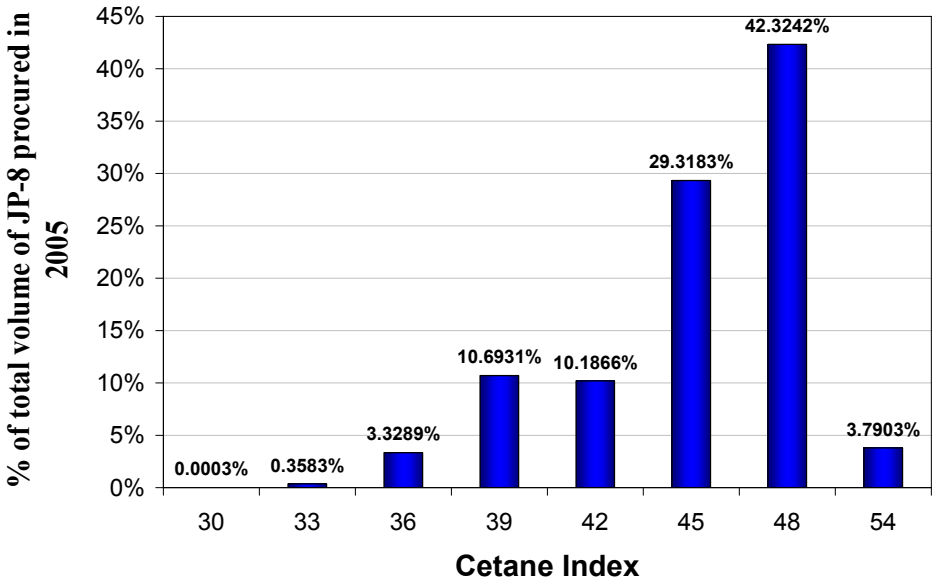


FIGURE C-7. Cetane index histogram JP-8 (2005) ***Courtesy of PQIS.***

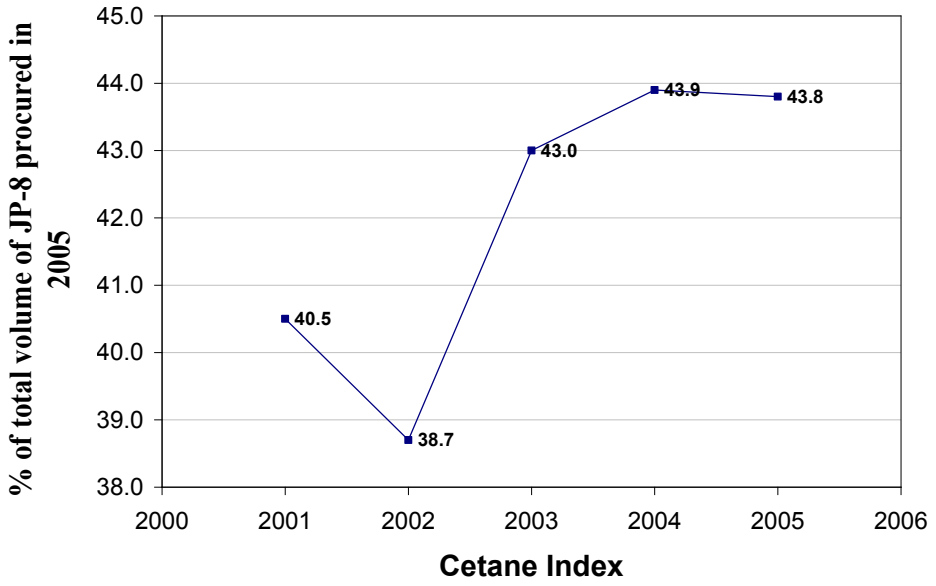


FIGURE C-8. Cetane index average JP-8 (year trend 2001-2005) ***Courtesy PQIS 2005.***

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TABLE C-XI. Calculated cetane index fuel property rating value.

	New Fuel
Y	>65
G	40 to 65
Y	37 to 39.9

TABLE C-XII. Reserved.

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C.10 CETANE NUMBER – SUBSET 2.

C.10.1 Definition.

Cetane number is a measure of the ignition performance of a diesel fuel oil obtained by comparing it to reference fuels in a standardized engine test.

C.10.2 Standard Test Method.

C.10.2.1 ASTM D613.

The cetane number of a diesel fuel oil is determined by comparing its combustion characteristics in a test engine with those for blends of reference fuels of known cetane number under standard operating conditions. This is accomplished using the bracketing handwheel procedure which varies the compression ratio (handwheel reading) for the sample and each of two bracketing reference fuels to obtain a specific ignition delay permitting interpolation of cetane number in terms of handwheel reading.*

C.10.3 Comments. Cetane number is a measure of the self ignition and ignition delay characteristics of the fuel. Very low cetane numbers, below 37, will result in difficult cold starting, cold smoke, and reduced life for most diesel engines and immediate structural failure of others. The entrance requirement specifies the calculated cetane index rather than the more conclusive cetane number, as determined in a special test engine. This was done for cost and time considerations. In Subset 1 the actual cetane number test is to be done. Interestingly, Zabetakis (1965) shows data that autoignition temperature was relatively constant at about 230°C for 4 diesel fuels that ranged from 41 to 68 cetane. Thus it appears that cetane number and autoignition temperature are not interchangeable tests for the purpose of ensuring fuel ignition properties remain within the experience base.

C.10.4 Data / Property Occurrence.

TABLE C-XIII. Cetane number fuel property rating value.

	New Fuel
Y	>65
G	40 to 65
Y	37 to 39.9

TABLE C-XIV. Reserved.

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C.11 COPPER STRIP CORROSION – ENTRANCE CRITERIA.

C.11.1 Definition. An ASTM test method used to determine the corrosiveness to copper of liquid hydrocarbons that have a vapor pressure no greater than 124 kPa (18 psi) at 37.8°C.

C.11.2 Standard Copper Strip Corrosion Test Methods.

C.11.2.1 Corrosivity of Copper by ASTM D130. A polished strip of copper is immersed into 30 mL of test fuel and placed into a pressure vessel where it is heated for 2 hours at 100°C. After washing, the strip appearance is compared to the ASTM copper strip corrosion standard for rating. Corrosion ratings range from 1 (no significant corrosion) to 4 (corrosion) with each numbered category having two or more lettered subdivisions that describe the copper strip after the test has been completed; consequently, any letter designation coupled with a number only provides extra information pertaining to the test. For example, a test result of "1a" corresponds to a description of "no significant corrosion with the copper strip exhibiting light orange color, almost the same as a freshly polished strip."

C.11.3 Comments. Copper is typically avoided in aircraft fuel systems, so this test is not designed to directly assess fuel system corrosion, but rather the tendency of a fuel to corrode metals in general.

C.11.4 Data / Property Occurrence.

TABLE C-XV. Copper strip corrosion fuel property rating value.

	New Fuel
G	1 (with any letter designation)
R	2, 3, or 4 (with any letter designation)

TABLE C-XVI. Reserved.

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C.12 DENSITY – ENTRANCE CRITERIA (versus Temperature – Subset 1, Thermal Expansion – Subset 2).

C.12.1 Definition. Density is the mass per unit volume relationship of fuels. Generally, specific gravity (also known as, " relative density") is the ratio of the fuel density to the density of water at 15.5 °C.

Thermal expansion is the change in volume of a fluid corresponding to an increase or decrease in the fluid temperature.

C.12.2 Standard Density Test Methods.

C.12.2.1 Density by Hydrometer by ASTM D1298. A hydrometer is floated in fuel in a cylinder or graduate and is spun to avoid wall contact. After the hydrometer comes to rest, fuel density is read on the scale in the hydrometer at the top of the fuel level. Simultaneously the fuel temperature is measured with a thermometer. Both density and temperature are reported. If desired, the density can be corrected to a standard temperature. Results are reported in kg/m^3.

C.12.2.2 Density by Digital Density Meter by ASTM D4052. A small volume of liquid sample is introduced into an oscillating sample tube. The oscillating frequency is established using water or other calibrating liquid. The change in the frequency caused by the change in the tube mass while testing is compared to the calibration data and determines the density of the sample. Results are reported in kg/m^3 and are measured at -20°C, 20°C, 40°C, and 60°C.*

C.12.2.3 No standard test method for Thermal Expansion. Thermal expansion is a derived property from temperature and density data. A temperature has to be established in order to set a reference volume and the corresponding volume change. The volume change may be reported as a multiple of volume with the reference temperature being 1.0. It may also be reported as a percent change. For example, MIL-DTL-83133F fuel expands about 5% as it is heated from 16°C (60°F) to 60°C (140°F).

C.12.3 Comments. Density at 15°C is an Entrance Criterion, and Density versus Temperature is a Subset 1 test because of its ties to numerous systems and subsystems including safety. Many fuel properties scale directly with density, such as dielectric constant and heat capacity. There are several groups researching both the lower and upper limits of density. There is an ongoing study by the CRC to research the history of the density limits in the jet fuel specifications. The JP-4 specification had a lower limit for specific gravity of 0.751. Often the density/specific gravity (SG) is reported in the units of "API gravity" (conversions listed below). Additionally, fuel density is roughly linear with temperature over typical operating temperature ranges. The specification density range for JP-8/Jet A/Jet A-1 is 775-840 kg/m3; the JP-5 specification range is 788-845 kg/m3. Figure C-10 displays the average density of JP-8 over a recent five-year period. Typical density data as a function of termperature is then shown on Figure C-11 [CRC, 1983].

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Conversions:

API gravity = (141.5/SG at 60 °F) - 131.5

SG at 60 °F = 141.5/(API gravity + 131.5)

In developing the jet fuel density versus temperature limits, a decrease of 731 micrograms per milliliter for every increase of 1 degree Celsius was selected for the slope of the graphs. This limitation was based on data from the World Fuel Sampling Program, CRC Report 647. A line with this slope was then drawn from the 15°C limitations to establish the usable limitations. The density values are valid from -40°C to 100°C, outside of this range no limitations have set.

With regards to thermal expansion, excess expansion can result in fuel tanks overflowing and spilling on the ground when fuel is heated by solar heating. The fact that the density-versus-temperature of different fuels may be parallel does not imply that the coefficient of thermal expansion is the same - the volume expansion is inversely related to density. Thus, lower density fuels like JP-4 have a higher coefficient of thermal expansion. Fuels that fall within the JP-4 to JP-5 range of densities will fall within the historical experience base for coefficient of thermal expansion. The thermal expansion figure below shows that MIL-DTL-83133F fuel expands about 5% as it is heated from 16°C (60°F) to 60°C (140°F). An upper thermal expansion boundary of JP-4 was used to help account for fuel tank overflowing. Little data exists on concerns for small thermal expansions therefore no limit was set.

C.12.4 Data / property occurrence @ 15°C.

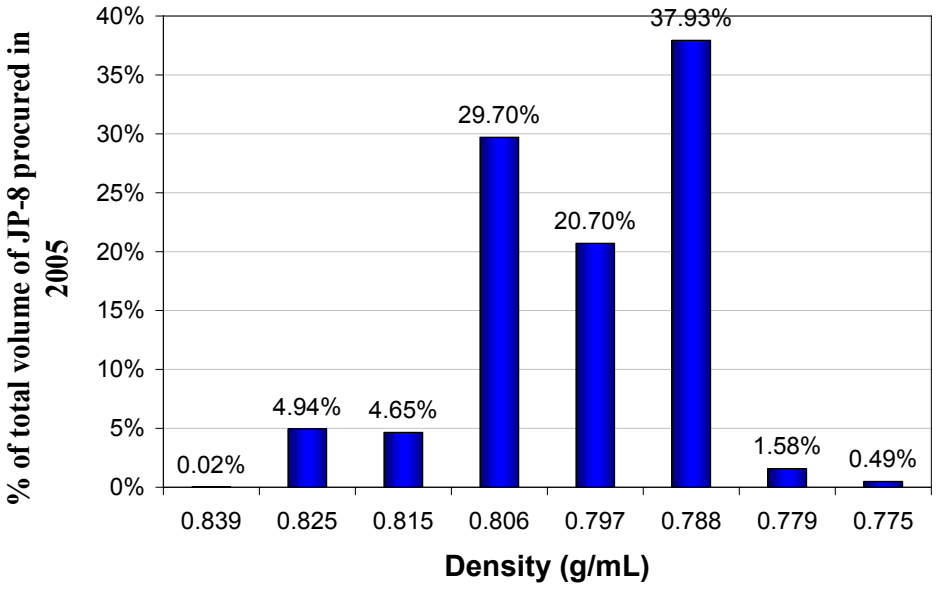


FIGURE C-9. Density histogram JP-8 (2005) ***Courtesy PQIS 2005.***

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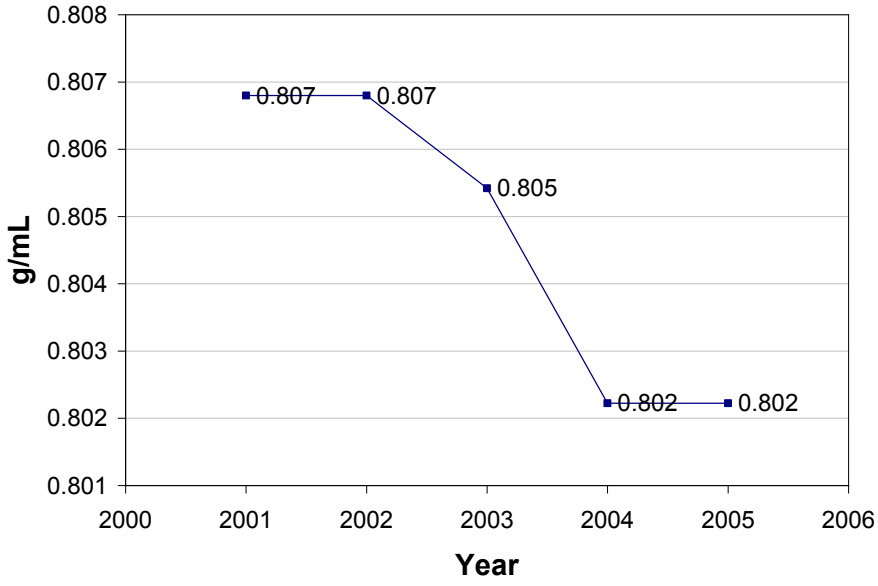


FIGURE C-10. Density average JP-8 (year trend 2001-2005) ***Courtesy PQIS 2005.***

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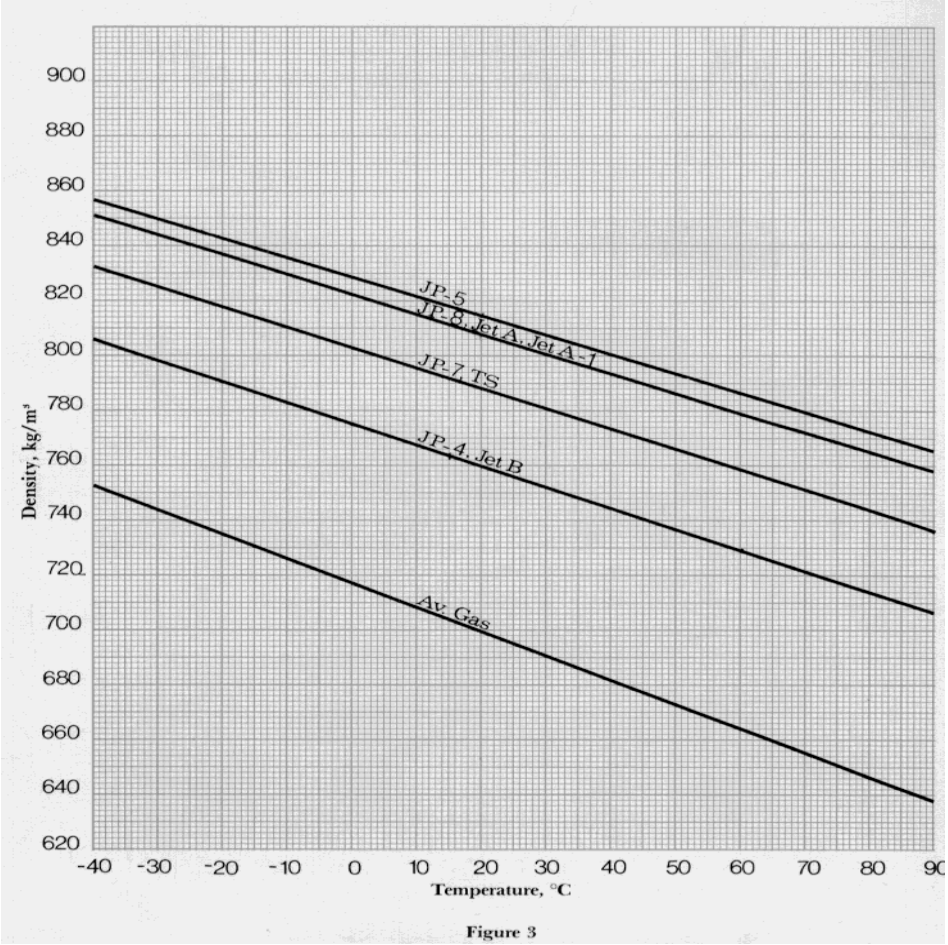


FIGURE C-11. Typical density as a function of temperature *[CRC, 1983].*

TABLE C-XVII. Density fuel property rating value at 15°C.

	New Fuel
R	>0.845 g/mL
Y	0.84 to 0.845 g/mL
G	0.775 to 0.839 g/mL
Y	<0.774 g/mL

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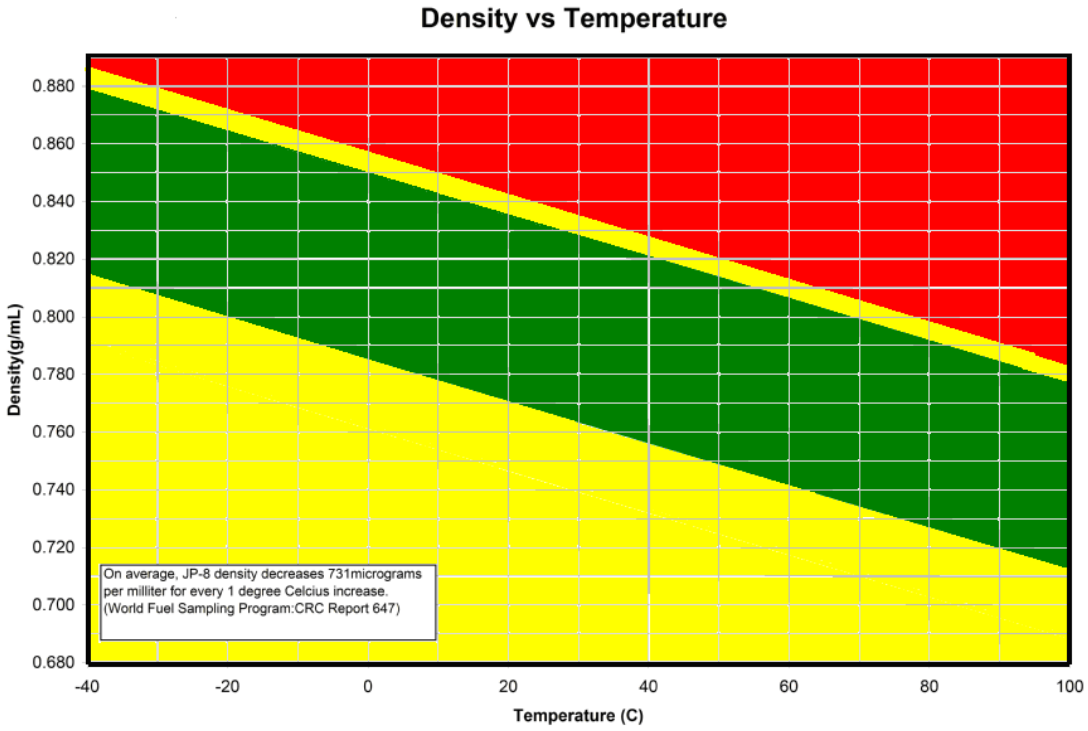


FIGURE C-12. Density versus temperature limitations.

TABLE C-XVIII. Density fuel property rating value at temperature, *T*.

Density versus Temperature	
R	Greater than [(-0.000731×T)+0.8560], g/mL
Y	Greater than Green and Less than [(-0.000731×T)+0.8560], g/mL
G	Less than [(-0.000731×T)+0.85, g/mL] and
G	Greater than [(-0.000731×T)+0.7859], g/mL
Y	Less than Green and Greater than [(-0.000731×T)+0.7620], g/mL

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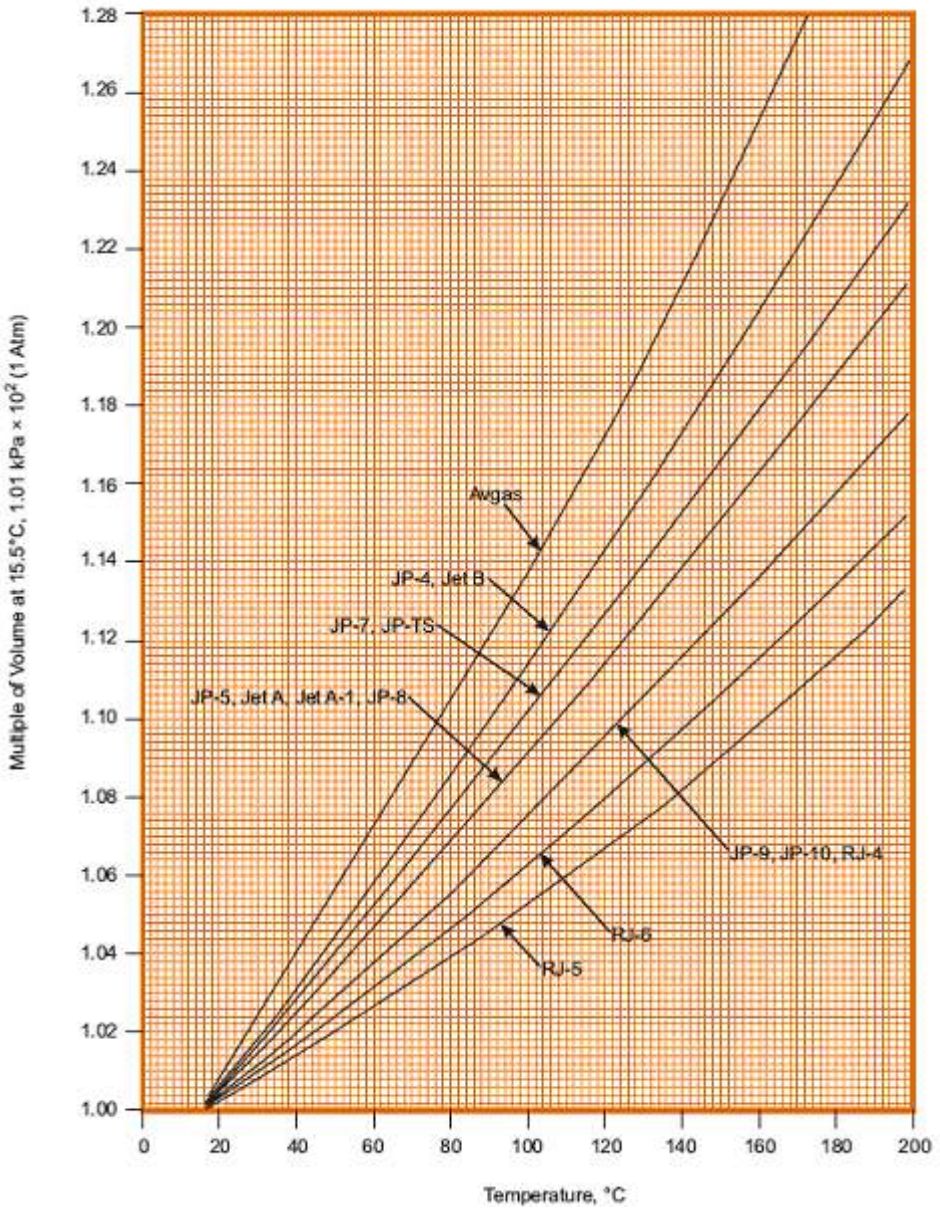


FIGURE C-13. Thermal expansion [CRC 2004].

TABLE C-XIX. Thermal expansion fuel property rating.

	New Fuel
Y	> JP-4
G	<JP-4

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C.13 DIELECTRIC CONSTANT – SUBSET 1 (versus Density, versus Temperature).

C.13.1 Definition. The dielectric constant of a fuel is the ratio of the electrical capacitance of a fuel to the electrical capacitance of air.

C.13.2 Standard Dielectric Constant Test Methods.

C.13.2.1 Dielectric Constant by ASTM D924. The electrical conductance of an air filled vessel is measured at a set frequency and temperature. The vessel is then filled with a fuel. The conductance of the fuel is then measured. The ratio of fuel conductance to air conductance is reported as the dielectric constant at that temperature and frequency.

C.13.3 Comments. Measurements of the dielectric constant of a fuel at various temperatures show that the dielectric constant is a linear function of temperature, decreasing with increasing temperature and varying with the applied frequency.

Evidence indicates that fuel dielectric constant can be scaled by density (Figure C-12), resulting is similar values for all jet fuels within the apparent accuracy of the technique. This data set includes isoparaffinic kerosene, so it appears that ensuring that fuel density range falls within the experience base will ensure that the dielectric constant also falls within the experience base. Changes in the electric frequency result in different values of dielectric constant; presently, there are no methods to correlate between dielectric constants observed at different electrical frequencies. Therefore only dielectric constants taken at the same frequency can be properly compared.

Figure C-11 (CRC, 2004) illustrates the relationship of dielectric constant versus temperature of various fuels for data collected at 400 Hz, an often-used frequency in commercial and military aircraft.

Dielectric constant data are recorded versus temperature, but usually presented as versus density. For evaluation purposes, the differences shown in dielectric constant versus temperature provide a better means for differentiating between acceptable and unacceptable fuels.

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C.13.4 Data / Property Occurrence.

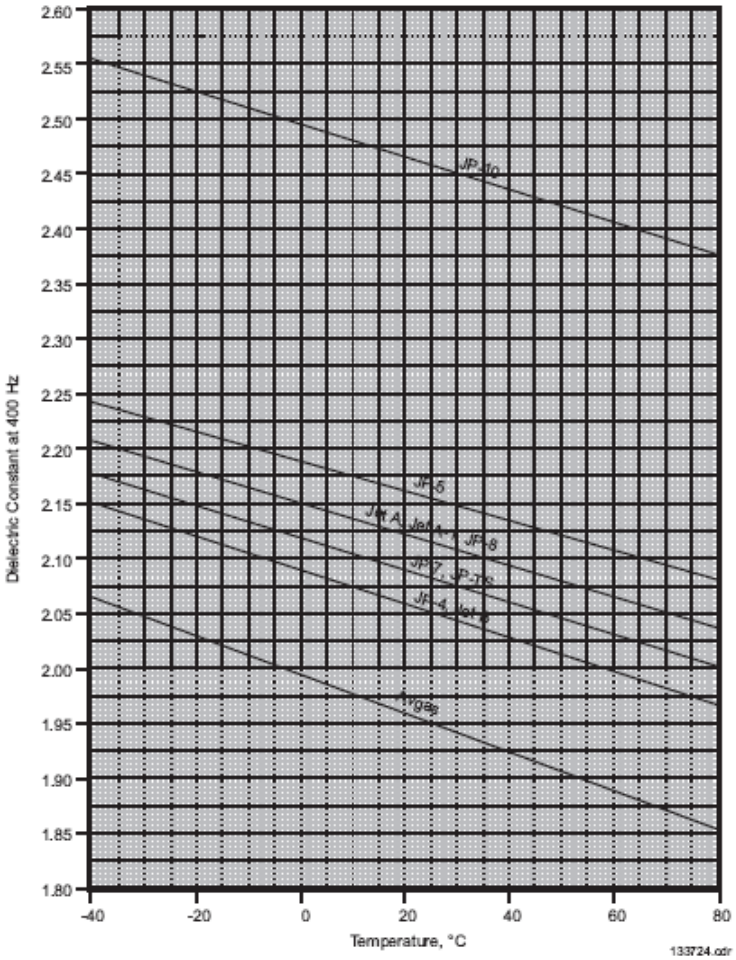


FIGURE C-14. Dielectric constant as a function of temperature ***[CRC, 2004].***

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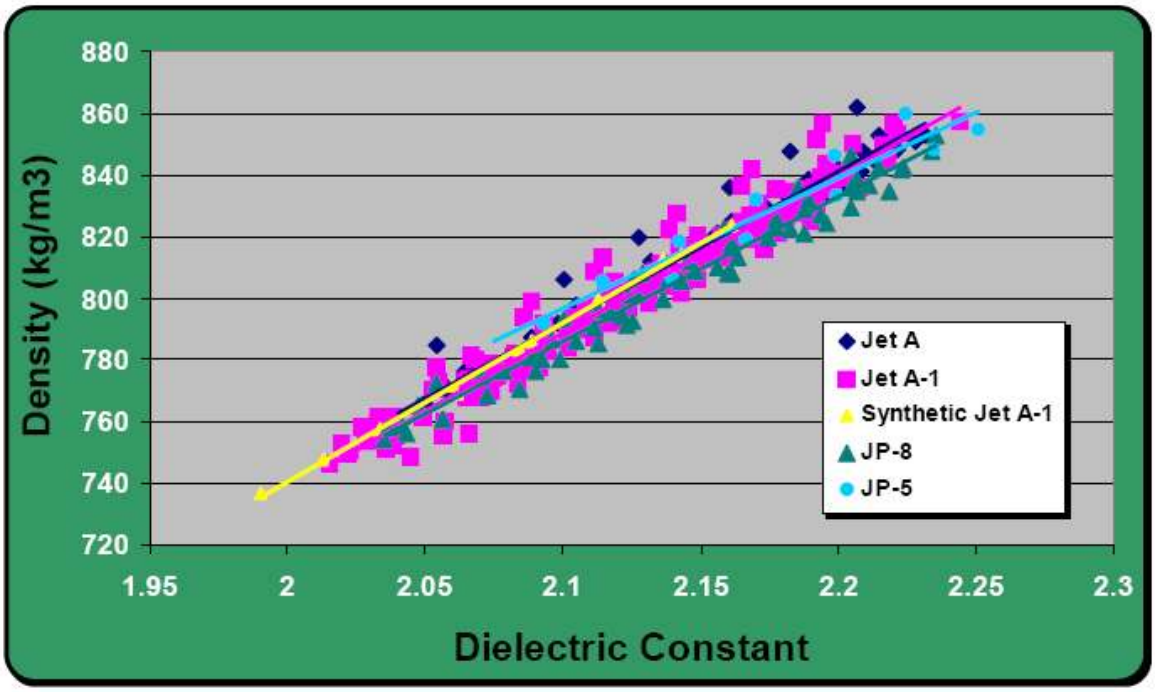


FIGURE C-15. Dielectric constant as a function of density ***[CRC World Fuel Survey, 2006].***

TABLE C-XX. Dielectric constant (as a function of T) fuel property rating value.

	New Fuel
Y	>JP-5 values
G	JP-8/JP-5 values
Y	<JP-4 values

TABLE C-XXI. Reserved.

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C.14 DISTILLATION CURVE – ENTRANCE CRITERIA.

C.14.1 Definition. Distillation curves are the range of temperatures over which a fuel boils at 1 bar. Typically, jet fuels are composed of a mixture of varying lengths of hydrocarbon chains causing a particular range of boiling temperatures. Lighter compounds (i.e. shorter hydrocarbon chains) boil initially and heavier compounds (i.e. longer hydrocarbon chains) boil later in the process. Varying the fractional composition and distribution of these hydrocarbon chains can affect the final product.

C.14.2 Standard Distillation Test Methods.

C.14.2.1 Distillation of Petroleum Products by ASTM D86. Fuel is heated at a constant rate in a flask and the vapors are boiled off. The vapors are condensed and collected. The distillation curve is the relationship between the percent of condensed vapor and the vapor temperature. Results are reported in ºC per percent recovered.

C.14.2.2 Distillation (Simulated) by ASTM D2887. Fuel is passed through a chromatographic column which separates hydrocarbons in boiling point order. Boiling temperatures are assigned from a calibration curve, obtained by running a known mixture of hydrocarbons under the same conditions, and a boiling point distribution is obtained. Results are reported in ºC at per percent recovered.

C.14.3 Comments. The current MIL-DTL-83133F specification is bounded only at the 10% (<205°C by D86) and final (<300°C) points, with the other points (initial, 20, 50, and 90%) listed as "report". The 10% point is closely correlated to the flash point. In contrast, the JP-4 specification evolved into specifying the 20%, 50%, and endpoint, primarily to prevent the use of low boiling materials which could be lost in storage [Barnett and Hibbard, 1956]. The possible introduction of narrow-boiling synthetic fuels has led to the discussion of additional definition in the distillation specification. A suggestion is presented below:

TABLE C-XXII. Suggested distillation specification requirement.

	Min	Max
Distillation temperature, °C*		
Initial boiling point	157	205
10 percent recovered (T10)		
20 percent recovered		
50 percent recovered (T50)	168	229
90 percent recovered (T90)	183	262
Final boiling point		300
T50 – T10	15	
T90 – T10	40	
Residue, vol percent		1.5
Loss, vol percent		1.5

*A condenser temperature of 0°C to 4°C (32° to 40°F) shall be used for the distillation by ASTM D86.

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C.14.4 Data / Property Occurrence.

TABLE C-XXIII. Reserved.

TABLE C-XXIV. Distillation curve fuel property rating value.

Initial BP	New Fuel
G	Report

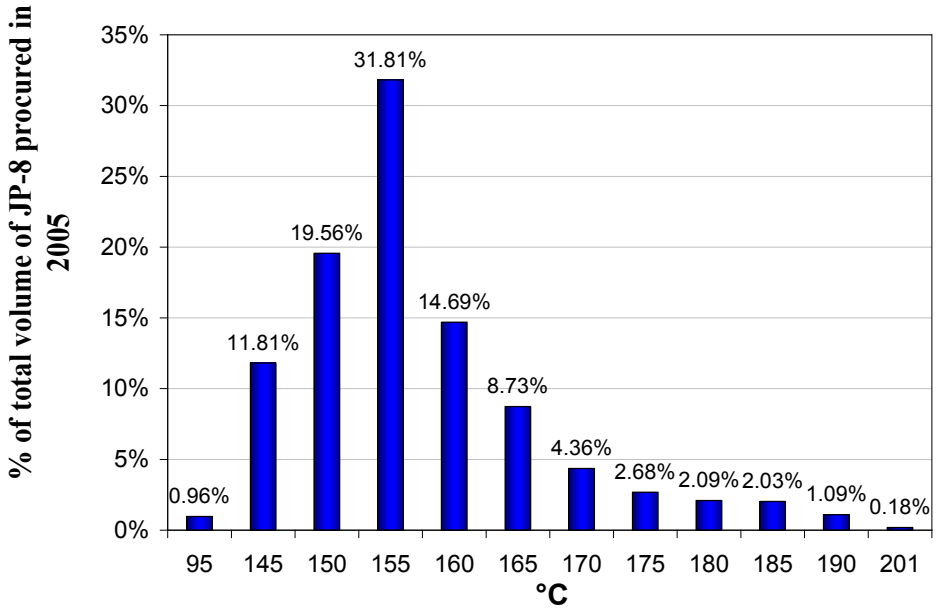


FIGURE C-16. Initial boiling point distillation histogram JP-8 (2005) ***Courtesy of PQIS.***

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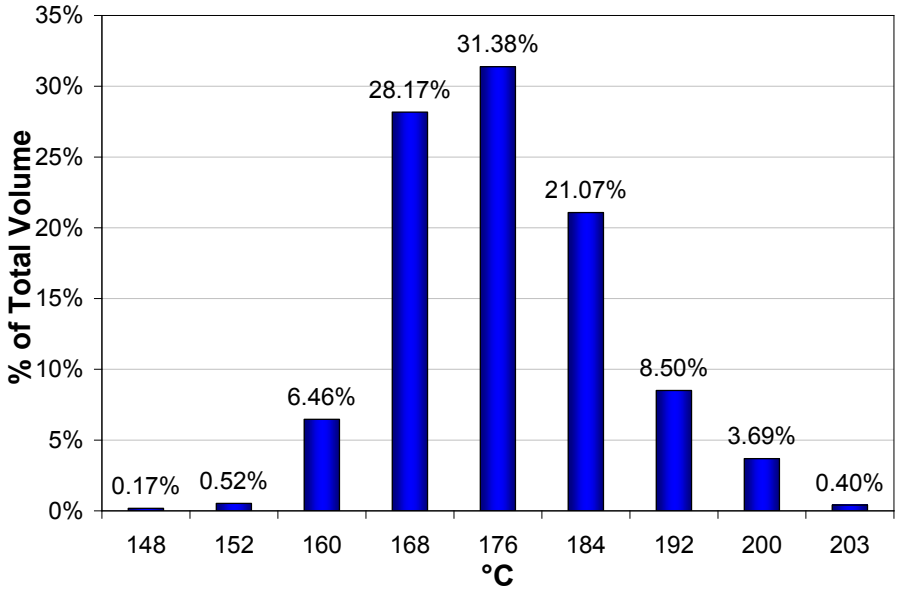


FIGURE C-17. Recovered distillation histogram JP-8 (2005) ***Courtesy of PQIS.***

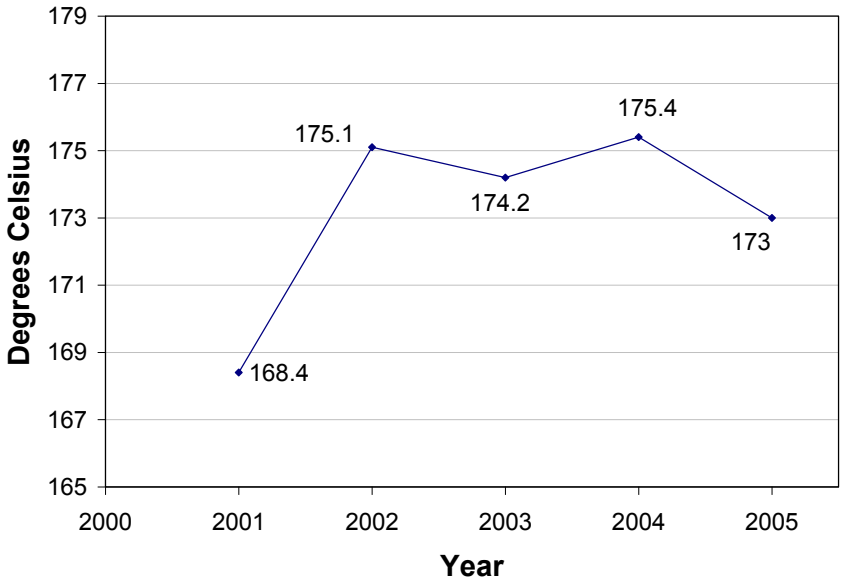


FIGURE C-18. 10% Recovered distillation average JP-8 (year trend 2001-2005) ***Courtesy PQIS 2005.***

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TABLE C-XXV. 10% Distillation curve fuel property rating value.

10%	New Fuel
Y	>205°C (>186°C ASTM D 2887)
G	205°C Max (186°C ASTM D 2887)

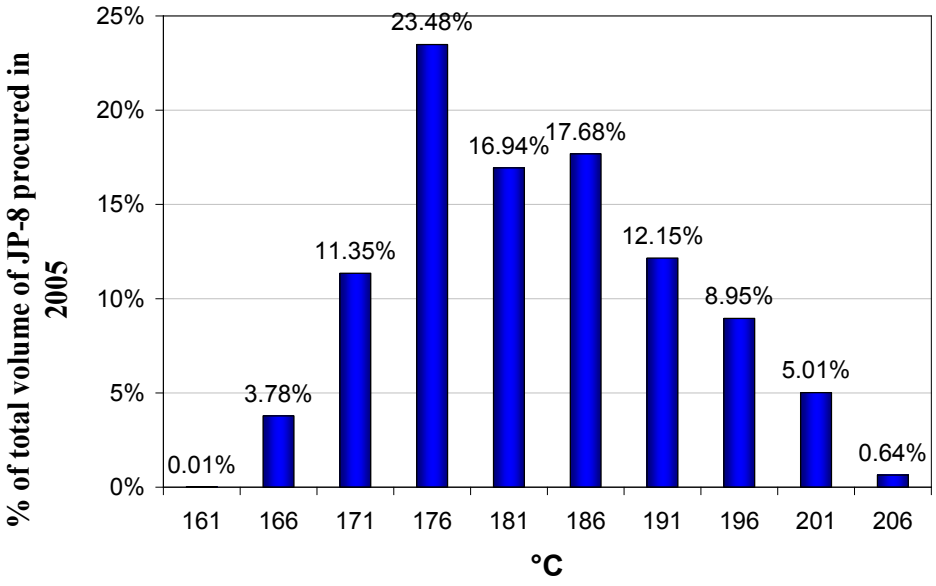


FIGURE C-19. 20% Recovered distillation histogram JP-8 (2005) ***Courtesy of PQIS.***

TABLE C-XXVI. 20% Distillation curve fuel property rating value.

20%	New Fuel
G	Report

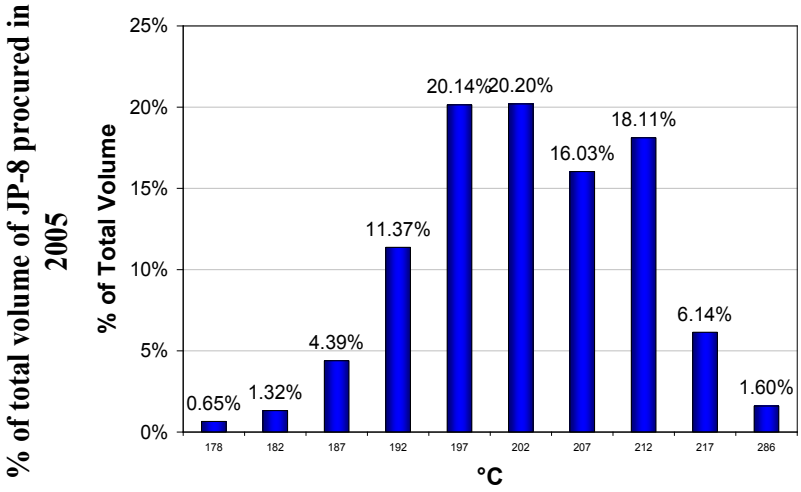


FIGURE C-20. 50% Recovered distillation histogram JP-8 (2005) ***Courtesy of PQIS.***

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TABLE C-XXVII. 50% Distillation curve fuel property rating value.

50%	New Fuel
G	Report

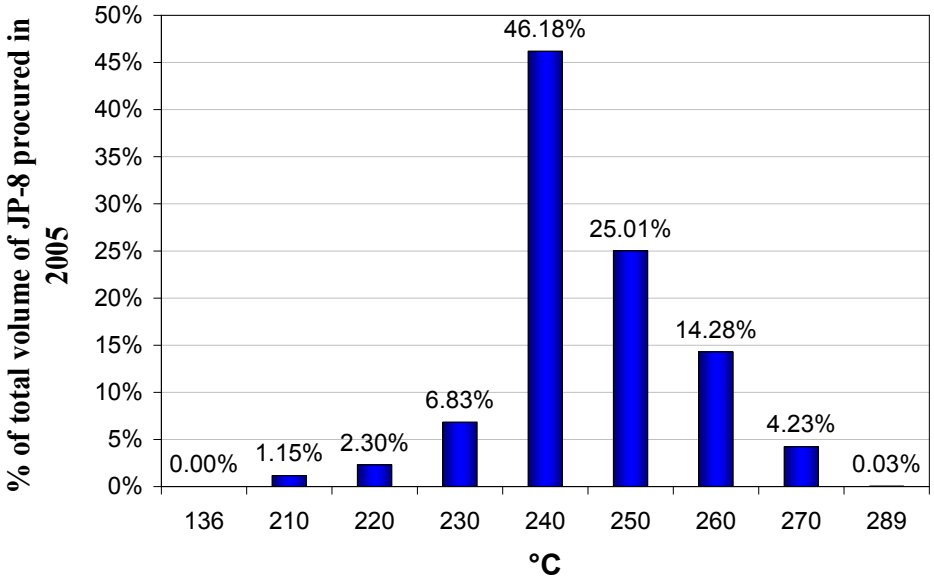


FIGURE C-21. 90% Recovered distillation histogram JP-8 (2005) ***Courtesy of PQIS.***

TABLE C-XXVIII. 90% Distillation curve fuel property rating value.

90%	New Fuel
G	Report

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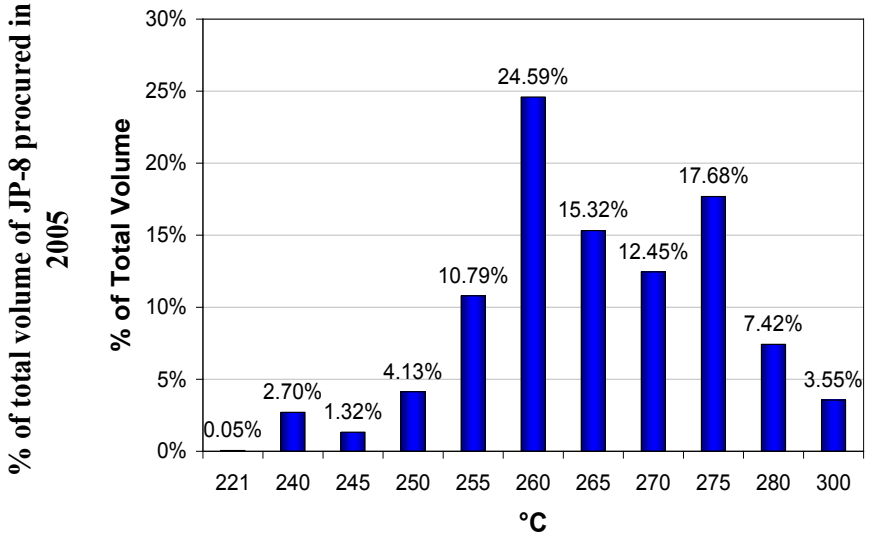


FIGURE C-22. Final boiling point distillation histogram JP-8 (2005) ***Courtesy of PQIS.***

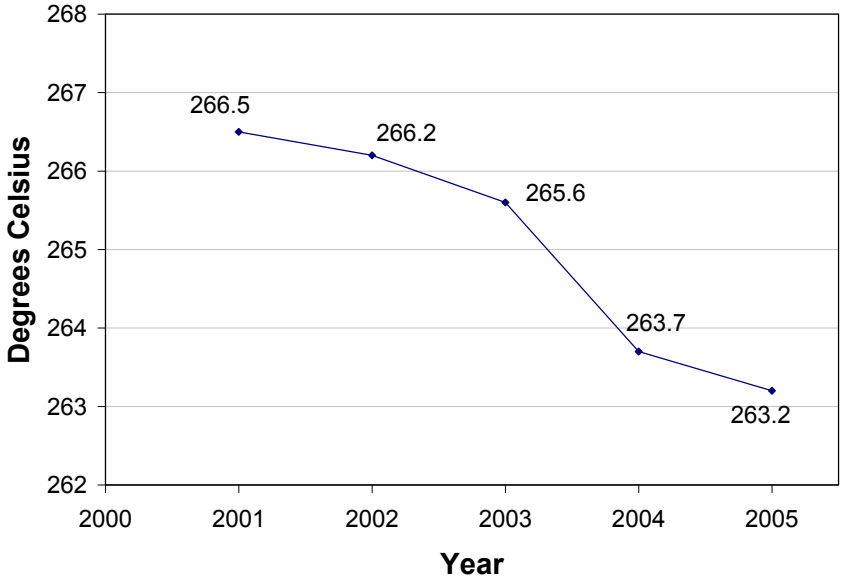


FIGURE C-23. Final boiling point distillation average JP-8 (year trend 2001-2005) ***Courtesy PQIS 2005.***

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TABLE C-XXIX. Distillation curve fuel property rating value.

Final	New Fuel
Y	>300°C (>330°C ASTM D 2887)
G	300°C Max (330°C ASTM D 2887)

TABLE C-XXX. Distillation curve fuel property rating value for residue.

Residue	New Fuel
Y	>1.5%
G	1.5% Max

TABLE C-XXXI. Distillation curve fuel property rating value for loss.

Loss	New Fuel
Y	>1.5%
G	1.5% Max

TABLE C XXXII. Distillation curve fuel property rating value for slope.

T50-T10	New Fuel
Y	<15
G	≥15

TABLE C-XXXIII. Distillation curve fuel property rating value for slope.

T90-T10	New Fuel
Y	<40
G	≥40

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C.15 ELECTRICAL CONDUCTIVITY AT STANDARD TEMPERATURE – ENTRANCE CRITERIA (Electrical Conductivity versus Temperature – Subset 2).

C.15.1 Definition. Electrical conductivity is a measurement of a substance used to determine at what rate it conducts an electric charge.

C.15.2 Standard Electrical Conductivity Test Methods.

C.15.2.1 Electrical Conductivity by ASTM D2624. A probe is immersed into the fuel sample and the conductivity is read directly on the conductivity meter. Depending on the manufacturer the probe may be attached to the meter or be at the end of a flexible cable. Conductivity is reported in pS/m. Sample temperature is also measured and reported.

C.15.3 Comments. The measurements and reported data for electrical conductivity can sometimes be very misleading. First, electrical conductivity measurements are very sensitive to temperature variation and contamination generating wide scatter in reported data. Additionally, fuels with low conductivity can typically be additized with an SDA to bring them with specification limits. Therefore, reported data and the subsequent analysis has to take into account if the fuels were additized and the effectiveness of SDA on the fuel's conductance.

C.15.4 Data / Property Occurrence.

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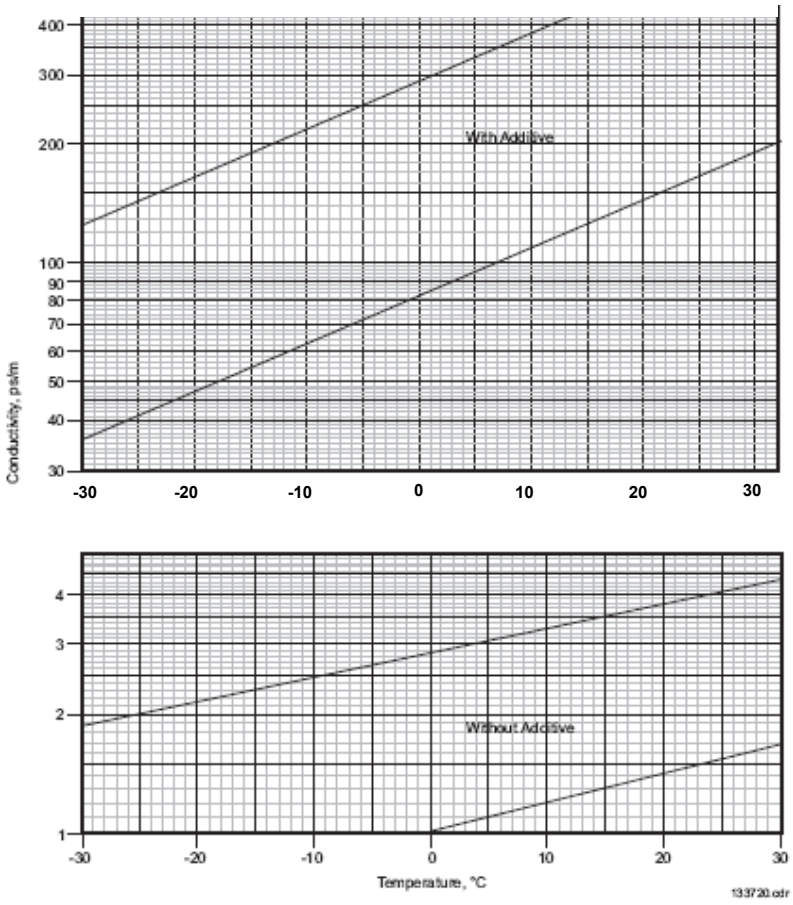


FIGURE C-24. Fuel conductivity as a function of temperature *[CRC, 2004].*

TABLE C-XXXIV. Electrical conductivity at standard T fuel property rating value.

	New Fuel @ 29.4°C
R	>900 pS/m
Y	600 to 900 pS/m
G	150 to 600 pS/m
Y	50 to 149.9 pS/m
R	<50 pS/m

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TABLE C-XXXV. Electrical conductivity versus temperature.

	New Fuel
G	Within limits of Figure C-21
Y	Outside limits of Figure C-21

C.16 ENTHALPY VERSUS TEMPERATURE (see Specific Heat) – SUBSET 3.

C.16.1 Definition.

Enthalpy is the heat energy required to bring a fuel from one reference state to another state. It is a function of the integral of the specific heat between the two states, and any latent heat of vaporization that was required in the interval. Enthalpy is quantified in terms of kJ/kg. Figure C-25 (CRC, 2004) is an enthalpy diagram for typical JP-5. In this figure, the saturated liquid curve represents the heat that can be absorbed in the liquid phase alone, and the saturated vapor curves depict the heat absorbed to completely vaporize the fuel. The intermediate area denotes partial vaporization, while the curves above this saturated vaporization line indicate super-heated vapor. The line of constant pressure provides the pressure relationship to determine the state of vaporization of the fuel for the addition of a given amount of heat.

C.16.2 Standard Enthalpy Test Methods.

C.16.2.1 TBD. Currently, Enthalpy test methods are under discussion and the recommended test method is listed as to be determined.

C.16.3 Comments. None.

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C.16.4 Data / Property Occurrence.

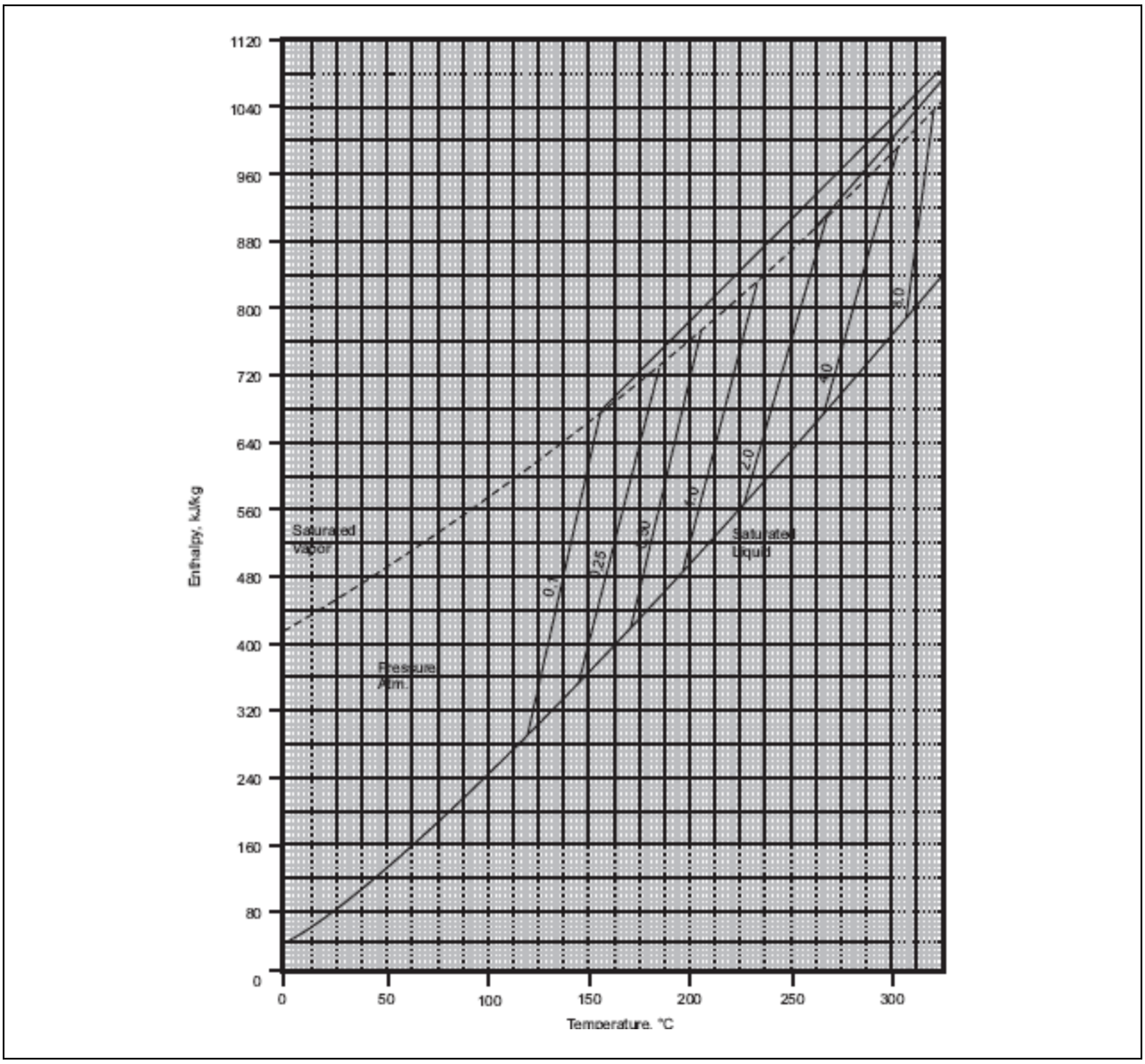


FIGURE C-25. JP-5 enthalpy data ***(Courtesy CRC).***

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TABLE C-XXXVI. Enthalpy versus temperature fuel property rating value.

	New Fuel
Y	>JP-5 values
G	JP-5 values
Y	<JP-5 values

TABLE C-XXXVII. Reserved.

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C.17 EXISTENT GUM – ENTRANCE CRITERIA

C.17.1 Definition. Existent Gum is the evaporation residue of hydrocarbon liquids.

C.17.2 Standard Existent Gum Test Methods

C.17.2.1 Gum Content (Existent) by ASTM D381. A weighed sample quantity is evaporated at 232°C (450°F) by blowing with superheated steam. After evaporation the residue is weighed and reported as existent gum. Solvent washing of the gum, as is done for motor gasoline, is not permitted. Gum content is reported as mg/100 mL.

C.17.3 Comments. This test is typically a 'strong indicator of contamination;' however, in the JP-8+100LT program, it was found that the low temperature additive produced a "fail" on existent gum, yet passed the JFTOT and combustion tests with no problems. In this case it was an indication that the very high content of high molecular weight compounds in the LT were not being vaporized. The gum test was originally designed to give an indication of the storage stability of the fuel in long-term storage. Test is an evolution of a test for deposits from gasoline degradation. From this information it should be noted that other high molecular weight additives will probably have a similar result as the LT additive.

C.17.4 Data / Property Occurrence.

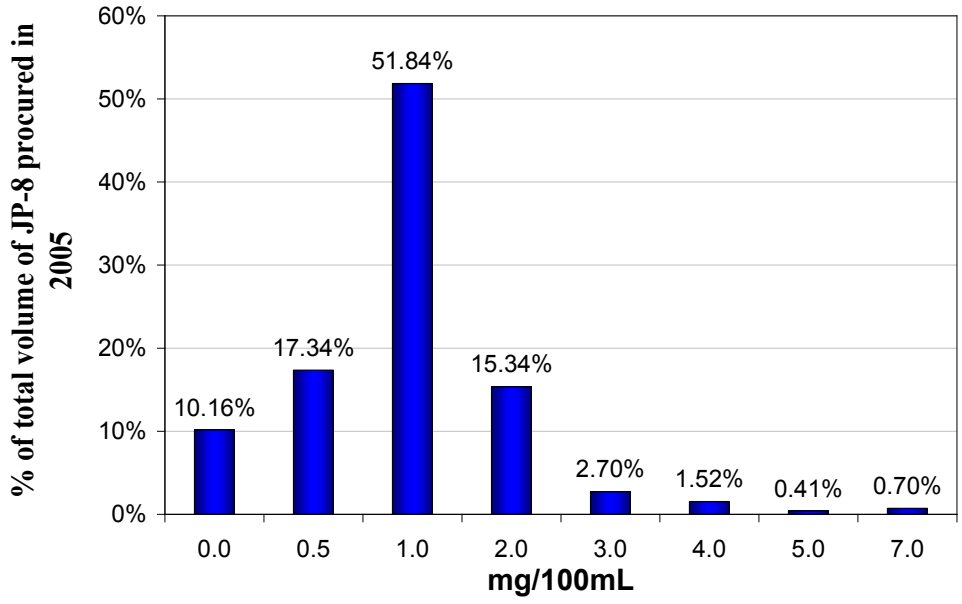


FIGURE C-26. Existent gum histogram JP-8 (2005) ***Courtesy of PQIS.***

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TABLE C-XXXVIII. Existent gum fuel property rating value.

	New Fuel
G	≤7 mg / 100 mL
R	>7 mg / 100 mL

TABLE C-XXXIX. Reserved.

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C.18 FILTRATION TIME – ENTRANCE CRITERIA

C.18.1 Definition. The filtration time test provides an indication of very fine particulate matter or some other unusual compound inherent in the fuel that could lead to the clogging of fuel filters and filter-coalescer cartridges by sticking to and accumulating on the filter media.

C.18.2 Standard Filtration Time Methods.

C.18.2.1 Filtration Time Test by MIL-DTL-83133F. In the laboratory, a vacuum, with a minimum rating of 50 mm (20 in) Hg, is used to pump one gallon of the sample fluid through a 0.8 micrometer membrane filter. The time required to filter the sample and the weight gain of the membrane are recorded.

C.18.3 Data / Property Occurrence.

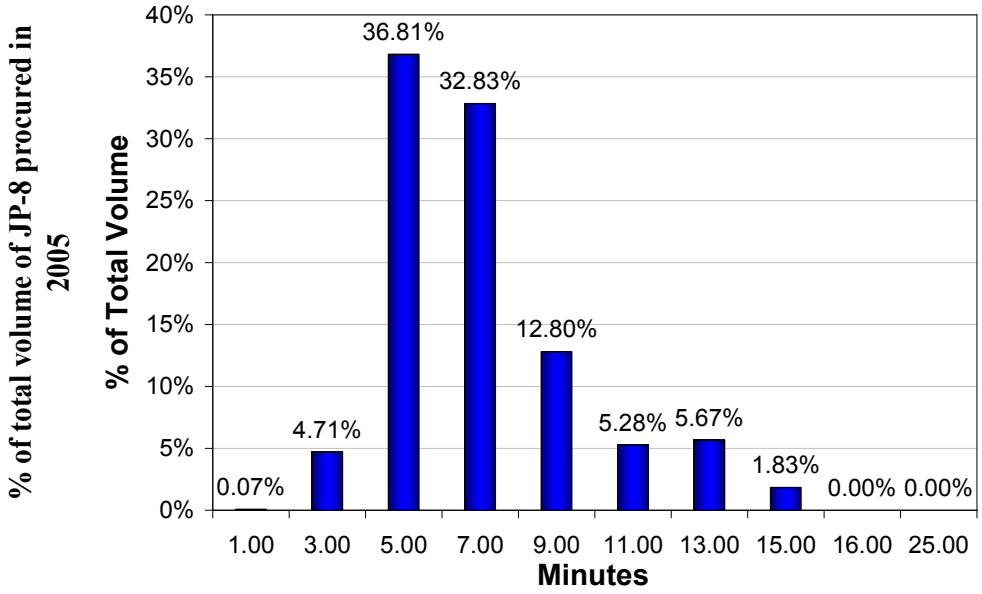


FIGURE C-27. Filtration time histogram JP-8 (2005) ***Courtesy of PQIS.***

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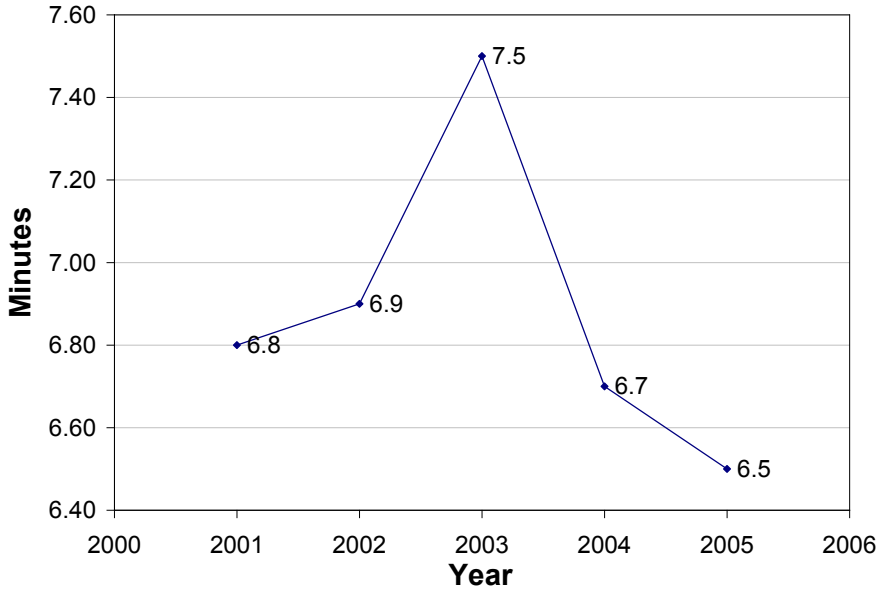


FIGURE C-28. Filtration time average JP-8 (year trend 2001-2005) ***Courtesy of PQIS 2005.***

TABLE C-XL. Filtration time fuel property rating value.

	New Fuel
G	Max: 15 minutes
R	>15 minutes

TABLE C-XLI. Reserved.

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C.19 FLAME SPEED TEST – SUBSET 1.

C.19.1 Definition. The speed of propagation of a flame front through a gaseous mixture (fuel and oxidizer) relative to a reference point; see Appendix F for pool fire information.

C.19.2 Standard Flame Speed Test Methods.

C.19.2.1 See Appendix F.

C.19.3 Data / Property Occurrence.

TABLE C-XLII. Flame speed test fuel property rating value.

	New Fuel
Y	>0.6 m/s
G	0.3 to 0.6 m/s
Y	<0.3 m/s

TABLE C-XLIII. Reserved.

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C.20 FLAMMABILITY LIMITS – SUBSET 1.

C.20.1 Definition. Self-sustained combustion occurs within certain ranges of fuel vapor/air ratios which are functions of temperature and pressure. Therefore at any given pressure, a fuel will have a lean flammability temperature limit (corresponding to the lean flammability of about 1vol% jet fuel in air), and at a higher temperature, a rich flammability limit (corresponding to about 5 vol%). Outside these limits, combustion will not occur if the system is in equilibrium with no spray or mist present. Figure C-29 (CRC, 2004) gives the flammability temperature limits for fuel versus altitude in meters. This classic plot has been in existence in various forms since at least 1946.

C.20.2 Standard Flammability Limits Test Methods.

C.20.2.1 ASTM E681. A uniform mixture of a gas or vapor with air is ignited in a closed vessel, and the upward and outward propagation of the flame away from the ignition source is noted by visual observation. The concentration of the flammable component is varied between trials until the composition that will just sustain propagation of the flame is determined.

C.20.3 Comments. It should be noted that Figure C-29 is based on the specification limits for fuel properties, such as the lean limit at sea level being the specification flash point limit. The typical flash points for MIL-DTL-83133F and MIL-DTL-5624 are not as different as shown in the figure. For example, the average flash points for MIL-DTL-83133F and MIL-DTL-5624 in 2005 were 47 and 63°C, respectively. It is not clear how much actual data underlies Figure C-29. Additionally it has been noted that repeatability of the test is poor, and further research into this area is recommended. Figure C-30 is an updated version of the figure from DOT-FAA-AR-9826 [1998]. There are some differences in flammability limits between fuels such as aviation gasoline, JP-4, and current kerosene fuels. For example, flammability limits have been reported as avgas (1.3-7.1 vol% fuel in air), JP-4 (1.3-8.2 vol%), JP-5 (0.6-4.5 vol%), JP-8/Jet A (0.6-4.7 vol%) [AFWAL-TR-85-2057, Aircraft Mishap Fire Pattern Investigations].

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C.20.3 Data / Property Occurrence.

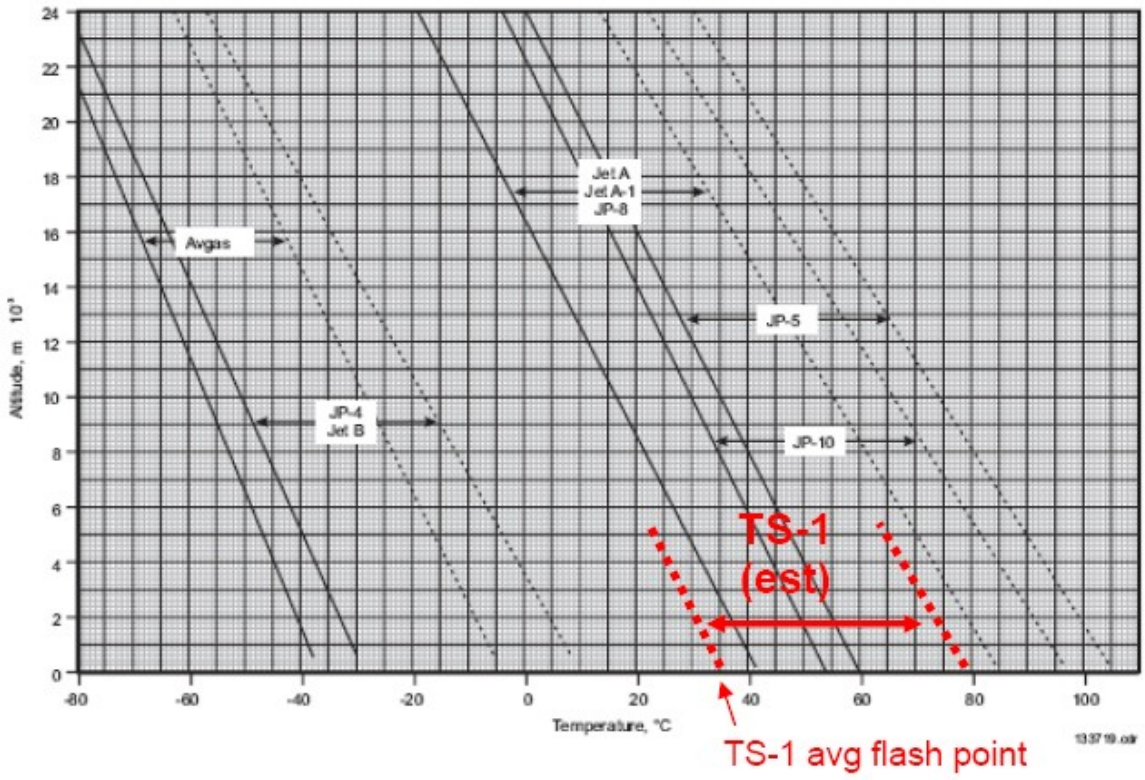


FIGURE C-29. Fuels flammability limits versus altitude (CRC, 2004).

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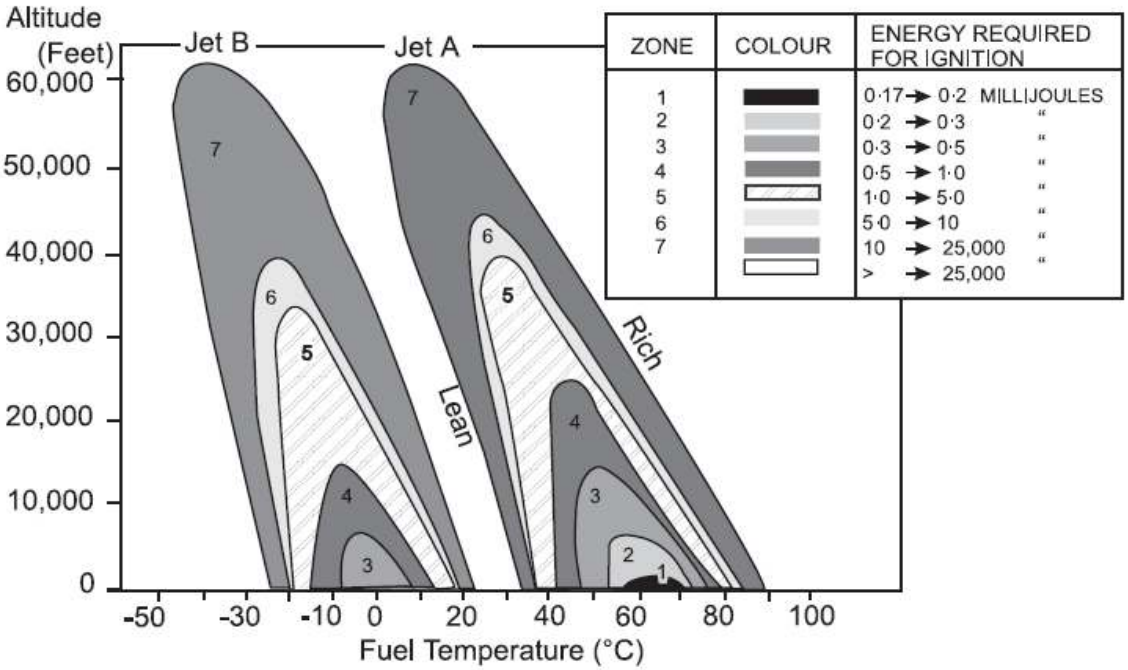


FIGURE C-30. Fuels flammability limits versus altitude [DOT, 1998].

TABLE C-XLIV. Flammability limits @ 25°C fuel property rating value.

	New Fuel
Y	> 4.7 vol %
G	0.6 to 4.7 vol %
Y	< 0.6 vol %

TABLE C-XLV. Reserved.

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C.21 FLASH POINT – ENTRANCE CRITERIA.

C.21.1 Definition. Flash point is the lowest temperature at which vapors evolving from a liquid can mix with air to form an ignitable vapor mixture. Flash point is most frequently used as a measure of the volatility or flammability hazard of combustible liquids, as the lower the flash point temperature the more volatile the liquid.

C.21.2 Standard Flash Point Test Methods.

C.21.2.1 Flash Point by ASTM D56 or ASTM D93. A sample is heated at a prescribed rate in closed container. Periodically the container is opened slightly and a small flame is introduced to try to ignite the vapors. The flash point is the temperature at which vapors ignite and then go out. Flash points are reported as ºC.

C.21.3 Comments. Flash point is an Entrance Criteria test because of its direct link to safety. The green range starts at MIL-DTL-83133F's value of 38°C, and the yellow range starts at MIL-DTL-5624's value of 60°C. On the lower limits, the military has experience using Russian TS-1, but we are unsure of where to draw the yellow line. Army units have had fires from cook stoves which used TS-1. JP-4's flash point is definitely red because this was a large driver in the JP-4 to JP-8 conversion. Also, there is a possible relation between upper limits of flash point which would link to problems igniting or re-igniting engines. Additionally for vehicles and ground support equipment, flash point is the temperature at which the fuel vapor over an open container will ignite. Experience has shown that by placing a requirement for the flash point temperature above a typical ambient temperature a reasonable level of safety can be maintained. Thus 28°C (82°F) as a minimum flash point is an entry requirement.

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C.21.4 Data / Property Occurrence.

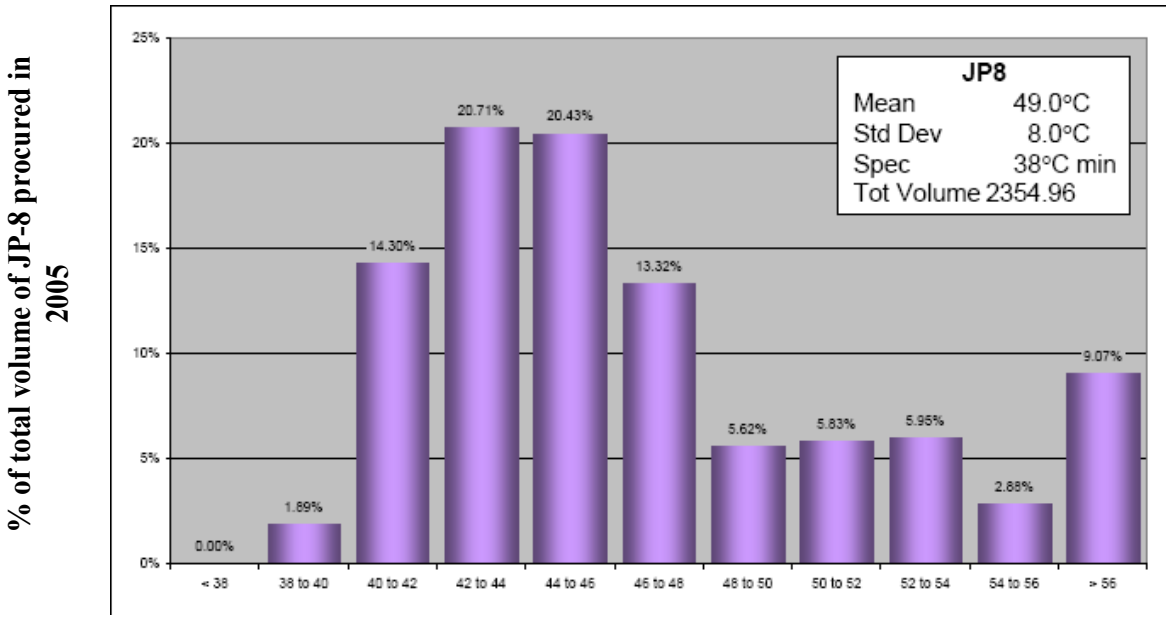


FIGURE C-31. Flash point histogram JP-8 (2005) ***Courtesy of PQIS.***

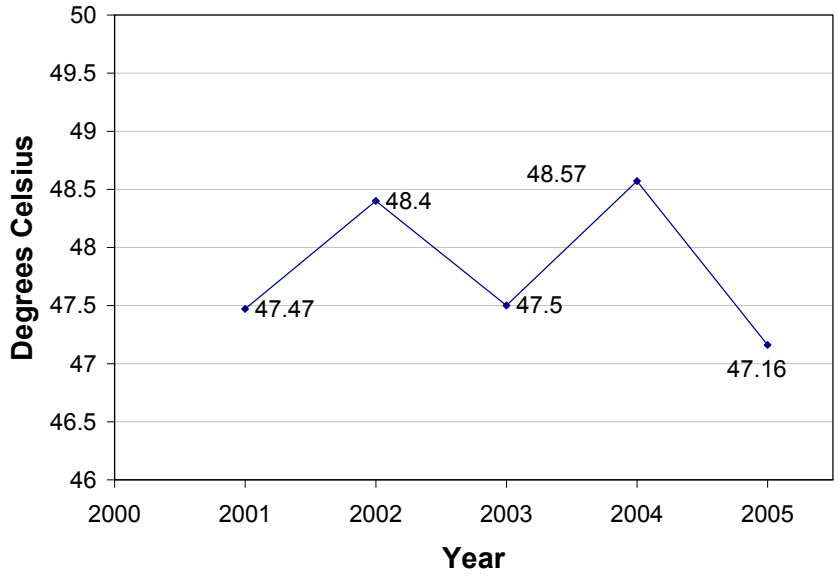


FIGURE C-32. Flash point average JP-8 (year trend 2001-2005) ***Courtesy PQIS 2005.***

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TABLE C-XLVI. Flash point fuel property rating value.

	New Fuel
Y	>68°C
G	38°C to 68°C
Y	28°C to 37.9°C
R	<28°C

TABLE C-XLVII. Reserved.

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C.22 FREEZING POINT – ENTRANCE CRITERIA.

C.22.1 Definition. Multicomponent fuels do not freeze (become solid) at a single temperature. As temperature is lowered, n-paraffins (waxes) begin to come out of solution. The typical method used requires a sample to be chilled until wax crystals appear; then, the sample is warmed until the last fuel wax crystal melts. This point is defined as the freeze point of jet fuels.

C.22.2 Standard Freeze Point Test Methods.

C.22.2.1 Freezing Point (Manual Method) by ASTM D2386. The sample is placed into a test tube which is in a vacuum flask containing a mixture of dry ice and isopropyl alcohol. The tube contains a thermometer and a stirring rod immersed in the sample. The fuel is cooled at prescribed rate until wax crystals appear. (Water/ice crystals are ignored.) The fuel is then warmed until the last crystal melts. The freezing point is the temperature at which the last crystal melts.

C.22.2.2 Freezing Point (Phase Change Method) by ASTM D5972. A 0.15mL sample is placed into apparatus with micropipette. A built-in cooler controls the sample cooling rate. Crystal appearance and disappearance is monitored by 130 light detectors focused on a concave, shallow lens. The test sequence is automatic and the freezing point is read directly on the instrument in °C.

C.22.3 Comments. Freezing point is an Entrance Criteria test because it is linked directly to operation of aircraft in freezing temperatures. Other low temperature property tests (viscosity, pour point) are inter-dependent with freeze point and will eventually be grouped together with freeze point. Freeze point can affect pump, valve and fuel nozzle operations primarily through the increase in viscosity seen at low temperatures, and from long-duration cold soaks resulting in fuel “hold-ups” in any surface where fuel can become frozen; e.g., components, tubes, tank walls.

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C.22.4 Data / Property Occurrence.

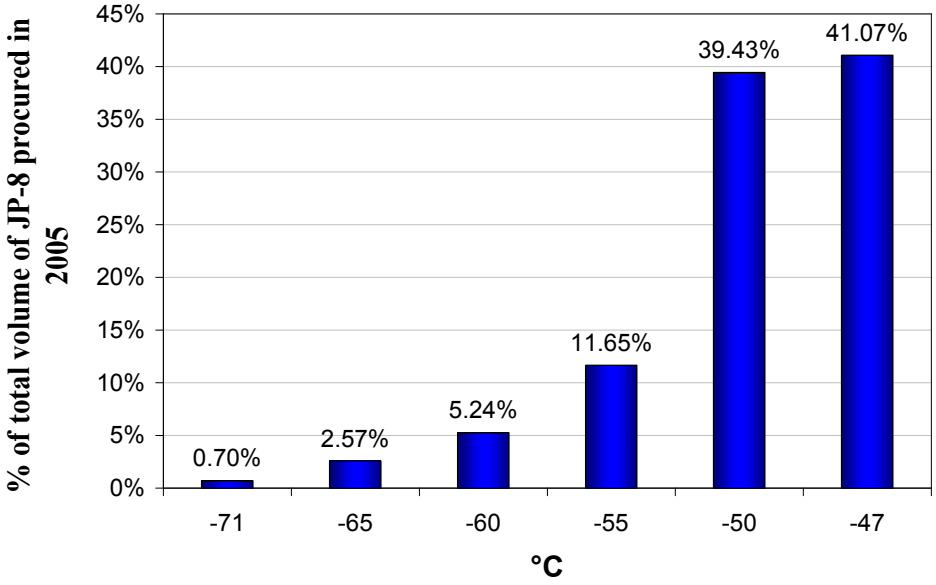


FIGURE C-33. Freezing point histogram JP-8 (2005) ***Courtesy of PQIS.***

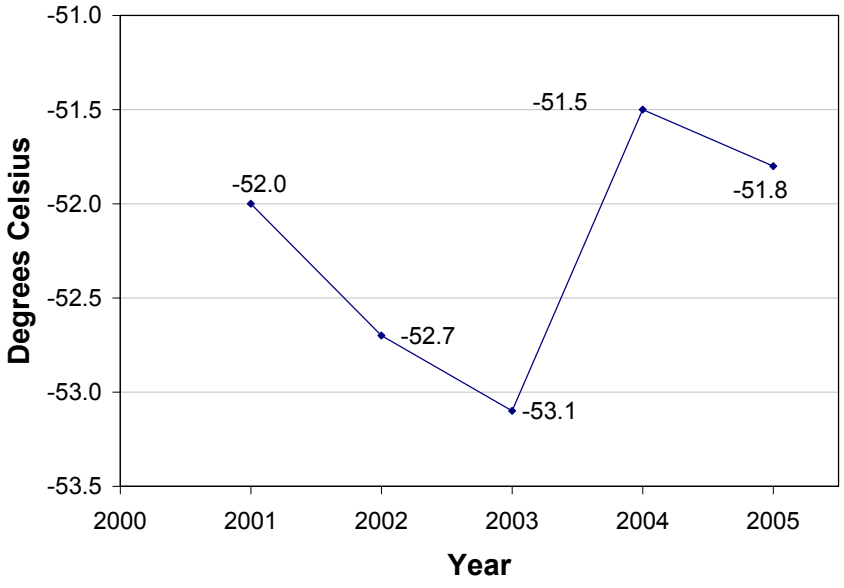


FIGURE C-34. Freezing point average JP-8 (year trend 2001-2005) ***Courtesy PQIS 2005.***

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TABLE C-XLVIII. Freezing point fuel property rating value.

	New Fuel
R	>-40°C
Y	-40°C to -46.9°C
G	<-47°C

TABLE C-XLIX. Reserved.

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C.23 FUEL SYSTEM ICING INHIBITOR (FSII) – ENTRANCE CRITERIA.

C.23.1 Definition. The United States Military uses additives within their liquid hydrocarbon fuels to aid in performance. One of these additives is Diethylene Glycol Monomethyl Ether, more commonly known as DiEGME, to help minimize icing within the liquid hydrocarbon. Ongoing studies are examining the potential for reducing the required amount of FSII.

C.23.2 Standard Icing Inhibitor Test Methods.

C.23.2.1 Fuel System Icing Inhibitor by ASTM D5006. The FSII in a measured volume of fuel is extracted with a fixed ratio of water. A few drops of the water extract are placed on the prism of a refractometer. In one refractometer, FSII content is read directly in volume percent. In a different refractometer a temperature correction is followed by minor calculations to obtain FSII content in volume percent.

C.23.3 Data / Property Occurrence.

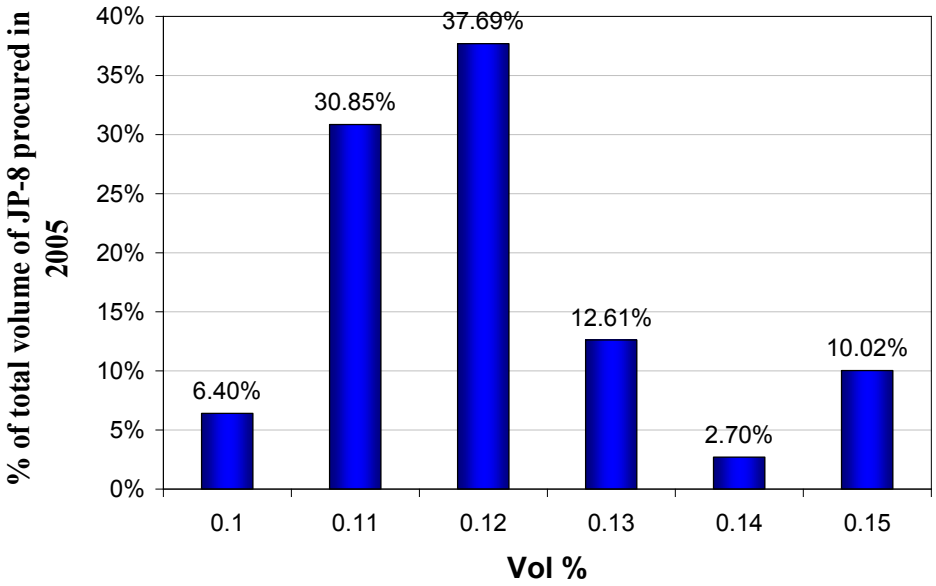


FIGURE C-35. FSII Histogram JP-8 (2005) ***Courtesy of PQIS.***

TABLE C-L. FSII fuel property rating value.

	New Fuel
Y	>0.15 vol %
G	0.1 to 0.15 vol %
Y	<0.1 vol %

TABLE C-LI. Reserved.

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C.24 GAS CHROMATOGRAPH (CHEMICAL DESCRIPTION) – ENTRANCE CRITERIA.

C.24.1 Definition. A gas chromatograph (GC) is an analytical chemistry instrument which helps to identify the chemical composition of substance. Regarding jet fuel samples, GC's can yield information on the hydrocarbon distribution, the chain branching, the hydrocarbon rings, and the aromatic compounds within a fuel. Despite its analytical powers, a GC does not positively identify every substance within a sample. Most trace components cannot be detected with a GC. A GC with a mass spectrometer detector can be used to separate jet fuels into compound classes [AIAA 2006-7972].

C.24.2 Standard GC Test Methods.

C.24.2.1 Distillation (Simulated) by ASTM D2887. Fuel is passed through a chromatographic column which separates hydrocarbons in boiling point order. Boiling temperatures are assigned from a calibration curve, obtained by running a known mixture of hydrocarbons under the same conditions, and a boiling point distribution is obtained. Results are reported in ºC at per percent recovered.

C.24.3 Comments. Distillation curves are the range of temperatures over which a fuel boils at 1 bar, as discussed in the distillation section above. The intent of this section is to use the GC "fingerprint" to ensure that the prospective fuel has an apparent composition that falls within the experience base. In combination with the boiling range limits, this is designed to ensure that the molecular distribution of species in the fuel is not unusually weighted (e.g., bimodal). For example, GCs of a Fischer-Tropsch synthetic jet fuel and a petroleum-derived JP-8 are shown on Figures C-36 and C-37, respectively. The even distribution of species (with molecular weight/boiling temperature generally increasing with residence time) is typical. In contrast, some biodiesel fuels consist of only 3-5 species and boil in a very narrow range. There may be combustion and other operational impacts for such a narrow-boiling distribution. A properly configured fuel will have a balanced distribution of carbon numbers without undue proportion of light or heavy fractions. Figure C-37 is an example of a typical GC distribution of JP-8 fuel and is considered a good model for any new fuel type. Fuels that have a narrow carbon distribution should be considered as being in the yellow range.

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C.24.4 Data / Property Occurrence.

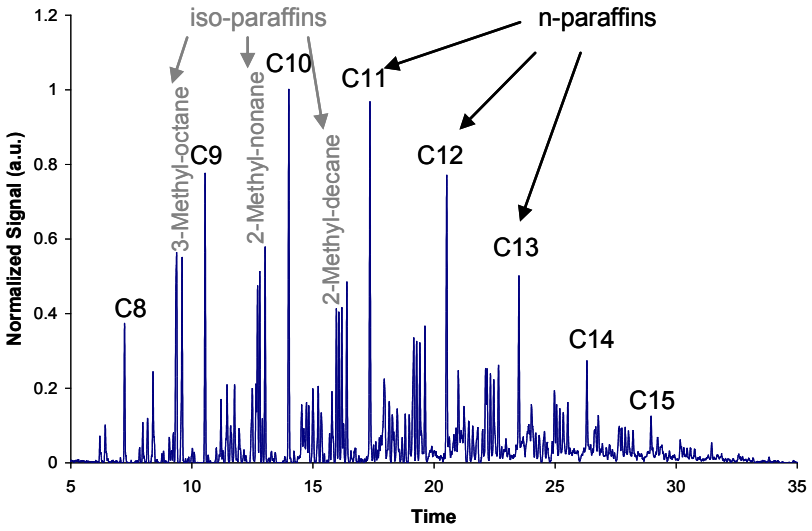


FIGURE C-36. Gas chromatograph of FT synthetic fuel (Courtesy AFRL, USAF).

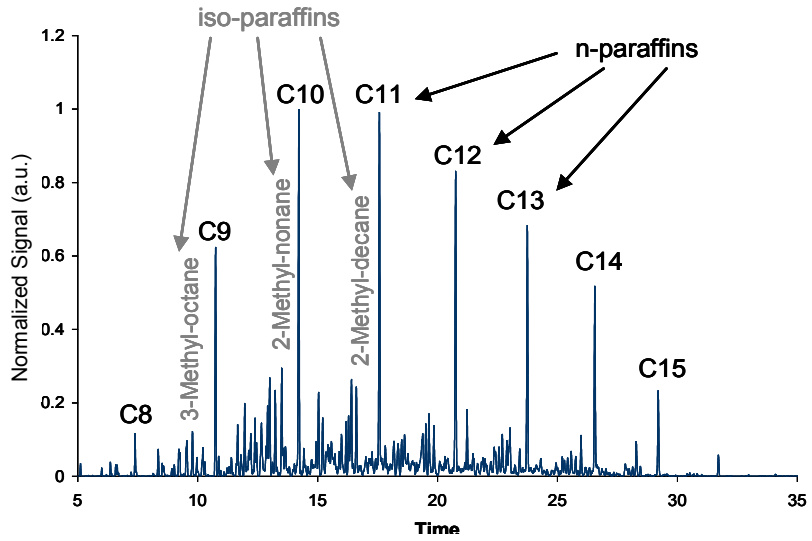


FIGURE C-37. Gas chromatograph of JP-8.

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TABLE C-LII. Gas chromatograph fuel property rating value.

	New Fuel
Y	Narrow carbon distribution
G	Figure C-34, Typical JP-8 carbon distribution

TABLE C-LIII. Reserved.

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C.25 HEAT OF COMBUSTION, NET – ENTRANCE CRITERIA.

C.25.1 Definition.

The net amount of heat energy released per unit mass when a substance is ignited with air. During combustion, some of the heat energy released vaporizes water created during the reaction and this loss is accounted for.

C.25.2 Net Heat of Combustion Test Methods.

C.25.2.1 Net Heat Content by Bomb Calorimeter by ASTM D4809. A weighed quantity is burned in a closed pressure vessel while immersed in a water bath. The total heat released by the substance is calculated from an accurate measurement of the rise in temperature of the water. Net specific energy is calculated by correcting for the condensation of water and sulfur reaction products. Net specific energy is reported in MJ/kg or BTU/lb. (The net specific energy can also be calculated by measuring other properties which have been shown to correlate with heat content. These properties include density, aniline point, boiling point, and aromatic content in ASTM D3338, or aniline point and density of ASTM D4529.)

C.25.3 Comments. More research required to establish upper limits from existing systems.

C.25.4 Data / Property Occurrence.

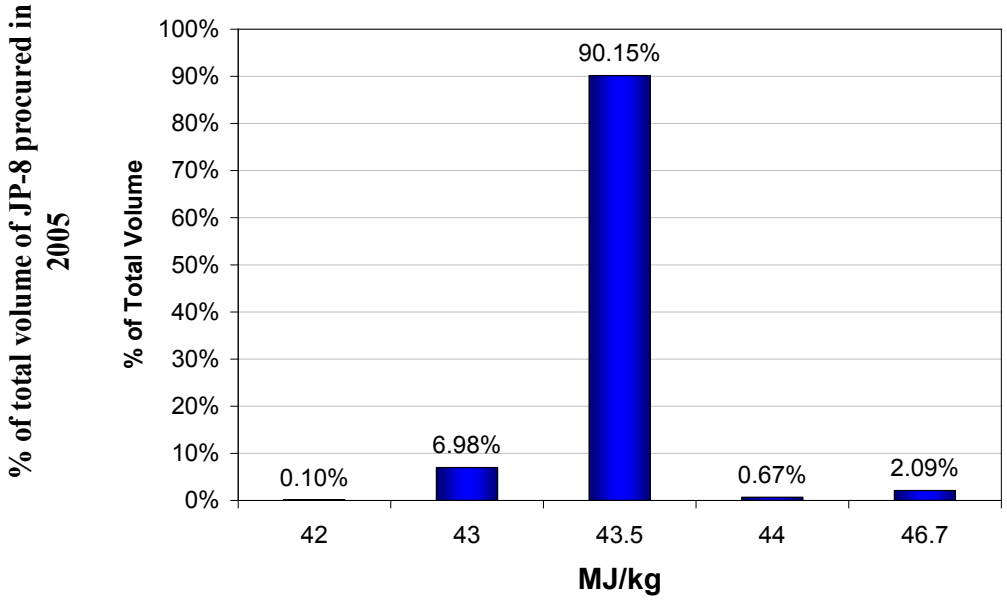


FIGURE C-38. Net heat of combustion histogram JP-8 (2005) ***Courtesy of PQIS.***

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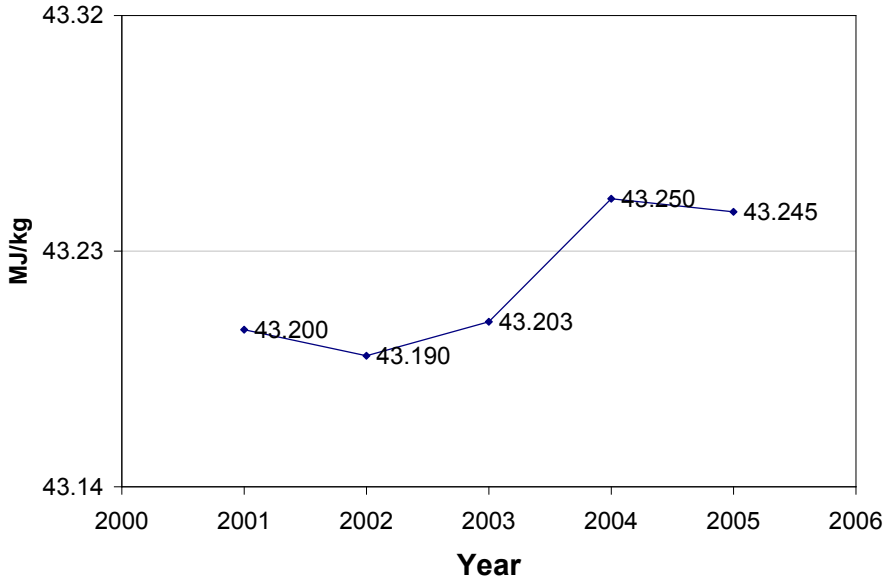


FIGURE C-39. Net heat of combustion average JP-8 (year trend 2001-2005) ***Courtesy PQIS 2005.***

TABLE C-LIV. Net heat of combustion fuel property rating value.

	New Fuel
Y	<42.6 MJ/kg
G	>42.6 MJ/kg

TABLE C-LV. Reserved.

C.26 HEAT OF VAPORIZATION, LATENT.

See C.46.

FIGURE C-40. Reserved.

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C.27 HOT SURFACE IGNITION – SUBSET 1.

C.27.1 Definition. Hot surface ignition temperature is the temperature of a material that will ignite a fuel upon contact. This property has obvious implications to safety involving fuel leaks. Criteria can be hard to set for hot surface ignition because surface geometry, closed or open test, air flow velocity, and residence time can all affect this property.

C.27.2 Standard Test Methods.

C.27.2.1 Federal Test Standard 791C Method 6053. The fuel is dripped onto an internally heated manifold until the fuel ignites.

C.27.2.2 ISO 20823 Hot Surface Temperature. The fuel is dripped onto an internally heated manifold until the fuel ignites.

C.27.3 Comments. Appendix F discusses more involved fire safety testing. In general the concern for new fuels would be hot surface ignition values outside of the general experience range for jet fuels (800-1200°F)

C.27.4 Data / Property Occurrence.

TABLE C-LVI. Hot surface ignition fuel property rating value.

	New Fuel
Y	>1200°F
G	1000°F to 1200°F
Y	800°F to 1000°F
R	<800°F

TABLE C-LVII. Reserved

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C.28 HYDROGEN CONTENT – ENTRANCE CRITERIA.

C.28.1 Definition. The amount of hydrogen contained within a hydrocarbon sample. The MIL-DTL-5624 and MIL-DTL-83133F specifications require >13.4 mass % hydrogen, while this requirement is absent in the Jet A and Jet A-1 specifications.

C.28.2 Hydrogen Content Test Methods.

C.28.2.1 Hydrogen Content by ASTM D3701. A sample is compared with a pure liquid hydrocarbon standard (n-dodecane) in a low resolution nuclear magnetic resonance spectrophotometer (NMR). Based on this comparison the instrument calculates and indicates the hydrogen content in mass percent. D3701 is the preferred test method.

C.28.2.2 Hydrogen Content by ASTM D3343. Hydrogen content is calculated by using an equation relating hydrogen content with distillation range, density and aromatic content. Hydrogen content is reported as percent by mass.

C.28.3 Comments. ASTM D3343 may be inaccurate outside of the range of fuels for which the equation was developed. Hydrogen content may be converted to fuel H/C (hydrogen/carbon) ratio using the below equation (which assumes the fuel contains only carbon and hydrogen). In extensive previous testing of engines with fuels of varying composition, it was found that engine smoke/soot emissions correlated most strongly with fuel hydrogen content. The ratio of hydrogen to carbon can be determined using the following equation derived from the mass balance:

$$\text{H/C Equation: } H/C = 12.0107*[(wt\%\ H)/(100.794*(1-(wt\%\ H/100)))]$$

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C.28.4 Data / Property Occurrence.

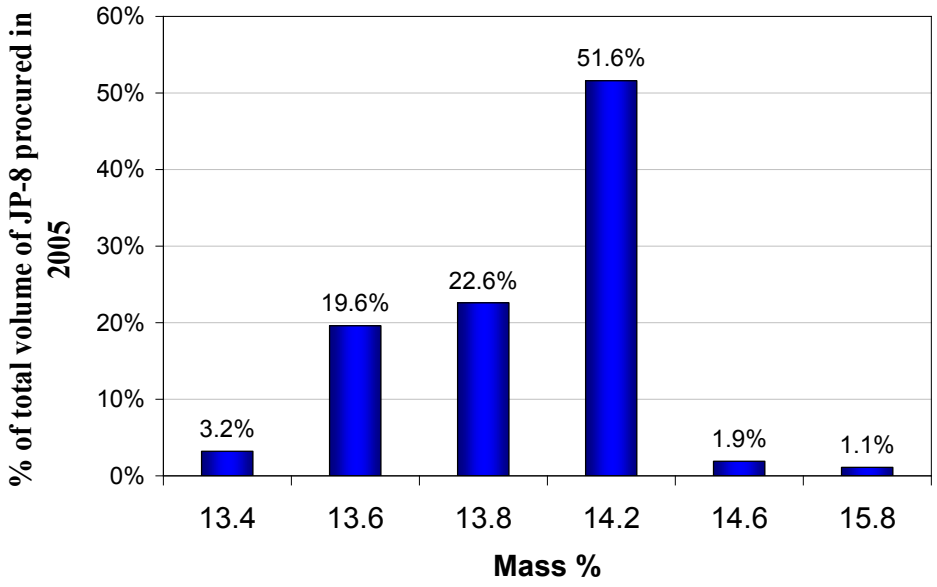


FIGURE C-41. Hydrogen content histogram JP-8 (2005) ***Courtesy of PQIS.***

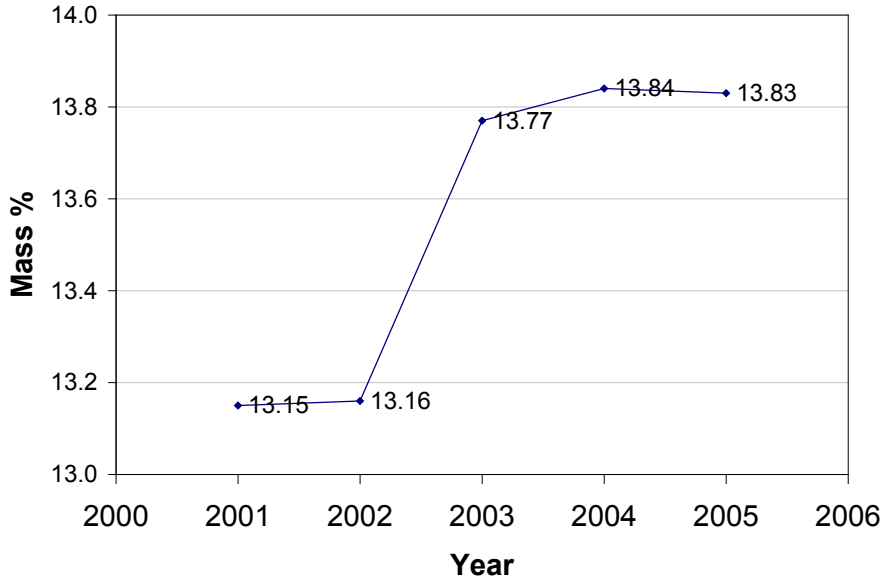


FIGURE C-42. Hydrogen content average JP-8 (year trend 2001-2005) ***Courtesy PQIS 2005.***

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TABLE C-LVIII. Hydrogen content fuel property rating value.

	New Fuel
Y	<13.4 mass %
G	>13.4 mass %

TABLE C-LIX. Reserved.

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C.29 LUBRICITY – SUBSET 1 CRITERIA.

C.29.1 Definition. Aviation turbine fuel serves as a lubricant in fuel pumps, engine controls, and servo valves. Straight-run fuels normally contain boundary lubricants in trace amounts. These are primarily polar compounds containing oxygen, nitrogen, or sulfur. Such polar compounds form thin films on metal surfaces, protect against corrosion, and provide boundary lubrication. Lack of lubrication can result in high friction and metal-to-metal contact leading to increased wear rates and possible scuffing. Severe refining conditions, such as hydrocracking, remove these natural lubricants from the fuel and severely hydrotreated or hydrocracked fuels are, therefore, more likely to have poor lubricity.

C.29.2 Standard Lubricity Test Methods.

C.29.2.1 Lubricity by ASTM D5001. In the Ball-on-Cylinder Lubricity Evaluator (BOCLE) test a fixed steel ball, under load, is pressed against a rotating cylinder, covered by fuel, for a predetermined period of time. Both the temperature and humidity of the air around the test section are controlled during the test. Fuel lubricity is based on the size of the resulting elliptical wear scar on the ball. The reported wear scar size, in millimeters, is the major axis plus the minor axis divided by two. Wear scar is reported in mm. Fuels giving high wear scars have poor lubricity or have been called hard fuels.

C.29.3 Comments. Lubricity problems are mitigated in JP-8 by the mandated addition of the CI/LI. These additives are qualified by MIL-PRF-25017.

C.29.4 Data / Property Occurrence.

TABLE C-LX. Lubricity fuel property rating value.

	New Fuel
R	>0.85 mm
Y	0.65 to 0.85 mm
G	0 to 0.65 mm

TABLE C-LXI. Reserved.

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C.30 MINIMUM SPARK IGNITION ENERGY – SUBSET 2.

C.30.1 Definition. According to the CRC Handbook, 2004, the minimum amount of energy required for a spark discharge to ignite an optimum hydrocarbon fuel/air mixture is in the 0.20 to 1 mJ range. The optimum fuel/air mixture is normally found at a point on the rich side near the stoichiometric point. As conditions depart from an ideal state, the energy requirements increase. Changing the fuel/air mixture, the electrode geometry, or the gap distance will change the amount of energy required for ignition. If the fuel is present in the form of a mist or spray, as opposed to a vapor, the ignition energy requirements will increase. On the other hand, an increase in the oxygen concentration of the air, such as found in aircraft ullage at altitude, will decrease the amount of energy required for ignition.

The minimum ignition energy for fuels is illustrated on Figures C-43 and C-44. The minimum spark ignition energy required to ignite a fuel varies with the temperature and is dependent upon the volatility of the fuel in question. For sprays, other variables include the configuration of the nozzle delivering the spray and pressure on the fuel which determines the droplet sizes in the spray.

C.30.2 Standard Spark Ignition Energy Test Methods.

C.30.2.1 TBD. Currently, Spark Ignition Energy test methods are under discussion and the recommended test method is listed as to be determined (TBD).

C.30.3 Comments. Given the lack of a standard test method and the difficulty of measuring the minimum spark ignition energy, the criteria is defined as "no easier to ignite than Jet A/JP-8". This definition is driven by safety. In terms of operability, an engine designer might prefer to have a fuel that is easier to ignite than current fuels—but that is not the constraint used in this handbook.

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C.30.3 Data / Property Occurrence.

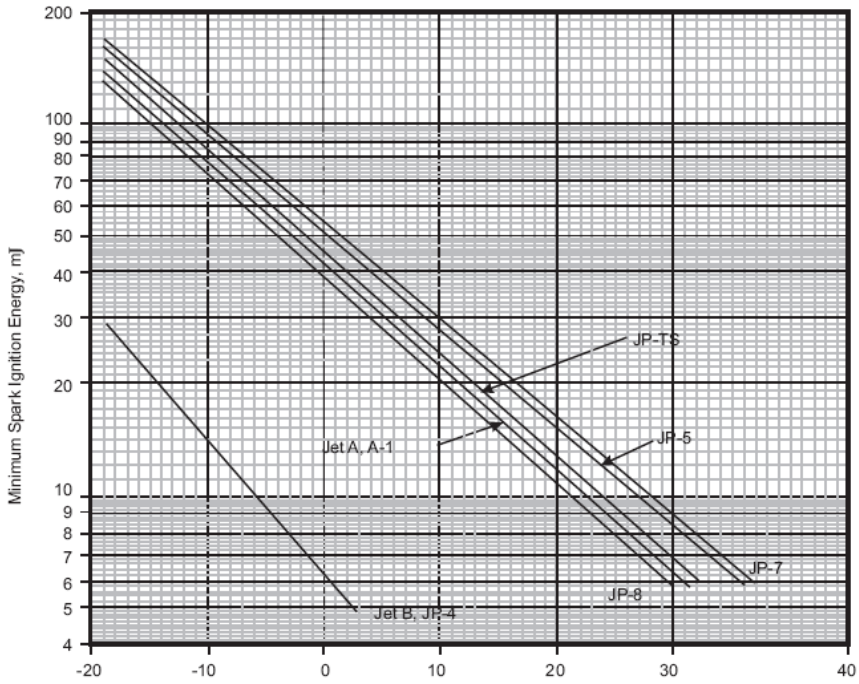


FIGURE C-43. Minimum spark ignition energy ***[CRC, 2004].***

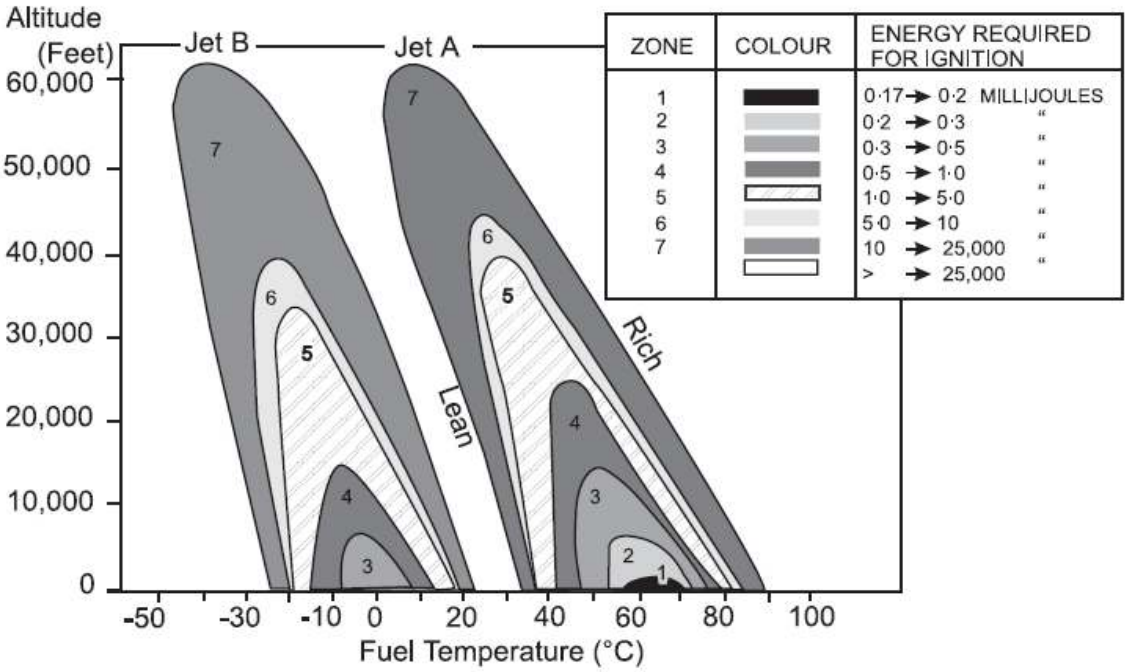


FIGURE C-44. Fuels flammability limits versus altitude [DOT, 1998].

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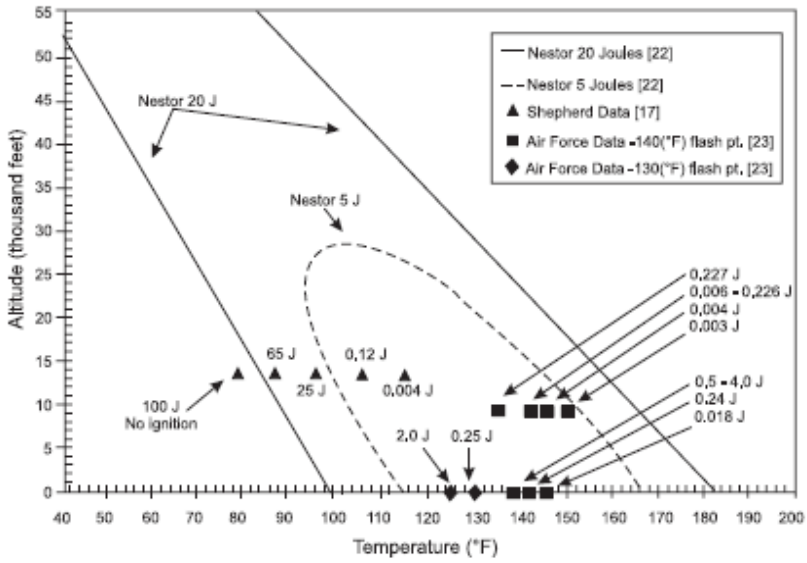


FIGURE C-45. Effect of temperature and altitude on Jet A flammability [DOT-FAA-AR-9826].

TABLE C-LXII. Minimum spark ignition energy fuel property rating value.

	New Fuel
R	<JP-4 values
Y	between JP-4 and JP-8 values
G	JP-5/JP-8 values
Y	>JP-8 values

TABLE C-LXIII. Reserved.

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C.31 NAPHTHALENES – ENTRANCE CRITERIA.

C.31.1 Definition. Naphthalene is an aromatic hydrocarbon, structurally consisting of two benzene rings fused together. Naphthalene with methyl group substituents (termed "naphthalenes") is often found in jet fuels at the per cent level. It is generally tested for in conjunction with smoke point, as aromatics can increase the smoke point of a liquid hydrocarbon.

C.31.2 Naphthalene Test Methods.

C.31.2.1 Naphthalenes Content by ASTM D1840. The naphthalene content is determined by ultraviolet absorption at a wave length of 285 nm. Naphthalene concentration is established by using the average response of a standard blend of naphthalene. Naphthalene content is reported in mass percent.

C.31.3 Comments. Napthalenes are effective at generating soot (which was the reason behind the limit in the specification), as well as being effective in swelling elastomers.

C.31.4 Data / Property Occurrence.

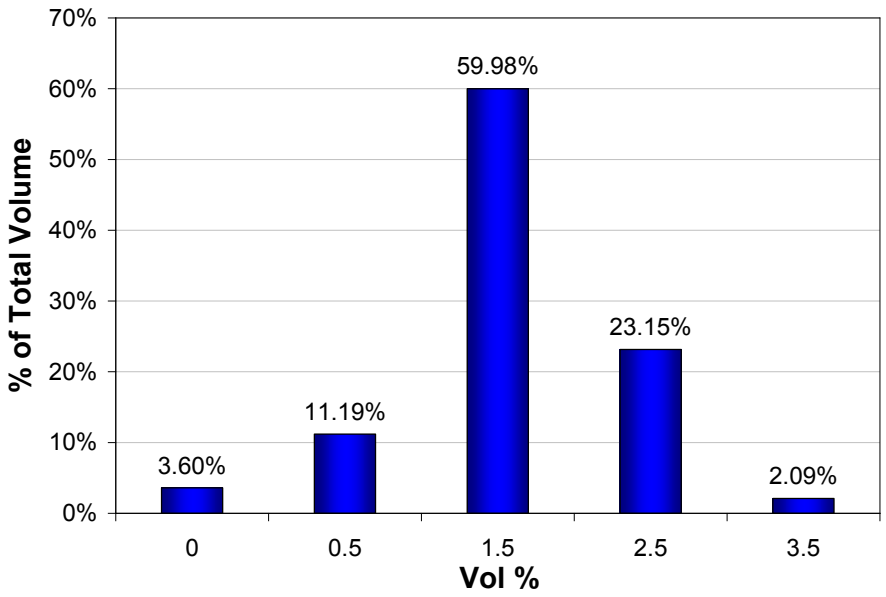


FIGURE C-46. Naphthalene histogram JP-8 (2005) ***Courtesy of PQIS.***

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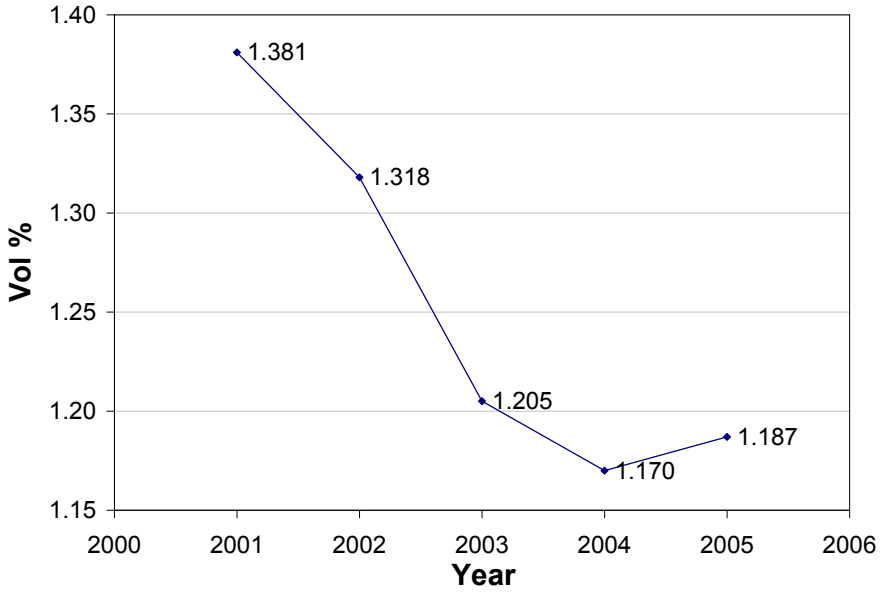


FIGURE C-47. Naphthalene average JP-8 (year trend 2001-2005) ***Courtesy PQIS 2005.***

TABLE C-LXIV. Naphthalenes fuel property rating value.

	New Fuel
G	<3%
R	>3%

TABLE C-LXV. Reserved.

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C.32 OSTWALD COEFFICIENT / GAS SOLUBILITY - SUBSET 2.

C.32.1 Definition. The solubility of gases in fuels is of high importance to the proper design of fuel systems and their components. High evolution of gases during climb can cause decreased pressure resulting in loss of fuel and, when pumping fuel, gas phase separation can occur leading to vapor lock or cavitation problems. The Ostwald Coefficient is the volume of gas dissolved in one volume of solvent. While several variables affect the solubility of gases in fuels including temperature, pressure, nature of fuel, and the gas itself, Ostwald Coefficient is independent of pressure, and the gas volume is measured at the conditions of the solution. Figure C-48 (CRC 2004) gives the typical solubility of CO_2, O_2, and N_2 in aviation fuels. Carbon dioxide, unlike the other gases, decreases in solubility with increasing temperature. The solubility of this gas is very high.

C.32.2 Standard Gas Solubility Test Methods.

C.32.2.1 Solubility of Gas by ASTM D2779. Correlations have been established by the National Aeronautics and Space Administration (NASA) (formerly National Advisory Committee on Aeronautics) in NACA Technical Note 3276 (1956). Their work was extended to include most of the data published since that time, and extrapolated by semi-empirical methods into regions where no data are available. The only data required are the density of liquid at 288 K (59°F) and the nature of the gas. These are used to estimate the Ostwald coefficient.*

*Reprinted, with permission from *ASTM D2779: Standard Test Method for Estimation of Solubility of Gases in Petroleum Liquids*, copyright ASTM International, 100 Barr Harbor Drive, West Conshohocken, PA 19428; www.astm.org.

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C.32.3 Data / Property Occurrence.

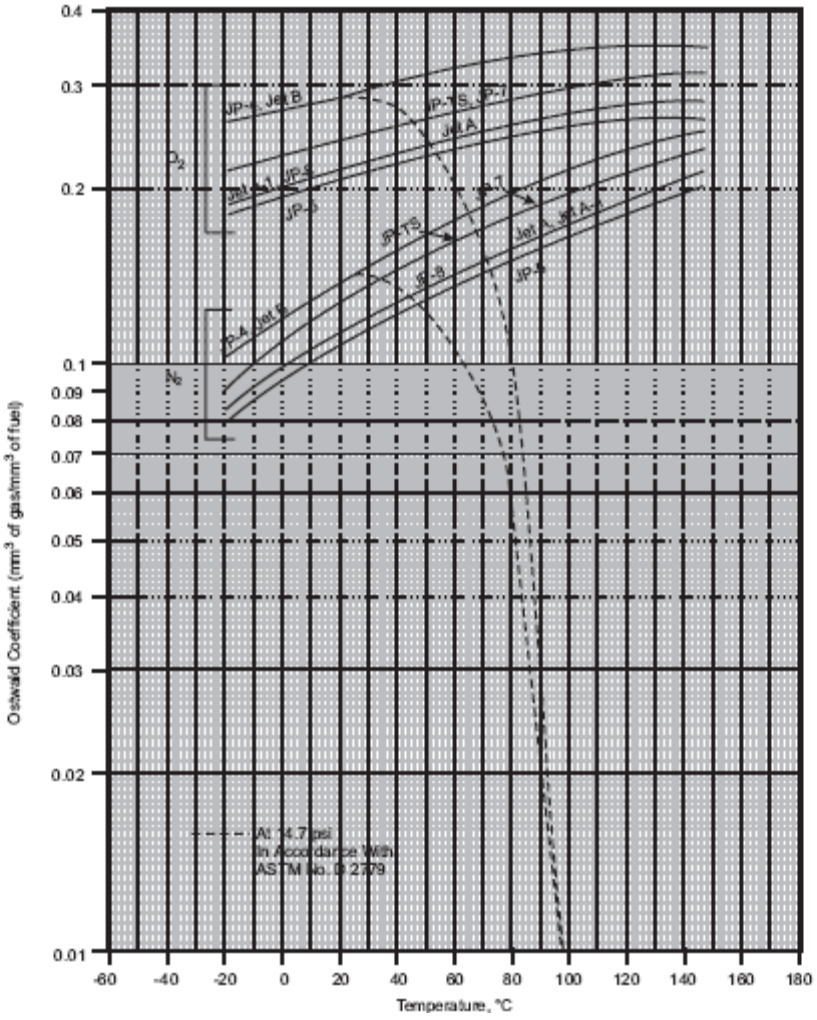


FIGURE C-48. Ostwald coefficient as a function of temperature ***[CRC, 2004].***

TABLE C-LXVI. Ostwald coefficient / gas solubility fuel property rating value.

	New Fuel
R	>JP-4 values
Y	JP-4 to JP-8 values
G	≤JP-5/JP-8 values

TABLE C-LXVII. Reserved.

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C.33 PARTICULATE MATTER – ENTRANCE CRITERIA.

C.33.1 Definition.

Particulate matter is any minute and separate particles contained within the hydrocarbon sample.

C.33.2 Standard Particulate Test Methods.

C.33.2.1 Particulates by Field Filtration by ASTM D2276.

A fixed volume of fuel (usually 4 L or 1 gal) is forced, under line pressure, through a membrane having a pore size of 0.8 micrometer. After drying the membrane, its color is rated by comparing it to a color standard. In a different version of the procedure the weight gain of the membrane, after field filtration, is determined in a laboratory. The weighing procedure is the same as in ASTM D5452, Particulates by Laboratory Filtration.

C.33.2.2 Particulates by Laboratory Filtration by ASTM D5452.

A known volume, usually 4 L, is filtered through two preweighed, matched weight membranes in series, having a pore size of 0.8 m, using a vacuum downstream of the membranes. The increase in weight of the upper membrane is determined after washing and drying of the membrane. Any change in weight of the second or control membrane is also determined. The particulate contaminant level is the difference in weight gain of the two membranes. Results are reported in mg/L or mg/gallon. The method can also be used to conduct the Time Filtration test.

C.33.3 Comments.

Setting limits on the tolerable size distribution and total mass of particulate contamination in fuel was at the very heart of a CRC effort (Av-04-04) initiated in FY05. This effort included a survey of engine OEMs and component OEMs and was being worked by Vic Hughes Associates Limited. As of June, 2006 no finalized agreement or final conclusion had yet been reached. Survey responses are varied and highly dependent on the particular OEM.

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C.33.4 Data / Property Occurrence.

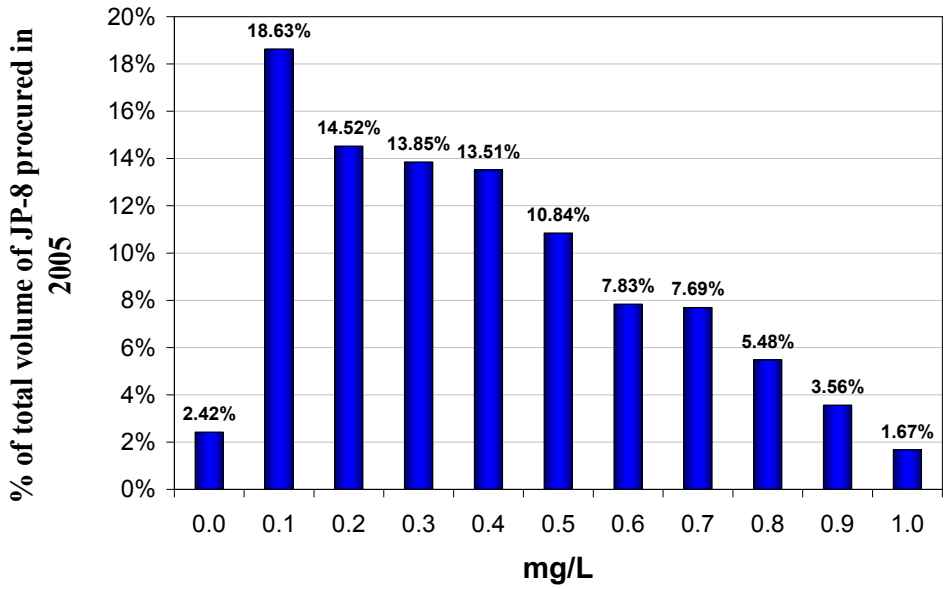


FIGURE C-49. Particulate matter histogram JP-8 (2005) ***Courtesy of PQIS.***

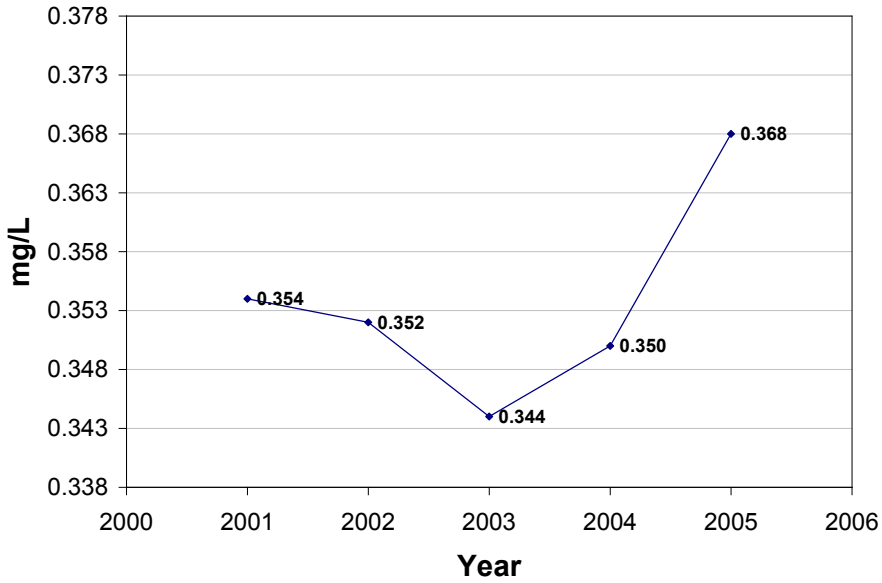


FIGURE C-50. Particulate matter average JP-8 (year trend 2001-2005) ***Courtesy PQIS 2005.***

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TABLE C-LXVIII. Particulate matter fuel property rating value.

	New Fuel
G	≤1 mg/L
R	>1 mg/L

TABLE C-LXIX. Reserved.

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C.34 POUR POINT – SUBSET 2.

C.34.1 Definition. Pour point is the lowest temperature at which a liquid will flow without a significant pressure behind it. Pour point is an indication of the pumping ability of a fuel at cold temperatures. In typical petroleum based fuels, failure to flow can be linked to the separation of wax from the fuel. However, it can also be caused by the high viscosity in the case of very viscous fuel oils.

C.34.2 Standard Test Methods.

C.34.2.1 Pour Point of Petroleum Products by ASTM D97. After preliminary heating, the sample is cooled at a specified rate and examined at intervals of 3°C for flow characteristics. The lowest temperature at which movement of the specimen is observed is recorded as the pour point.*

C.34.3 Comments. The pour point is lower than the freeze point, which is the point where the last visible crystal of fuel disappears upon warming. Pour point is also related to the dramatic increase in viscosity-versus -temperature curves.

C.34.4 Data / Property Occurrence.

TABLE C-LXX. Pour point fuel property rating value.

	New Fuel
Y	>-61°C
G	<-61°C

TABLE C-LXXI. Reserved.

*Reprinted, with permission from *ASTM D97: Standard Test Method for Pour Point of Petroleum Products*, copyright ASTM International, 100 Barr Harbor Drive, West Conshohocken, PA 19428; www.astm.org.

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C.35 SAYBOLT COLOR – ENTRANCE CRITERIA.

C.35.1 Definition. A color based indicator that fuel is required to obtain as a quality test of both the manufacturing process and the fuel.

C.35.2 Standard Saybolt Color Test Methods.

C.35.2.1 Color by Saybolt Chromometer by ASTM D156. Fuel is placed into a sample tube and its color is observed by looking through the length of the sample tube and comparing the color to the standard in an adjacent tube. (Both tubes are visible in the eye piece.) The sample height is decreased until the sample color matches the color of the standard in the other tube. The reported fuel color is based on the final height of the fuel column and the particular standard used. Results are reported as values ranging from +30 (water white) to -16 (straw color).

C.35.3 Comments. Saybolt color helps fuels personnel in the logistical fuel supply chain identify potential problems most commonly linked to contamination. Other fuels contamination and particulate tests that have to be run on new fuels give yield more quantifiable information.

C.35.4 Data / Property Occurrence.

TABLE C-LXXII. Saybolt color fuel property rating value.

	New Fuel
G	Report

TABLE C-LXXIII. Reserved.

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C.36 SMOKE POINT – ENTRANCE CRITERIA.

C.36.1 Definition. The smoke point of a substance is the point at which a wick-fed flame begins to generate visible smoke. It provides an indication of the relative smoke-producing properties of a liquid hydrocarbon in a diffusion-controlled flame, and the smoke point is related to the hydrocarbon composition. Generally the more aromatic the sample is, the smokier the flame becomes. The smoke point (and Luminometer number with which it can be correlated) is quantitatively related to the potential radiant heat transfer from the combustion products of the sample.

C.36.2 Smoke Point Test Methods.

C.36.2.1 Smoke Point by ASTM D1322. A sample is placed into a wick-fed kerosene lamp and the wick is ignited. A scale marked in millimeters is behind the flame. The wick is raised until the tip of the flame starts to smoke. The wick is then lowered until smoke disappears. The smoke point is the maximum height of the flame at which no smoke is noted. Smoke point is reported in mm.

C.36.3 Data / Property Occurrence.

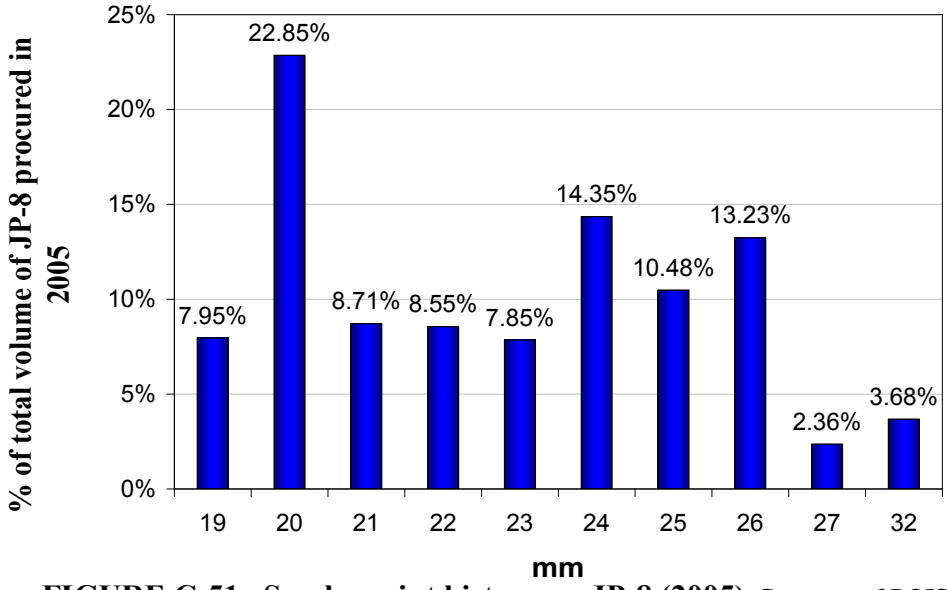


FIGURE C-51. Smoke point histogram JP-8 (2005) ***Courtesy of PQIS.***

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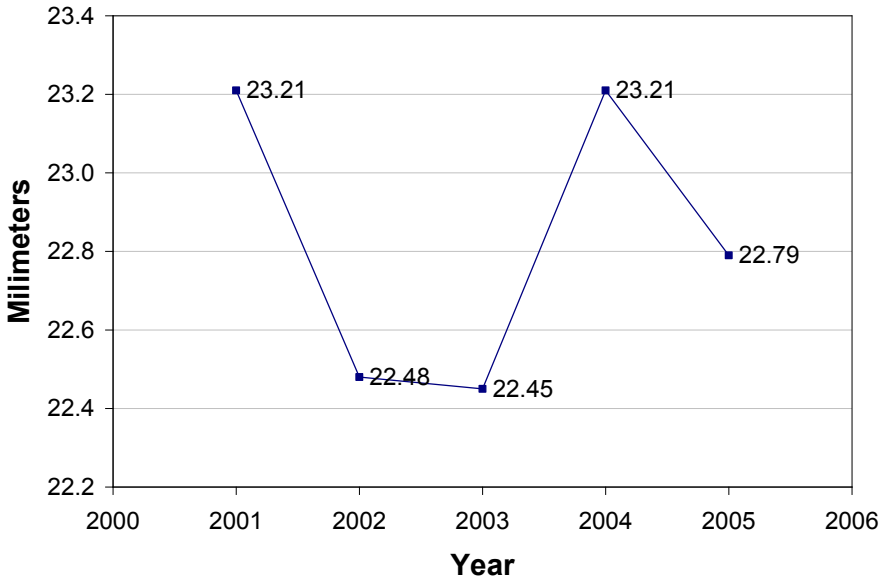


FIGURE C-52. Smoke point average JP-8 (year trend 2001-2005) ***Courtesy PQIS 2005.***

TABLE C-LXXIV. Smoke point fuel property rating value.

	New Fuel
Y	<25 mm/min; <19 mm w/ napthalenes
G	≥25 mm/min; ≥19 mm w/ napthalenes

TABLE C-LXXV. Reserved.

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C.37 SPECIFIC HEAT (AS A FUNCTION OF TEMPERATURE) – SUBSET 1.

C.37.1 Definition. The specific heat of a fuel is the amount of heat-energy transferred into or out of a unit mass of the fuel when increasing or decreasing its temperature. In fuel system analysis, specific heats are used in the calculation of heat transfer, using the fuel as a coolant or as a heat sink.

C.37.2 Standard Specific Heat Test Methods.

C.37.2.1 Currently Calculated. Figure C-53 shows the specific heat as a function of temperature [CRC, 2004]. These data were derived from experimental and calculated methods. The experimental data were developed using a differential scanning calorimeter, while the bulk of the determinations were calculated from a correlation published by J. B. Maxwell using averaged fuel gravity and distillation data.

C.37.3 Comments. Typically, the specific heat at constant pressure (C_p) is the property of interest. The energy (enthalpy) absorbed during a heat transfer process that increases a volume of fuel by a temperature increment (ÄT) is the product of C_p and ÄT. The specific heats (and enthalpies) of most jet fuels are very similar. Generalized aviation fuel correlations typically include only temperature and density [Barnett and Hibbard, 1956], so controlling fuel density within the experience base should maintain the specific heat within the experience base also. The specific heat is a weak function of density, typically approximated as proportional to density to the negative 0.5 power.

C.37.4 Data / Property Occurrence.

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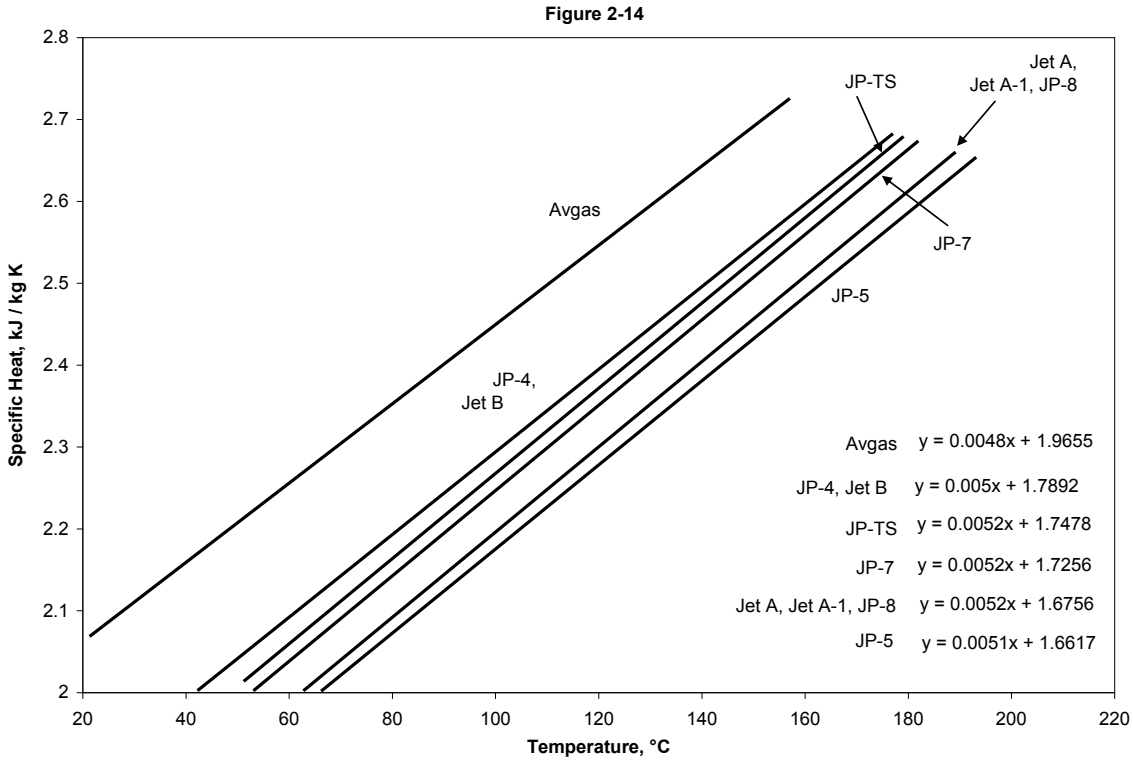


FIGURE C-53. Specific heat as a function of temperature *[CRC,2004].*

TABLE C-LXXVI. Specific heat as a function of temperature fuel property rating value.

	New Fuel
R	<JP-5 values
Y	>JP-8 values
G	JP-5/JP-8 values

TABLE C-LXXVII. Reserved.

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C.38 STORAGE STABILITY – SUBSET 1.

C.38.1 Definition.

The ability of a substance to remain chemically unaltered while in a storage environment.

C.38.2 Storage Stability Test Methods.

C.38.2.1 MIL-STD-3004. MIL-F-16884 defines the Storage Stability Requirement for F-76 Naval Distillate Fuel. Based on this and MIL-STD-3004 (Quality Surveillance for Fuels, Lubricants and Related Products), the ASTM-D-5304 test method (Assessing Distillate Fuel Storage Stability by Oxygen Overpressure) was developed to evaluate the storage stability of fuel.

C.38.3 Comments. Conventional jet fuels do not have a storage stability requirement in their specifications. Storage stability of fuel is important but difficult to test in a reasonable time frame. A fuel that is not stable during storage will form gums and other insoluble material, which can affect fuel and engine system operation and durability. The JP-7 specification (MIL-DTL-38219D) required a one-year storage test in a drum at 130°F. The JP-7 had to meet the specification requirements at the end of three-month intervals during this period. There is no corresponding requirement for MIL-DTL-83133F fuel - which is not surprising given the differences between the two fuels. JP-7 was typically purchased from a single manufacturer and stored at just a few bases. A corresponding storage stability test regime for MIL-DTL-83133F fuel would require hundreds of drums being stored (and storage stability problems identified long after the fuel was burned). Several organizations have identified accelerated storage stability tests, where tests are run at higher temperatures and/or elevated oxygen concentrations to decrease the test duration to days rather than years.

C.38.4 Data / Property Occurrence.

TABLE C-LXXVIII. Storage Stability fuel property rating value.

	New Fuel
G	Report

TABLE C-LXXIX. Reserved.

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C.39 SULFUR, MERCAPTAN – ENTRANCE CRITERIA.

C.39.1 Definition.: Mercaptan sulfur, also referred to as thiols or sulfides, is a class of organic compounds that contain a sulfur-hydrogen group (SH) bound to a hydrocarbon chain (R) to form R-SH. The sulfur group in mercaptans increase the reactivity of the compound and this can lead to a corrosive attack on fuel wetted metal components within a system.

C.39.2 Standard Mercaptan Test Methods.

C.39.2.1 Mercaptan Sulfur by ASTM D3227. The fuel sample is dissolved in an alcoholic sodium acetate titration solvent and titrated potentiometrically with silver nitrate solution, mercaptan sulfur content is then reported in mass percent.

C.39.2.2 Doctor Test, ASTM D4952. This is a test in which the sample is shaken with sodium plumbite solution to which a small amount of powdered sulfur is added. The presence mercaptans or hydrogen sulfide will result in discoloration of the sulfur at the interface or discoloration of either liquid phase. Results are reported as pass or fail.

C.39.3 Comments. Mercaptan sulfur is a test directly relating to durability of aircraft and logistics.

C.39.4 Data / Property Occurrence.

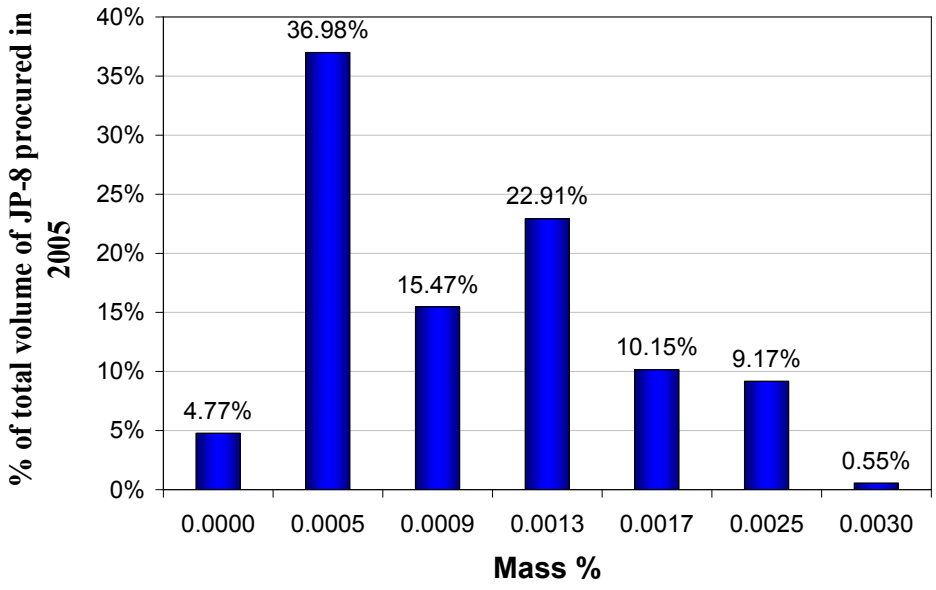


FIGURE C-54. Mercaptan sulfur histogram JP-8 (2005) ***Courtesy of PQIS.***

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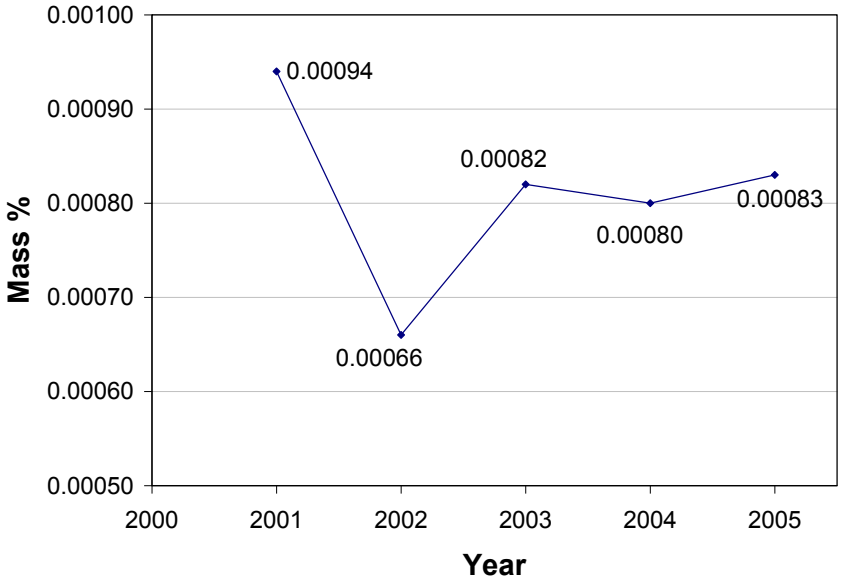


FIGURE C-55. Mercaptan sulfur average JP-8 (year trend 2001-2005) ***Courtesy PQIS 2005.***

TABLE C-LXXX. Mercaptan sulfur fuel property rating value.

	New Fuel
R	>0.003 mass %
Y	0.0021 to 0.003 mass %
G	0.00 to 0.002 mass %

TABLE C-LXXXI. Reserved.

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C.40 SULFUR, TOTAL – ENTRANCE CRITERIA.

C.40.1 Definition. Total sulfur for fuels include any and all forms sulfur, either elemental or in a compound. This includes but is not limited to mercaptan sulfur (thiol, R-S-H, where R is a carbon atom), hydrogen sulfide, free sulfur, sulfides (R-S-R,), disulfides (R-S-S-R), and thiophenes (sulfur incorporated into carbon ring structures). All fuels derived from the distillation of petroleum contain some form of sulfur unless some type of sulfur removal process like hydrotreating is employed.

C.40.2 Standard Sulfur Test Methods.

C.40.2.1 Sulfur Content ASTM D129, ASTM D1266, ASTM D2622, ASTM D3120, ASTM D4294, or ASTM D5453. A variety of methods are available for the quantitative determination of sulfur content. These include combustion methods in which the resultant sulfur oxides are measured, measurement of the X-ray fluorescence of the sulfur compounds or measurement of the fluorescence of sulfur oxides exposed to ultraviolet radiation. Sulfur content is reported in mass percent.

C.40.3 Comments. Further work and review of historical data is required to determine if there is a yellow limit on the amount of sulfur allowable. There may be a lower limit linked to the lubricity of the fuel but that relationship has yet to be verified. Diesel and gasoline have much more stringent sulfur limits currently than jet fuel. There is a Current National Security Issue (waiver) for using jet fuel in diesel-powered vehicles.

C.40.4 Data / Property Occurrence.

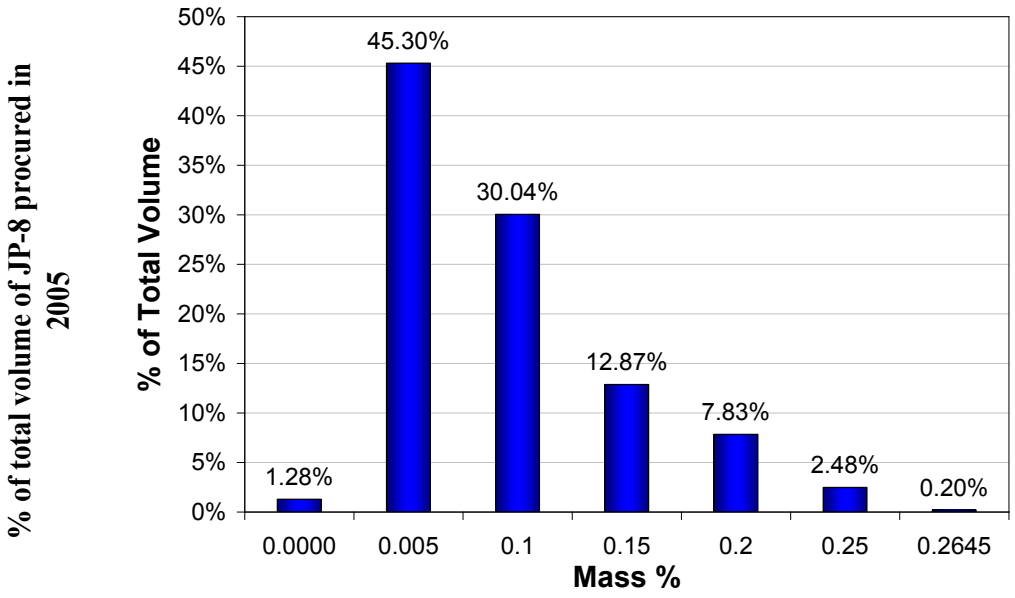


FIGURE C-56. Total sulfur histogram JP-8 (2005) ***Courtesy of PQIS.***

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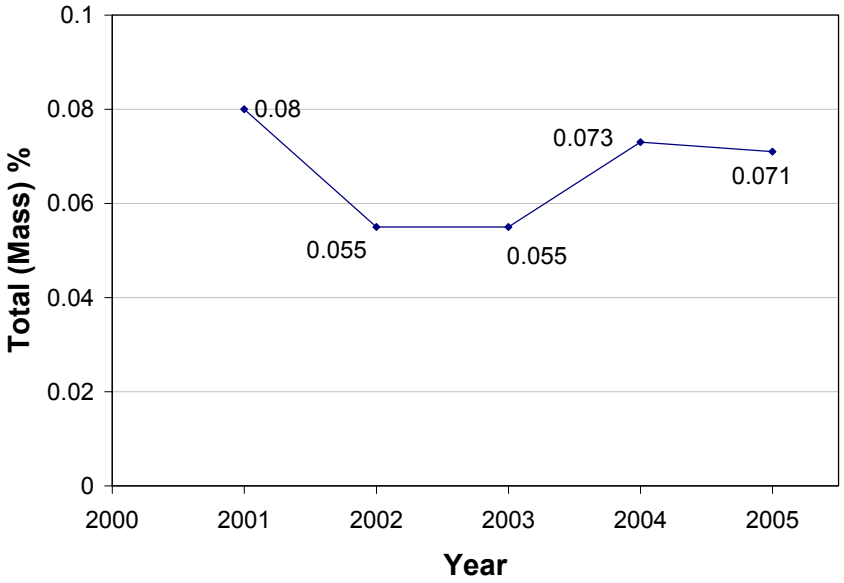


FIGURE C-57. Total sulfur average JP-8 (year trend 2001-2005) ***Courtesy PQIS 2005.***

TABLE C-LXXXII. Total sulfur fuel property rating value.

	New Fuel
G	0 to 0.3 mass %
R	>0.3 mass %

TABLE C-LXXXIII. Reserved.

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C.41 SURFACE TENSION VERSUS TEMPERATURE – SUBSET 1.

C.41.1 Definition. The specific free energy of a liquid surface at interface with another fluid is surface tension. Values for surface tension are usually given when the surface of the liquid is in contact with air. Of importance in gas evolution and solubility, it has a pronounced effect on atomization characteristics of fuels. Fluids with large cohesive forces among molecules, like those found in water, exhibit high surface tensions. Non-polar fluids such as hydrocarbons have lower internal cohesive forces and lower surface tensions. Surface tensions decrease toward zero as temperature increases, and cohesive forces are overcome until, at the fluid's critical temperature, surface tension ceases to exist. Surface tension can be estimated by using the Ramsey and Shields correlation if density, molecular weight, and the critical temperature of the fluid are known. The surface tension data for fuels on Figure C-58 of the CRC 2004 Handbook have been estimated from the Ramsey and Shields correlation, and the figure shows the reduction of surface tension caused by increasing temperature. Impurities, in particular surfactants, have a very strong effect, causing a reduction in surface tension. In such situations, a direct measure of surface tension is necessary to obtain meaningful data.

C.41.2 Standard Surface Tension Test Methods.

C.41.2.1 Surface Tension by ASTM D971. Standard Test Method for Interfacial Tension of Oil against Water by the Ring. Method is commonly used for jet fuel. This method is, typically, used for mineral oils. However, since a standard test method for the surface tension of aviation fuels does not exist; it is occasionally used for this purpose. Surface tension in this method is a function of the force required for a platinum ring to be pulled through a fuel/water interface, the densities of the fuel and water, and the dimensions of the ring (CRC World Fuel Survey). Additionally, data reported below deviates from values in CRC's fuel handbook because the values were calculated using Ramsey and Shields correlation and do not directly correlate to the experimental results below.

C.41.2.2 Surface Tension by ASTM D1331. An alternative method for the measurement of surface tension is ASTM D1331, "Standard Test Methods for Surface and Interfacial Tension of Solutions of Surface-Active Agents". This method uses a precision tensiometer to measure the force required to pull a platinum ring through the test fluid.

C.41.3 Comments. Surface tension is currently not measured or controlled in jet fuel specifications. The temperature range to be measured is -10 to 40°C.

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C.41.4 Data / Property Occurrence.

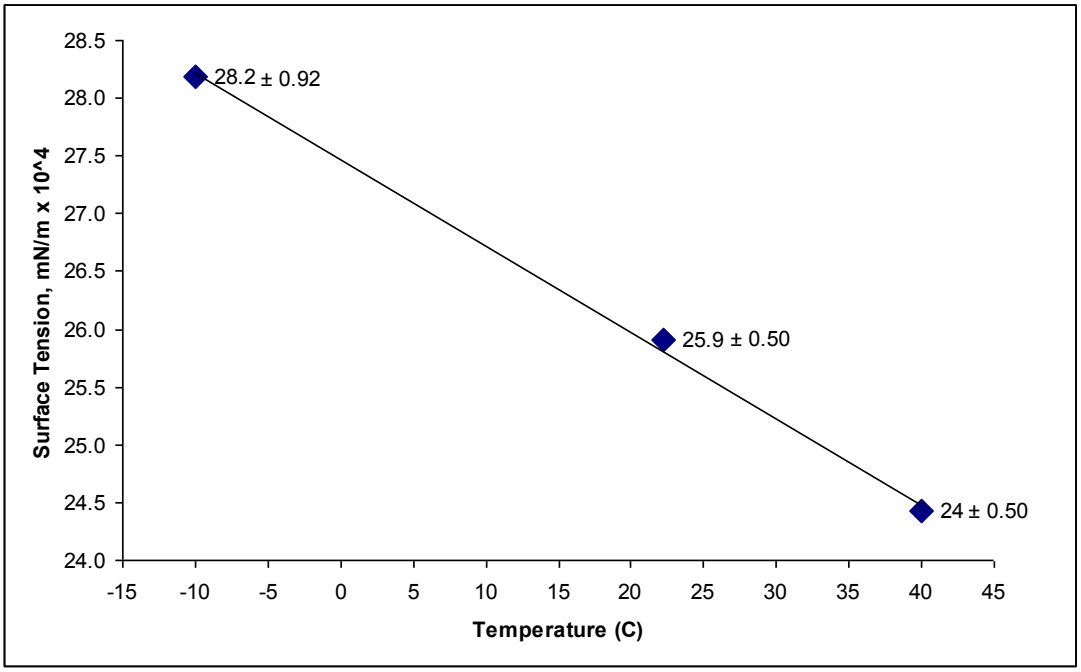


FIGURE C-58. World average jet fuel surface tension. ***Courtesy CRC World Fuel Survey.***

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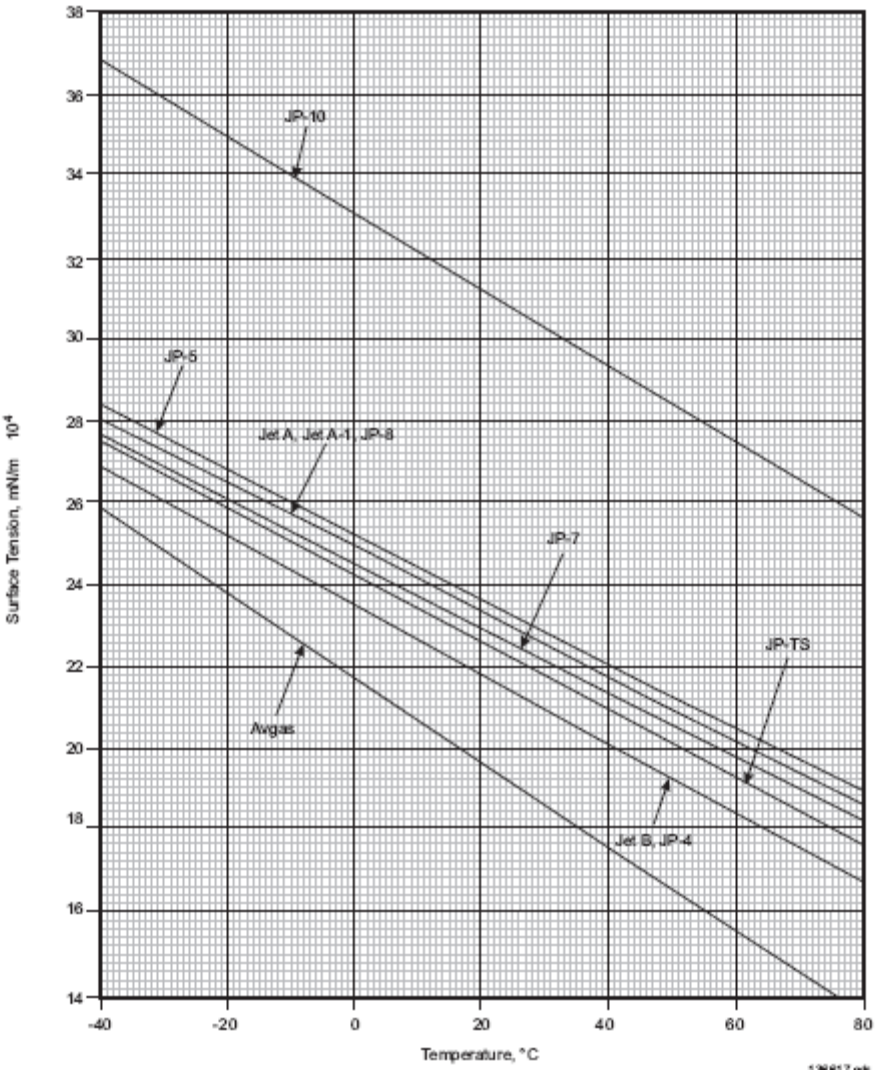


FIGURE C-59. Surface tension as a function of temperature *[CRC, 2004].*

TABLE C-LXXXIV. Surface tension fuel property rating value at 25°C.

	New Fuel
R	>JP-5 values
G	JP-5/JP-8 values
Y	JP-4 values
R	<JP-4 values

TABLE C-LXXXV. Reserved.

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C.42 THERMAL CONDUCTIVITY VERSUS TEMPERATURE – SUBSET 1.

C.42.1 Definition. The thermal conductivity of a fuel is the property that controls the rate at which heat can flow by conduction through that fuel and is expressed as watts per meter Kelvin (W/m K). It is used extensively in heat-transfer calculations when fuel temperature is elevated in heat exchangers, used as a heat sink, when fuel is heated or cooled in flight or on the ground, or whenever there is a temperature gradient within the fuel.

C.42.2 Standard Thermal Conductivity Test Methods.

C.42.2.1 Thermal Conductivity by ASTM D2717. In ASTM D2717, "Standard Test Method for Thermal Conductivity of Liquids," thermal conductivity is measured in a borosilicate glass tube (cell) with a calibrated platinum resistance thermometer. Thermal conductivity is determined by measurement of the temperature gradient produced across the liquid sample by a known amount of energy introduced into the cell by electrically heating the platinum element.

C.42.2.2 Transient Hot-Wire Test Method. In the transient hot-wire technique (essentially similar to ASTM D2717), small diameter wires are immersed in the fluid and used simultaneously as electrical resistance heaters and as resistance thermometers to measure the resulting temperature rise due to the resistance heating. The hot-wire cells are designed to approximate a simple 1-dimensional transient line-source of heat in an infinite medium as closely as possible to minimize corrections for the actual geometry. Two hot wires of differing length are operated in a differential mode to eliminate axial conduction effects due to the large diameter leads attached to the ends of each hot wire. Based on the transient line-source model, the thermal conductivity can be found from the slope of the measured linear temperature rise as a function of elapsed time with a typical uncertainty of less than 1%. The thermal diffusivity can be found from the intercept of this same linear temperature rise curve with a typical uncertainty of less than 10%. Independent measurements of the thermal diffusivity of fluids can also be made by dynamic light scattering (DLS) measurements at low angles with a typical uncertainty of 2%. (courtesy of National Institute of Standards and Technology (NIST), Physical and Chemical Properties Division, Experimental Properties of Fluids Group, Mail Stop 838.07, 325 Broadway, Boulder CO 80305-3328)

C.42.3 Comments. Thermal conductivity is currently not measured or controlled in jet fuel specifications. The measurements are difficult to perform accurately, hence the variations in the data from edition-to-edition in the CRC Handbook. More research is ongoing. The thermal conductivity of typical jet fuels is very similar, thus the CRC Handbook shows a single line for all jet fuels, as reproduced on Figure C-60. Thermal conductivity data is needed in the temperature range from 30-200°C.

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C.42.4 Data / Property Occurrence.

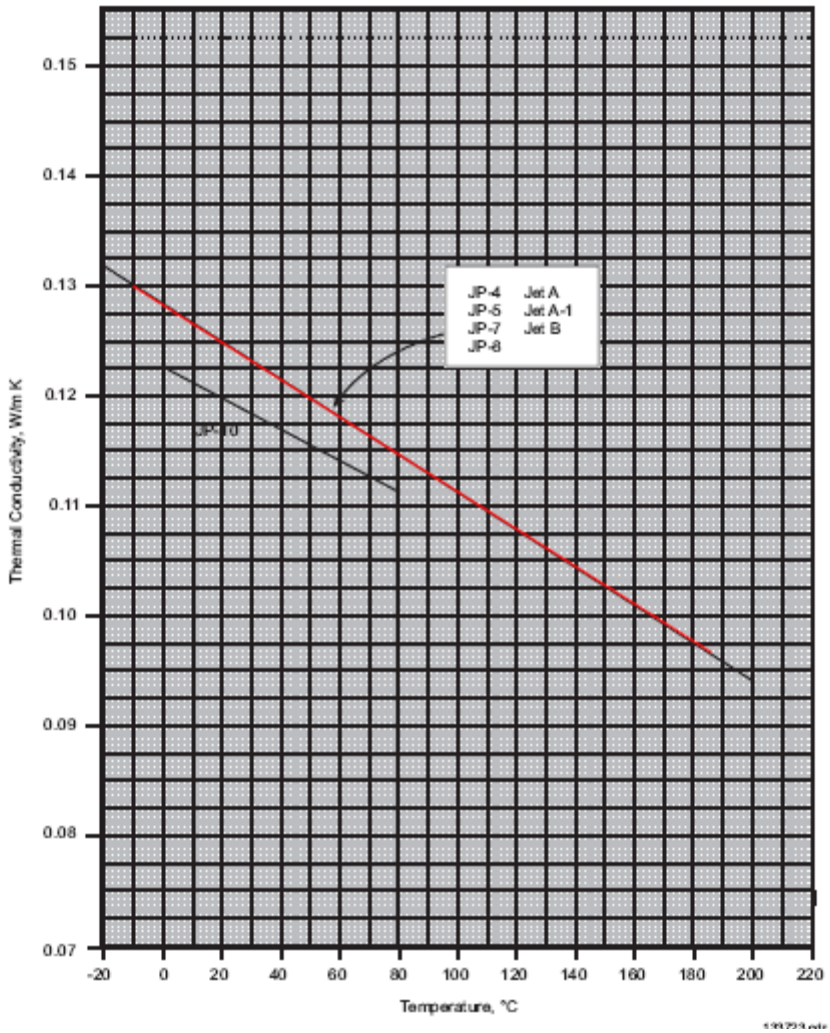


FIGURE C-60. Thermal conductivity as a function of temperature ***[CRC, 2004].***

TABLE C-LXXXVI. Thermal conductivity fuel property rating value.

	New Fuel
Y	<JP-8/JP-5 values
G	JP-8/JP-5 values

TABLE C-LXXXVII. Reserved.

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C.43 THERMAL EXPANSION (COEFFICIENT OF) – SUBSET 2

See C.12.

FIGURE C-61. Reserved.

TABLE C-LXXXVIII. Reserved.

TABLE C-LXXXIX. Reserved.

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C.44 THERMAL STABILITY – ENTRANCE CRITERIA.

C.44.1 Definition: Thermal Stability is the measure of a substance's ability to handle increased temperatures without compromising the substance's chemical integrity.

C.44.2 Standard Thermal Stability Test Methods.

C.44.2.1 Jet Fuel Thermal Oxidation Tester (JFTOT) by ASTM D3241. Nitrogen Pressure System. The 450 mL of fuel is placed into a closed system that is pressurized by nitrogen to 500 psi (3,500 kPa). The fuel flows over a heated aluminum tube into a filter screen, through a cooler and then a metering pump which regulates the flow rate. The maximum tube temperature is the test control variable. The fuel makes one pass through the system. A fuel's oxidation resistance is measured by the color and extent of deposits on the tube and the pressure drop across the filter at the end of the test. Color ratings range from 0 to 4, with 0 being a clean tube, while 4 represents dark deposits. Pressure drop is reported in mm of Hg.

C.44.2.2 Hydraulic System. In this later version of the method fuel is pushed through the system by a hydraulic piston driven by a screw. The test section, operating conditions and the evaluation of test results are identical to the nitrogen pressure system.

C.44.3 Comments. The specification thermal stability test is a quality control, "go/no go" test. Note that the PQIS data reproduced below shows that fuels typically are not close to failing the test on ÄP. Thus the JFTOT results are not particularly informative. An alternative method for assessing thermal stability is running the test at increasing temperatures until the fuel fails the test criteria. The temperature at which the fuel fails the test is termed its "break point". The break point distribution for a recent world fuel survey is shown below. Typically, the temperature limit of jet fuels (JP-5/8, Jet A/Jet A-1) during use is considered to be 300-325°F. The USAF developed a thermal stability additive to raise this temperature limit 100°F in the 1990s - hence it was termed "JP-8+100". Additionally, many other thermal stability tests have been performed. The draft ASTM "Standard Practice – Guideline for the Qualification and Approval of New Aviation Turbine Fuels and Fuel Additives" recommends that the JFTOT tube deposit be assessed by the standard visual technique and the alternate ellipsometer technique to minimize the risk that changes in alternative fuel deposit appearance would cause them to be overlooked.

C.44.4 Data / Property Occurrence.

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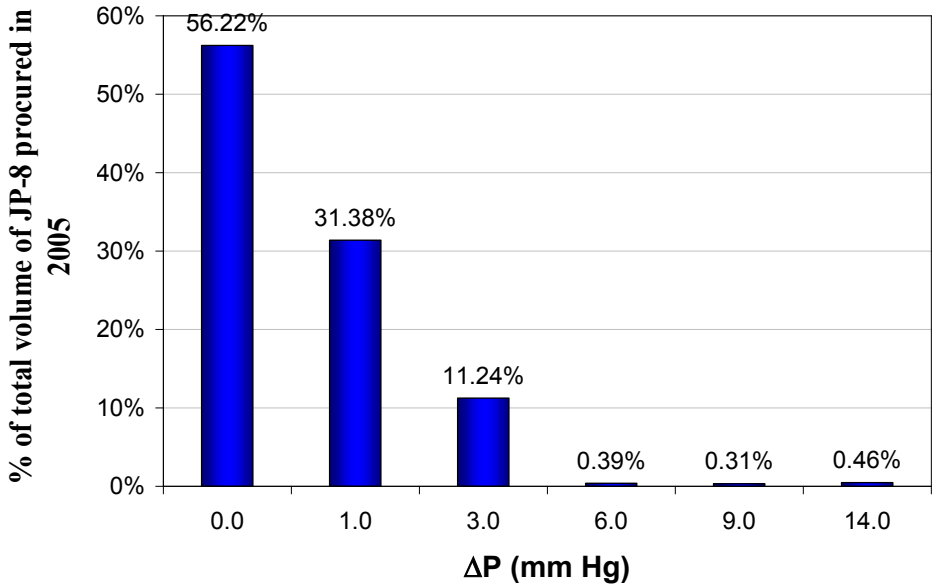


FIGURE C-62. Thermal stability (JFTOT ΔP) histogram JP-8 (2005) ***Courtesy of PQIS.***

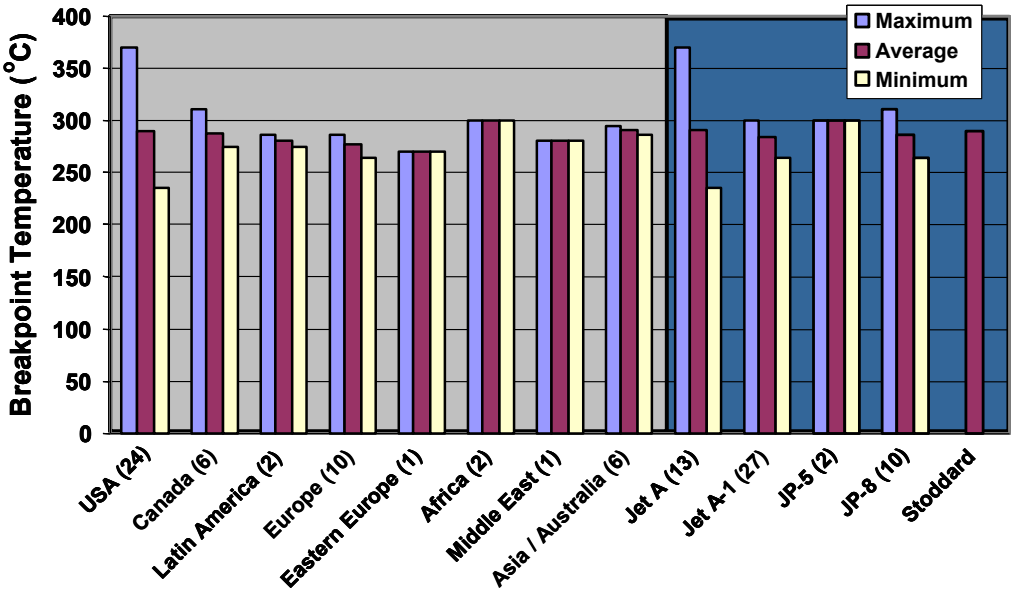


FIGURE C-63. Jet fuel break point distribution ***[CRC World Fuel Survey, 2006].***

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TABLE C-XC. Thermal stability fuel property rating value.

ΔP	New Fuel
G	≤25 mm Hg
R	>25 mm Hg

TABLE C-XCI. Thermal stability fuel property rating value.

Tube rating	New Fuel
G	1 to 2
R	>3

TABLE C-XCII. Reserved.

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C.45 TRACE ELEMENTS – SUBSET 1.

C.45.1 Definition. The category of Trace Elements includes trace metals and trace organics, as described below. Other trace contaminants are described elsewhere (sulfur, acid number), as are dissolved gases. The trace metals of concern are Cu, Zn, Fe, V, Ca, Li, Pb, K, Na, P, Mg, Ni, Si, and Mn, as described below in C.45.2. The trace organic species of concern are the oxygen-containing species carbonyls, alcohols, esters, and phenols, as described in C.45.3 below.

C.45.2 Trace Metals.

C.45.2.1 Determination of Trace Elements in Middle Distillate Fuels by Inductively Coupled Plasma Atomic Emission Spectrometry (ICP_AES) by ASTM D7111. Calibration standards are prepared by mixing organometallic standard materials in kerosene. An internal standard material is added to the calibration standards and fuel samples. The calibration standards and the fuel samples are aspirated into the ICP-AES instrument. The concentrations of the elements in the fuel are calculated by comparing emission intensity ratios of the fuel and calibration standards to the internal standard.*

C.45.2.2 Trace Metals Comments. There are a number of trace metals that can be found in jet fuels that can have deleterious effects on the fuel/engine systems. For example, one set of trace contaminants that were harmful to engines consisted of V (vanadium), Na (sodium), K (potassium), Pb (lead), and Ca (calcium) [Boyce, M. P., Gas Turbine Engine Handbook, Elsevier, 2006]. Na, K, and S (sulfur) lead to "hot corrosion" of turbine blade materials. Note that sodium would be prevalent in sea spray (from salt, sodium chloride), obviously an issue for jet engines operating near the oceans. Copper (Cu) is most notorious fuel contaminant that leads to thermal stability problems (deposit formation) [Hazlett, R. N., *Thermal Oxidation Stability of Aviation Turbine Fuels*, ASTM Monograph 1, American Society for Testing and Materials, Philadelphia, PA, 1991]. Zinc (Zn), is also a noted reactive metal and is typically tested for in most USAF jet fuels.

C.45.3 Trace Organic Species.

C.45.3.1 - Determination of trace organic species. Trace organic species are typically measured by gas chromatographic or other sensitive techniques, as outlined in methods such as EPA 8015B, EPA 8260C, EPA 8270C, UOP624, and UOP 626. Some of the trace organic species would also be detected in other tests, such as acid number and existent gum.

C.45.3.2 Trace Organics Comments. Trace organic species are mostly a concern from the standpoint of their effect on fuel thermal stability [Hazlett].

C.45.4 Data / Property Occurrences. Data for trace metals and trace organics is fairly sparse. Some data is listed in Hazlett.

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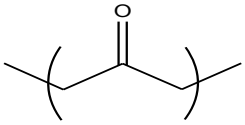
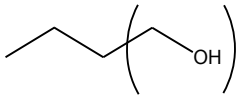
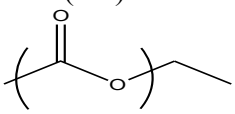
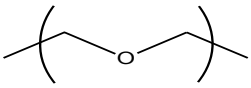
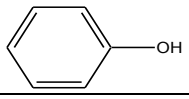
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TABLE C-XCIII. Trace metals property rating value.

Copper (Cu)	<20 ppb	G
	>20 ppb	Y
Zinc (Zn)	<50 ppb	G
	>50 ppb	Y
Iron (Fe)	<50 ppb	G
	>50 ppb	Y
Vanadium (V)	<50 ppb	G
	>50 ppb	Y
Calcium (Ca)	<50 ppb	G
	>50 ppb	Y
Lithium (Li)	<50 ppb	G
	>50 ppb	Y
Lead (Pb)	<50 ppb	G
	>50 ppb	Y
Potassium (K)	<50 ppb	G
	>50 ppb	Y
Sodium (Na)	<50 ppb	G
	>50 ppb	Y
Phosphorus (P)	<50 ppb	G
	>50 ppb	Y
Magnesium (Mg)	<50 ppb	G
	>50 ppb	Y
Nickel (Ni)	<50 ppb	G
	>50 ppb	Y
Silicon (Si)	<50 ppb	G
	>50 ppb	Y
Manganese (Mn)	<50 ppb	G
	>50 ppb	Y
Total measured metals	<100 ppb	G
	>100 ppb	Y

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TABLE C-XCIV. Trace organics property rating value.

Trace organic	Chemical functional group	Limit	
Carbonyls	-C=O 	G	Report
Alcohols	-C-O-H 	G	<5 ppm
		Y	>5 ppm
Esters/ Ethers	-C(=O)-O-C-  /-C-O-C- 	G	<50 ppm
		Y	>50 ppm
Phenols	aromatic ring – OH 	G	<50 ppm
		Y	>50 ppm

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C.46 VAPOR PRESSURE, TRUE VERSUS TEMPERATURE - SUBSET 1.

C.46.1 Definition. Vapor pressure is the pressure exerted by the vapor of a liquid when in equilibrium with the liquid at a specific temperature.

The heat of vaporization (ΔH_{vap}), often termed the latent heat of vaporization or evaporation or just latent heat, is the amount of heat added to vaporize a unit weight of a liquid at a constant pressure below the critical point. For multicomponent fluids such as jet fuel, heat of vaporization can be calculated from vapor pressure versus temperature data.

C.46.2 Standard Vapor Pressure Test Methods.

C.46.2.1 Vapor Pressure by ASTM D5191. A volume of chilled, air-saturated sample is introduced into a thermostatically controlled, evacuated test chamber with a moveable piston that expands the volume after sample introduction. After introduction into the test chamber, the test specimen is allowed to reach thermal equilibrium at the specified test temperature. The resulting rise in pressure in the chamber is the vapor pressure.

C.46.2.2 Reid Vapor Pressure by ASTM D323. The Reid vapor pressure is the pressure exerted by a fuel when heated to a specified temperature in a pressure vessel with a vapor-to-liquid ratio of 4:1. Prior to the test the sample is saturated with water. Reid Vapor Pressure includes the partial pressures of air and water vapor.

Reid vapor pressure deviates from a liquid's vapor pressure by 1 to 5%. Correlations exist to convert between vapor pressure and Reid vapor pressure. Frequently, this test method may be more commonly found at certain test labs.

C.46.2.3 Heat of Vaporization by ASTM E2071. Using vapor pressure versus temperature data, the heat of vaporization can be correlated via the Clausius Clapeyron equation as described in the following equation

$$\ln\left(P^{vap}\right) = -\frac{\Delta H^{vap}}{RT} + C$$

Pvap = Vapor Pressure,

ΔHvap = Heat of Vaporization,

R = Ideal Gas Constant,

T = Absolute Temperature, and

C = Constant of Integration.

By plotting the natural log of the vapor pressure versus the inverse of the absolute temperature, the slope of the line created is the average heat of vaporization divided by the ideal gas constant. If the line created is curved or otherwise non-linear, then the heat of vaporization changes as

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temperature changes and the heat of vaporization have to be evaluated over the desired temperature range.

C.46.3 Comments. For safety considerations when dealing with vehicles and ground support equipment, gravity feed of the fuel from the tank to the engine is not permitted in ground support equipment. A pump is necessary to lift the fuel from the tank. Since the vapor pressure of MIL-DTL-83133F fuel is so low the lifting of the fuel never presented a consideration of vapor lock in the design of the fuel system. The lift pump is usually located at the engine rather than close to the fuel tank. Had vapor lock been a consideration the pump would have been located close to the tank, or as is the standard practice in automotive design, placed inside the tank. A maximum vapor pressure specification for the fuel which would prevent vapor lock for the SE&V is not known at this time and will be the subject of further research. From experience using Russian TS1, vapor pressure maximum of 0.16 psi at 40 °C will not cause vapor lock in the current fleet.

Vapor pressure is not currently specified for JP-5, JP-8, Jet A, Jet A-1, or TS-1. JP-4 does have specified vapor pressure limits of 14-21 kPa at 38°C. Operational experience in 2005, has shown that TS-1's vapor pressure was higher than JP-8 and caused some difficulties, therefore it should be an upper limit vapor pressure. Unfortunately, thorough data on the vapor pressure of TS-1 is not presently available so JP-4 will be considered the upper limit until such time as TS-1 vapor pressure data is established.

Additionally, a 'round robin' on wide cut, kerosene and other fuels has been completed by S/C 14 of ASTM with the goal of being able to measure fuel vapor pressure over a range of temperatures from 25°C up to 70°C using the triple expansion method of VP measurement and in support of the special certification fuel described in ASTM D7223.

Currently, the latent heat of vaporization is not controlled or monitored by either military or commercial specifications. The risk of deviations from typical jet fuel values has not been examined. Therefore, the heat of vaporization should be reported without limitation until better data is available.

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C.46.3 Data / Property Occurrence.

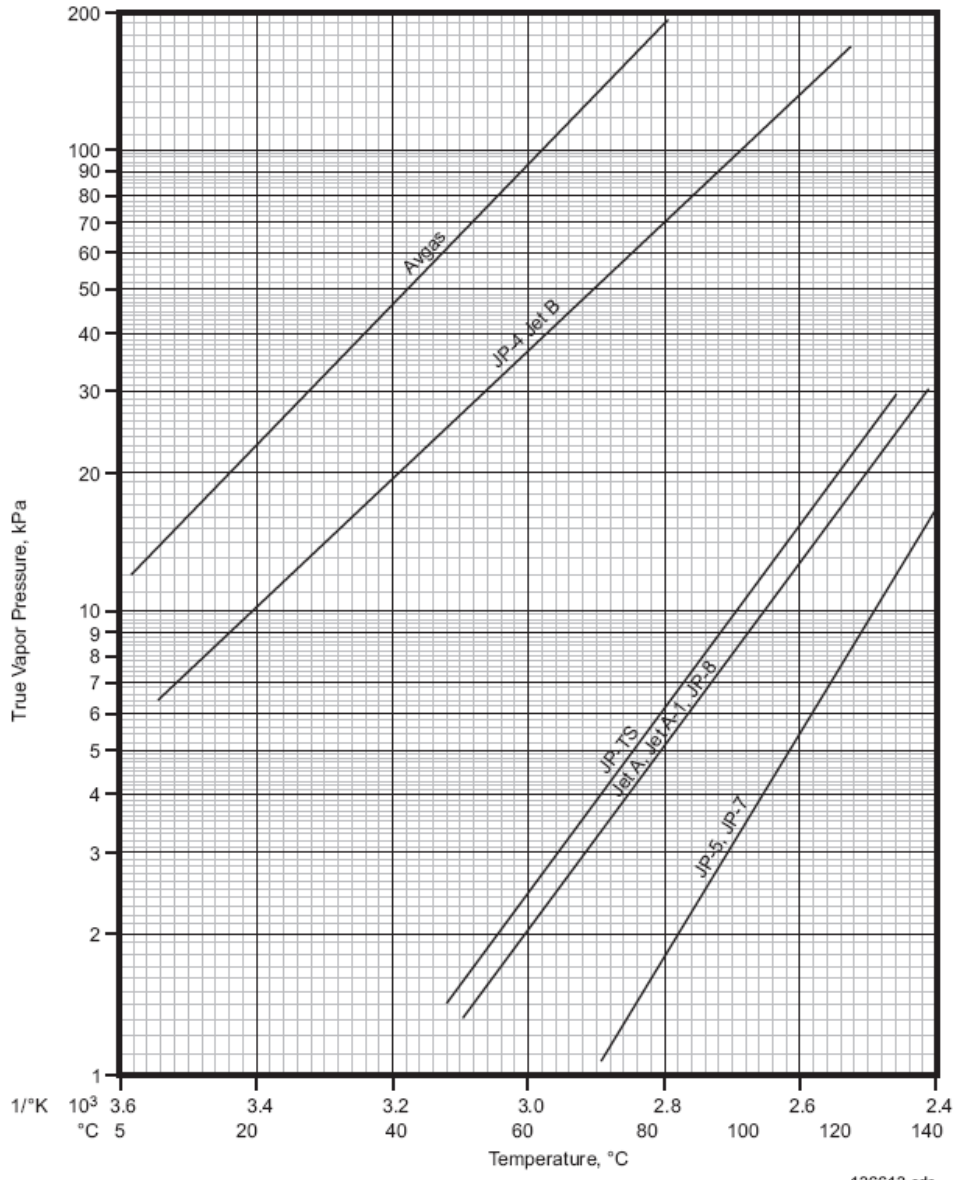


FIGURE C-64. Vapor pressure as a function of temperature [CRC, 2004].

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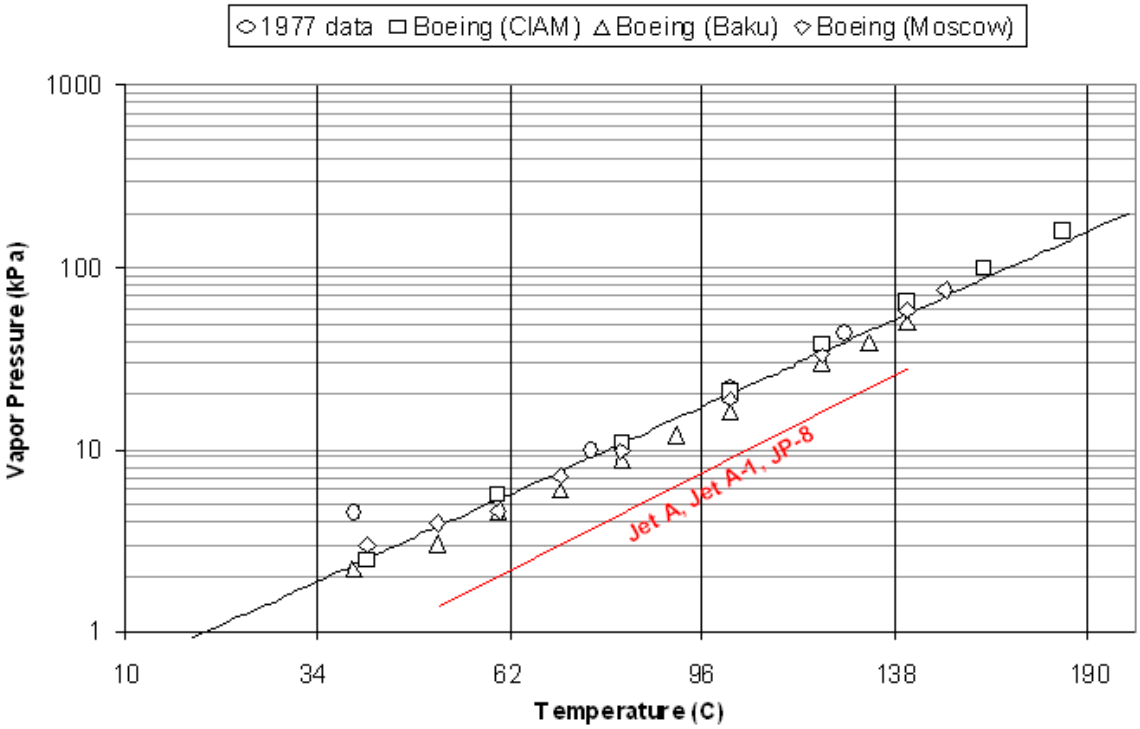


FIGURE C-65. TS-1 Vapor pressure as a function of temperature ***[courtesy Boeing].***

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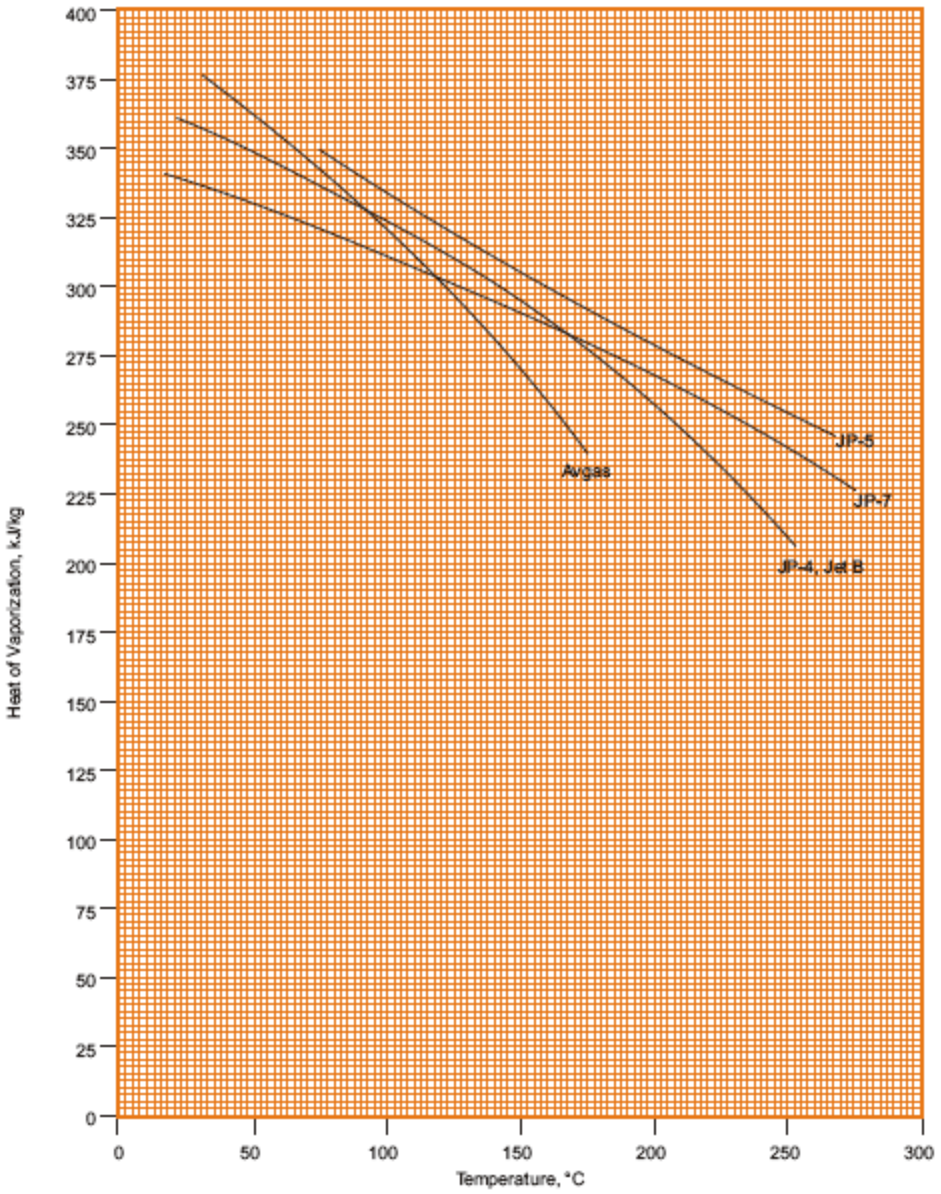


FIGURE C-66. Heat of vaporization as a function of temperature ***[CRC, 2004].***

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TABLE C-XCV. Vapor pressure versus temperature fuel property rating value.

	New Fuel
R	> TS-1 values
Y	< TS-1 values and > JP-8 values
G	≤ JP-8 values

TABLE C-XCVI. Reserved.

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C.46A Velocity of Sound (versus T) – Subset 2.

C.46A.1 Definition. The velocity of sound in a fuel is used in the design of some fuel gauging systems.

C.46A.2 Standard Velocity of Sound Test Methods. Standard test methods are not available.

C.46A.3 Comments. Data are available in the CRC World Fuel Survey, where it is noted that velocity of sound is proportional to density (sound moves faster in denser fuels). The data presented on Figure C-63 is not completely consistent with that statement, where the separation due to density is much less than that seen when the velocity of sound is plotted versus density (Figure 5.3.2 in the World Survey). A standard device often used is a Mapco/Nusonics 6080 analyzer [WL-TR-95-2158]. See also C.8. The relationship between Bulk Modulus and Velocity of Sound is given by the following equations:

Velocity of Sound = Square Root of (Bulk Modulus / Density),

or

Bulk Modulus = Density x (Velocity of Sound Squared).

C.46A.4 Data/Property Occurrence.

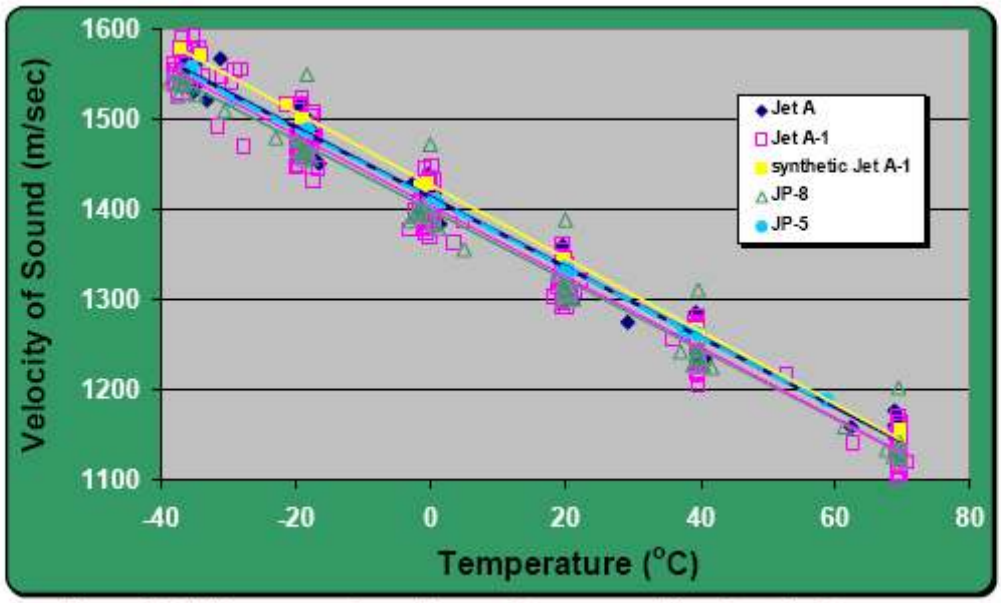


FIGURE C-67. Velocity of sound versus temperature ***[CRC World Fuel Survey, 2006].***

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C.47 VISCOSITY AT -20°C – ENTRANCE CRITERIA (Viscosity versus Temperature – Subset 1).

C.47.1 Definition. The viscosity of a fluid is a measure of its internal resistance to motion caused by cohesive forces among the fluid molecules. Absolute viscosity is the shear stress at a point divided by the velocity gradient at that point, and the unit of absolute viscosity is the Pascal second (P/s). In practice, absolute viscosity is used in conjunction with density, particularly in the calculation of Reynolds number. This relationship between viscosity and density is defined as kinematic viscosity, the ratio of the absolute viscosity of a fluid to the density with both properties measured at the same temperature and pressure, usually expressed as centistoke (cSt), where one cSt is equivalent to one mm^2/sec. Since viscosity varies inversely with temperature, lowering the temperature of the fuel has the effect of increasing its viscosity.

C.47.2 Standard Viscosity Test Methods.

C.47.2.1 Viscosity by ASTM D445. A fixed volume of fuel flows through capillaries of specific diameters and lengths at standard temperatures. The viscosity or resistance to flow is calculated from the flow time and the capillary constant. Viscosity is reported in mm^2/s or cSt.

C.47.3 Comments. Fluid viscosity is critical to proper equipment operations. For aircraft the greatest concern is typically the viscosity at lower temperatures. Aircraft are exposed to extremely low temperatures at high altitudes. As the fuel cools, its viscosity begins to exponentially increase as it approaches the freezing point. Therefore a viscosity ceiling of 8.0 centistokes cannot be exceeded at temperatures equal to or warmer than -20° C. Honeywell has indicated that exceeding 12 cSt at -40° C will impair APU cold start.

Military SE&V frequently use jet fuel in their diesel compression ignition engines. Within these applications, fuels can have a low viscosity at higher temperatures and thereby damage parts that use the fuel as a lubricant. For unlimited use, viscosity should be consistent with ASTM D975, greater than or equal to 1.3 centistokes at 40° C. Viscosities below 1.2 centistokes could show an increasing degree of damage over time. Viscosities below 1.0 may limit durability to such a degree that failure could occur in a very short time. (These values apply to SE&V procured after 2007.).

For new fuels, viscosity should be measured at (as a minimum) -20° C, 25° C, 40° C, and 60° C. Viscosity should be measured at -40° C if possible.

C.47.4 Data / Property Occurrence.

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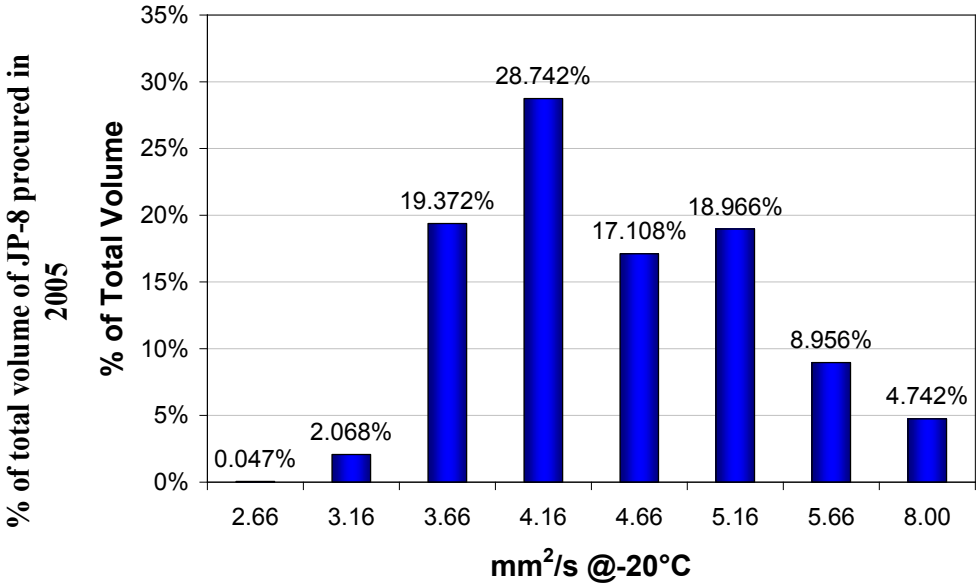


FIGURE C-68. Viscosity (@-20°C) histogram JP-8 (2005) ***Courtesy of PQIS.***

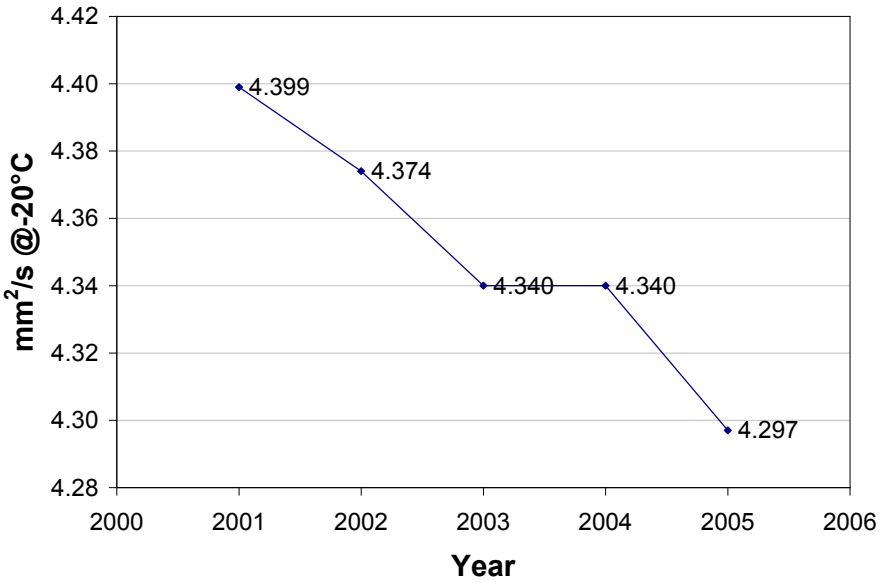


FIGURE C-69. Viscosity (@-20°C) average JP-8 (year trend 2001-2005) ***Courtesy PQIS 2005.***

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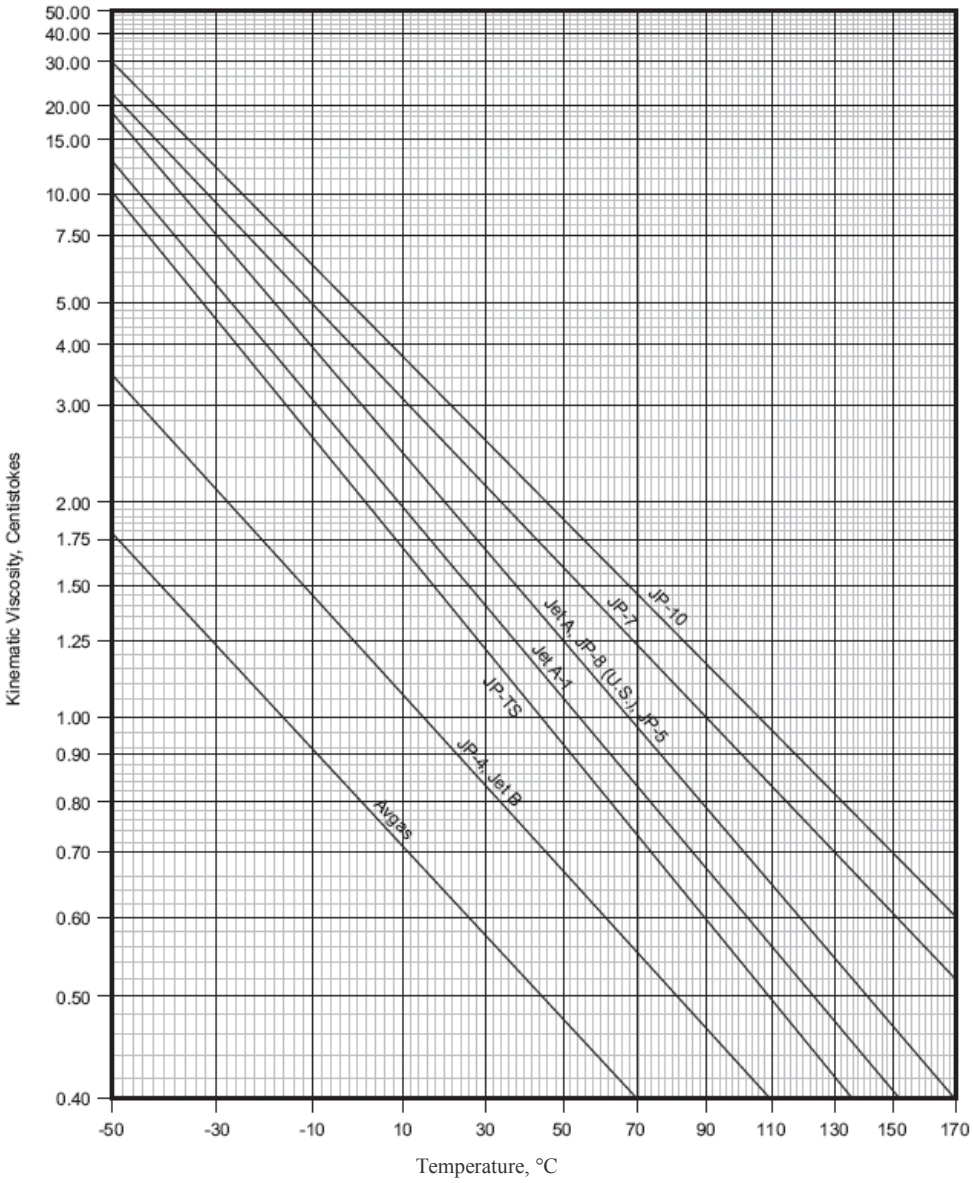


FIGURE C-70. Viscosity as a function of temperature ***[CRC, 2004].***

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TABLE C-XCVII. Viscosity fuel property rating value: aircraft.

-20° C	New Fuel
R	> 8 cSt
G	< 8 cSt

TABLE C-XCVIII. Viscosity fuel property rating value: support equipment & vehicles.

40° C	New Fuel
R	<1.3 cSt
G	≥1.3 cSt

TABLE C-XCIX. Reserved.

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C.48 WATER REACTION INTERFACE RATING – ENTRANCE CRITERIA.

C.48.1 Definition. Water Reaction Interface is a rating method that covers the determination of the presence of water-miscible components in hydrocarbon liquids, and the effect of these components on volume change and on the fuel-water interface.

C.48.2 Standard Water Interface Rating Test Methods.

C.48.2.1 Water Reaction by ASTM D1094 acidity by ASTM D3242. Twenty mL of buffered water is added to 80 mL of test fuel in a graduated glass cylinder and is shaken by hand. After 5 minutes of settling any volume change of either the fuel or the water is reported. For jet fuel the appearance of the interface between fuel and water is rated by comparison to a series of written descriptions which are numbered 1 for the cleanest and 4 for the dirtiest. A letter designation may be assigned to the interface rating to provide additional test result descriptions such as 1B. A pass is defined as a 1 with any letter designation.

C.48.3 Data / Property Occurrence.

TABLE C-C. Water reaction interface rating fuel property rating value.

	New Fuel
Y	>1B
G	1B

TABLE C-CI. Reserved.

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C.49 WATER SEPARATION INDEX – ENTRANCE CRITERIA.

C.49.1 Definition. This is an ASTM test method that addresses the ability of hydrocarbon liquids to release entrained or emulsified water when passed through fiberglass coalescing material.

C.49.2 Standard Water Separation Index Test Methods.

C.49.2.1 Water Separation Characteristics by Portable Separometer by ASTM D3948. A fuel/water emulsion is created in a disposable syringe with a high speed mixer. The emulsion is then pushed through a special fiber glass filter intended to strip the water from the emulsion. The presence of water in the filtrate is then determined by exposing the filtrate to a light beam and measuring the resultant light scatter due to water droplets. Results are given in MSEP units, with a fuel having an MSEP rating of 100 showing no light dispersion due to free water, while a rating of 0 indicates a filtered fuel heavily contaminated with free water.

C.49.3 Data / Property Occurrence.

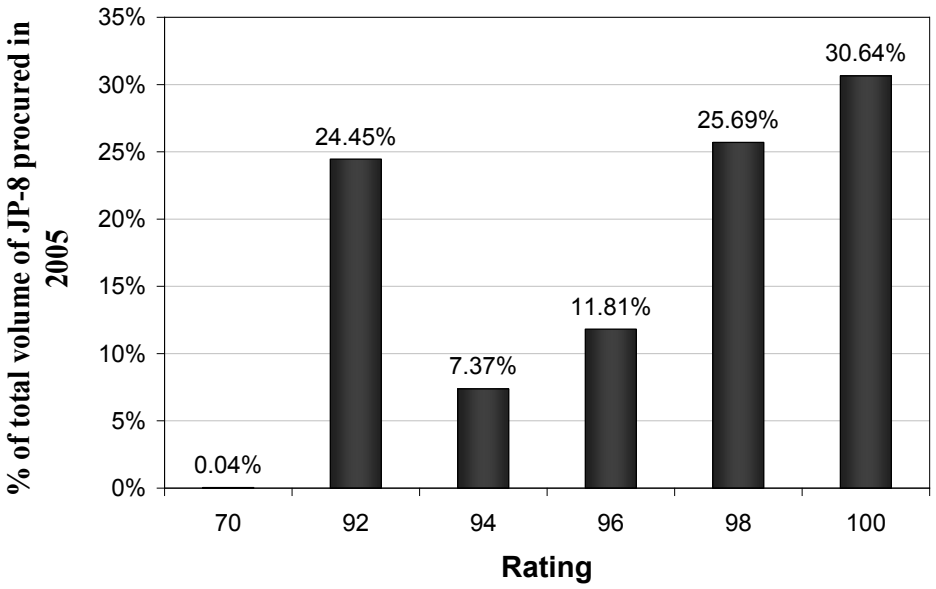


FIGURE C-71. Water separation index histogram JP-8 (2005) ***Courtesy of PQIS.***

TABLE C-CII. Water separation index fuel property rating value.

	New Fuel
G	Report

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TABLE C-CIII. Note 9 of Table 2 of MIL-DTL-83133F: "The minimum microseparometer rating using a Micro-Separometer (MSEP) shall be as follows:

JP-8 Additives	MSEP Rating, min.	Limit	
Antioxidant (AO), Metal Deactivator (MDA)	90	>90	G
		<90	Y
AO, MDA, and Fuel System Icing Inhibitor (FSII)	85	>85	G
		<85	Y
AO, MDA, and Corrosion Inhibitor/Lubricity Improver (CI/LI)	80	>80	G
		<80	Y
AO, MDA, FSII and CI/LI	70	>70	G
		<70	Y"

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C.50 WATER SOLUBILITY – SUBSET 1.

C.50.1 Definition. The quantity of water dissolved in aircraft fuels is determined by the partial pressure of water in the vapor space above the fuel. When this vapor space is saturated with water at a given temperature; i.e., 100 percent relative humidity, the water dissolved in fuel at equilibrium will reflect the saturation values shown on Figures C-72, and C-73; CRC, 2004; PQIS, 2005.

C.50.2 Standard Water Solubility Test Methods.

C.50.2.1 Water Solubility by ASTM D6304. An aliquot is injected into the titration vessel of a coulometric Karl Fischer apparatus in which iodine for the Karl Fisher reaction is generated coulometrically at the anode. When all of the water has been titrated, excess iodine is detected by an electrometric end point detector and the titration is terminated. Based on the stoichiometry of the reaction, 1 mol of iodine reacts with 1 mol of water; thus, the quantity of water is proportional to the total integrated current according to Faraday's Law.

C.50.3 Comments. At relative humidity values less than 100 percent, the amount of water dissolved in fuel will be correspondingly less than saturation values in accordance with Henry's Law. Because water solubility is sensitive to temperature, a drop of 10°C in water-saturated fuel's temperature will create from 15 to 25 ppm of undissolved or "free water." It is difficult to visually detect "free water" at levels below 30 ppm. Several go/no-go tests, including the Shell Water Detector, can be used to detect levels in the range of 15 to 30 ppm. For quantitative measurement in the range of 1 to 60 ppm, ASTM D3240 is frequently used.

C.50.4 Data / Property Occurrence.

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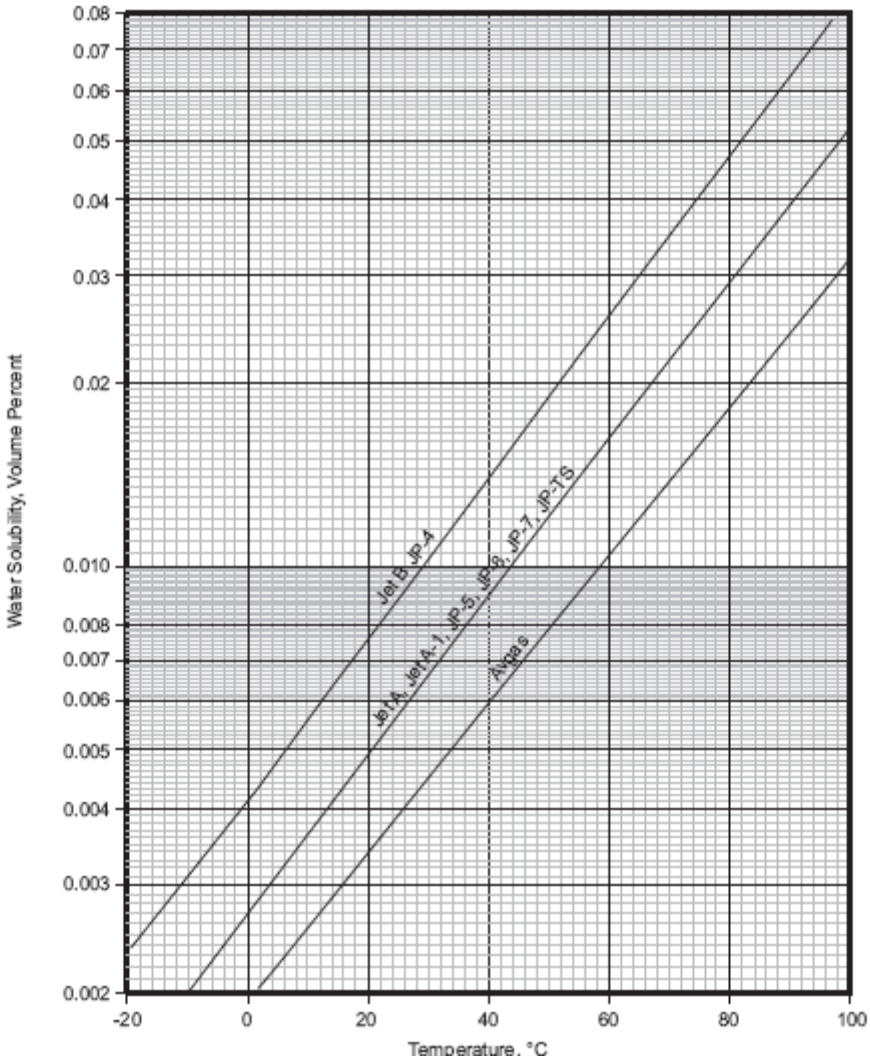


FIGURE C-72. Water solubility versus temperature for aircraft fuels *(CRC, 2004).*

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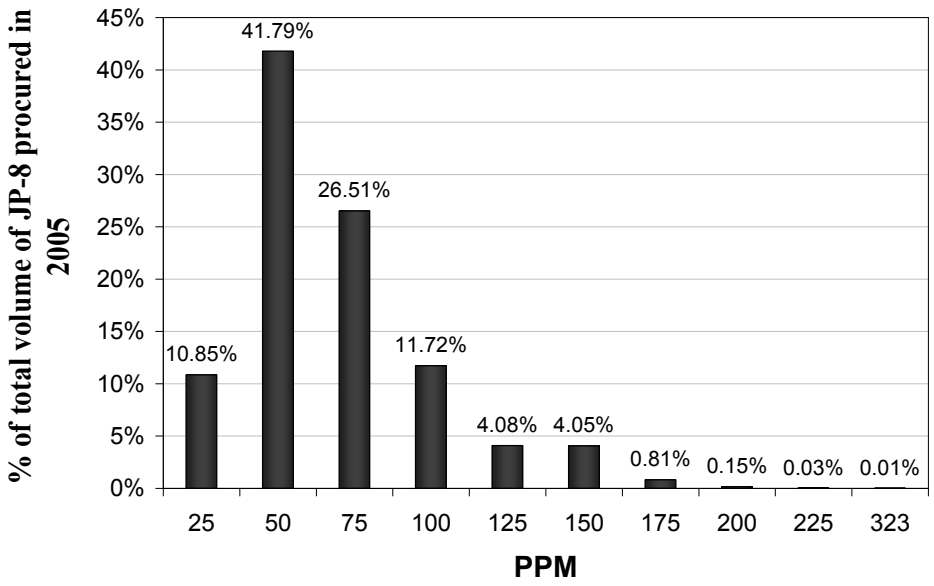


FIGURE C-73. Water solubility histogram JP-8 (2005) ***Courtesy of PQIS.***

TABLE C-CIV. Water solubility fuel property rating value.

	New Fuel
Y	>JP-8/JP-5 values
G	JP-8/JP-5 values

TABLE C-CV. Reserved.

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APPENDIX D

EVALUATING THE COMPATIBILITY OF ADDITIVES OR ALTERNATIVE FUELS WITH FUEL SYSTEM MATERIALS

D.1 SCOPE.

D.1.1 General.

This Appendix describes the recommended practice for testing and approving the material compatibility of fuel additives or alternative fuels with the materials found in aircraft fuel tanks, fuel systems, engines, ground supply vehicles, and the supply chain.

D.1.2 Entrance Criteria and Subset Testing.

The Air Force SMEs evaluated all fuel properties and characteristics based on the requirements decomposition process that correlated requirements to safety, performance, durability and supportability, as described in Appendix C. This Appendix defines the subset to which each material compatibility characteristic belongs and the significance for its selection. It also includes component tests to be conducted to further evaluate the material compatibility characteristics of the candidate fuel requiring further investigation, as deemed appropriate by the Alternative Fuels Certification Office (AFCO) at Wright-Patterson AFB.

D.2 APPLICABLE DOCUMENTS.

D.2.1 General.

The documents listed below are not necessarily all of the documents referenced herein, but are those needed to understand the information provided by this Appendix.

D.2.2 Government documents.

D.2.2.1 Specifications, standards, and handbooks.

The following specifications, standards, and handbooks form a part of this document to the extent specified herein.

FEDERAL SPECIFICATIONS

A-A-3097	Adhesives, Cyanoacrylate, Rapid Room Temperature-Curing, Solventless
QQ-S-571	Solder, Electronic (96 to 485 Deg. C)
MMM-A-132	Adhesives, Heat Resistant, Airframe Structural, Metal to Metal

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FEDERAL STANDARDS

FED-STD-141	Paint, Varnish, Lacquer and Related Materials: Methods of Inspection, Sampling and Testing

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-370	Hose and Hose Assemblies, Nonmetallic: Elastomeric, Liquid Fuel
MIL-H-4495	Hose Assembly, Rubber, Aerial Refueling
MIL-DTL-5541	Chemical Conversion Coatings on Aluminum and Aluminum Alloys
MIL-T-5578	Tank, Fuel, Aircraft, Self-Sealing
MIL-S-6709	Steel, Chrome-Molybdenum-Aluminum; Bars, Rods, Billets and Forging Stock (for Nitriding) (Aircraft Quality)
MIL-R-6855	Rubber, Synthetic, Sheets, Strips, Molded or Extruded Shapes
MIL-I-7444	Insulation Sleeving, Electrical, Flexible
MIL-B-7883	Brazing of Steels, Copper, Copper Alloys, Nickel Alloys, Aluminum and Aluminum Alloys
MIL-PRF-8516	Sealing Compound, Synthetic Rubber, Electric Connectors and Electric Systems, Chemically Cured
MIL-A-8625	Anodic Coatings for Aluminum and Aluminum Alloys
MIL-H-17902	Hose, End Fittings and Hose Assemblies, Synthetic Rubber, Aircraft Fuels
MIL-S-22473	Sealing, Locking, and Retaining Compounds: (Single Component)
MIL-DTL-24441	Paint, Epoxy-Polyamide, General Specification for
MIL-P-25732	Packing, Preformed, Petroleum Hydraulic Fluid Resistant, Limited Service at 275 Deg. F (135 Deg. C)
MIL-DTL-25988	Rubber, Fluorosilicone Elastomer, Oil and Fuel Resistant, Sheets, Strips, Molded Parts, and Extruded Shapes
MIL-DTL-26521	Hose Assembly, Nonmetallic, Fuel, Collapsible, Low Temperature with Non-Reusable Couplings

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MIL-PRF-46010	Lubricant, Solid Film, Heat Cured, Corrosion Inhibiting
MIL-R-46082	Retaining Compounds, Single Component, Anaerobic MIL-PRF-81733 Sealing and Coating Compound, Corrosion Inhibitive
MIL-C-83019	Coating, Polyurethane, For Protection of Integral Fuel Tank Sealing Compound
MIL-DTL-83054	Baffle and Inerting Material, Aircraft Fuel Tank
MIL-DTL-83133	Turbine Fuel, Aviation, Kerosene Type, NATO F-34 (JP-8), NATO F-35, and JP-8 + 100 (NATO F-37)
MIL-S-85334	Sealing Compound, Noncuring, Low Consistency, Silicone, Groove Injection, for Integral Fuel Tanks
DOD-L-85645	Lubricant, Dry Film, Molecular Bonded
MIL-PRF-87260	Foam Material, Explosion Suppression, Inherently Electrostatically Conductive, for Aircraft Fuel Tanks

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

D.2.3 Non-Government publications.

The following documents form a part of this document to the extent specified herein.

ASTM INTERNATIONAL

ASTM A240	Standard Specification for Chromium and Chromium-Nickel Stainless Steel Plate, Sheet, and Strip for Pressure Vessels and for General Applications
ASTM B36	Standard Specification for Brass Plate, Sheet, Strip, and Rolled Bar
ASTM B93	Standard Specification for Magnesium Alloys in Ingot Form for Sand Castings, Permanent Mold Castings, and Die Castings
ASTM D257	Standard Test Methods for DC Resistance or Conductance of Insulating Materials
ASTM D395	Standard Test Methods for Rubber Property—Compression Set

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ASTM D412	Standard Test Methods for Vulcanized Rubber and Thermoplastic Elastomers—Tension
ASTM D471	Standard Test Method for Rubber Property-Effect of Liquids
ASTM D790	Standard Test Methods for Flexural Properties of Unreinforced and Reinforced Plastics and Electrical Insulating Materials
ASTM D1002	Standard Test Method for Apparent Shear Strength of Single-Lap-Joint Adhesively Bonded Metal Specimens by Tension Loading (Metal-to-Metal)
ASTM D1414	Standard Test Methods for Rubber O-Rings
ASTM D2240	Standard Test Method for Rubber Property—Durometer Hardness
ASTM D2624	Standard Test Methods for Electrical Conductivity of Aviation and Distillate Fuels
ASTM D3359-02	Standard Test Methods for Measuring Adhesion by Tape Test
ASTM D3363-05	Standard Test Method for Film Hardness by Pencil Test
ASTM D4066	Standard Classification System for Nylon Injection and Extrusion Materials (PA)
ASTM D4308	Standard Test Method for Electrical Conductivity of Liquid Hydrocarbons by Precision Meter
ASTM D5363	Standard Specification for Anaerobic Single-Component Adhesives (AN)

(Copies of these documents are available online at www.astm.org or from ASTM International, 100 Barr Harbor Drive, P.O. Box C700, West Conshohocken, PA 19428-2959. Approved users may access the documents at www.ihs.com.)

SAE INTERNATIONAL

SAE-AMS-2410	Plating, Silver Nickel Strike, High Bake
SAE-AMS-2427	Aluminum Coating Ion Vapor Deposition
SAE-AMS-3215	Acrylonitrile Butadiene (NBR) Rubber Aromatic Fuel Resistant 65 - 75

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SAE-AMS-3265	Sealing Compound, Polysulfide (T) Rubber, Fuel Resistant, Nonchromated Corrosion Inhibiting for Intermittent Use to 360MDF (182MDC)
SAE-AMS-3276	Sealing Compound, Integral Fuel Tanks and General Purpose, Intermittent Use to 360 Deg. F (182 Deg. C)
SAE-AMS-3277	Sealing Compound, Polythioether Rubber Fast Curing Integral Fuel Tanks And General Purpose, Intermittent Use to 400 MDF (204 MDC)
SAE-AMS-3278	Sealing and Coating Compound: Polyurethane (PUR) Fuel Resistant High Tensile Strength / Elongation for Integral Fuel Tanks / Fuel Cavities / General Purpose
SAE-AMS-3279	Sealing Compound, Sprayable, for Integral Fuel Tanks and Fuel Cell Cavities, for Intermittent Use to 350MDF (177MDC)
SAE-AMS-3281	Sealing Compound, Polysulfide (T) Synthetic Rubber for Integral Fuel Tank and Fuel Cell Cavities Low Density (1.20 To 1.35 Sp Gr), for Intermittent Use to 360 Deg F (182 Deg C)
SAE-AMS-3283	Sealing Compound, Polysulfide Non-Curing, Groove Injection Temperature and Fuel Resistant
SAE-AMS-3361	Silicone Potting Compound, Elastomeric, Two-Part, General Purpose, 140 to 400 Poise (15 to 40 PaMks) Viscosity
SAE-AMS-3375	Adhesive/Sealant, Fluorosilicone, Aromatic Fuel Resistant, One-Part Room Temperature Vulcanizing
SAE-AMS-3376	Sealing Compound, Non-Curing, Groove Injection Temperature and Fuel Resistant
SAE-AMS-4017	Aluminum Alloy, Sheet and Plate, 2.5Mg 0.25Cr, (5052-H34), Strain-Hardened, Half-Hard, and Stabilized
SAE-AMS-4027	Aluminum Alloy, Sheet and Plate, 1.0Mg - 0.60Si - 0.28Cu - 0.20Cr, (6061; -T6 Sheet, - T651 Plate) Solution and Precipitation Heat Treated
SAE-AMS-4029	Aluminum Alloy Sheet and Plate 4.5Cu - 0.85Si - 0.80Mn - 0.50Mg (2014; -T6 Sheet, -T651 Plate) Solution and Precipitation Heat Treated

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SAE-AMS-4037	Aluminum Alloy Sheet and Plate 4.4Cu - 1.5Mg - 0.60Mn (2024;-T3 Flat Sheet, -T351 Plate) Solution Heat Treated
SAE-AMS-4107	Aluminum Alloy Die Forgings 6.2Zn - 2.3Cu - 2.2Mg - 0.12Zr (7050-T74) Solution Heat Treated and Overaged
SAE-AMS-4260	Aluminum Alloy, Investment Castings, 7.0Si - 0.32Mg (356.0-T6), Solution and Precipitation Heat Treated
SAE-AMS-4750	Solder, Tin-Lead, 45Sn-55Pb
SAE-AMS-4751	Tin - Lead Alloy Eutectic, 63Sn - 37Pb (NONCURRENT)
SAE-AMS-5504	Steel, Corrosion and Heat Resistant, Sheet, Strip, and Plate 12.5Cr (SAE 51410) Annealed
SAE-AMS-5525	Steel, Corrosion and Heat Resistant, Sheet, Strip, and Plate 15Cr - 25.5Ni - 1.2Mo - 2.1Ti - 0.006B - 0.30V 1800 Deg. F (982 Deg. C) Solution Heat Treated
SAE-AMS-5604	Steel, Corrosion Resistant, Sheet, Strip, and Plate 16.5Cr - 4.0Ni - 4.0Cu - 0.30Cb Solution Heat Treated, Precipitation Hardenable
SAE-AMS-5613	Steel, Corrosion and Heat Resistant, Bars, Wire, Forgings, Tubing, and Rings 12.5Cr (SAE 51410) Annealed
SAE-AMS-5643	Steel, Corrosion Resistant, Bars, Wire, Forgings, Tubing, and Rings 16Cr - 4.0Ni - 0.30Cb - 4.0 Cu Solution Heat Treated, Precipitation Hardenable
SAE-AMS-5688	Steel, Corrosion Resistant, Wire 18Cr - 9.0Ni (SAE 30302) Spring Temper
SAE-AMS-5737	Steel, Corrosion and Heat Resistant, Bars, Wire, Forgings, and Tubing 15Cr - 25.5Ni - 1.2Mo - 2.1Ti - 0.006B - 0.30V Consumable Electrode Melted, 1650 Deg. F (899 Deg. C) Solution and Precipitation Heat Treated
SAE-AMS-6277	Steel Bars, Forgings, and Tubing 0.50Cr - 0.55Ni - 0.20Mo (0.18 - 0.23C) (SAE 8620) Vacuum Arc or Electroslag Remelted

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SAE-AMS-6345	Steel, Sheet, Strip and Plate 0.95Cr-0.20Mo (0.28 - 0.33C) (SAE 4130) Normalized or Otherwise Heat Treated
SAE-AMS-6415	Steel, Bars, Forgings, and Tubing 0.80Cr - 1.8Ni - 0.25Mo (0.38 - 0.43C) (SAE 4340)
SAE-AMS-6444	Steel Bars, Forgings, and Tubing 1.45Cr (0.98 - 1.10C) (SAE 52100) Premium Aircraft-Quality, Consumable Electrode Vacuum Melted
SAE-AMS-6470	Steel, Nitriding, Bars, Forgings, and Tubing 1.6Cr - 0.35Mo - 1.13Al (0.38 -0.43C)
SAE-AMS-7257	Rings, Sealing, Perfluorocarbon (FFKM) Rubber High Temperature Fluid Resistant 70 – 80
SAE-AMS-7271	Rings, Sealing, Butadiene-Acrylonitrile (NBR) Rubber Fuel and Low Temperature Resistant 60 – 70
SAE-AMS-7276	Rings, Sealing, Fluorocarbon (FKM) Rubber High-Temperature-Fluid Resistant Low Compression Set 70 – 80
SAE-AMS-7902	Beryllium Sheet and Plate, 98Be
SAE-AMS-C-27725	Coating, Corrosion Preventative, Polyurethane for Aircraft Integral Fuel Tanks for Use to 250 Deg. F. (121 Deg. C.)
SAE-AMS-DTL-23053/5	Insulation Sleeving, Electrical, Heat Shrinkable, Polyolefin, Flexible, Crosslinked
SAE-AMS-P-5315	Butadiene - Acrylonitrile (NBR) Rubber for Fuel-Resistant Seals 60 to 70
SAE-AMS-P-83461	Packing, Preformed, Petroleum Hydraulic Fluid Resistant, Improved Performance at 275 Deg. F (135 Deg. C)
SAE-AMS-QQ-A-250/1	Aluminum 1100, Plate and Sheet
SAE-AMS-QQ-A-250/11	Aluminum Alloy 6061, Plate and Sheet
SAE-AMS-QQ-A-250/12	Aluminum Alloy 7075, Plate and Sheet
SAE-AMS-QQ-P-416	Plating, Cadmium (Electrodeposited)

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SAE-AMS-R-25988	Rubber, Fluorosilicone Elastomer, Oil-and-Fuel-Resistant, Sheets, Strips, Molded Parts, and Extruded Shapes
SAE-AMS-R-83485	Rubber, Fluorocarbon Elastomer, Improved Performance at Low Temperatures
SAE-AMS-S-4383	Sealing Compound, Topcoat, Fuel Tank, Buna-N Type
SAE-AMS-S-8802	Sealing Compound, Temperature Resistant, Integral Fuel Tanks and Fuel Cell Cavities, High Adhesion
SAE-AMS-S-22141	Steel, Investment Castings Alloy and Carbon
SAE-AMS-T-9046	Titanium and Titanium Alloy, Sheet, Strip, and Plate
SAE AS5127/1	Test Methods for Aerospace Sealants Two-Component Synthetic Rubber Compounds

(Copies of these documents are available online at www.sae.org or from SAE International, 400 Commonwealth Drive, Warrendale, PA 15096-0001.)

BOEING MATERIAL SPECIFICATIONS (BMS)

BMS 5-267	Fuel Tank Coating
BMS 10-20	Corrosion Resistant Finish for Integral Fuel Tanks
BMS 10-39	Fuel and Moisture Resistant Finish for Fuel Tanks

(Copies of these documents may be obtained by calling (937) 255-8050, DSN 785-8050.)

D.3 DEFINITIONS, TERMS AND ACRONYMS.

Alternative Fuels Certification Office (AFCO)

Air Force Research Laboratory (AFRL)

Air Force Research Laboratory, Materials and Manufacturing Directorate (AFRL/RXSA)

Boeing Material Specification (BMS)

Integrated Product Team (IPT)

D.4 BACKGROUND.

Over the years many material compatibility programs have been performed on various fuels and fuel additives; however, none of these programs were standardized. In 1994, the Air Force Research Laboratory, Materials and Manufacturing Directorate (AFRL/RXSA) was asked

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to conduct material compatibility testing on JP-8+100 additives. After a survey of fuel tank, fuel system and engine materials, a list of 256 different materials were tested. As a result of the JP-8+100 program, short lists of metallic and nonmetallic materials were compiled for future testing. These two short lists were intended to be representative or worst case products from each type of material. For example, many different polysulfide sealants are used in fuel tanks, but a representative manganese dioxide cured product and a representative chromate cured product were selected for the short list. Soak temperatures and durations, test methods, and acceptance criteria were also called out in the short lists. Since the JP-8+100 program, other fuels and additives have been successfully tested for material compatibility using the short lists.

In 2006, an Integrated Product Team (IPT) was assembled to standardize and centralize the process for certifying new fuels. This IPT was charged with reviewing the current process and recommending a standardized process for certification. This materials compatibility appendix is part of that effort. It reflects not only aircraft and engine materials, but also vehicles, ground support and supply chain materials. The short lists were revisited and updated, evaluation criteria became better defined, and a Subset program was documented. Subset 1 is the revised short list testing, Subset 2 defines further testing required should there be a failure during Subset 1, and Subset 3 is large scale functional testing and/or flight testing based on the results of Subset 2.

D.5 TESTING APPROACH AND ENTRANCE CRITERIA.

Entrance Criteria for materials testing is a chemical description of the fuel. Based upon this chemical description AFRL/RXSA will conduct an analysis to determine if Subset 1 testing is necessary. Subset 1 is laboratory scale testing which compares representative metallic and nonmetallic materials after soaking in a baseline fuel and after soaking in the new fuel or additized fuel to be tested. It is designed as a first level screening test to provide an indication of any compatibility problems. If all tests pass then the risk level of the new additive or fuel is minimal. If there are material compatibility concerns (see D.6.10 for evaluation criteria), then a Subset 2 program is required that is designed to further investigate those material families that failed. Subset 2 involves a complete testing of all the materials in the family of materials that failed, an analysis of the root cause of failure and possibly component or system level tests. A Subset 3 program based on the findings of Subset 2 could be conducted to further reduce risk and determine compatibility. Subset 3 may consist of large scale functional testing and/or flight testing.

D.6 SUBSET 1 TESTING.

D.6.1 Objective. Determine the effect of a candidate additive or an alternative fuel on fuel system materials, ground handling equipment and supply chain materials.

D.6.2 Additives. The additive supplier will recommend a concentration to be tested. An evaluation will be made to determine the maximum possible concentration of the additive in the fuel, the water layer and the vapor. Fuel system additives are to be evaluated at 4 times the concentration being sought for approval or the maximum concentration, whichever is greater.

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D.6.3 Alternative Fuels. Alternative fuels will be characterized for chemical consistency. If the chemical consistency causes variation in properties between batches, high and low boundaries for each property will be determined. At a minimum, sulfur content, acid number, aromatic content, flash point and conductivity will be determined. If there is a significant variation in any property, the alternative fuel will be tested at the anticipated maximum and minimum of the affected properties.

D.6.4 Test Temperatures. Materials are to be tested at the highest temperature to which it will be subjected for its specific application within an aircraft and engine fuel system. Testing at temperatures beyond these maximums result in diminished baseline material performance and significantly reduced test sensitivity for the additive evaluation. The appropriate test temperature for each material is listed in Tables D-II and D-III.

D.6.5 Baseline Test Fluids. Two baseline test fluids are approved for determining compatibility of a candidate additive with fuel system materials. Either of the two test fluids may be used. It is not required that both be evaluated. A JP-8 conforming to MIL-DTL-83133F and having an aromatic content between 20%-25% may be used. Also, the jet reference fuel (JRF) as formulated in Table D-I may be used. The JRF is a blend developed by the Air Force Research Laboratory (AFRL) to be representative of modern kerosene-type fuels.

JRF is formulated by blending the following technical grade constituents in the volumes shown in Table D-I.

TABLE D-I. Jet reference fuel.

Formulation*				Property Analyses (Typical)		
Component	**Name**	**Volume %**		**Property**		**Jet A-1 Specification**
Paraffins	Exxsol D-40	37		Aromatics	25.8 vol. %	25 vol. % max.
	Exxsol D-80	37		Olefins	0.9 vol. %	No Requirement
Aromatics **(spec: <25 vol%)**	Aromatic Blend (total)	25.0		Flash Point	55°C	38° min.
	Aromatic 100	(7.5)		Freeze Point	-55°C	-47°C max.
	Aromatic 150	(15.0)		Naphthalenes	2.3 vol. %	3.0 vol. % max.
	Aromatic 200	(2.5)		API Gravity	42.2	37 min. – 51 max.
Sulfur **(spec: <0.3 wt%)**	tert-Butyl Disulfide	0.73 (0.3 mass%)		Total Sulfur	0.31 mass %	0.30 mass % max.
Mercaptan **(spec: <0.002 wt%)**	Decanethiol	0.010 (0.002 mass%)		Mercaptan Sulfur	0.002 mass %	0.003 mass % max.
Icing Inhibitor **(spec: 0.1-0.15 vol%)**	DiEGME	0.15		Fuel System Icing Inhibitor	0.14 vol. %	No Requirement
Corrosion Inhibitor	Unicor J	0.0017 (16 mg/L)		Lubricity Test (BOCLE) Wear Scar	0.59 mm	No Requirement

* Formulation established assuming zero aromatics in Exxsol D-40/D-80 and additives, as well as assuming zero sulfur in paraffins, aromatics, and additives. Ssome adjustment in the formulation is being considered to address the limitations of these assumptions as indicated in the property analysis column.

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D.6.6 Additive Testing. Candidate additives will be evaluated by adding the required concentration (D.6.2) to at least one of the two test fluids. A control test will be performed using the test fluid without the additive to establish a baseline for comparison. The material tests to be performed are shown in Tables D-II and D-III.

D.6.7 Alternative Fuel Testing. Control testing will be performed using a baseline test fluid (D.6.5) to establish a baseline for comparison. Alternative fuels as determined in D.6.3 will be tested. The material tests to be performed are shown in Tables D-II and D-III.

D.6.8 List of Materials. Table D-II is a list of representative nonmetallics and Table D-III is a list of representative metals used in the airframe and gas turbine engine fuel systems. Table D-II and Table D-III also show the properties to be tested for each material class, the temperature at which the test will be performed, and the pass/fail criteria.

D.6.9 Testing Procedure.

D.6.9.1 Nonmetallic Material Compatibility Tests Should be Conducted as Listed in TABLE D-II. All sealant peel strength test panels will be aluminum AMS 4045 panels, sulfuric acid anodized per AMS 2471, and coated with AMS-C-27725 II, Class B corrosion preventive coating. For the duration of the aging process of specimens, a fuel change-out will occur after each 14 day period.

D.6.9.2 Metallic Material Tests are as Follow:

1.) Corrosion
2.) Light-Optical Evaluation
3.) Microstructural Evaluation.

D.6.9.3 Metallic Specimens have to be Weighed During the Temperature Aging Process. The weight gain/loss measurements have to be taken at 4 different intervals (i.e., before/after temperature aging at 7 days and 28 days).

D.6.10 Evaluation Criteria.

D.6.10.1 The Test Materials will be Evaluated Using the Pass/Fail Criterion Provided for Each Test of a Given Material.

D.6.10.2 The Pass/Fail Criteria for Nonmetallics are Listed in Table D-II. The logical pass/fail criterion is to look at significant variations between the baseline fuel and the new fuel or additive results. The allowable variations from the baseline fuel for nonmetallic materials are based upon the precision and bias in the test method. Also, with most of the materials there are test requirements expressed as maximum or minimum values. The test requirements are taken from the material specification when applicable. If there is no material specification or the specification does not have a fuel soak requirement, then the criteria is taken from the requirements for similar materials based on previous test programs.

If the test results are within the allowable variation and pass the test requirements then the risk level of the new additive or fuel for the particular material is minimal. If the test results

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are not within the allowable variation and the test requirements are not met then a Subset 2 program is required that is designed to further investigate that material family. If one of the evaluation criteria is met but the other is not met, then the magnitude and effects of the failure have to be evaluated to determine if Subset 2 testing is necessary.

D.6.10.3 The Pass/Fail Criteria for Metallics is as Follows: No signs of corrosion.

D.7 SUBSET 2 TESTING.

D.7.1 Objective. Perform additional testing to further investigate failures from Subset 1. This Subset requires an assessment of Subset 1 failures and a test plan to evaluate risks discovered under Subset 1. The assessment should identify risks of using the new additive or fuel to material durability, subsystem functionality and flight worthiness. A comprehensive test plan that assesses each risk should be executed under Subset 2**.**

D.7.2 Additives. The same concentration of additive used in Subset 1 will be used in Subset 2. The batch of fuel used for Subset 2 will be the same as that used in Subset 1, whenever possible.

D.7.3 Alternative Fuels. The batch of fuel used for Subset 2 will be the same as that used in Subset 1, whenever possible.

D.7.4 Complete Materials Testing. For each material that failed in Subset 1, that complete material family will be tested. For example if one polysulfide sealant failed, then all polysulfide sealants will be tested. The complete list of materials is found in Table D-IV. The results of these tests will be used to evaluate the extent of compatibility.

D.7.5 Related Materials Testing. The cause of failures from Subset 1 will be evaluated. Some common failure causes are lack of swell, hardening, loss of flexibility, reversion due to polymer chain scission, acid attack, sulfur or mercaptan attack, corrosion, etc. The root cause of each Subset 1 failure will be determined. Those materials that are not in the same family, but that may be prone to similar attack will be tested for the appropriate physical properties.

Based upon the results of Subset 1 and/or the results of Complete Material Testing and Related Material Testing in Subset 2, a determination will be made if functional testing is required to assess risks. The functional testing would correspond to the failed material physical property. For example, if lack of swell was a cause of failure in o-rings, then functional testing would be required to determine if fuel leaks are likely. Flowing rig fuel coupling tests, static seal tests, pump seal tests and fuel control valve tests are examples of functional tests that can be used to assess the risk of fuel leaks. These functional tests should address subsystem or system level concerns.

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D.8 SUBSET 3 TESTING.

D.8.1 Objective. Perform system level or flight testing to further investigate concerns from Subset 2. This Subset requires an assessment of Subset 2 results and a test plan developed to evaluate risks not resolved under Subset 2. The assessment should identify risks of using the new additive or fuel to system functionality. A comprehensive test plan that assesses each risk should be executed under Subset 3.

D.8.2 Additives. The same concentration of additive used in Subset 1 and 2 will be used in Subset 3. The batch of fuel used will be the same for Subset 3 as was used for Subset 1 and 2, whenever possible.

D.8.3 Alternative Fuels. The batch of fuel used for Subset 3 will be the same as that used in Subset 1 and 2, whenever possible.

D.8.4 System Level Testing. If concerns still exist after Subset 2, then system level functional tests may be performed to assess the risks. A test plan that addresses each risk will be executed. For example, if sealant tensile strength has failed in Subset 1 and dynamic cycling tests are inconclusive or marginal in Subset 2, then large scale integral fuel tank testing that simulates tank structure, thermal cycling and lifetime stress/strain criteria could be accomplished.

D.8.5 Flight Tests. Flight testing could also be accomplished to assess the risks and concerns from Subset 2. For example if sealant functionality and/or durability are still concerns, flight testing can be used to assess those concerns. The flight testing should be of sufficient duration and flight envelope to evaluate the worst case scenario. Flight testing should have predetermined pass/fail criteria.

D.9 REFERENCES.

AFRL/MLS 06-103	Evaluation of Material Compatibility of Fischer-Tropsch Fuel
AFRL-PR-WP-TR-2000-2015	Fuel and Fuel System Materials Compatibility Test Program for a JP-8 +100 Fuel Additive

(Copies of these documents are available online at The Defense Technical Information Center Website and are available from DTIC Headquarters, 8725 John J. Kingman Rd., Ft. Belvoir VA 22060-6218; telephone (800) 225-3842.)

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TABLE D-II. Nonmetallic materials, tests, and test temperatures.

Material	Description	Spec	Soak Temperature/ Duration	Test	Test Procedure	Evaluation Criteria	
						Test Requirements	Allowable Variation from Baseline
Adhesive	Vinyl Phenolic	MMM-A-132 Type 1, Class 3	200 °F / 28 days	Lap Shear	ASTM D1002	> 1500 psi	250 psi decrease
Adhesive	Epoxy Resin	~	200 °F / 28 days	Lap Shear	ASTM D1002	> 1500 psi	250 psi decrease
Adhesive	Nitrile Phenolic	MMM-A-132 Type 1, Class 2	200 °F / 28 days	Lap Shear	ASTM D1002	> 1500 psi	250 psi decrease
Adhesive	Epoxy Paste	MMM-A-132 Type 1, Class 3	200 °F / 28 days	Lap Shear	ASTM D1002	> 1500 psi	250 psi decrease
Adhesive	Nitrile Epoxy Film	MMM-A-132 Type 1, Class 2	200 °F / 28 days	Lap Shear	ASTM D1002	> 1500 psi	250 psi decrease
Adhesive	Methacrylate	ASTM D5363 Group 4, Class 1, Grade 1	200 °F / 28 days	Static Shear	MIL-R-46082, Method A	> 1200 psi	250 psi decrease
Bladder (Inner Liner)	Nitrile	~	160 °F / 28 days	Tensile Strength Elongation Volume Swell	ASTM D412 ASTM D412 ASTM D471	> 1500 psi > 300% < 25%	200 psi decrease 40% decrease ± 5%
Bladder (Inner Liner)	Polyurethane	~	200 °F / 28 days	Tensile Strength Elongation Volume Swell	ASTM D412 ASTM D412 ASTM D471	> 1500 psi > 300% < 25%	200 psi decrease 40% decrease ± 5%
Bladder (Self Sealing)	Nitrile	MIL-T-5578	RT / 30 mins	Volume Swell	ASTM D471	~	± 5%
Coating	Nitrile	SAE-AMS-S-4383	200 °F / 28 days	Hardness (Pencil) Tape Adhesion	ASTM D3363-05 FED-STD-141	≥ unaged Pass.	1 pt decrease
Coating	Polyurethane	SAE-AMS-C-27725 Type II	200 °F / 28 days	Hardness (Pencil) Tape Adhesion	ASTM D3363-05 FED-STD-141	≥ unaged Pass	1 pt decrease
Coating	Epoxy	BMS 10-20	200 °F / 28 days	Hardness (Pencil) Tape Adhesion	ASTM D3363-05 FED-STD-141	≥ unaged Pass	1 pt decrease
Coating	Epoxy	BMS 10-39	200 °F / 28 days	Hardness (Pencil) Tape Adhesion	ASTM D3363-05 FED-STD-141	≥ unaged Pass	1 pt decrease
Bulk Tank Coating	Epoxy-Polyamide	MIL-DTL-24441	120 °F / 28 days	Hardness (Pencil) Tape Adhesion	ASTM D3363-05 FED-STD-141	≥ unaged Pass	1 pt decrease
Sealant	Polysulfide Dichromate Cured	SAE-AMS-S-8802 Type I, Class B-2	200 °F / 28 days	Peel Strength Hardness, Shore A Tensile Strength Elongation Volume Swell	SAE AS5127/1 ASTM D2240 ASTM D412 ASTM D412 ASTM D471	> 20 lb/in / 100% cohes. > 35 pts > 200 psi > 150% 0% - 20%	8 lb./in. decrease ± 5 pts 35 psi decrease 25% decrease 5% increase

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TABLE D-II. Nonmetallic materials, tests, and test temperatures – Continued.

Material	Description	Spec	Soak Temperature/ Duration	Test	Test Procedure	Evaluation Criteria	
						Test Requirements	Allowable Variation from Baseline
Sealant	Polysulfide Manganese Cured	SAE-AMS-S-8802 Type II, Class B-2	200 °F / 28 days	Peel Strength Hardness, Shore A Tensile Strength Elongation Volume Swell	SAE AS5127/1 ASTM D2240 ASTM D412 ASTM D412 ASTM D471	> 20 lb/in / 100% cohes. > 35 pts > 200 psi > 150% 0% - 20%	8 lb./in. decrease ± 5 pts 35 psi decrease 25% decrease 5% increase
Sealant	Fluorosilicone	SAE-AMS-3375	200 °F / 28 days	Peel Strength Hardness, Shore A Tensile Strength Elongation Volume Swell	SAE AS5127/1 ASTM D2240 ASTM D412 ASTM D412 ASTM D471	> 10 lb/in / 100% cohes. > 35 pts > 200 psi > 150% 0% - 20%	8 lb./in. decrease ± 5 pts 35 psi decrease 25% decrease 5% increase
Sealant	Polyurethane	SAE-AMS-3279	200 °F / 28 days	Peel Strength Hardness, Shore A Tensile Strength Elongation Volume Swell	SAE AS5127/1 ASTM D2240 ASTM D412 ASTM D412 ASTM D471	> 20 lb/in / 100% cohes. > 35 pts > 200 psi > 150% 0% - 30%	8 lb./in. decrease ± 5 pts 35 psi decrease 25% decrease 5% increase
Sealant	Polythioether	SAE-AMS-3277 Class B-2	200 °F / 28 days	Peel Strength Hardness, Shore A Tensile Strength Elongation Volume Swell	SAE AS5127/1 ASTM D2240 ASTM D412 ASTM D412 ASTM D471	> 20 lb/in / 100% cohes. > 35 pts > 200 psi > 150% 0% - 25%	8 lb./in. decrease ± 5 pts 35 psi decrease 25% decrease 5% increase
Sealant	Polysulfide Lightweight	SAE-AMS-3281	200 °F / 28 days	Peel Strength Hardness, Shore A Tensile Strength Elongation Volume Swell	SAE AS5127/1 ASTM D2240 ASTM D412 ASTM D412 ASTM D471	> 20 lb/in / 100% cohes. > 35 pts > 200 psi > 150% 0% - 20%	8 lb./in. decrease ± 5 pts 35 psi decrease 25% decrease 5% increase
Sealant (Groove Injection)	Polysulfide	SAE-AMS-3283	160 °F / 28 days	Volume Swell	ASTM D471	1% to 12%	± 5%
Sealant (Groove Injection)	Fluorosilicone	MIL-S-85334	160 °F / 28 days	Volume Swell	ASTM D471	1% to 12%	± 5%
Composite, Epoxy Graphite	AS4/3501-6	~	200 °F / 28 days	Laminar Shear	ASTM D790	> 5000 psi	500 psi decrease

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TABLE D-II. Nonmetallic materials, tests, and test temperatures – Continued.

Material	Description	Spec	Soak Temperature/ Duration	Test	Test Procedure	Evaluation Criteria	
						Test Requirements	Allowable Variation from Baseline
Composite, Graphite Bismaliemide	IM7/5250-4	~	200 °F / 28 days	Laminar Shear	ASTM D790	> 5000 psi	500 psi decrease
Composite, Epoxy Graphite	IM7/977-3	~	200 °F / 28 days	Laminar Shear	ASTM D790	> 5000 psi	500 psi decrease
Composite, Epoxy Graphite	IM7/8552	~	200 °F / 28 days	Laminar Shear	ASTM D790	> 5000 psi	500 psi decrease
Foam	Polyurethane	MIL-PRF-87260	200 °F / 28 days	Tensile Strength Elongation Resistivity	ASTM D412 ASTM D412 ASTM D257	> 10 psi > 100% < 1.0E12 Ohm-cm	5 psi decrease 15% decrease
Gasket, O-Ring	Nitrile	SAE-AMS-P-5315	160 °F / 28 days	Hardness, Shore M Tensile Strength Elongation Compression Set Volume Swell	ASTM D2240 ASTM D1414 ASTM D1414 ASTM D395 ASTM D471	± 5 pts from unaged > 1000 psi > 200% < 50% 0% to 25%	± 5 pts 125 psi decrease 35% decrease 5% increase ± 10%
Gasket, O-Ring	Fluorosilicone	SAE-AMS-R-25988, Type I, Class 1, Grade 70	225 °F / 28 days	Hardness, Shore M Tensile Strength Elongation Compression Set Volume Swell	ASTM D2240 ASTM D1414 ASTM D1414 ASTM D395 ASTM D471	- 20 pts from unaged > 500 psi > 125% < 65% 0% to 25%	± 5 pts 125 psi decrease 35% decrease 5% increase ± 10%
Gasket, O-Ring	Fluorocarbon	SAE-AMS-7276	325 °F / 28 days	Hardness, Shore M Tensile Strength Elongation Compression Set Volume Swell	ASTM D2240 ASTM D1414 ASTM D1414 ASTM D395 ASTM D471	± 5 pts from unaged > 1000 psi > 150% < 60% 0% to 10%	± 5 pts 125 psi decrease 35% decrease 5% increase ± 10%
Gasket	Low Temperature Fluorocarbon	SAE-AMS-R-83485 Type I	325 °F / 28 days	Hardness, Shore M Tensile Strength Elongation Compression Set Volume Swell	ASTM D2240 ASTM D1414 ASTM D1414 ASTM D395 ASTM D471	± 5 pts from unaged > 1000 psi > 150% < 60% 0% to 10%	± 5 pts 125 psi decrease 35% decrease 5% increase ± 10%

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TABLE D-II. Nonmetallic materials, tests, and test temperatures – Continued.

Material	Description	Spec	Soak Temperature/ Duration	Test	Test Procedure	Evaluation Criteria	
						Test Requirements	Allowable Variation from Baseline
Hose (Aerial Refueling)	Acrylic Nitrile	MIL-H-4495	160 °F / 28 days	Tensile Strength Elongation Hardness, Shore A Volume Swell	ASTM D412 ASTM D412 ASTM D2240 ASTM D471	> 1200 psi > 150% ± 5 pts from unaged < 8%	125 psi decrease 25% decrease ± 5 pts ± 5%
Hose (Ground Refueling)	Epichloro-hydrin	MIL-DTL-26521	160 °F / 28 days	Tensile Strength Elongation Hardness, Shore A Volume Swell	ASTM D412 ASTM D412 ASTM D2240 ASTM D471	> 1500 psi > 300% ± 5 pts from unaged < 8%	125 psi decrease 25% decrease ± 5 pts ± 5%
Teflon (Film)	Teflon	~	160 °F / 28 days	Tensile Strength Elongation	ASTM D412 ASTM D412	> 500 psi > 25%	150 psi decrease 15% decrease
Nylon (Film)	Nylon	~	160 °F / 28 days	Tensile Strength Elongation	ASTM D412 ASTM D412	> 500 psi > 25%	850 psi decrease 5% decrease
Polyethylene (Film)	Polyethylene	~	160 °F / 28 days	Tensile Strength Elongation	ASTM D412 ASTM D412	> 500 psi > 25%	250 psi decrease 50% decrease
Kapton (Film)	Kapton	~	200 °F / 28 days	Tensile Strength Elongation	ASTM D412 ASTM D412	> 500 psi > 25%	1800 psi decrease 5% decrease
Potting Compound	Polysulfide	MIL-PRF-8516, Cure B	160 °F / 28 days	Hardness, Shore A Tensile Strength Elongation Peel Strength Volume Swell	ASTM D2240 ASTM D412 ASTM D412 SAE AS5127/1 ASTM D471	> 20 pts >100 psi > 150% > 10 lb/in / 100% cohes. > - 20%	± 5 pts 35 psi decrease 25% decrease 8 lb./in. decrease ± 10%

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TABLE D-III. Metallic materials, tests, and test temperatures.

Material	Material Specification	Coating Specification	Soak Temp
7075 T6 Aluminum Chromic Acid Anodize Type I	SAE-AMS-QQ-A-250/12	MIL-A-8625, Type I	200º F
7075-T6 Sulfuric Acid Anodize Type IIB	SAE-AMS-QQ-A-250/12	MIL-A-8625, Type II B	200º F
7075-T6 Chromate Conversion Coated Class IA	SAE-AMS-QQ-A-250/12	MIL-DTL-5541, Class 1A	200º F
7050-T74	SAE-AMS-4107	N/A	200º F
2024-T3 Bare	SAE-AMS-4037	N/A	200º F
6061-T6 Bare	SAE-AMS-4027	N/A	200º F
5052-H34 Bare	SAE-AMS-4017	N/A	200º F
356 T6 Cast Aluminum	SAE-AMS-4260	N/A	200º F
AZ91 T6	ASTM B93	N/A	200º F
CU/NI 90/10		N/A	200º F
Sn 60 Pb 40 Solder		N/A	200º F
304 SS	ASTM A240	N/A	325º F
17-4 pH	SAE-AMS-5604	N/A	325º F
440 SS	ASTM A240	N/A	325º F
TI 8Al -1V -1MO	SAE-AMS-T-9046	N/A	325º F
TI CP 70	SAE-AMS-T-9046	N/A	325º F
TI 3AL – 2.5V	SAE-AMS-T-9046	N/A	325º F
4130 IVD Coating	SAE-AMS-6345	SAE-AMS-2427	325º F
Alloy Steel Fastener MS24694 HL21PN20-16	SAE-AMS-6415	SAE-AMS-QQ-P-416, Type II, Class 2	325º F
A286 Fastener MS24694 HL49GU20-16	SAE-AMS-5737	Sliver Plate SAE-AMS-2410	325º F
CPM 10V		N/A	325º F
INCO 625		N/A	325º F
INCO 718		N/A	325º F
Nitralloy 135	SAE-AMS-S-22141	N/A	325º F
IN 200 Ni		N/A	325º F
Monel 400		N/A	325º F
Waspalloy		N/A	325º F
Lead	SAE-AMS-4751	N/A	325º F
268 Brass Sheet	ASTM B36	N/A	325º F
TAP MS 285		N/A	325º F
Mag Wire Type I		N/A	325º F

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TABLE D-IV. Complete list of materials.

I.D. No.	AIRCRAFT USE	MATERIAL DESIGNATION	MATERIAL TYPE
I.A.1	Adhesive	Epoxy/Polyamide EC3569, BR-127	Epoxy/Polyamide
I.A.2	Adhesive	FM 47 Vinyl Phenolic, BR-127	Vinyl Phenolic
I.A.3	Adhesive	AF 126-2 Nitrile Mod. Epoxy, BR-127	Nitrile
I.A.4	Adhesive	AF 143-2 Mod. Hi. Temp. Epoxy	Epoxy
I.A.5 (I.P.1)	Adhesive	Epon 828/DTA Un. Mod. Epoxy	Epoxy
I.A.6	Adhesive	FM 73W/BR-127 Primer	Nitrile Epoxy
I.A.7	Adhesive	AF-10E/EC 1290, Primer Scotchweld	Primer Scotchweld
I.A.8	Adhesive	AF-10 W/EC 3950, Primer Scotchweld	Primer Scotchweld
I.A.9 (I.C.1)	Adhesive	EC 776 Coating Explosion Suppression Foam Adhesive, SAE-AMS-S-4383	Nitrile
I.A.10	Adhesive	EA 9446	Acrylic
I.A.11.1	Adhesive	Fusor 309 (1:1 mix)	Epoxy
I.A.11.2	Adhesive	Fusor 309 (2:1 mix)	Epoxy
I.A.12	Adhesive	Henkel EA9309.1NA, Epoxy	Epoxy
I.A.13	Adhesive	Henkel EA9394	Epoxy
I.A.14	Adhesive	Loctite 609 (Methacrylate)	Methacrylate
I.A.15	Adhesive	Loctite 495 (Cyanoacrylate)	Cyanoacrylate
I.B.1	Fuel Bladder	AMFUEL, PS-598 Innerliner	Nitrile
I.B.2	Fuel Bladder	AMFUEL, U5200B, Innerliner	Nitrile
I.B.3	Fuel Bladder	AMFUEL, PU-339, Innerliner	Polyurethane
I.B.4	Fuel Bladder	Engineered Fabrics, P/N 51956 Innerliner	Nitrile
I.B.5	Fuel Bladder	Engineered Fabrics, P/N 5904C Innerliner	Polyurethane
I.B.6	Fuel Bladder	Goodyear 26950, Self Sealing	Nitrile
I.B.7	Fuel Bladder	Goodyear 51956, Innerliner	Nitrile
I.B.8	Fuel Bladder	Goodyear 80C29, Innerliner	Urethane
I.B.9	Fuel Bladder	Goodyear 80C39, Innerliner	Nitrile
I.B.10	Fuel Bladder	(Repair Material) Goodyear 80C29	Polyurethane
I.B.11	Fuel Bladder	Engineered Fabrics T/N 3572N Cloth	Nylon (36”x60”)
I.B.12	Fuel Bladder	Engineered Fabrics T/N 491 Cloth	Polyester (42”x48”)
I.B.13	Fuel Bladder	Amfuel Cloth PN C121	Nylon cloth
I.B.14	Fuel Bladder	Amfuel Cloth PN C130	Nylon cloth
I.B.15	Fuel Bladder	Amfuel 1316-1A, Self Sealing	Nitrile
I.B.16	Fuel Bladder	Engineered Fabrics P/N 320-4-49274/FTL-107, Self Sealing	Polyurethane
I.C.1 (I.A.9)	Int. Fuel Tank Coating	EC 776, 3M, SAE-AMS-S-4383	Nitrile
I.C.2	Int. Fuel Tank Coating	Coating, SAE-AMS-C-27725	Polyurethane
I.C.3	Int. Fuel Tank Coating	Coating, BMS 10-20	Epoxy

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TABLE D-IV. Complete list of materials – Continued.

I.D. No.	AIRCRAFT USE	MATERIAL DESIGNATION	MATERIAL TYPE
I.C.4 (I.D.2)	Int. Fuel Tank Coating	PR1440B2 Pro-Seal 890, BMS 5-267, SAE-AMS-S-8802, Type 2	Manganese Cured Polysulfide
I.C.5	Int. Fuel Tank Coating	PR2911 MMS 425 New Spray/PreCoat-PR2904S-2	Polyurethane
I.C.6	Int. Fuel Tank Coating	MIL-C-83019	Polyurethane
I.C.7	Int. Fuel Tank Coating	Akzo Nobel Aerospace Coatings, product code 454-4-1/CA-109	Epoxy
I.C.8	Ground Tank Fuel Storage	Note: Test at 100° F 3 part epoxy system MIL-DTL-24441 A-36 plate steel, lapweld /20 Form 150 Type III /30 Form 151 Type IV /31 Form 152 Type IV 6010 carbon steel	Epoxy Polyamide 2-4 mil thick 8-10 mil max thick
I.D.1	Int. Fuel Tank Sealant	PR 1422 Type I, B2 SAE-AMS-S-8802, Type I	Dichromate Cured Polysulfide
I.D.2 (I.C.4)	Int. Fuel Tank Sealant	PR1440 (PS 890) SAE-AMS-S-8802, Type 2	Manganese Cured Polysulfide
I.D.3	Int. Fuel Tank Sealant	PR1750, B2, SAE-AMS-3276	Polysulfide
I.D.4	Int. Fuel Tank Sealant	PR1221, B2, SAE-AMS-3278	Polyurethane
I.D.5	Int. Fuel Tank Sealant	Q4-2817, W 1200 Primer SAE-AMS-3375	Fluorosilicone
I.D.6	Int. Fuel Tank Sealant	PR2911, SAE-AMS-3279	Polyurethane
I.D.7	Int. Fuel Tank Sealant	PR1828, B2, SAE-AMS-3277	Polythioether
I.D.8	Int. Fuel Tank Sealant	PR1776, SAE-AMS-3281	Polysulfide
I.D.9	Int. Fuel Tank Sealant	PR1775 B2, SAE-AMS-3265	Polysulfide
I.D.10	Int. Fuel Tank Sealant	P/S 870 B-2, MIL-PRF-81733	Polysulfide
I.D.11	Int. Fuel Tank Sealant	PR705, SAE-AMS-3283, Groove Injection	Polysulfide
I.D.12	Int. Fuel Tank Sealant	Q4-2805, MIL-S-85334, Groove Injection	Fluorosilicone
I.D.13	Int. Fuel Tank Sealant	DC 94031, MIL-S-85334, Groove Injection	Fluorosilicone
I.D.14	Int. Fuel Tank Sealant	SAE-AMS-3376, Groove Injection	Fluorosilicone
I.D.15	Int. Fuel Tank Sealant	G651, Groove Injection	Cyanosilicone
I.E.1	Composite	Composite, AS 4/3501-6	Epoxy Graphite
I.E.2	Composite	Composite, IM 7/5250-4	Graphite Bismaliemide

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TABLE D-IV. Complete list of materials – Continued.

I.D. No.	AIRCRAFT USE	MATERIAL DESIGNATION	MATERIAL TYPE
I.E.3	Composite	Composite, AS7/8551-7A	Epoxy Graphite
I.E.4	Composite	Composite, IM7/977-3	Epoxy Graphite
I.E.5	Composite	Composite, IM7/8552	Epoxy Graphite
I.E.6	Vent Lines	Composite	Fiberglass
I.E.7	Isolator Tube	Composite	Epoxy Resin
I.F.1 I.F.1.1 I.F.1.2	Fuel Filter 11/18/97 11/18/97	 AC-B683F-2435 AC-B253F-2435Y1, 1/4	 F-100 Eng. F-110 Eng.
I.F.2	Fuel Filter 14 Aug '97	AC-9985F-10	T-700 Eng.
I.F.3	Fuel Tank Explosion Suppression	Foam, Fomex Yellow Type II, MIL-DTL-83054	Polyurethane (Ester)
I.F.4	Fuel Tank Explosion Suppression	Foam, Fomex Blue IV, MIL-DTL-83054	Polyurethane (Ether)
I.F.5	Fuel Tank Explosion Suppression	Foam (ESM), Fomex, Charcoal Gray, Class I MIL-PRF-87260	Polyurethane (Ether)
I.F.6	Fuel Tank Explosion Suppression	Foam Crest Charcoal Gray, Class II, MIL-PRF-87260	Polyurethane (Ether)
I.F.7	Fuel Tank Explosion Suppression	Foam Fomex Charcoal Gray, Class II, MIL-PRF-87260	Polyurethane (Ether)
I.F.8	Fuel Tank Explosion Suppression	Foam Crest Yellow, Type II, Non-conductive, MIL-DTL-83054	Polyurethane (Ester)
I.F.9	Fuel Tank Explosion Suppression	Beige (tan), Type II, Non-conductive, MIL-DTL-83054	Polyester (Ester)
I.G.1	O-Ring	O-Ring, N-756 Parker, SAE-AMS-P-83461 (Hydraulic)	Nitrile
I.G.2	O-Ring	O-Ring, N304-75 Parker MIL-P-25732 (Hydraulic)	Nitrile
I.G.3	O-Ring	O-Ring, N602-70 Parker, SAE-AMS-P-5315	Nitrile
I.G.4	O-Ring	O-Ring, N506-65 Parker, SAE-AMS-7271 / MS9201	Nitrile
I.G.5 (II.G.2)	O-Ring	O-Ring, L677-70 Parker, MIL-DTL-25988	Fluorosilicone
I.G.6 (II.G.9)	O-Ring	O-Ring, V747 Viton Parker, SAE-AMS-7276	Fluorocarbon
I.G.7 (II.G.3)	O-Ring	O-Ring, Viton (GLT) Parker, SAE-AMS-R-83485	Fluorocarbon

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TABLE D-IV. Complete list of materials – Continued.

I.D. No.	AIRCRAFT USE	MATERIAL DESIGNATION	MATERIAL TYPE
I.G.8 (II.G.4)	O-Ring	O-Ring, Kalrez 92344G, Dupont, SAE-AMS-7257	Perfluoroelastomer
I.G.9	O-Ring	O-Ring, #74-2, CIS8715 Coast-Craft, ABE3, F1	Type S Nitrile
I.G.10 (II.G.1)	O-Ring	O-Ring, EX2000 Bendix, MIL-DTL-25988	Fluorosilicone
I.G.11 (II.G.10)	Seal	Washer, PN 212147, JT8 PO-652, Argo-Tech, PN 21247	Urethane
I.G.12 (II.G.11)	Seal	Tang, JT90, Parker Compound/ P4662A90, ArgoTech, PN 212351	Urethane
I.G.13 (I.O.5)	Cork Seal	Cork P/N 30-155-5-1 Parker	Cork
I.G.14	Door Seal	Parker N406-60, MIL-R-6855, Class 1, Grade 60	Nitrile
II.G.1 (I.G.10)	Engine Plumbing	O-Ring, ES2000/953591 Bendix MIL-DTL-25988	Fluorosilicone
II.G.2 (I.G.5)	Engine Plumbing	O-Ring, Parker L677 MIL-DTL-25988	Fluorosilicone
II.G.3 (I.G.7)	Engine Plumbing	O-Ring, Parker PN/VO835 GLT SAE-AMS-R-83485 (Low Temp.)	Fluorocarbon
II.G.4 (I.G.8)	Engine Plumbing	O-Ring, DuPont Kalrez 93-244G SAE-AMS-7257	Perfluoroelastomer
II.G.5	Engine Plumbing	O-Ring, ESS928, Bendix Jonal MIL-DTL-25988	Fluorosilicone
II.G.6	Engine Plumbing	O-Ring, GTC-777, SAE-AMS-R-83485	Fluorocarbon
II.G.7	Engine Plumbing	O-Ring, GTC 409, MIL-DTL-25988	Fluorosilicone
II.G.8	Engine Plumbing	O-Ring, GTC-505 FFKM, SAE-AMS-7257	Perfluoroelastomer
II.G.9 (I.G.6)	Engine Plumbing	O-Rings, V747 Viton Parker SAE-AMS-7276	Fluorocarbon
II.G.10 (I.G.11)	Plumbing Gasket	Washer, PN 212147, JT8 PO-652, Argo-Tech, PN 21247	Urethane (See I.G.11)
II.G.11 (I.G.12)	Plumbing Gasket	Tang, JT90, Parker Compound/P4662A90, Argo-Tech PN 212351	(see I.G.12)
II.G.12	Plumbing Gasket	O-Ring, GTC-778, SAE-AMS-R-83485	Fluorocarbon (Improved 777)
II.G.13	Plumbing Gasket	O-Ring, GTC-B-95, MIL-DTL-25988	Fluorosilicone 677
II.G.14	Plumbing Gasket	O-Ring, Stillman P/N TH-1384 MIL-DTL-25988	Fluorosilicone (TeflonTM)
II.G.15	Plumbing Gasket	O-Ring, Parker P/N L 1186-80 MIL-DTL-25988	Fluorosilicone (TeflonTM)

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TABLE D-IV. Complete list of materials – Continued.

I.D. No.	AIRCRAFT USE	MATERIAL DESIGNATION	MATERIAL TYPE
I.H.1	Hose	Self-Sealing, AR-184	
I.H.2	Hose Aerial Refueling Tanker	PN AC 603-01 Durodyne, MIL-H-4495	Acrylic/Nitrile
I.H.3	Hose (Ground Refueling)	MIL-PRF-370 PN AC 646-01 Durodyne Ground Refueling	Nitrile
I.H.4	Hose (Navy Aircraft Carrier)	PN AC 6611-06 MIL-H-17902 Durodyne Ground Refueling System	Nitrile
I.H.5	Hose (Ground Refueling)	PN EC 614-01 Durodyne MIL-DTL-26521	Epichlorohydrin
I.I.1	Insulation/ Electrical Wire /Clamps/Misc.	Teflon	TFE (Teflon) (Film)
I.I.2	Insulation/ Electrical Wire / Clamps/Misc.	Zytel 101, DuPont ASTM D4066	Nylon 101 Film OLD Film NEW Film
I.I.3	Insulation/ Electrical Wire /Clamps/Misc.	Polyethylene Film	Polyethylene (HDP) (Film)
I.I.4	Insulation/ Electrical Wire /Clamps/Misc.	UPILEX	Kapton (Film)
I.I.5	Insulation/ Electrical Wire /Clamps/Misc.	Marmon clamp	KKK-125 (Pacific Molded)
I.I.6	Insulation/ Electrical Wire /Clamps/Misc.	MIL-I-7444, Type 1	Vinyl Plastic
I.I.7	Fuel Line Clamps & Electrical Ties	Kynar	Kynar
I.I.8	Conduit Clamp	Kirkhill TA, SAE-AMS-3215	Nitrile
I.I.9	Tube Clamp Cushions	SAE-AMS-DTL-23053/5	Polyolefin
I.I.10	Bladder Tanks	See I.B.11, 12, 13, 14	Nylon Cloth
I.I.11	Engine Fuel Control Stepper Motor	Magnetic Wire Insulation, Type I	HML Varnish
I.I.12	Wire Insulation	Teflon / Kapton	Hybrid Teflon / Kapton (Wire)
I.I.13	Wire Bundle Wrap	Shrink Wrap	
I.I.14	Wire Insulation	Teflon Insulation, Wire Insulation	Wire
I.I.15	Wire Insulation	Nylon Insulation, Wire Insulation	Wire
I.I.15.1	Wire	Nylon Wire, Coax Center	Wire

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TABLE D-IV. Complete list of materials – Continued.

I.D. No.	AIRCRAFT USE	MATERIAL DESIGNATION	MATERIAL TYPE
I.J.1	Joining Material	2219-T87 (AL), Welded	UNS A 92319 4191D9 (AMS)
I.J.2	Joining Material	6AL-4V (Ti), Welded	Match Fill
I.J.3	Joining Material	3AL-2.5V (Ti), Welded	Match Fill
I.J.4	Joining Material	Inco 718 (Ni), Welded	Match Fill
I.J.5	Joining Material	Inco 625 (Ni), Welded	Match Fill
I.J.6	Joining Material	321 (SS), Welded	Match Fill
I.J.7	Joining Material	IN200/201 (Ni), Welded	Match Fill
I.J.8	Joining Material	IN200/201 (Ni), Welded	BNI (5 or 6)
I.J.9	Joining Material	Waspalloy (Ni), Brazed	AMS 4786 Au
I.J.10	Joining Material	321 SS, Brazed	B Ag (5 or 6)
I.J.11	Joining Material	QQ-S-571, SN60 (Tin 60%, Lead 40%), B-36-21A	Tin & Lead (Solder Spots)
I.J.12	Joining Material	6061 T-6 MIL-B-7883, Type V, Grade B, Dip Braze	4145 or 4147 fill
I.J.13	Joining Material	Ti, Cu, Ni Braze P & W	Ti, Cu, Ni
I.J.14	Joining Material	6061-T6 Welded with 4043 filler	Aluminum
I.J.15	Joining Material	5052 H-34 Welded w/ 6061T6 w/ 5356 Filler	Aluminum
I.J.16	Joining Material	Sn 95, Sb 05 Base Material, B 36-21A	Copper w/Solder Spots
I.K.1	Airframe, Coatings	Cover Ink Stamp, EC 776, Top Coating SAE-AMS-QQ-A-250/11	(1 per test fuel) Shaw Aerospace
I.K.2	Airframe, Coatings	Dry Film Lubricant, Dicronite DoD-L-85645	Dicronite
I.K.3	Airframe, Coatings	Dry Thread Lubricant	Graphite
I.K.4	Airframe, Coatings	Name Plate, SAE-AMS-QQ-A-250/1, Color A11136 (Fed Std-596)	Shaw Aerospace
I.K.5	Airframe, Coatings	Dry Film Lubricant	Molybdenum Disulfide
I.K.6.1	Airframe, Coatings		Aluminum Varnish
I.K.6.2	Airframe, Coatings	Resin: No 48-C-31, ES #11110 Midland Div.	
I.K.6.3	Airframe, Coatings	Reducer: LAMNERX500, Spec. No. 66-C-28, ES #11110 Midland Div.	
I.K.7	Airframe, Coatings	Pump, Carbon Bearing, #6001 (CR Plate)	SS, 410, RC 26-34, SAE-AMS-5613
I.K.8.1	Airframe, Coatings	Pump, Carbon Bearing, Pure Carbon Co. PG18RCH	PureBon OP-658 (Carbon)

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TABLE D-IV. Complete list of materials – Continued.

I.D. No.	AIRCRAFT USE	MATERIAL DESIGNATION	MATERIAL TYPE
I.K.8.2	Airframe, Coatings	Pump, Carbon Bearing, Pure Carbon Co. P658RCH	Bearings
I.K.8.3	Airframe, Coatings	Pump, Carbon Bearing, Pure Carbon Co. P5N2	Bearings
I.K.9	Airframe, Coatings	Seal, MIL-PRF-46010, Type I, Micro-Seal Green Tweed	Sliding Seal
I.K.10.1	Airframe, Qty. Probe	B. F. Goodrich Probe P/N 391002-250	Coating
I.K.10.2	Airframe, Qty. Probe	B. F. Goodrich Electronics Fuel Quantity Probe P/N 391002-250	Coating
I.K.11	Airframe, Qty. Probe	Ragan Data Systems, Probe P/N 75-108-2F	Coating
I.K.12	Airframe, Qty, Probe	Fuel Quantity Probe, Ametek Aerospace Products CH-5851-L	Polyphenylene Sulfide 40% glass filled
I.L.1	Locking Devices	Threadlock, MIL-S-22473 Grade A or AV, Loctite	Cyanoacrylate
I.L.2	Locking Devices	Threadlock, MIL-S-22473, (Red)	Cyanoacrylate
I.L.3	Locking Devices	Threadlock, MIL-S-22473 (Brown)	Cyanoacrylate
I.L.4	Locking Devices	Lockwire, See Metals Category (I.M.19/II.M.10)	SAE-AMS-5688 wire (30302)
I.M.1	Airframe, Tank, & Plumbing	5052-0 Bare	Aluminum
I.M.2	Airframe, Tank, & Plumbing	6061-T4 Bare	Aluminum
I.M.3	Airframe, Tank, & Plumbing	6061-T6 Bare	Aluminum
I.M.4	Airframe, Tank, & Plumbing	7075-T6 Chromic Acid Anodize	Aluminum
I.M.5	Airframe, Tank, & Plumbing	7075-T6 Alodine/200	Aluminum
I.M.6	Airframe, Tank, & Plumbing	7075-T6 Bare	Aluminum
I.M.7	Airframe, Tank, & Plumbing	2024-T3 Bare	Aluminum
I.M.8	Airframe, Tank, & Plumbing	2219-T87 Bare	Aluminum
I.M.9	Airframe, Tank, & Plumbing	3003 Bare	Aluminum
I.M.10 (II.M.17)	Airframe, Tank, & Plumbing	C-355-T6	Aluminum
I.M.11 (II.M.18)	Airframe, Tank, & Plumbing	C-356-T6	Aluminum

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TABLE D-IV. Complete list of materials – Continued.

I.D. No.	AIRCRAFT USE	MATERIAL DESIGNATION	MATERIAL TYPE
I.M.12	Airframe, Tank, & Plumbing	7050-T74	Aluminum
I.M.13 (II.M.13)	Airframe, Tank, & Plumbing	316	Stainless Steel
I.M.14 (II.M.14)	Airframe, Tank, & Plumbing	321	Stainless Steel
I.M.15 (II.M.12)	Airframe, Tank, & Plumbing	304	Stainless Steel
I.M.16 (II.M.6)	Airframe, Tank, & Plumbing	INCO 718	Nickel
I.M.17 (II.M.11)	Airframe, Tank, & Plumbing	440C	Stainless Steel
I.M.18 (II.M.8)	Airframe, Tank, & Plumbing	347	Stainless Steel
I.M.19 (II.M.10)	Airframe, Tank, & Plumbing	30302, SAE-AMS-5688 (Wire) (Lockwire)	Stainless Steel
I.M.20 (II.M.22)	Airframe, Tank, & Plumbing	17-4 PH SAE-AMS-5604/5643	Stainless Steel
I.M.21	Airframe, Tank, & Plumbing	1010 Cadmium Plate (Class 2)	Ferrous
I.M.22	Airframe, Tank, & Plumbing	1010 Zinc	Ferrous
I.M.23	Airframe, Tank, & Plumbing	4130 Cadmium Plate (Class II, Type 2, Gold)	Ferrous
I.M.24 (II.M.1)	Airframe, Tank, & Plumbing	6AL-4V	Titanium
I.M.25	Airframe, Tank, & Plumbing	950 Bronze Aluminum	Copper/AL
I.M.26.1	Airframe, Tank, & Plumbing	Naval Brass	Copper/Nickel - 70/30
I.M.26.2	Airframe, Tank, & Plumbing	Naval Brass	Copper/Nickel - 90/10
I.M.27	Airframe, Tank, & Plumbing	Brass, Sheet 268 Substitute 260	Copper
I.M.28	Airframe, Tank, & Plumbing	Lead, SAE-AMS-4751/4750	Lead
I.M.29	Airframe, Tank, & Plumbing	Barium, Ferrite (Shaw Aerospace)	Barium
I.M.30	Airframe, Tank, & Plumbing	Neo-dymium (Shaw Aerospace)	(1 per fuel)
I.M.31	Airframe, Tank, & Plumbing	Brass Sheet, B36-91A	Copper

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TABLE D-IV. Complete list of materials – Continued.

I.D. No.	AIRCRAFT USE	MATERIAL DESIGNATION	MATERIAL TYPE
I.M.32	Airframe, Tank, & Plumbing	1010 Bare	Ferrous
I.M.33	Airframe, Tank, & Plumbing	B-29 (Shaw Aerospace) P/N 79-1527-RM Spec ASTM	Soft Lead
I.M.34 (II.M.25)	Airframe, Tank, & Plumbing	Monel 400, Sheet	Nickel/Copper
I.M.35	Airframe, Tank, & Plumbing	15-5 PH	Ferrous Cr, Ni, Cu
I.M.36	Airframe, Tank, & Plumbing	5052-H34	Aluminum
I.M.37	Airframe, Tank, & Plumbing	1045 Bare	Ferrous
I.M.38	Airframe, Tank, & Plumbing	Magnesium AZ91 T-6 (Substitute AZ31-H24)	Magnesium
I.M.39	Airframe, Tank, & Plumbing	4130 Bare	Ferrous, Steel
I.M.40	Airframe, Tank, & Plumbing	Sn 95, Sb 05	Solder (0.020)
I.M.41	Airframe, Tank, & Plumbing	2014-T6, SAE-AMS-4029	Aluminum
I.M.42	Airframe, Tank, & Plumbing	4340 , SAE-AMS-6415, 280KSI Tensile	Steel Bar Stock
II.M.1 (I.M.24)	Eng. Fuel lines & Components	6AL-4V	Titanium
II.M.2	Eng. Fuel lines & Components	3AL-2.5V (Tubing)	Titanium
II.M.3	Eng. Fuel lines & Components	Hastalloy	Nickel
II.M.4	Eng. Fuel lines & Components	Waspalloy	Nickel
II.M.5	Eng. Fuel lines & Components	INCO 625	Nickel
II.M.6 (I.M.16)	Eng. Fuel lines & Components	INCO 718	Nickel
II.M.7	Eng. Fuel lines & Components	Stellite 30	Chromium/ Carbide
II.M.8 (I.M.18)	Eng. Fuel lines & Components	347	Stainless Steel
II.M.9	Eng. Fuel lines & Components	Greek Ascolloy (30302)	Ferrous

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TABLE D-IV. Complete list of materials – Continued.

I.D. No.	AIRCRAFT USE	MATERIAL DESIGNATION	MATERIAL TYPE
II.M.10 (I.M.19)	Eng. Fuel lines & Components	SAE-AMS-5688 (S.S. Wire) (30302)	Ferrous
II.M.11 (I.M.17)	Eng. Fuel lines & Components	440C	Stainless Steel
II.M.12 (I.M.15)	Eng. Fuel lines & Components	304	Stainless Steel
II.M.13 (I.M.13)	Eng. Fuel lines & Components	316	Stainless Steel
II.M.14 (I.M.14)	Eng. Fuel lines & Components	321	Stainless Steel
II.M.15	Eng. Fuel lines & Components	ASI 51410 SS (SAE-AMS-5504)	Stainless Steel
II.M.16	Eng. Fuel lines & Components	CPM 10-V	Powder Metallurgy rolled Fe, V, Cr, C, Mn, Si, T, S, Mo
II.M.17 (I.M.10)	Eng. Fuel lines & Components	C-355 T6	Aluminum
II.M.18 (I.M.11)	Eng. Fuel lines & Components	C-356 T6	Aluminum
II.M.19	Eng. Fuel lines & Components	A-286 SAE-AMS-5525 Silver Plate (2410)	Ferrous
II.M.20	Eng. Fuel lines & Components	135 Modified (MIL-S-6709, SAE-AMS-6470)	Nitralloy
II.M.21.1	Eng. Fuel lines & Components	Bronze, Leaded (Tap MS 285) .1) Saw Cut, Cut up Bearing	Copper
II.M.21.2	Eng. Fuel lines & Components	.2) Polished Cylinder (Argo-Tech)	Polished Cylinder Dry Lub End
II.M.21.3	Eng. Fuel lines & Components	.3) Coated Cylinder (Indium) (Argo-Tech “A”)	Indium Cyl. Surf. Dry Lub End
II.M.21.4	Eng. Fuel lines & Components	.4) Coated Cylinder (Indium) (Argo-Tech “B”)	Indium All Cu Surf. Dry Lub End
II.M.22 (I.M.20)	Eng. Fuel Line & Components	17-4 PH Stainless Steel SAE-AMS-5604	Ferrous (S.S.)
II.M.23	Eng. Fuel Line & Components	IN 200 Nickel	Nickel
II.M.24	Eng. Fuel lines & Components	Augmentor Spray Bar P & W	Stainless Steel Nr,Ci,Co,Au Braze Nozzles
II.M.25 (I.M.34)	Eng. Fuel lines & Components	Monel 400, Sheet	Nickel Copper

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TABLE D-IV. Complete list of materials – Continued.

I.D. No.	AIRCRAFT USE	MATERIAL DESIGNATION	MATERIAL TYPE
II.M.26	Eng. Fuel lines & Components	Incoloy 909	Ni, Co, Fe
II.M.27	Eng. Fuel lines & Components	Titanium 6-2-4-2, (4919C) Sheet	Titanium
II.M.28	Eng. Fuel lines & Components	Haynes 188	Co, Cr, Ni
II.M.29	Eng. Fuel lines & Components	Haynes 214	Ni, Cr, Fe, Al
II.M.30.1	Eng. Fuel lines & Components	SAE-AMS-7902 AlBeMet 162 Reactive Material Sheet & Plate, Beryllium Alloy	.1) as cast alloy (310)
II.M.30.2	Eng. Fuel lines & Components		.2) investment cast high strength alloy with machined surfaces (157)
II.M.30.3	Eng. Fuel lines & Components		.3) AM 162 rolled Standard grind finish
II.M.31	Eng. Fuel lines & Components	UNS C17200 Be Cu Spring	Cu, Be
II.M.32	Eng. Fuel lines & Components	DB Inconel 718 Diffusion Bonded	Ni,Cr
II.M.33	Eng. Fuel lines & Components	Si C Reinforced Ti, MMC	Titanium, MMC
II.M.34	Eng. Fuel lines & Components	8 Al-1V-1 Mo	Titanium
II.M.35	Eng. Fuel lines & Components	Ion Vapor Deposit IVD onto 4130	4130 Steel, Fe, Cr, Mo
II.M.36	Eng. Fuel lines & Components	52100 SAE-AMS-6444	Steel
II.M.37	Eng. Fuel lines & Components	8620 SAE-AMS-6277	Steel
II.M.38	Eng. Fuel lines & Components	303 Stainless	Steel
II.M.39	Eng. Fuel lines & Components	TI-CP-70	Titanium
I.O.1	Float	HR Textron Inc.	Unicellular Buna-N
I.O.2	Float	HR Textron Inc., Foam Molders Inc.	Polyurethane Unicellular
I.O.3	Float	HR Textron Inc.	Polyurethane

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TABLE D-IV. Complete list of materials – Continued.

I.D. No.	AIRCRAFT USE	MATERIAL DESIGNATION	MATERIAL TYPE
I.O.4	Float	XAR Industries Inc.	
I.O.5 (I.G.13)	Float	Parker 30-155-5-1	Cork
I.P.1 (I.A.5)	Potting Compound	Epon 828 / DTA Unmodified Epoxy (See I.A.5)	Epoxy
I.P.2.1	Potting Compound	Chem Seal, CS3100, MIL-PRF-8516, Cure B	Polysulfide
I.P.2.2	Potting Compound		Electrical Connector Application
I.P.3	Potting Compound	SAE-AMS-3361, Fluorosilicone	Fluorosilicone
I.P.4	Potting Compound	Urethane	Urethane

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APPENDIX E

TOXICITY TEST PROTOCOL AND ESOH BASELINE INFORMATION

E.1 SCOPE.

E.1.1 General.

This appendix is providing a toxicity testing review and a baseline of known information regarding JP-8 jet fuel. JP-8 baseline knowledge is detailed in E.3. When candidate fuels and additives are tested, certified results will be incorporated into the baseline knowledge.

Toxicology is the study of the adverse effects of chemical, physical or biological agents on living organisms and the ecosystem, including the prevention and amelioration of such adverse effects. ESOH refers to Environmental Safety and Occupational Health. When sufficient toxicity data are available, environmental and occupational exposure standards can be developed that are protective of ecosystems (the environment) and workers (occupational medicine and health). ESOH should be addressed during development of new fuels and weapon systems, not after they are fielded.

E.1.2 Entrance Criteria and Subset Testing.

Air Force SMEs evaluated all fuel properties and characteristics that were decomposed from system requirements via the systems engineering process, as described in Appendix A. Consequently from this decomposition, Appendix E defines the entrance criteria, subset testing and supporting rationale for all of the properties / characteristics related to personnel and environmental health and safety.

E.1.3 Testing Rationale.

The following indexing table provides an overview of the toxicity tests used for fuel evaluation, in the format of the testing performed on candidate fuels and additives. The table also includes the rationale for the proposed tests and the location of the baseline information for JP-8 contained within this appendix.

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TABLE E-1. Toxicity test overview index.

Test Name	Purpose	Related Ref. JP-8 Baseline Test
Minimum Tests Used for candidate fuels and additives		
In vitro genotoxicity bacterial reverse mutation test mammalian cell gene mutation test	Screen for possible mutagens and carcinogens using bacteria using human lymphocytes	E.3.2.8
Dermal irritation	Determine potential irritant effects	E.3.2.1.2
Acute inhalation	Identify effects from single 4 hr exposure	E.3.2.1.1
Inhalation rangefinder (2-wk)	Screen for signs of toxicity and gross pathology	Sets dosage for 90 day test
In vivo genotoxicity – micronucleus	Screen for possible mutagens (part of 2-wk)	E.3.2.8
90-day inhalation toxicity with additional chem analysis	Repeated dose study required for HHA fully characterize vapor versus aerosol in fuel	E.3.2.4
Additional Tests for candidate fuels and additives		
HHA Development	Health Hazard Assessment for above studies	N/A
Immunotoxicity	Assess effects on immune system	E.3.2.5
Sensory Irritation (Alarie)	Compare respiratory tract irritation with JP-8	E.3.2.1.2
Genetic Biomarkers	Screen for genetic biomarkers of exposure	JP-8 Baseline TBD
Additional Tests for candidate fuels and additives		
90-day inhalation toxicity additional histopathology test for alpha 2 microglobulin measure enzyme induction (P450)	When warranted by previous test results satisfy predicted regulatory changes satisfy review panel recommendations and support mode of action for predicted kidney toxicity in rats support histopathology and model development	E.3.2.4
Optional Effort to Support Testing		
Model Development	Physiologically Based PharmacoKinetic (PBPK) model to predict blend effects and for future alternative fuels and additives	JP-8 Baseline TBD

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E.2 APPLICABLE DOCUMENTS.

E.2.1 General.

The documents listed below are comprehensive toxicity reviews for JP-8. A more complete list of references can be found in E.5.

E.2.2 Other Government documents, drawings, and publications.

The following other Government documents, drawings, and publications form a part of this document to the extent specified herein.

ATSDR 1998 Toxicological Profile for JP-5 and JP-8. Agency for Toxic Substances and Disease Registry, Atlanta, GA http://www.atsdr.cdc.gov/toxpro2.html

COT 1996 Permissible Exposure Levels for Selected Military Fuel Vapors. National Research Council Committee on Toxicology, ed. Washington, D.C.: National Academy Press.

COT 2003 Toxicologic Assessment of Jet-Propulsion Fuel 8. National Research Council Committee on Toxicology, ed. Washington, D.C.: National Academy Press.

Ritchie, G. D., Still, K. R., Rossi, J., Bekkedal, M. Y. V., Bobb, A. J., and Arfsten, D. P. 2003. Biological and health effects of exposure to kerosene-based jet fuels and performance additives. J. Toxicol. Environ. Health B. 6:357-451

(Copies of these documents are available online at The Defense Technical Information Center Website and are available from DTIC Headquarters, 8725 John J. Kingman Rd., Ft. Belvoir VA 22060-6218; telephone (800) 225-3842.)

E.3 BASELINE INFORMATION. (Knowledge of JP-8 toxicity form the baseline to which candidate fuels and additives are assessed.)

E.3.1 Regulatory Issues.

In 1996, the National Research Council (NRC), Committee on Toxicology (COT), Subcommittee on Permissible Exposure Levels for Military Fuels was convened on behalf of the Navy to review their interim guidelines for fuels, including JP-8. The NRC 1996 report identified major data gaps in human occupational studies of possible JP-8 toxicity (National Research Council, 1996). The expert panel proposed occupational JP-8 exposure standards as 8-h time weighted averages with a threshold limit value (TLV) of 350 mg/m^3 and a 15-min, short-term exposure limit (STEL) of 1000 mg/m^3. The Air Force accepted these proposed values as interim permissible exposure limits. In addition, the Centers for Disease Control and Prevention, Agency for Toxic Substances and Disease Registry (CDC-ATSDR), prepared an extensive document, the Toxicology Profile for Jet Fuels (JP-5 and JP-8) (ATSDR, 1998). This report

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further concluded that possible jet fuel toxicity and underlying physiological mechanisms are not well defined or understood.

More recently, the U.S. Environmental Protection Agency (EPA), Office of Prevention, Pesticides and Toxic Substances (OPPTS), National Advisory Committee for Acute Exposure Guidelines for Hazardous Substances (NAC-AEGL), officially recognized significant data gaps in the toxicology profiles submitted by the ATSDR. The NAC-AEGL still has not been able to set acute exposure guidelines for JP-8 because of this lack of information. In 2000, an American Conference of Governmental Industrial Hygienists (ACGIH) Draft Notice of Intended Changes (NIC) was filed to recommend reduction of the current 8-h threshold limit value (TLV)–time-weighted average (TWA) standards for both kerosene and diesel fuel, from 350 mg/m^3 to 100 mg/m^3. At the same time, the NRC COT convened a second expert panel at the request of the Air Force to review the occupational JP-8 exposure standards of 350 mg/m^3. The findings of this panel were recently published and represent a comprehensive investigation of JP-8 toxicity. The NRC panel found JP-8 to be potentially toxic to the immune system, respiratory tract and nervous system at exposure concentrations near the interim value of 350 mg/m^3 (National Research Council, 2003). The ACGIH proposed a threshold limit value for kerosene and jet fuels in 2003, as a total hydrocarbon vapor, of 200 mg/m^3. Based on the ACGIH and NRC, the Air Force adopted TLV-TWA of 200 mg/m^3 as the permissible exposure limit for JP-8. In 2004, the ACGIH lowered the TLV-TWA to 100 mg/m^3. National Institutes for Occupational Safety and Health (NIOSH) currently shares this value as a recommended exposure level (REL) TWA for kerosene.

E.3.1.1 References.

ACGIH, 2003. Kerosene, American Conference of Governmental Industrial Hygienists, Cincinnati, OH.

ACGIH, 2004. Kerosene, American Conference of Governmental Industrial Hygienists, Cincinnati, OH.

NIOSH, 2003. National Institutes for Occupational Safety and Health Pocket Guide to Chemical Hazards, Publication No. 2004-103.

(Copies of these documents are available online at The Defense Technical Information Center Website and are available from DTIC Headquarters, 8725 John J. Kingman Rd., Ft. Belvoir VA 22060-6218; telephone (800) 225-3842.)

E.3.2 Summary of Baseline Fuel Toxicity.

The USAF has been using JP-8 as its standard fuel since the mid-1980s; therefore, all data for toxicity will be compared to JP-8 as this fuel has the largest database for currently (2007) used jet fuels. JP-8 toxicity studies are still being released. Periodically, further investigation of the effects of interest for JP-8 is necessary in order to update the baseline and compare to new alternative fuels. The following is a brief summary of JP-8 effects for studies most appropriate to a new or current risk assessment.

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E.3.2.1 Acute Toxicity.

E.3.2.1.1 Lethality.

(Acute inhalation - Inhalation rangefinder (2-wk)**:** Lethality tests indicate that JP-8 is considered slightly toxic given the concentrations tested. Male and female F344 rats were gavaged with neat jet fuel and monitored for 14 days. The oral dose in rats that resulted in lethality of 50% of the tested population (LD_{50}) was greater than 5.0 g/kg (highest dose tested) for both JP-8 and JP-8+100. Patches on the back of male and female New Zealand white rats were clipped and neat jet fuel was applied evenly and occluded for 24 hours; the rabbits were monitored for 14 days. The dermal LD_{50} in rats was >2.0 g/kg (highest dose tested) for both fuels. Similarly, in the acute inhalation tests, male and female F344 rats were exposed (whole-body) to vapor only or vapor plus aerosol fuel for 4 hours and monitored for 14 days. The inhaled vapor concentration LC_{50} was >3.43 mg/L, while the vapor and aerosol combined LC_{50} was >4.39 mg/L, the highest concentrations tested in both fuels (Wolfe *et al.*, 1996).

E.3.2.1.2 Irritation.

(Dermal irritation - Sensory Irritation, Alarie - Genetic Biomarkers): JP-8 was evaluated in an acute toxicity test battery including the primary eye irritation, skin irritation and dermal sensitization protocols by Draize. JP-8 was found to be non-irritating in the rabbit primary eye irritation test. New Zealand white rabbits were given a topical anesthetic in both eyes; one eye was exposed to 0.1 mL JP-8 and the other used for control response comparison (Smith *et al.*, 1981, Kinkead *et al.*, 1992). Results in the corresponding rabbit skin irritation test ranged from non-irritating to slightly irritating and back to non-irritating. In each test, 0.5 mL JP-8 was applied to clipped skin on the backs of New Zealand white rabbits, occluded and evaluated at multiple time points. Similarly in the guinea pig skin sensitization test, results ranged from non-sensitizing to a weak dermal sensitizer and back to non-sensitizing. In this test, Hartly albino guinea pigs were dermally exposed to 0.1 mL JP-8 on 4 separate occasions within 10 days; after a 2-week induction period, a dermal challenge of 0.1 ml JP-8 was applied to a different skin location and the response scored (Smith *et al.*, 1981, Kinkead *et al.*, 1992, Wolfe *et al.*, 1996, respectively). JP-8+100 packages were all negative in the skin irritation and sensitization tests (Wolfe *et al.*, 1996). Minor differences in how the different studies were conducted may have resulted in slightly different outcomes. However, murine local lymph node assays also indicate that JP-8, but not JP-8+100, is a weak dermal sensitizer. In this assay, 25 µL JP-8 was applied to the ear flaps of female CBA/Ca mice for 3 consecutive days; allergic response was measured by increased uptake of [^{3}H]-methyl thymidine into auricular lymph node cells compared to concurrent controls (Kanikkannnan *et al.*, 2000).

Respiratory tract sensory irritation was studied for both JP-8 and JP-8+100 in male Swiss-Webster mice. Groups of mice were exposed head-only to JP-8 (681, 708, 1090, 1837 or 3565 mg/m^3) or JP-8+100 (777, 1519 or 2356 mg/m^3) for 30 minutes. The calculated concentration at which the respiratory rate decreased 50% (RD_{50}) was 2876 and 1629 mg/m^3 for each fuel, respectively (Whitman and Hinz, 2001).

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E.3.2.2 Dermal Toxicity.

Few dermal systemic toxicity tests, aside from the acute irritation tests above, have been performed with JP-8 itself. A good review of the dermal toxicity of petroleum distillates closely related to JP-8 can be found in McDougal and Rogers (2004). Much work has been done on the absorption and local effects of JP-8, including several *in vitro* studies, as outlined in the reference list. It is clear from the body of knowledge combined with reports from military personnel working with JP-8 that the fuel does cause considerable skin irritation over continued use and exposure.

One subchronic dermal study of JP-8 was conducted by Baker and coauthors in 1999. Dermal histological changes were investigated in male F344 rats. A daily un-occluded dermal exposure to 0.156 mL JP-8, JP+100 or JP-4 for 4 weeks was followed by a three week recovery period. Proliferative, degenerative and inflammatory changes were significantly greater in the fuel exposed skin versus non-exposed control skin sites on the same animal immediately post-exposure, but fuel treatment results did not differ from each other. Following the recovery period, the dermal histology of all the exposed skin sites had returned to control scores.

E.3.2.3 Hearing Loss.

JP-8 exposure increases hearing damage from noise in rats. Male Long-Evans rats were exposed, nose-only, to 1000 ppm JP-8 for 4-hours. A subset was then exposed to noise (97 or 102 dB) for 1 or 4 hours. Exposures were performed either singularly or repeated for 5 consecutive days. JP-8 alone did not affect hearing but JP-8 followed by noise resulted in small but consistent disruption of outer hair cell function and hair cell loss greater than for noise alone. The effect was only partially reversible with time (4 weeks) (Fechter *et al.*, 2007). These results are corroborated with an occupational study among jet fuel and noise exposed military workers. JP-8 estimated exposures below 350 mg/m^3 (the permissible exposure limit (PEL) at the time the study was performed) for 3 or 12 years, combined with noisy working environments, were associated with increased odds of hearing loss. Odds of hearing loss also increased with time on the job (Kaufman *et al.*, 2005).

E.3.2.4 Subchronic Studies.

(90-day inhalation toxicity with additional chemical analysis – histopathology test for alpha 2 microglobulin – and - measure enzyme induction, P450)**:** Subchronic 90-day studies were conducted through inhalation and oral pathways. Male and female F344 rats and C57BL/6 mice were exposed (whole-body) to JP-8 vapor (0, 500 or 1000 mg/m^3) continuously over 90 days. Decreased bodyweight occurred in male rats, along with renal hydrocarbon nephropathy, a male rat specific response to hydrocarbons that is not related to human health. Male mice were significantly decreased in survivability over the 21-month recovery period due to complications from fighting. No additional treatment related adverse effects were found (Mattie *et al.*, 1991). In a subsequent study, male Sprague-Dawley rats were dosed with 0, 750, 1500 or 3000 mg/kg neat JP-8 by gavage daily for 90 days. Decreased bodyweight and hydrocarbon nephropathy occurred in all JP-8 dosed rat groups. Additional effects included gastritis, perianal dermatitis and increased liver enzymes (AST and ALT). JP-8 was concluded to have minimal toxic effects outside the male rat specific nephropathy (Mattie et al., 1995).

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E.3.2.5 Immune.

(Immunotoxicity): Immunological parameters, host resistance and thyroid hormones were evaluated by Keil *et al*., (2003) in F1 mice exposed *in utero* to JP-8. C57BL/6 pregnant dams (mated with C3H/HeJ males) were gavaged daily on gestation days 6–15 with JP-8 in a vehicle of olive oil at 0, 1000 or 2000 mg/kg. At weaning (3 weeks of age), no significant differences were observed in body, liver, spleen or thymus weight, splenic and thymic cellularity, splenic CD4/CD8 lymphocyte subpopulations, or T-cell proliferation. Yet, lymphocytic proliferative responses to B-cell mitogens were suppressed in the 2000 mg/kg treatment group. In addition, thymic CD4-/CD8+ cells were significantly increased. By adulthood (8 weeks of age), lymphocyte proliferative responses and the alteration in thymic CD4-/CD8+ cells had returned to normal. However, splenic weight and thymic cellularity were altered, and the IgM plaque forming cell response was suppressed by 46% and 81% in the 1000 and 2000 mg/kg treatment groups, respectively. Furthermore, a 38% decrease was detected in the total T_4 serum hormone level at 2000 mg/kg. In F1 adults, no significant alterations were observed in natural killer cell activity, T-cell lymphocyte proliferation, bone marrow cellularity and proliferative responses, complete blood counts, peritoneal and splenic cellularity, liver, kidney, or thymus weight, macrophage phagocytosis or nitric oxide production, splenic CD4/CD8 lymphocyte subpopulations, or total T_3 serum hormone levels. Host resistance models in treated F1 adults demonstrated that immunological responses were normal after challenge with *Listeria monocytogenes*, but heightened susceptibility to B16F10 tumor challenge was seen at both treatment levels. This study demonstrates that prenatal exposure to JP-8 can target the developing murine fetus and result in impaired immune function and altered T_4 levels in adulthood.

A study by Keil *et al*. (2004) examined the effects of JP-8 on humoral and cell-mediated and hematological parameters. A suite of immunotoxicological endpoints was evaluated in adult female B6C3F1 mice gavaged with JP-8 (in an olive oil vehicle) ranging from 250–2500mg/kg/d for 14 d. One day following the last exposure, significant increases in liver mass were detected beginning at exposure levels of 1000 mg/kg/d, while thymic mass was decreased at exposure levels of 1500mg/kg/d and above. Decreases in thymic cellularity, however, were only observed at exposure levels of 2000 mg/kg/d and above. Mean corpuscular volume was increased (1500–2500 mg/kg/d), while the hematocrit, hemoglobin concentration, and red blood cell count were decreased only at the 2500 mg/kg/d exposure level. Natural killer cell (NK) activity and T- and B-cell proliferation were not altered. Decreases in the plaque forming cell (PFC) response were dose responsive at levels of 500 mg/kg/d and greater, while unexpectedly, serum levels of anti-SRBC immunoglobulin M (IgM) were not altered. Alterations were detected in thymic and splenic CD4/8 subpopulations, and proliferative responses of bone marrow progenitor cells were enhanced in mice exposed to 2000 mg/kg/d of JP-8. This study showed that humoral immune function is impaired with lower exposure levels of JP-8 than are required to affect primary and secondary immune organ weights and cellularities, CD4/8 subpopulations and hematological endpoints (Keil *et al*., 2004).

E.3.2.6 Neurobehavioral.

Neurobehavioral effects have been assessed in adult rats following JP-8 vapor inhalation. Changes in behavioral response were observed in two studies where rats were exposed to 0, 500

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or 1000 mg/m^3 for 6 hours/day 5 days a week for 6 weeks. JP-8 inhalation affected performance on very specific tasks and did not cause a generalized deficit. When animals were subjected to different operant tasks with varying levels of complexity, the low and high exposure groups scored the same as control animals on all tests except for the most complex tasks. In these two operant tests, group differences emerged; low dose animals demonstrated better performance than high dose animals while neither group performed differently from controls (Ritchie *et al*., 2001). In a second study using the same exposure methods, animals were tested in a large battery of neurobehavior tasks. No exposure group differences were found in acoustic startle responses, forelimb grip strength, nociception, social interaction, the forced swim test, spontaneous locomotor activity, passive avoidance or Morris watermaze performance. However, differences were found in a test for behavioral sensitization. The appetitive stimulus approach sensitization (ARAS) measures the time an animal spends proximal to an appetitive stimulus versus a neutral stimulus. Animals exposed to JP-8 spent more time than control animals investigating the appetitive stimulus, suggesting behavioral sensitization and altered neural pathways related to the dopaminergic system (Rossi *et al*., 2001). Overall, the data suggest very specific, versus generalized, neurobehavioral effects of JP-8 vapor exposure in adult rats.

As a continuation of the reproductive study by Mattie *et al*. (2000), the pups from the female study were assessed for potential developmental neurobehavioral deficiencies after exposure *in utero* and during lactation to JP-8 (Mattie *et al*., 2001). Litters were standardized to four male and four female pups at PND 4; all eight pups in a litter were tested for surface righting and negative geotaxis. JP-8 did not affect age of onset for surface righting reflex in pups tested on PND 4. Negative geotaxis abilities, tested on PNDs 5 through 8, developed at the same age for pups in all JP-8 groups; however, all females met the criterion sooner than males. Development of motor coordination related to swimming was tested in one male and one female pup from each litter every other day from PNDs 6 through 20. A dose-related difference in composite scores for swimming abilities was observed on PNDs 8 and 14, indicating a delay in development of coordinated motor movements related to the swimming task. On PND 8, pup scores from all doses were ≥20% lower than control scores. On PND 14, composite swimming scores were 8% lower in the 750 and 1500 mg/kg-day dose groups versus controls. Pups were tested in an M swimming maze on PNDs 70 and 77. JP-8 did not affect the number of trials to criterion on either test date; on PND 77, male pups met the criterion of five errorless trials in fewer attempts than females. The effects of JP-8 exposure found in results from developmental neurobehavioral testing indicated developmental delays in pups caused by exposure of dams to JP-8 in utero and during lactation may be associated with key developmental milestones in the developing cerebellum of the pups. Although recovery occurred, a delay in motor development has potential adverse impacts. The lowest dose of 325 mg/kg-day would be the lowest observed adverse effect level (LOAEL) for developmental neurobehavioral effects (Mattie *et al*., 2001).

A study by Smith et al. (1997) used the postural stability technique to investigate the neurological effects of cumulative low-level exposure to JP-8 jet fuel vapors in aircraft maintenance personnel. All subjects performed two sets of four 30-second postural sway tests. The results of mean cumulative exposure levels (in parts per million ± standard error of the mean) were as follows: naphthas, 1308 ± 292; benzene, 21.2 ± 5.7; toluene, 23.8 ± 6.1; and m-, o-, p-xylenes, 22.7 ± 5.4. Covariate adjusted regression analysis of the exposed group data showed a statistically significant association ($p<0.05$) between the solvents (benzene, toluene and xylenes) and increased postural sway response. For all solvent exposures, the “eyes closed,

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on foam" test provided the strongest association between sway length and JP-8 benzene (r2 range, 0.45 to 0.52), implying subtle influence on vestibular/proprioception functionalities.

E.3.2.7 Developmental/Reproductive.

A developmental toxicity study indicated that JP-8 is not a teratogen in the rat (Cooper and Mattie, 1996). Female rats were dosed with 0, 500, 1000, 1500 or 2000 mg/kg neat JP-8 daily by gavage on days 6 through 15 of gestation. Maternal body weights significantly decreased in the 1000, 1500 and 2000 mg/kg-day dose groups while fetal weights decreased in the 1500 and 2000 mg/kg-day groups. Fetal malformations and variations did not differ significantly between control and treatment groups (Cooper and Mattie, 1996).

Two reproductive studies were performed as part of an aforementioned multi-step investigation (Mattie *et al*., 1995). JP-8 was shown not to be a reproductive toxicant in rats. In the first study, male rats were given 0, 750, 1500 or 3000 mg/kg neat JP-8 daily by gavage for 70 days prior to mating with naïve females to assess fertility and sperm parameters. After 70 days of dosing, body weights in the 3000 mg/kg group were over 30% lower than control weights. There were no significant changes for pregnancy rate, gestation length or sperm parameters as compared to control values (Mattie *et al*., 2000).

In the second reproductive study, general toxicity, fertility and reproductive endpoints were assessed in female rats dosed with neat JP-8 (0, 325, 750 or 1500 mg/kg) daily by gavage for a total of 21 weeks (90-days plus mating with naïve males, gestation and lactation). Results of general toxicity revealed a significant dose-dependent decrease in body weights of the female rats. Significant organ weight ratio increases were seen for the liver:body, liver:brain and kidney:brain weights. Corresponding histopathologic changes and increases in liver enzymes (ALT, AST) were not observed although there was an increase in liver weight. Significant pathological changes were limited to squamous hyperplasia of the stomach and perianal dermatitis. There were no statistically significant changes from control values for gestation length, pregnancy rate and numbers of pups per litter. There was a trend for decreased pup weight with increasing dose from postnatal days 4 through 21 with the 1500 mg/kg pups statistically and biologically significantly lower on these days. Recovery occurred by 90 days. Based on the results of both reproductive studies, the no observed adverse effect level (NOAEL) for JP-8 reproductive and development effects is 750 mg/kg with 1500 mg/kg as a LOAEL based on decreased pup weights (Mattie *et al*., 2000).

E.3.2.8 Mutagenicity/Oncogenicity.

(*In vitro* genotoxicity - bacterial reverse mutation test - mammalian cell gene mutation test – and - *In vivo* genotoxicity – micronucleus): Brusick and Matheson (1978) tested JP-8 for mutagencity in a number of test systems. JP-8 was not mutagenic for Salmonella in the Ames Bacterial Reverse Mutation Test. The chemical was toxic to most of the bacteria strains at concentrations above 1 µL per plate. In the Mouse Lymphoma Assay, JP-8 did not induce gene mutation in mouse cells. The material was moderately toxic in this assay at 0.16 µl/mL. JP-8 induced levels of 3H-thymidine incorporation in the Unscheduled DNA Synthesis Assay which was significant. The increase in activity of the WI-38 cells was moderate and the effect plateaued and was not dose related. The dose of 5.0 µL/mL was beginning to show clear

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evidence of cytotoxicity. These data suggest that the material could interact with DNA producing nonspecific lesions. The Dominant Lethal Assays showed that JP-8 was only moderately toxic for mice and rats. The dose levels used for mice were 0.13, 0.4 and 1.3 mL/kg per day for 5 days. The dose levels employed for rats were 0.1, 0.3 and 1.0 mL/kg per day for 5 days. Mouse and rat test results for JP-8 were negative. None of the parameters measured in either study showed compound-induced effects. The positive control values for this study were clearly elevated but were not as high as usual. No evidence for mutagenicity was evident in the test battery and the indications for mutagenic and carcinogenic potential for JP-8 are minimal at best. There is no suggestion of significant genetic risk associated with this material according to Brusick and Matheson (1978).

Jet A was tested in a number of test systems by Hazleton Laboratories America, Inc. in 1979. Jet A was not mutagenic for Salmonella in the Ames Bacterial Reverse Mutation Test. However, Jet A was shown to induce mutations in the presence of metabolic activation when tested in the Mouse Lymphoma Assay. Jet A was also tested in an *in vivo* Bone Marrow Cytogenicity Study that is now called the Mammalian Bone Marrow Chromosome Aberration Test. Jet A has the ability to produce structural alterations (chromosomal aberrations) in the bone marrow cells of rats exposed by inhalation to 100 and 400 ppm (methane equivalent). Rats exposed to 100 ppm received 19 exposures (6 hours per day) and rats exposed to 400 ppm received 5 exposures (6 hours per day).

Mice were treated dermally with either a single or multiple applications of JP-8 and Jet A fuels in the Mammalian Micronucleus Test. Peripheral blood and bone marrow smears were prepared to examine the incidence of micronuclei (MN) in polychromatic erythrocytes (PCEs). In all experiments, using several different exposure regimens, no statistically significant increase in the incidence of MN was observed in the bone marrow and/or peripheral blood of mice treated with JP-8 or Jet-A when compared with those of untreated control animals (Vijayalaxmi *et al*., 2006).

E.4 TOXICITY TEST REVIEW AND PROTOCOL.

E.4.1 Entrance Criteria.

A Material Safety Data Sheet (MSDS) will be provided with the alternative fuel to be evaluated. A MSDS provides basic safety information for the handling and safe use of a chemical or material. If exposure standards or regulations are known, then they will be available in the MSDS. A thorough MSDS actually provides toxicity data summaries and references but often this information is not provided or is not available. New candidate fuels may not have exposure standards or regulations yet and if there are any toxicity data, it will most likely be limited.

E.4.2 Subset 1: Toxicology and Environmental Evaluation and Tests.

E.4.2.1 Toxicity Evaluation.

Initial evaluation of a new fuel involves identifying components of the fuel, performing literature searches on the fuel and its components for any known toxicity data, and identifying

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research that is necessary to complete the development of exposure standards. This handbook is trying to formalize the process for ESOH review for the DoD. In the Air Force, ESOH reviews have been performed by the Air Force Institute for Occupational Health (AFIOH), the Air Force Research Laboratory (AFRL) or an outside contractor. See E.4.5. for ESOH contacts for assistance in developing the toxicity testing program.

Based on the initial evaluation identified in the above paragraph, acute toxicity studies, genotoxicity tests and a range-finder study for a long-term toxicity test should be conducted with comparisons to JP-8. For the new fuel, conduct a 90-day toxicity test with doses based on the 2-week rangefinder study. This 90-day study is the minimum study required for development of exposure standards. Additional studies may be recommended based on the results of the 90-day study and the health hazard assessment.

A Health Hazard Assessment (HHA) should be conducted using an exposure assessment and the above toxicity data. The HHA would result in a recommendation for exposure limits (standards) for the safe use of the alternative fuel. The recommended exposure limits could become AFOSH standards until they are promulgated as federal standards by OSHA, EPA or ACGIH.

E.4.2.2 Industrial Hygiene (IH) Review (Bioenvironmental Engineering (BEE)). This involves identifying potential exposure hazards based on the toxicity evaluation and recommending interim personal protection (PPE) or engineering controls (IH controls) to prevent exposure to personnel. In the Air Force, this has been performed by the Air Force Institute for Occupational Health or by the base Bioenvironmental Engineering Office. IH sampling can be performed by AFIOH and/or base Bioenvionmental Engineering (BEE) office with results reviewed by AFIOH and recommendation made for safe handling of the fuel.

E.4.2.3 Environmental Review.

This involves reviewing ecotoxicity data, fate and transport data and potential pathways of exposure. In the Air Force, this has been performed by the Air Force Institute for Occupational Health. Below are listed comprehensive resources for this type of assessment.

Johnson, Mark S.; Ruppert, William H.; Taylor, Patrick J.; Packer, Bonnie; Watts, Byrd, Edward F. C.; Hurley, Margaret M.; McQuaid, Michael J.; Rice, Betsy M. McAtee, Matthew J. Assessing the Potential Environmental Consequences of a New Energetic Material: A Phased Approach. U.S. Army Center for Health Promotion and Preventative Medicine (USACHPPM), Technical Report 87-XE-03N3-01 (December 2007).

Standard Guide for Assessing the Environmental and Human Health Impacts of New Energetic Compounds – ASTM E2552, 15 March 08.

Wentsel, R. S., La Point, T. W., Simini, M., Checkai, R. T., Ludwig, D., and Brewer, L. W. 1996. Tri-Service procedural guidelines for ecological risk assessments. Volume 1. U.S. Army Edgewood Research, Development and Engineering Center, Aberdeen Proving Ground, MD.

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See Total Petroleum Hydrocarbon Working Group (TPHCWG) references in E.5 For issues related to establishing soil cleanup levels protective of human health at contaminated sites.

E.4.2.4 Decision points.

The section E.4.2 evaluation and review results will be compared to the JP-8 results, and if the candidate fuel provides equal or less toxic results as JP-8, the candidate fuel poses minimal risk to personnel and the environment. If the candidate fuel is much more toxic than JP-8, then this should be documented and the candidate fuel could be rejected and the process terminated. If the candidate fuel is slightly different than JP-8, then the candidate fuel would be recommended for additional testing. At this time, the Air Force can proceed with testing but will need to use increased IH controls and PPE until a revised Health Hazard Assessment (HHA) is complete.

E.4.3 Subset 2: Toxicology and Environmental Evaluation and Tests.

E.4.3.1 Toxicity Testing. N/A

E.4.3.2 Exposure Assessment. N/A

E.4.3.3 Environmental. N/A

E.4.4 Subset 3: Toxicology and Environmental Evaluation and Tests.

E.4.4.1 Toxicity Testing.

Conduct additional studies that were recommended based on the results of the 90-day study and health hazard assessment.

E.4.4.2 Exposure Assessment.

The Health Hazard Assessment should be reviewed or revised using additional exposure assessment and toxicity data. This would result in verification or an update of exposure limits (standards) for safe use of the alternative fuel.

E.4.4.3 Environmental Effects.

Conduct additional studies that were recommended based on the results of Subset 1.

E.4.5 ESOH Points of Contact.

E.4.5.1 Toxicity Testing.

Air Force Research Laboratory, Human Effectiveness Directorate, Biosciences and Protection Division, Applied Biotechnology Branch (AFRL/RHPB, Wright-Patterson AFB OH)

Naval Health Research Center, Environmental Health Effects Laboratory (NHRC/EHEL)

U.S. Army Center for Health Promotion and Preventive Medicine (USACHPPM)

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E.4.5.2 Environmental.

Tri-Service Environmental Working Group

E.4.5.3 Exposure Assessment.

IH sampling

Air Force Institute for Operational Health (AFIOH) - AFIOH/RSH

Navy and Marine Corps Public Health Center (NMCPHC)

USACHPPM

E.4.5.4 Health Hazard Assessment.

Proposing occupational exposure limits

AFIOH/RSRE and AFRL/RHPB

NMCPHC

USACHPPM

E.5 REFERENCES.

E.5.1 JP-8 Toxicity.

The following is a listing of JP-8 specific and jet fuel relevant references maintained by AFRL/RHPB, Wright-Patterson AFB OH.

E.5.1.1 Acute.

Clark, C. R., Ferguson, P. W., Katchen, M. A., Dennis, M. W., and Craig, D. K. 1989. Comparative acute toxicity of shale and petroleum derived distillates. Toxicol. Ind. Health. 5:1005-1016.

Gibson, R. L., Shanklin, S. L., and Warner, R. L. 2001. Health events comparison: Risk assessment of acute exposure to jet fuel. In: JP-8 Final Risk Assessment. R. L. Gibson, ed. Lubbock, TX: The Institute of Environmental and Human Health. Ch. 125-129.

Kanikkannan, N., Jackson, T., Sudhan, S. M., and Singh, M. 2000. Evaluation of skin sensitization potential of jet fuels by murine local lymph node assay. Toxicol. Lett. 116:165-170.

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Kobayashi, A. and Kikukawa, A. 2000. Increased formaldehyde in jet engine exhaust with changes to JP-8, lower temperature, and lower humidity irritates eyes and respiratory tract. Aviat. Space. Environ. Med. 71:396-399.

MacEwen, J. D. and Vernot, E. H. 1985. Investigation of the 1-hour emergency exposure limit of JP-5. Toxic Hazards Research Unit annual technical report: 1985. Harry G. Armstrong Aerospace Medical Research Laboratory, Wright-Patterson AFB OH. AAMRL-TR-85-058.

Smith, L. H., Haschek, W. M., and Witschi, H. 1981. Acute toxicity of selected crude and refined shale oil- and petroleum-derived substances. In: Health effects investigation of oil shale development. W. H. Griest, M. R. Guerin, and D. L. Coffin, eds. Ann Arbor, MI: Ann Arbor Science Publishers, Inc. Ch. 11. pp. 141-160.

Vernot, E. H., Drew, R. T., and Kane, M. L. 1990. Acute toxicological evaluation of jet fuel A. Acute Toxicity Data. 1:29-30.

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Edwards, D. A., Andriot, M. D., Amoruso, M. A., Tummey, A. C., Bevan, C. J., Tvelt, A., Hayes, L. A., Youngren, S. H., and Nakles, D. V. 1997. Total Petroleum Hydrocarbon Criteria Working Group Series. Volume 4: Development of Fraction Specific Reference Doses (RfDs) and Reference Concentrations (RfCs) for Total Petroleum Hydrocarbons (TPH). Amherst, MA: Amherst Scientific Publishers.

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Weisman, W., ed. 1998. Total Petroleum Hydrocarbon Criteria Working Group Series. Volume 1: Analysis of Petroleum Hydrocarbons in Environmental Media. Amherst, MA: Amherst Scientific Publishers.

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(Copies of these documents are maintained by The Human Effectiveness Directorate, AFRL/RHPB, Wright-Patterson AFB OH 45433, (937) 904-5938, DSN 674-9570.)

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APPENDIX F

FIRE PROTECTION AND SURVIVABILITY/VULNERABILITY

F.1 SCOPE.

F.1.1 General. The following guidelines provide the process for evaluation and final recommendation for approval of a new fuel from a fire safety and aircraft vulnerability perspective. Fire detection, extinguishing, suppression, and protection technologies for JP-8 are considered as the baseline when evaluating a candidate alternative fuel. Regardless how close the candidate fuels' characteristics and behavior are to JP-8, the candidate fuel has to be evaluated against established standards to show that no changes are necessary in equipment, material, or procedures for fire detection, extinguishing, suppression, and protection. Five basic areas have to be evaluated: (1) fire detection on-aircraft, (2) fire extinguishing / suppression on-aircraft, (3) ground fire detection (including fuel storage and aircraft hangars) where fuel is present, (4) ground fire extinguishing/suppression, and (5) vapor detection for maintainers and fuel handlers' safety. When testing candidate alternate fuels for detection, extinguishing, suppression, and protection, it is critical to test neat candidate fuel as well as blended forms for ground operations. Even though candidate fuel may not be used operationally in a neat form, testing phases with un-blended fuel poses unique risk requiring the same standards be applied to ensure neat fuel does not pose hazards beyond the capabilities in place for detection, extinguishing, suppression, and protection.

F.1.2 Entrance Criteria and Subset Testing. Air Force SMEs evaluated all fuel properties and characteristics based on the requirements decomposition process that correlated requirements to safety, performance, durability and supportability, as described in Appendix A. Consequently, Appendix F defines which subset each fire safety and aircraft vulnerability compatibility characteristic belongs to and the significance for its selection. Appendix F also includes component tests that should be conducted to further evaluate the candidate fuel/fuel additive for those fire safety and aircraft vulnerability compatibility characteristics requiring further investigation, as deemed appropriate by the Alternative Fuel Certification Office (AFCO) at Wright-Patterson AFB.

F.2 APPLICABLE DOCUMENTS.

F.2.1 General.

The documents listed below are not necessarily all of the documents referenced herein, but are those needed to understand the information provided by this Appendix.

F.2.2 Government documents.

F.2.2.1 Specifications, standards, and handbooks.

The following specifications, standards, and handbooks form a part of this document to the extent specified herein.

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FEDERAL STANDARDS

FED-STD-791	Lubricants, Liquid Fuels, and Related Products, Testing Method of

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-5578	Tanks, Fuel, Aircraft, Self-Sealing
MIL-F-24385	Fire Extinguishing Agent, Aqueous Film-Forming Foam (AFFF) Liquid Concentrate, For Fresh and Sea Water Survivability, Aircraft, Non-nuclear, General Criteria, Vol. 1, Dept. of Defense, MIL-HDBK-336-1, Washington, DC, 25 Oct 1982
MIL-DTL-27422	Tank, Fuel, Crash-Resistant, Ballistic-Tolerant, Aircraft
MIL-PRF-87260	Foam Material, Explosion Suppression, Inherently Electrostatically Conductive, for Aircraft Fuel Tanks

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-882	Standard Practice for System Safety, U.S. Dept. of Defense, 10 Feb. 2000
MIL-STD-2105	Hazard Assessment Tests for Non-Nuclear Munitions, Dept. of Defense, Washington, DC, 12 Jan 1994

F.2.3 Non-Government publications.

The following documents form a part of this document to the extent specified herein.

ASTM INTERNATIONAL

ASTM E659-78	Standard Test Method for Autoignition

(A copy of this document may be ordered online at www.astm.org; and from ASTM International, 100 Barr Harbor Drive, PO Box C700, West Conshohocken PA 19428-2959; (610) 832-9585. Approved users may access the document online at www.ihs.com.)

"The Fundamentals of Aircraft Combat Survivability Analysis and Design", Second Edition, Ball, Robert E., Ph.D., AIAA Education Series, Blacksburg, VA, 2003

"Hot Surface Ignition and Aircraft Safety Criteria", Clodfelter, Robert G., SAE 901950, October 1990

"Survivability, Safety, and Reliability Analyses Integration Process," Dotseth, W. D., World Aviation Congress, Paper 97WAC-16, Society of Automotive Engineers and AIAA, Oct 1997.

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"Combat Survivability—By Design, Not Just By Chance," Mower, D. W., and Levy, R.B., AIAA Student Journal, Vol. 34, No. 4, 1997

"Enhancing Aircraft Survivability: A Vulnerability Perspective," National Defense Industrial Association, Arlington, VA 1997

"Reticulated Polyurethane Safety Foam Explosion Suppressant Material for Fuel Systems and Dry Bays, "Society of Automotive Engineers, SAE AIR 4170, Nov 1998

"Aviation Fuels with Improved Fire Safety," A Proceeding, Committee on Aviation Fuels with Improved Fire Safety, National Academy Press, Washington, DC, 1997

"Fuel System Protection Methods," Aircraft Fire Safety, Wyeth, H. W. G., AGARD Lecture Series No. 123, AGARD, Neuilly Sur Seine, France

"Fire Safety in Military Aircraft Fuel Systems, " Aviation Fuels with Improved Fire Safety, Clodfelter, R. G., National Academy Press, Washington, DC, 1997

"Vaporization of JP-8 Jet Fuel in a Standard Aircraft Fuel Tank Under Varying Ambient Conditions," Ochs, R.I., and Polymeropoulos, C. E., Society of Automotive Engineers, SAE 2006-01-2445, August 2006

"Limiting Oxygen Concentration of Aviation Fuels," Society of Automotive Engineers, Summer Steven M., SAE 2006-01-2446. August 2006

"Assessment of JP-8 as a Replacement Fuel for the Air Force Standard Jet Fuel JP-4 Part 1, Assessment of JP-8/JP-4 Fuel in a Noncombat Environment,"AFAPL-TR-74-71, Part 1, June 1975

"A Review of the Flammability Hazard of Jet A Fuel Vapor in Civil Transport Aircraft Fuel Tanks," US. Dept. of Transportation, FAA, DOT/FAA/AR-98/26, June 1998

Automatic Fire Detectors; National Fire Protection Association Handbook; Cote, Arthur (editor); Seventeenth Edition; July 1991

American National Standard for Radiant Energy-Sensing Fire Detectors for Automatic Fire Alarm Signaling ANSI/FM Approvals 3260; June 2004.

(Copies of these documents are available online at The Defense Technical Information Center Website and are available from DTIC Headquarters, 8725 John J. Kingman Rd., Ft. Belvoir VA 22060-6218; telephone (800) 225-3842.)

F.3 ENTRANCE CRITERIA.

Prior to beginning this process the candidate fuel will meet the entrance criteria defined in B.4.1.

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F.4 SUBSET 1 CRITERIA.

Subset 1 is composed of fuel properties and characteristics that affect fire protection, fire detection and protection technology for ground fires, on-aircraft fire detection and other safety-related items including vapor detection and the flame speed relating to the candidate fuel. The baseline fire detection sensor characteristics and flame speed criteria for these characteristics are provided in Table F-I.

TABLE F-I. Baseline fire detection sensors and flame speed criteria.

Fire Safety Property and Significance	Nominal Baseline Fuel Value
Detection limit of Vapor Detectors – Ability of current vapor detectors to protect personnel entering spaces that contained fuel	10% Lower Explosive Limit
Detection Threshold of flame when seen by UV/IR detectors – Ability of currently used flame detectors to detect a fire	Example: Flame from 5 inch diameter pan, 6 feet away, detected in less than 5 seconds. UV detection is between 185- 245 nanometers and IR is 960 – 5000 nanometers.
Visibility of flame – How easily is the flame seen under different conditions, basic safety consideration	Easily visible under variety of conditions. Visible range is 400 - 720 nanometers.
Flame speed – Rate at which flame front moves across a pool of fuel, important for ground safety	~100 feet/minute

F.4 Overall Subset 1 Test Objectives. Determine if the new candidate fuel is compatible with technologies used for (1) fire detection on-aircraft, (2) fire detection on-ground (including fuel storage and aircraft hangars) where fuel is present, (3) ground fire extinguishing/suppression, and (4) vapor detection for maintainers and fuel handlers' safety involving the candidate fuel.

F.4.1 Flammability Limits. The basic flammability limits of the fuel and standard test methods are described in Appendix C. The flammable (explosive) temperature range is an extremely important characteristic from a fire fighting perspective due to the fact that the gas or vapor concentration will burn or explode if an ignition source is introduced. The limiting concentrations are commonly called the "Lower Explosive or Flammable Limit" (LEL/LFL) and the "Upper Explosive or Flammable Limit" (UEL/UFL). Below the lower explosive or flammable limit the mixture of fuel and oxygen is too lean to ignite and above the upper

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explosive or flammable limit the fuel/oxygen mixture is too rich to ignite. The lower and upper explosion concentration limits for aviation fuels currently being used in the U.S. are provided on Figures C-29 and C-30. Data has to be generated for any new / alternative fuel used in the aircraft.

F.4.1.1 Test Plan. Standard flammability test procedures for aviation fuels are described in ASTM E681. For the aviation industry, flammability limit results are typically presented over a temperature range for a given pressure environment that is related to realistic aircraft operating conditions. Additional testing will be performed for an alternative fuel using the following test procedure. The flammability limits of a fuel will be established using a spherical, 20 L, stainless steel combustion chamber. The chamber will contain electrical igniter rods at fixed spacing to ignite the fuel vapors. Furthermore, optical access and instrumentation ports will confirm the presence of an ignition as well as record the pressure, temperature, hydrocarbon concentration and oxygen concentration throughout the test. Prior to testing, the chamber pressure will be reduced to near vacuum conditions and then filled with an air mixture designed to mimic atmospheric conditions at a specific pressure altitude. The sample will be introduced to the chamber. The test chamber is placed inside a heated enclosure that circulates temperature controlled air on the outside of the combustion chamber, thereby heating the fuel sample due to conduction through the cylinder walls. An electrical spark igniter of fixed spark gap is triggered and the presence of an ignition is verified visually and with a measured pressure change in the vapor space. The conditions in the chamber are recorded as well as verification of an ignition. The remaining liquid fuel is drained and disposed in the appropriate fuel disposal location and the combustion products removed with the vacuum pump prior to cleaning the igniter rods and the chamber. Tests are conducted at upper and lower fuel temperature limits to determine the fuel rich and fuel lean conditions, respectively, at different pressure environments and a graphical range of flammable fuel temperature conditions is recorded.

F.4.1.2 Expected Outcome. The basic flammability limits of the fuel will be established for a range of conditions to assure that the fuel is comparable to the baseline (JP-8) see C.20.

F.4.2 Fuel Vapor Detector Compatibility. Vapor Detector Compatibility should be determined by testing equipment currently in use against established standards.

F.4.2.1 Test Objective. Determine if the current vapor detectors will accurately sense if a flammable fuel vapor exists.

F.4.2.2 Test Plan. Tests will be conducted in a controlled environment at the upper and lower fuel temperature limits at fuel rich and fuel lean conditions, respectively, at different pressures and record the detectors response.

F.4.2.3 Expected Outcome. Data for interpreting vapor detector readings will be generated in order to give guidance to ground safety and other support personnel who use these detectors to ensure a safe work environment.

F.4.3 Flame Visibility. Flame visibility by ultraviolet/infrared fire detectors should be determined based on standard practices and guidance provided by; American National

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Standard for Radiant Energy-Sensing Fire Detectors for Automatic Fire Alarm Signaling ANSI/FM Approvals 3260; June 2004, Automatic Fire Detectors; National Fire Protection Association Handbook; Cote, Arthur (editor); Seventeenth Edition; July 1991, and per the test plan described below.

F.4.3.1 Test Objective. The purpose of this test series is to determine if existing technologies used for detecting JP-8 flames will work for candidate fuels. The existing detectors have to alarm with the candidate fuel blends at the same thresholds (fire size, distance, and time) as they do with the baseline JP-8. Secondly, the test has to confirm that the flames are visible to the human eye. Since there are numerous flame detector manufacturers, each with proprietary algorithms designed to avoid false alarms by using flicker rates, data on flicker rate will also be collected and made available. These flicker data are considered essential since only detectors representative of the different methods will be tested rather than every detector in the Air Force inventory.

F.4.3.2 Test Plan. There are several ways of detecting visible flame. One method is to examine the air for smoke using ionization or a photo-detector. Smoke detectors have advantages in looking for fires that are in the smoldering stage and/or in large spaces by placing the detectors within the heating and air conditioning ductwork. Another detection method is to look for heat buildup, which is a slower but very robust and false-alarm resistant method. There can also be an advantage if fire is not the only heat source of concern. Overheat detectors in aircraft engine bays serve to detect fires as well as potentially dangerous bleed air duct ruptures. However, if speed is critical, generally the quickest method to detect Class B fires (liquid fuels) is to 'see' the flame. Flames are visible across a wide spectrum. Most fuels, excepting hydrogen and some alcohols, generate flames that can be seen within the spectrum visible to the human eye, (~400 – ~700 nm). Flames also emit in the ultra-violet (UV) and infrared (IR) spectrum. Of course many objects emit in the visible, UV and IR spectra, e.g. sunlight. Therefore there need to be methods to distinguish actual flames from background sources. Fortunately, as hydrocarbons burn their flames generate many byproducts of combustion such as water vapor and carbon dioxide (CO_2). By looking for emissions at certain wavelengths, flame detectors can use algorithms to determine if a flame is present. These algorithms may look for a signal to reach a certain threshold or they may look for flicker at frequencies in the 5 – 30Hz range to distinguish a flame from a static light source, e.g. sunlight. Since the detectors may also use multiple frequencies and compare ratios among those frequencies, the testing will depend on the type of detector is used in the device. No matter the technique, the goal is to detect a fire quickly while avoiding nuisance false alarms.

F.4.3.3 Expected Outcome. Measure of confidence of the ability and responsiveness of currently used fire detectors to detect a flame from the candidate fuel under consideration will be determined.

F.4.4 Flame Speed. Flame speed across pooled fuel is an important consideration from a ground fire-fighting perspective.

F.4.4.1 Test Objective. Determine the rate of progression of the flame front across a pool of the candidate fuel in a tightly controlled environment.

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F.4.4.2 Test Plan. Measurements of the rate at which the flame front moves across a pool of fuel will be measured in a wind free environment using a stop watch.

F.4.4.3 Expected Outcome. Knowledge if the candidate fuel poses a greater hazard to ground personnel or in a post-crash scenario because the flame spread rate is much greater than the baseline fuel. The flame speed of the candidate fuel will be compared with the baseline JP-8 fuel.

F.4.5 Flash Point. The flash point for aviation fuels, a required fuel property specified in Appendix C, specifies the temperature at which a fuel releases a sufficient amount of vapors to yield an ignition in air corrected to a pressure of 760 mm Hg in the presence of an ignition source. As a result, flash point is a key physical property that is often reported in Material Safety Data Sheets (MSDS) and other chemical property handbooks for the safe handling and transport of a fuel. The flash point is an important parameter for determining the explosion and fire hazard, or volatility, of the specimen. These tests should be performed at varying pressures and temperatures.

F.4.5.1 Test Objective. Determine the minimum liquid fuel temperature required to generate an ignitable mixture of fuel vapor and air immediately above the liquid fuel.

F.4.5.2 Test Plan. For aviation fuels, the recognized methods for the determination of flash temperature according to the ASTM D1655 are the Tag Closed Cup (ASTM D56), Pensky-Martin Closed Cup (ASTM D93), and Small Scale Closed Cup (ASTM D3828). The Small Scale Closed Cup Method has been an established ASTM method since 1979 and is considered an “equilibrium” method since it allows for the fuel vapors to diffuse into the vapor space at a set temperature. However, it does not have the extensive library of results for comparison that is available to the Tag method and is not widely employed in current fuel testing laboratories. The Tag Closed Cup Method, ASTM D56, is the preferred flash point method due to its well documented history and prevalent use in current national laboratories.

F.4.5.3 Expected Outcomes. The data recorded will be fuel temperatures tested and barometric pressure. Data generated will be temperature increases in vapor air mixture, flash presence, and altitude corrected flash point. These data will then be compared to a concurrently generated baseline aviation fuel.

F.4.6 Auto Ignition Temperature. The auto ignition temperature for aviation fuels is an important parameter for determining the explosion and fire hazard, or volatility, of the fuel as specified in Appendix C.

F.4.6.1 Test Objective. The objective of this test is to determine the minimum fuel temperature, or self-ignition temperature, required to generate an ignitable mixture of fuel vapor in air at atmospheric pressure in the absence of an ignition source.

F.4.6.2 Test Plan. Determination of the auto ignition temperature of a fuel sample will adhere to the Auto Ignition Temperature of Liquid Chemicals (ASTM E659) testing standards using an automated auto ignition apparatus specifically constructed for this method. In a darkened test facility (lights are shuttered off), an enclosed flask is heated and maintained at a testing temperature. An electrically heated furnace and temperature control system

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maintains a high temperature test environment. An internal flask of sufficient thermal conductivity is used to hold the sample and provide a suitable testing environment. Surrounding the flask is a lining of aluminum foil with the top of the flask insulated within the temperature controlled furnace. Once the flask is heated and maintained at the desired test temperature, a 100-µL sample of liquid fuel is carefully measured and extracted using a hypodermic syringe and injected into the borosilicate flask. The vapor space in the flask is visually monitored for the presence of an ignition for a specified length of time. At the completion of the test, the combustion gases are purged from the flask and the sample is then safely discarded. For each test temperature, an "ignition" or "no ignition" is determined.

F.4.6.3 Expected Outcomes. Data gathered and recorded are fuel amount, fuel temperature, ambient pressure and temperature and temperature rise. These data will then be compared to a concurrently generated baseline JP-8 fuel. A successful test is determined by visual confirmation of a flame within the test article.

F.4.7 Heat of Combustion. The heat of combustion for aviation fuels is a standard test required by the fuel specification covered in Appendix C.

F.4.7.1 Test Objective. Measure the heat of combustion of the alternative fuel based on established standards.

F.4.7.2 Test Plan. The standard test procedure employs a bomb calorimeter to measure the heat release of a fuel sample. Determination of the heat of combustion of aviation turbine fuels by oxygen bomb calorimetry is the preferred test method ASTM D1655-07, which specifies aviation fuel properties testing. The standard indicates that the heat of combustion should be determined using the standards of either ASTM D240 or ASTM D4809, which apply to oxygen bomb calorimetry. The active standard D4809 was designed specifically for use with aviation fuels, where the difference between repeated measurements should be on the order of 0.2%, and was therefore followed in the current procedure.

F.4.7.3 Expected Outcome. The **r**esulting energy released measurements from the combustion of the alternative fuel sample will be compared to the baseline JP-8 and provide valuable data to update the design of fire detectors and fire fighting agents if required.

F.4.8 Ground Fire Suppression Agents. To aid Air Force firefighters' response to an incident involving alternative fuels, evaluations of fire fighting agent performance against alternative fuel fires needs to be conducted.

F.4.8.1 Test Objective. Determine if Mil-Spec MIL-F-24385F, Fire Extinguishing Agent, Aqueous Film-Forming Foam Liquid Concentrate, for Fresh and Sea Water, is capable of extinguishing the alternative fuel fires.

F.4.8.2 Test Plan. Air Force fire departments typically use Aqueous Film Forming Foam (AFFF) to extinguish fuel spill fires involving petroleum based fuel, such as jet fuel (JP-8), diesel fuel, or gasoline The Fire suppression agent tests will be performed using JP-8 (baseline fuels), and the candidate fuels. This test series will be performed in accordance with parameters set forth in the MIL-F-24385F, Section 4.7.13 for the twenty-eight-square-foot Fire Test. This test program will evaluate, on a medium-scale, the fire suppression capabilities of commercially available AFFF fire fighting agent with potential for Air Force applications for alternative fuel fire extinguishment. Fire suppression agent will be evaluated for ability to

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suppress a fuel-on-water fire. Evaluation will include extinguishment effectiveness, flame knockdown time, extinguishment time and burn back time.

F.4.8.3 Expected Outcome. These tests will determine if the alternative fuel fires can be extinguished by the fire suppression agent in a manner equal to or better than JP-8. Alternative fuels that are not able to be extinguished in a manner equal to or better than JP-8, or which would require changes in firefighting equipment, materials, or procedures, will be unacceptable.

F.5 SUBSET 2 CRITERIA.

Subset 2 is composed of tests to determine the (1) Minimum Hot Surface Ignition Temperature under Turbulent Airflow and (2) Minimum Ignition Energy of the candidate fuel and on-aircraft fire suppression equipment. This evaluation is critical for personnel ground safety and for classification of surfaces that require insulation to prevent a hot surface ignition on board military vehicles. These parameters will play a key role in determining the vulnerability of the fuel tank and the overall survivability of the aircraft.

F.5.0 Overall Subset 2 Test Objectives. Determine if current requirements for minimum hot surface ignition and minimum ignition energy are acceptable for the candidate fuel.

F.5.1 Safety Related Fuel Characteristics.

TABLE F-II. Baseline Fuel Values and Candidate Fuel Criteria

Fuel Property and Significance	Baseline Fuel Value
Minimum Hot Surface Ignition Temperature Under Turbulent Airflow – Lowest Surface Temperature at which a fluid will ignite.* Criterion is used to determine which surfaces require insulation.	650 °F - 1100 °F
Minimum Ignition Energy (at altitude) – Least amount of energy required to start a sustained reaction in a mixture of fuel vapor and air.	0.25 mJ

* Varies by Test Method

F.5.2 Minimum Hot Surface Ignition Temperature under Turbulent Air Flow.

The minimum hot surface ignition temperature or hot manifold ignition temperature is a key safety parameter to the aviation industry due to the range of environments where fuel delivery and usage occurs within close proximity to heated surfaces. The minimum hot surface ignition temperature is defined as the lowest external surface temperature that can yield an ignition of a

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resident hydrocarbon fuel. Hot surface Ignition should be evaluated per test plan described below.

F.5.2.1 Test Objective. Determine the minimum temperature of a surface at which the candidate fuel will ignite. Historically, any aircraft component that has a hot surface has required a risk assessment be conducted to determine whether the component is to be treated as an ignition zone or a fire zone. If a component is determined to be a fire zone, then some form of mitigation is required.

F.5.2.2 Test Plan. The hot surface ignition temperature of a fuel is to be tested on a flat, uniformly heated circular plate which resides in an optically conducive, octagon-shaped enclosure. A fuel conditioning and delivery system regulates the temperature of the fuel as well as the fuel release condition from the nozzle. Three separate delivery modes (spray, continuous stream, steady drip) and two different fuel temperatures (ambient and heated) are tested in three precisely controlled airflow environments. The three airflow environments are documented with a high accuracy velocity, turbulence and temperature measurement system prior to conducting hot surface ignition tests. For the ignition tests, a set amount of fuel (10 mL) is delivered to the uniformly heated surface at a fixed flow rate. The probability of ignition at a range of fuel temperatures is determined for each test condition through ten repeated instances for a given fuel and environmental condition. The resulting statistical probability value is reported graphically and provided along with the facility documentation to allow comparison between multiple test facilities and different fuel specimens.

F.5.2.3 Expected Outcome. An evaluation of the candidate fuel as to whether procedures need to be altered or if surface temperature-based requirements are no longer valid or if aircraft fire zones need to be reclassified for fire hazard purposes.

F.5.3 Minimum Ignition Energy. The minimum ignition energy required for a spark discharge to ignite a hydrocarbon fuel/air mixture is defined in Appendix C for the currently-used jet fuels. The Minimum Ignition energy for the proposed alternative fuel will be determined using an Atomization Characterization test described below.

F.5.3.1 Test Objective. Determine the minimum energy required to ignite a flammable mixture of air/fuel for a range of fuel vapor concentrations.

F.5.3.2 Minimum Ignition Energy Test Plan. The minimum ignition energy (MIE) of a fuel will be tested in a spherical, 20 L, stainless steel combustion chamber. The chamber will contain electrical igniter rods at variable spacing to ignite the fuel vapors. Furthermore, optical access and instrumentation ports will confirm the presence of an ignition as well as record the pressure, temperature, hydrocarbon concentration and oxygen concentration throughout the test. Prior to testing, the chamber will be evacuated and then filled to the desired pressure altitude conditions using the Fuel Analysis Laboratory bottle farm. The sample will be introduced to the chamber. The chamber is placed inside a heated enclosure that circulates temperature controlled air on the outside of the combustion chamber. The distance between the igniter rods and the voltage on the DC electrical power supply are set to yield the ignition testing conditions specified in the test matrix. The electrical igniter is triggered and the presence of an ignition is verified visually and with a measured pressure rise in the vapor space. The conditions in the chamber and the electrical signal on the igniter

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circuit are recorded as well as verification of an ignition. The fuel is evacuated and the combustion products removed with the vacuum pump prior to cleaning the igniter rods and the chamber. For each pressure altitude examined, testing is conducted over a flammable chamber temperature range and a series of ignition energies to determine the minimum amount of energy required for ignition at the different fuel temperature and pressure altitude environments. A graphical range of the minimum ignition energy for the range of fuel temperature conditions at each examined pressure altitude is reported.

F.5.3.3 Expected Outcome. The expected outcome is a comparison between data generated from baseline JP-8 and the candidate fuel. The data gathered will provide a measure of the ignition energy versus changing fuel vapor temperature, under a constant pressure. This assumes a constant fuel input pressure with a nozzle that remains the same and controls the fuel spray and droplet size. These later variables can be changed while keeping the fuel vapor temperature and pressure constant to get a different ignition energy value.

F.5.4 Aircraft Engine Fire Suppressant Agents. The effectiveness of fire suppressant agents in extinguishing a given combustion of alternative fuels for an aircraft engine test scenario will be determined.

F.5.4.1 Test Objective. Measure the effectiveness of fire suppressant agent in an engine fire scenario involving alternative fuel.

F.5.4.2 Test Plan. These tests will be performed in a strategically designed fixture using fire suppression testing techniques and incorporating airflow as appropriate. Input data include ignition energy, amount of fuel, fire suppressant agent, bottle nozzle location and nozzle type. These tests are system specific and may be conducted at the discretion of the Single Manager.

F.5.4.3 Expected Outcomes. Gathered data consist of time to extinguish fire and amount of fire suppressant agent needed.

F.6 SUBSET 3 CRITERIA.

Subset 3 is composed of tests to provide additional safety, survivability, and fire protection related information. These tests are usually performed by the Single Managers to select and fund, based on their individual aircraft, ground vehicle, equipment, or facility in order to identify and reduce the potential risk, validate if risk mitigation will work, or determine the best choice of mitigation for their configuration. Historically, these tests have been highly individualized per airframe, customized per combat threats and environments, and scope has been limited to a single aircraft configuration captured in time. These tests involve direct simultaneous testing and comparison to the baseline fuel and the new candidate fuel.

F.6.0 Overall Subset 3 Test Objectives. Determine parameters for the candidate fuel necessary to answer additional safety and survivability related questions. These data can be used directly by decision makers as well as feed into other analysis tools to help answer other questions such as the impact on aircraft vulnerability.

F.6.1 Component Level Aircraft Fire Protection Technology. The expected outcome is similar to system level testing, but without the complication and sophistication of a

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system level test. Performed at the aircraft system manager's discretion, the candidate fuel will be tested with the aircraft system fire protection technology to determine its effectiveness in eliminating a candidate fuel fire.

F.6.1.1 Test Objective. The objective is (a) to determine the effectiveness of fire protection technology when eliminating a candidate fuel fire, in a mockup or surrogate type of environment and (b) to verify that existing technology will still function properly when using a candidate fuel in the fire environment. The component level fire protection technology testing will assess and evaluate the fire protection technology, associated with the aircraft system of the single manager, and the candidate fuel. There needs to be compatibility between the candidate fuel and the fire protection technology present, whatever that system might be. This is important because fire protection technologies are sized and optimized to work in a specific environment and extinguish JP-8 fires. By changing to the candidate fuel a key variable has been changed and needs to be evaluated in a cost-effective manner.

F.6.1.2 Test Plan. The specific test plan will need to be developed for each aircraft subsystem test being performed. The test article for component level testing is much scaled down when compared to system level testing. The test article for this level of investigation would be a steel mock up, duplicating environment and the space being protected, by the fire protection technology. Examples of this environment include, but are not limited to wing dry bay spaces next to fuel tanks, engine nacelles, cargo bays, and fuselage dry bays that capture flammable fuels and hydraulic fluids. This is going to be highly dependent on the aircraft system being evaluated. The steel test article will then have a production fire protection technology system installed, as it is on the aircraft system. The resulting test article will be simple and easy to fabricate, and cost effective compared to procuring a system level test article. This technology needs to be in the same location, same orientation, and have the same fire extinguishing amounts like the production unit. Enough fire protection units need to be purchased to have a number of repetitions to testing. Any unplanned deviations from this replication will result in invalid test data. The fuel and ignition source need to be introduced into the test article in a realistic capacity. The fire will have to be introduced into the test article by a fuel spray nozzle, which simulates the damage from a combat event or mechanical problem. The fire protection technology was designed to extinguish a fire in a certain scenario, and this scenario needs to be duplicated during the test. Past research indicates a fuel spray between 2-8 gallons per minute is a realistic flow rate comparable to the flows resulting from combat damage. Some test articles will require a modest amount of airflow. It has been shown that airflow has a bearing on the fire propagation and fire protection technology effectiveness. Research will need to be accomplished to determine the type of flow that exists within the real system and corresponding bays. These data are best retrieved from the manufacturer through the single manager of the aircraft system in question. These flow conditions are then duplicated in the test article. There are many methods to reproduce airflow in a test article. This can be accomplished by fans or by pressurized bottles, to name a few. By not including the airflow in the test article, the technology may be biased to either work or not work. The bottom line is the tester may be generating erroneous data if airflow is present in the actual system but not part of the evaluation testing. Once the test article has been fabricated, repetitive fire tests are possible on the component level with the production fire protection technology. The tests need to be run a number of times, with at least five successful tests in a row.

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F.6.1.3 Expected Outcome. The data collected will include test verification and a video record of a candidate fuel fire being extinguished by the aircraft system specific fire protection technology. A test is successful when the fire is extinguished by the fire protection system. From previous experience, 10 tests are planned in an attempt to achieve at least 5 successful tests in a row. This isn't to be confused with 5 successful tests out of 10. Once the former test record has been achieved, the data provides good statistical confidence to say the system works with the candidate fuel. If no baseline JP-8 data exist, the system will have to be tested with the baseline fuel.

F.6.2 Aircraft Fire Protection from Ballistic Threats. Component level tests should be performed for risk reduction prior to conducting advanced systems level tests. The effectiveness of various fire protection technologies and damage assessments from ballistic threats are scenario and aircraft system specific and are usually performed by the Aircraft Single Managers based on Live Fire Test requirements for that system.

F.6.2.1 Test Objective. Determine the relative effectiveness of several fire protection technologies that are designed to protect an aircraft from ballistic ignited fires with the candidate fuel.

F.6.2.2 Test Plan. These tests are aircraft and system specific and will need to be developed for each aircraft and subsystem tested.

F.6.2.3 Expected Outcome. Data that measure any change in effectiveness of currently used technologies designed to protect aircraft fuel tanks and dry bays from fire and explosion as well as minimize fuel loss.

F.6.3 Ballistic Testing of Self-Sealing Fuel Bladders.

F.6.3.1 Test Objective. Self-sealing fuel system components typically incorporate a fuel resistant inner liner, a fuel activated self-sealing layer, and an outer environmental (abrasion) layer. The self-sealing layer has to swell rapidly when it comes into contact with fuel, The standard performance test for self-sealing bladders is a ballistic test which is required to validate the results of the volume swell tests.

F.6.3.2 Test Purpose. The purpose of this test is to determine if self-sealing ballistic test cells will seal sufficiently after ballistic testing when filled with the candidate fuel.

F.6.3.3 Test Plan. Testing will be accomplished in accordance with MIL-T-5578C, Phase I, protection level A (Sections 4.6.5.3 – 4.6.5.4). The test cells will be unpressurized when testing is performed. The gunfire will be at service velocity. Vendor provided self-sealing bladders designed to MIL-T-5578C, Phase I Test Cells (Type II, Class B, Style I) will be tested to the Protection Level A with the candidate alternative fuel.

F.6.3.4 Expected Outcome. The pass/fail criteria for the self-sealing bladders as specified in MIL-T-5578C will be determined.

F.6.4 Hydrodynamic Ram. Testing for this parameter is not mandatory and should be decided by the aircraft Single Manager. If the alternative fuel parameters are very similar to JP-8 then this test can and should be bypassed. The parameters to compare with the baseline fuel are specific gravity, bulk modulus, thermal expansion, and viscosity. If these parameters

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compare closely with JP-8, then no hydrodynamic ram testing needs to be accomplished on the candidate fuel because of a lack of key differences. How much variation in the four parameters and how that impacts the hydrodynamic ram effect, isn't known. If a noticeable variation exists in one of the four mentioned parameters, then the testing should proceed at the Single Manager's discretion.

F.6.1 Test Objective. Measurements of the overpressure generated during the pressure wave cycle in a fuel sample contained in strategically designed container or tank, using standard vulnerability test methods, may be accomplished for specific threat projectiles.

F.6.1.2 Test Plan. These are system specific tests that can be accomplished at a component level versus a more involved system level test using aircraft specific parameters and geometry. The amount of over pressure and tank damage generated by a given projectile will depend on the threat type, impact velocity, projectile trajectory and impact location in the tank. A component level test is usually recommended for this type of testing. This would consist of a steel construction tank that measures 5' × 3' × 3' required to hold the appropriate amount of fuel and sturdy enough to withstand the induced ram pressures. The tank is constructed with an easily replaceable projectile entrance and exit panels which would be replaced after every shot. Ram pressures will be measured for both the 12.7mm AP and 23mm AP projectiles fired at their design service velocity. Five shots per ammunition type should be fired at a minimum. These repetitions will bring credibility to the data gathering and overall data set. Test instrumentation should consist of Kistler pressure transducers (PTs) and a data acquisition system capable of 10,000 Hz range for a sampling rate. The minimum number of PTs required in the cluster is four. Three 2 kpsi capable PTs and one 10 kpsi PT are recommended to gather the required data for each shot. These PTs have to be spaced and suspended around the shot line location for the best data pick up in the tank. A system level test, using full scale components, may also be performed following the same basic procedures.

F.6.4.3 Expected Outcomes. The overpressure will be measured in the fuel container or tank. These data will then be compared to archive data generated for a baseline aviation fuel.

F.6.5 Fuel Tank Inerting. Fuel tank inerting with nitrogen enriched air (NEA) is used for explosion prevention on select aircraft . These tests are system specific and may be conducted at the discretion of the Single Manager.

F.6.5.1 Test Objective. Determine if fuel tank inerting requirements (percent oxygen allowed) are different for candidate fuels under consideration as compared to baseline JP-8.

F.6.5.2 Test Plan. These tests are system specific and may be designed and conducted at the discretion of the Single Manager.

F.6.5.3 Expected Outcome. The expected outcome is comparison data between the baseline and the candidate fuel. The data will consist of overpressure (psi) versus % oxygen. Other data collected will include time to peak pressure. These data will determine if existing inerting criteria and systems would provide an equivalent level of protection to what is seen with the existing aircraft systems currently using the baseline fuel.

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APPENDIX G

AIRCRAFT PROPULSION FUELS CERTIFICATON PROCESS

G.1 SCOPE.

G.1.1 General. This Appendix provides information to aid in the determination of a fuel's acceptability for use in aircraft propulsion systems. It is intended to provide the Alternative Fuel Certification Office (AFCO) and Single Manager (SM) a methodology to identify, evaluate and mitigate safety, performance, durability and supportability risks associated with candidate fuels used in propulsion systems. Additional component and system level tests are suggested to help identify and reduce any associated risk.

G.1.2 Entrance Criteria and Subsets. The Aircraft Propulsion SMEs have evaluated all fuel properties and characteristics based on the requirements decomposition process described in Appendix A. The results of this analysis were used to develop propulsion process Entrance Criteria and three Subsets of propulsion properties. The Entrance Criteria contains a list of fuel properties and characteristics that the candidate fuel has to satisfy. Subset 1 includes those properties that can affect safe operation of the propulsion system, Subset 2 includes those properties that can affect the performance of the propulsion system and Subset 3 includes those properties that can affect the durability and supportability of the propulsion system. These properties are documented in tables for each of the Subsets.

G.1.3 Fuel Functions on Aircraft Propulsion Systems.

G.1.3.1 Fuel Functions. Aircraft Propulsion systems use fuel to accomplish four main functions: 1) provide performance, 2) lubricate wear surfaces and bearings, 3) provide fueldraulic muscle for actuation devices and 4) remove excessive heat. Each of these functions depends on different properties of the fuel. In order to properly evaluate a candidate fuel one needs to consider each of these functions and weigh their relative importance to proper operation of the aircraft propulsion system.

G.1.3.1.1 Performance. Fuel has to burn to start and operate a propulsion system to provide the amount of thrust necessary to operate the aircraft. There are several steps in the overall combustion process. The fuel has to be atomized and vaporized when passing through the fuel nozzles into the combustor. In the combustor, it is vaporized, ignited and burned to provide the needed heat release and thus required thrust. In order to get the fuel to the combustor it has to be pumped from the aircraft tank and metered according to an established set of schedules. Performance relies on fuel properties such as flammability, viscosity, lubricity and density.

G.1.3.1.2 Wear Surface Lubrication. Many propulsion system components rely on fuel lubrication for proper operation and to minimize wear and degradation. Pumps, actuators, hydromechanical controls and servo valves all contain fuel wetted bearings and surfaces that depend heavily on proper fuel lubrication. Proper fuel lubrication relies on fuel properties such as viscosity, lubricity and density.

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G.1.3.1.3 Fueldraulics. Aircraft Propulsion system control and thrust scheduling involves many moving parts and depends on fuel driven actuators to move them. Most of today's modern propulsion systems rely on the fuel as a medium for producing the hydraulic muscle (fueldraulics) for these actuators. Fueldraulics relies on fuel properties such as density, bulk modulus and viscosity.

G.1.3.1.4 Heat Removal. One of the biggest propulsion system durability drivers is the ability to remove and dispose of excessive heat. This "thermal management" capability is a major problem because fuel can only absorb so much heat and only so much fuel can be burned through the combustor. Whatever fuel is not burned, is either recirculated to the aircraft tank or recirculated within the fuel system itself, resulting in continual heat loading on the fuel. This high heat loading can result in a number of issues including coking, varnishing and the overheating of electronic components. The ability of the fuel to remove excessive heat relies on fuel properties such as specific heat and thermal stability.

G.2 APPLICABLE DOCUMENTS.

G.2.1 General.

The documents listed below are not necessarily all of the documents reference herein, but are those needed to understand the information provided by this Appendix.

G.2.2 Government documents.

G.2.2.1 Specifications, standards, and handbooks.

The following specifications, standards, and handbooks form a part of this document to the extent specified herein.

DEPARTMENT OF DEFENSE SPECIFICATIONS

JSSG-2007	Joint Service Specification Guide, Engines, Aircraft, Turbine, dated 29 Jan 04

(Copies of this document are available online at http://assist.daps.dla.mil/quicksearch/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

G.2.2.2 Other Government documents, drawings, and publications.

The following other Government documents, drawings, and publications form a part of this document to the extent specified herein.

PCOE - BP-03-18, Rev. A	Propulsion Best Practice, Propulsion/Engine Test Planning, dated 11 June 05

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PCOE - BP-01-14	Propulsion Best Practice, Lead the Fleet Pacer/ACI Programs, dated 7 Nov 01
PCOE - BP-02-17	Propulsion Best Practice, Accelerated Mission Testing (AMT), dated 1 Nov 02

(Copies of these documents are available online at The Defense Technical Information Center Website and are available from DTIC Headquarters, 8725 John J. Kingman Rd., Ft. Belvoir VA 22060-6218; telephone (800) 225-3842.)

G.3 DEFINITIONS.

Alternative Fuel Certification Office (AFCO). The government team responsible for the implementation and coordination of the fuel certification process.

G.4 THE PROPULSION SYSTEM FUEL EVALUATION PROCESS.

G.4.1 Process Overview. There is an inherent, time-phasing associated with the steps of this process. The Entrance Criteria, Subset 1, Subset 2 and Subset 3 properties are evaluated in order, at different points in time, using the same methodology.

G.4.2 Process Methodology. The first step requires the Propulsion Entrance Criteria properties to be evaluated. The results of this analysis are provided to the AFCO who determines if there is sufficient justification to proceed. If the AFCO determines the process is to continue, the next step requires Subset 1 properties to be evaluated. These results are used to accomplish a Risk Assessment Process (RAP). The RAP will generate three possible recommendations for the fuel: 1) rejection, 2) acceptance or 3) pursue additional risk reduction activities. The appropriate recommendation is made to the AFCO who will give permission to proceed further in the process. If the AFCO determines the process is to be continued, the same steps applied to Subset 1 are repeated for Subsets 2 and 3, including RAPs for each. Each time a RAP is performed, it will be based on an increasing body of knowledge. In addition to the increasing body of knowledge from Subset evaluations, there will also be knowledge obtained from any risk reduction activities. Each time, the RAP will recommend rejecting, accepting or pursuing additional risk reduction activities. The following sections provide more details on the process.

G.4.3 Entrance Criteria Evaluation. The first step involves evaluation of the Entrance Criteria using Appendix B. The Entrance Criteria contains the basic information required to determine if there is sufficient justification to initiate the propulsion evaluation process. The Entrance Criteria for the propulsion system evaluation process is provided in Table G-I. The results of this analysis are provided to the AFCO who determines if there is sufficient justification to proceed. Additional information on Entrance Criteria is provided in Appendix B.

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TABLE G-I. Propulsion system entrance criteria.

Fuel Property and Significance	Test Method	Pass Criteria
Flash Point, affects combustibility. It is also a leading factor determining fire safety in fuel handling.	Appendix C	Temperature Range: 28°C to 68°C
Freezing Point, affects low temperature fuel behavior. It can cause issues with pumps and nozzle operations.	Appendix C	Max -40°C
Viscosity @ -20 °C, affects pumpability over the operating temperature range. It also relates to droplet size in sprays produced by burner nozzles.	Appendix C	Max 8.0 cSt

G.4.4 Subset 1. Subset 1 properties are those that can affect the system safety and performance. The propulsion system Subset 1 properties are provided in Table G-II.

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TABLE G-II. Propulsion System Subset 1 Properties.

Fuel Property/Characteristic and Significance	Test Method and Evaluation Criteria
Viscosity versus Temperature (-40 ^{0}C to +90 ^{0}C), affects pumping fuel over the operating temperature range. Relates to droplet size in sprays produced by burner nozzles.	Appendix C
Surface Tension, affects the droplet size when the fuel is atomized, which affects combustion and air starting.	Appendix C
Vapor Pressure (Reid vapor pressure), affects vaporization, which affects combustion and air starting. Also affects vapor lock effects on pumps.	Appendix C
Heat of Combustion, Net, affects the amount of heat released per unit quantity of fuel which affects Specific Fuel Consumption (SFC)	Appendix C
Latent Heat of Vaporization, is the ability to change the fuel from a liquid to a gas. Affects vaporization, which affects combustion and air starting.	Appendix C
Flammability Limits, affect ignition characteristics which affect air starting.	Appendix C
Trace Elements, could result in excessive deposition or erosion of hot section components and materials, resulting in their failure.	Appendix C
Density versus Temperature (-40° C to +90° C), affects thermal expansion, flow calculations and metering devices.	Appendix C

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G.4.5 Subset 2. Subset 2 properties are those that can affect the performance of the propulsion system. Many of the properties listed in Subset 1 (Safety) also affect propulsion system performance and are not repeated in Subset 2. The propulsion system Subset 2 properties are provided in Table G-III.

TABLE G-III. Propulsion System Subset 2 Properties.

Fuel Property/Characteristic and Significance	Test Method and Evaluation Criteria
Bulk Modulus versus Temperature (-40 ^{0}C to +90 ^{0}C), affects the compressibility of the fuel which affects the ability to accurately pump the fuel. Also affects overall system stability.	Appendix C

G.4.6 Subset 3. Subset 3 properties are those that can affect the durability and supportability of the propulsion system. The propulsion system Subset 3 properties are provided in Table G-IV.

TABLE G-IV. Propulsion System Subset 3 Properties.

Fuel Property/Characteristic and Significance	Test Method
Enthalpy versus Temperature (0 ^{0}C to +250 ^{0}C), affects the ability to cool fuel wetted components.	Appendix C
Thermal Stability, affects cooling, coking and varnishing of fuel wetted components.	Appendix C
Lubricity, affects the lubrication of moving parts such as pumps and hydromechanical controls.	Appendix C
Thermal Conductivity, affects the suitability of a fuel as a primary heat sink which affects the ability to cool fuel wetted components.	Appendix C
Specific Heat versus Temperature (-40 ^{0}C to +150 ^{0}C), affects the amount of heat a fuel can absorb which affects the ability to cool fuel wetted components.	Appendix C

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G.5 AIRCRAFT PROPULSION SYSTEM RISK ASSESSMENT PROCESS (RAP) CONSIDERATIONS.

Section 5.2 defines the overall Risk Assessment Process. Several observations can be made by examining the overall process:

The risk to be assessed is the risk associated with certifying a candidate fuel for use in a fleet.

There are at least three entities that conduct risk assessments: The AFCO, the Single Manager, and the High Level Decision Authority.

Both the AFCO and Single Manger RAP conduct a series of assessment, each one done when another subset of fuel properties become available.

The entity most likely to conduct a detailed risk assessment for a propulsion system is the Single Manager.

The Higher Level Decision Authority can override the recommendation of one Single Manager and impose fuel use for the Service.

It is important that the Single Manager include the Original Equipment Manufacture on the risk assessment team. The Single Manager may need to conduct tests on a given propulsion system before a recommendation for certification can be given. Examples of such tests are given in the next section of this appendix. Many of these tests can be expensive, and lengthy. The SM needs to ensure that the OEM applies knowledge from all product lines in minimizing the number of tests needed for a particular system. Consideration should be given to develop collaborative efforts between the AFCO, other Single Managers, and OEMs for particularly difficult tests (durability tests typically fall into this category).

For those cases when the Higher Level Decision Authority directs the use of a fuel, the Single Manager should use the risk assessment data and team to define any needed mitigating actions.

G.6 RISK ASSESSMENT TESTING.

G.6.1 Summary. This Section provides a list of potential risk assessment activities that may be used at any stage of the fuel certification process. Any time a list is created, there is a possibility that it may not be complete, hence this list should be used as a guide and consideration given to any additional recommended activities that are not on this list As noted earlier, Single Managers should try to minimize the testing needed for their individual propulsion system by collaborating with the AFCO, other Single Managers, and OEM's.

G.6.2 Flame Tube Test. The RAP may indicate the existence of unacceptably high amounts of trace elements in the candidate fuel. These elements may be harmful to engine hot section components. During the combustion process, these elements may become liberated and deposit themselves in critical areas such as cooling holes and passageways. Some of these elements may also result in high amounts or erosion or delamination. A flame tube test is one way to evaluate the existence and effects of a fuel's trace elements

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G.6.3 Combustor Rig Testing. The RAP would take into account known fuel properties, the influence of additives, trace element concentrations, batch-to-batch variability, compatibility with non-metals, and compatibility of combustion products with hot section materials. The results of this assessment may indicate one or more combustor rigs, such as a fuel injector coking rig, sector rig, hot section materials test rig, or full annual rig.

G.6.3.1 Fuel Injector Coking Rig Description. Coking can cause serious problems with propulsion systems. In its worst form coking in fuel nozzles can cause a mal-distribution in fuel, resulting in combustor hot spots that could cause structural failure and possible loss of aircraft. The results of the RAP may recommend this type of test to reduce any perceived risk of coking. While this type of testing is sometimes accomplished using a JFTOT rig, the decision on what type rig or where to conduct this testing should be decided by collaboration between the AFCO and affected Single Managers.

G.6.3.2 Combustor Sector Rig. A combustor sector rig may be used to address any concerns associated with ignition, re-light, or lean blowout. Although these characteristics could be evaluated in a full scale engine test, a combustor sector rig would potentially offer evaluations over a wider range of conditions at a lower cost.

G.6.3.3 Full Annular Rig. Full annular rigs can evaluate overall combustor performance, emissions and pattern and profile factors. Although full scale engines could be instrumented to obtain the same information, the full annular rig would offer evaluations over a wider range of conditions at a lower cost.

G.6.4 Sea Level and Simulated Altitude Engine Testing. Determining the propulsion system's ability to meet safety, performance and durability requirements is critical to mission accomplishment. Sea level and simulated altitude engine testing has been determined to be a very reliable evaluation method. These tests can be accomplished under ambient conditions, or in a test facility to simulate atmospheric conditions at altitude. The data obtained can be used to evaluate performance (thrust and specific fuel consumption), lean blow out margin, transient times and starting. The extent of testing depends on several different criteria including mission applicability, fuel property categorization and risk acceptance tolerance levels. Great care needs to be taken in selecting which engine or engine families will be tested and what data is needed to support fuel use on other non-tested engine families. Proper test planning is critical to a successful and economically viable sea level and simulated altitude engine test program. Propulsion system engineers, Propulsion Best Practice PCOE - BP-03-18, Rev. A and the JSSG-2007 should be consulted when developing engine test plans.

G.6.5 Engine Flight Testing. Based on the results of the RAP, it may be necessary to conduct some level of engine flight testing. Propulsion flight testing for alternative fuel use would likely be focused on performance and operability characteristics. USAF engine flight testing is usually conducted at the Air Force Flight Test Center at Edwards AFB CA. The depth of engine flight testing depends on several different criteria including mission applicability, fuel property categorization, risk acceptance tolerance levels and the results from any sea level engine testing. Great care needs to be taken in selecting which engine or engine families will be flight tested and what data is needed to support fuel use on other non-flight tested engine families. Proper test planning is critical to a successful and economically viable

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engine flight test program. Propulsion system engineers and JSSG-2007 should be consulted when developing engine flight test plans.

G.6.6 Field Service Evaluations (FSE). An FSE is a proven way to obtain engine performance, operability and durability data under real world operational conditions. It provides a way to evaluate alternative fuels for "unknown-unknown" risks prior to full field implementation. In addition, an FSE can provide invaluable data on maintenance and sustainment (supportability) impacts. FSEs can be conducted at almost any operational base but require significant coordination and a willingness of the warfighter community to accept the extra responsibility and workload associated with them. In addition, the selected FSE base will have to be able to handle the logistical impacts of carrying the fuel under evaluation. Propulsion system engineers, Propulsion Best Practice PCOE - BP-01-14 and JSSG-2007 should be consulted when developing an FSE plan. Ultimately, the warfighter accepts or rejects any FSE recommendations.

G.6.7 Engine Test Durability Qualification and Accelerated Mission Testing (AMT). The RAP may indicate concerns associated with durability. Durability tests could be conducted to clear an engine for varying durations, depending on need. For example, a short duration durability test could be conducted to clear an engine for a limited flight test program or Field Service Evaluation. In other cases, known durability risks could indicate a longer term test to characterize hot section life or define inspection intervals. Propulsion system engineers, Best Practice PCOE - BP-02-17 and JSSG-2007 should be consulted when developing an engine durability (AMT) test plan.

G.6.8 Engine Subsystem Component Testing. The RAP may indicate a need to conduct individual engine component or subsystem testing. These types of tests provide the ability to identify and isolate potential fuel related risks on individual components such as pumps, valves, actuators, heat exchangers and hydromechanical controls. These tests can be accomplished using component test stands or system (wet) rigs.

G.6.9 Inspections. The RAP may indicate the need to visually verify the condition of engine components while under test or field operation. Routine or Special inspections are a proven way to observe and gather this information and assess risk. While inspections provide visual verifications, it should be noted that some are very labor intensive and require skilled personnel in order to obtain the most accurate information.

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APPENDIX H

AEROSPACE FUELS INFRASTRUCTURE REQUIREMENTS

H.1 SCOPE.

H.1.1 General. The Aerospace Fuels Infrastructure (AFI) includes the systems and procedures for the storage, handling, and distribution of aviation fuels within the US Air Force. This portion of the aerospace fuel certification process includes all of the AFI related certification requirements that define the desired level of safety, performance, durability, supportability, interoperability, etc., that need to be maintained by a candidate fuel. By meeting or exceeding these requirements, the candidate will have demonstrated a propensity to impose the least possible economic burden on the Air Force fuels infrastructure.

H.1.2 Entrance Criteria and Subset Testing. Air Force SMEs have evaluated all fuel properties and characteristics based on a requirements-decomposition process that categorizes requirements in terms of safety, performance, durability and supportability, as described in Appendix A. Appendix H identifies the subset to which each infrastructure compatibility characteristic belongs and the basis for its selection. Appendix H also includes component tests that could be conducted to further evaluate the candidate fuel for those AFI compatibility characteristics as deemed appropriate by the Alternative Fuel Certification Office (AFCO).

H.1.3 Background. Although the US Air Force operates under a DoD directive to minimize the number and complexity of their aviation fuels, the supply chain portion of the AFI remains fairly complex. Fuel is shipped to Air Force bases using one or more transportation modes that include ocean going tanker, barge, pipeline and rail car. In the simplest of supply chains, fuel is shipped directly from the refinery but for most bases fuel received from intermediate terminals operated by the Defense Energy Support Center. In addition, “intoplane” fueling contracts provide fuel servicing of Air Force aircraft at airports both within and outside of the US. In many foreign countries, this represents the sole source of fuel and provides the foundation for fuel support to rapidly deploying forces. All fuels entering this relatively complex system of fuel distribution and storage will come into contact with the variety of handling systems and materials that constitute the AFI. All candidate fuels have to demonstrate a compatibility with these AFI systems and materials prior to certification

H.2 APPLICABLE DOCUMENTS.

H.2.1 General. The documents listed below are not necessarily all of the documents referenced herein but are considered to be those which are most important in providing the user an understanding of the information provided by this Appendix.

H.2.2 Government documents.

H.2.2.1 Specifications, standards, and handbooks.

The following specifications, standards, and handbooks form a part of this document to the extent specified herein.

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DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-24441	Paint, Epoxy-Polyamide, General Specification for
MIL-T-52983	Tanks, Fabric, Collapsible
MIL-DTL-83133	Turbine Fuels, Aviation, Kerosene Types, NATO F-34 (JP-8), NATO F-35, and JP-8+100

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

H.2.2.2 Other Government documents, drawings, and publications.

The following other Government documents, drawings, and publications form a part of this document to the extent specified herein.

UNITED FACILITIES GUIDE SPECIFICATIONS

UFGS-1519N	Aviation Fuel Distribution and Dispensing
UFGS-09973	Interior Coating of Welded Steel Petroleum Fuel Tanks
UFGS-13202	Fuel Storage Systems

(Copies of these documents are available online at The Defense Technical Information Center Website and are available from DTIC Headquarters, 8725 John J. Kingman Rd., Ft. Belvoir VA 22060-6218; telephone (800) 225-3842. They are also available at www.wbdg.org.)

H.2.3 Non-Government publications.

The following documents form a part of this document to the extent specified herein.

AMERICAN PETROLEUM INSTITUTE

API Specification 6D	Specifications for Pipeline Valves
API Standard 1529	Aviation Fueling Hose
API/IP 1540	Design, Construction and Operation of Aviation Fueling Facilities
API/IP 1581	Specification and Qualification Procedures for Aviation Jet Fuel Filter/Separators API Specification 5L, Line Pipe

(Copies of these documents are available at www.api.org.)

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ASTM INTERNATIONAL

ASTM D1903	Standard Test Method for Coefficient of Thermal expansion of Electrical Insulating Liquids of Petroleum Origin, and Askarels
ASTM D2717	Standard Test Method for Thermal Conductivity of Liquids
ASTM F146	Standard Test Methods for Fluid Resistance of Gasket Materials
ASME B16.21	Nonmetallic Flat Gaskets for Pipe Flanges
ASME B16.9	Factory-Made Wrought Buttwelding Fittings

(Copies of these documents may be ordered online at www.astm.org; and from ASTM International, 100 Barr Harbor Drive, PO Box C700, West Conshohocken PA 19428-2959; (610) 832-9585. Approved users may access the documents online at www.ihs.com.)

H.3 ENTRANCE CRITERIA.

The AFI Entrance Criteria is composed of all fuel property test requirements specified in MIL-DTL-83133F and described in Appendix B.

H.4 SUBSET 1 TESTING.

H.4.1 Single-Element Filter/Separator Performance. Fuel Filter/Separators (F/S) are an integral part of the Air Force fuel distribution/handling system. The ability of these subsystems to extract and separate free water from the fuel as the fuel flows through the distribution system is a critical part of the overall system performance. The candidate fuel cannot lead to a lower performance of F/S qualified to API/IP Specification 1581 when compared to the performance of these F/S when using JP-8 fuel. This may be accomplished through testing with F/S hardware or may be done by fuel-similarity considerations. If the AFCO deems that testing is required in this subset, representative F/S hardware will be selected based on manufacturer and type and will be tested in accordance with the single-element testing sections of API/IP Specification 1581 using the test fuel in a category M configuration at 100% rated flow

H.4.2 Subset 1 Criteria Evaluation. The AFCO will determine if the candidate fuel meets or fails to meet the AFI requirements described above. The results of the AFI Subset 1 evaluation will be documented by the AFCO.

H.5 SUBSET 2 TESTING.

H.5.1 Electrostatic Charge Dissipation. Since the electrical conductivity of hydrocarbon fuels is minimal, the risk of static electric charge accumulation and the potential for a subsequent catastrophic discharge event taking place during the transfer of fuel can be

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significant. This is equally valid for all transfers including from the point of manufacture, upon up-loading, and upon off-loading and receipt. The measurement of the change in the electrical conductivity of the candidate fuel due to the addition of Static Dissipater Additive (SDA) is required at a single temperature as part of the entrance criteria in H.3. However, the change in this conductivity as a function of temperature is a characteristic of the candidate fuel that has to be determined in order to maintain consistent fuel handing operations. For this reason, the electrical conductivity of the candidate fuel after treating with SDA per MIL-DTL-83133F has to be determined and reported as a function of temperature as discussed in Appendices B and C.

H.5.2 Aviation Fueling Hose and Hose Assemblies. The candidate fuel will meet or exceed the requirements and limits specified in API/IP 1529 for Volume Increase, Fuel-Soluble Matter, Adhesion, Fuel Contamination, Fuel Discoloration (see sections 4.3.3, 4.3.4, 4.4.5, 4.4.10, 4.4.11). In these tests, the candidate fuel will be used as the test fuel.

H.5.3 Aviation Fuel Storage and Transport Interface Materials. The candidate fuel will not adversely affect the ability of functional materials to provide leak-free interfaces between the various components of the fuel distribution and storage systems. This includes fuel resistant gaskets and sealing interfaces. The candidate fuel will perform equivalent to or superior to the performance demonstrated by JP-8 when tested by the method described in section 6 of ASTM F146. This includes Volume Change, Tensile Strength, Compressibility, and Flexibility.

H.5.4 Collapsible Aviation Fuel Storage Tank Materials. The candidate fuel will not adversely affect the function of collapsible fuel storage tanks as described in MIL-T-52983, which is inactive for new designs but remains in effect for equipment currently in use. The candidate fuel will be used as the “immersion test fluid” as required by this specification and will meet or exceed all requirements related to Tensile Strength Retention, Elongation Retention, Volume Swell, Breaking Strength, and Coating Adhesion.

H.5.5 Fuel Distribution Utilities. The AFCO will ensure that the thermal expansion and thermal conductivity of the candidate fuel will have been measured by standard methods ASTM D1903 and ASTM D2717 respectively. These fuel properties can be found in Appendix B. These properties, along with others which have been addressed as part of the general entrance criteria, are important factors that have to be known in order to operate bulk fuel storage containers and pipelines properly.

H.5.6 Subset 2 Criteria Evaluation. The AFCO will determine if the candidate fuel exhibits an acceptable thermal expansion and thermal conductivity. The results of this second subset testing and evaluation will be documented by the AFCO.

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H.6 SUBSET 3 TESTING.

H.6.1 Aviation Fuel Storage Tank Coatings. The candidate fuel will meet or exceed the requirements specified in MIL-DTL-24441 when tested as specified in sections 3.6.3/4.5.10, 3.6.8/4.4.15, and 3.6.9/4.5.11. This includes measures for Flexibility, Solvent resistance, and Immersion testing. In the case of Solvent resistance, the candidate fuel will serve as the test solvent.

H.6.2 Full-Scale Filter/Separator Performance. As mentioned in H.4.1, F/S are an integral component of the Air Force fuel distribution/handling system and it is important for the candidate fuel not to alter the performance of F/S qualified to API/IP Specification 1581 as compared to the performance of these F/S when using JP-8 fuel. The AFCO will determine if a full-scale system component test is necessary to demonstrate this system component performance. If testing is deemed necessary under this subset, representative F/S hardware will be selected based on manufacturer and type and will be tested in accordance with the full-scale testing sections of API/IP Specification 1581 using the test fuel in a category M configuration at 100% rated flow.

H.6.3 Automatic Storage Tank Gauging. The candidate fuel will not adversely affect the operation of automatic storage tank gauging. This will be demonstrated first by similarity analysis in which the pertinent physicochemical properties of the candidate fuel will be compared to those of JP-8. This comparison will demonstrate that automatic gauging systems will operate in the candidate fuel to the same the same level of performance as that found when operating in JP-8. Secondly, the performance of any automatic gauging systems used in the storage of the candidate fuel during the certification process will be examined closely and will also demonstrate capability.

H.6.4 Subset 3 Criteria Evaluation. The AFCO will determine if the performance of the candidate fuel is acceptable with regard to filter/separator systems as described above. The results of this Subset 3 testing and evaluation will be documented by the AFCO.

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APPENDIX I

RISK ASSESSMENT RESULTS

I.1 SCOPE.

I.1.1 General. This appendix provides information obtained from the various risk assessments conducted for each candidate fuel. This information includes but is not limited to risk assessment reference documents and the results from candidate fuel risk assessments including property subset categorizations, hazard risk indexes and property risk impacts. Appendix I also contains risk evaluation activities including component analyses and test descriptions/results and risk mitigation actions including fuel additives, periodic inspections and flight restrictions. Appendix I is intended to be a living document, updated as each risk assessment is accomplished. Periodic updates to this handbook will be the vehicle by which this information is documented. Appendix I will contain a summary of the above information and will provide contacts from which additional information may be obtained.

I.2 APPLICABLE DOCUMENTS.

I.2.1 General.

The documents listed below are not necessarily all of the documents referenced herein, but are those needed to understand the information provided by this Appendix.

I.2.2 Government documents.

I.2.2.1 Specifications, standards, and handbooks.

The following specifications, standards, and handbooks form a part of this document to the extent specified herein.

I.2.2.2 Other Government documents, drawings, and publications.

The following other Government documents, drawings, and publications form a part of this document to the extent specified herein.

I.2.3 Non-Government publications.

The following documents form a part of this document to the extent specified herein.

I.3 RISK ASSESSMENTS RESULTS.

I.3.1 Candidate Fuel X. TBD

I.3.2 Candidate Fuel X. TBD

I.3.3 Candidate Fuel X. TBD

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I.4 RISK ASSESSMENT LESSONS LEARNED.

I.4.1 Candidate Fuel X. TBD

I.4.2 Candidate Fuel X. TBD

I.4.3 Candidate Fuel X. TBD

I.5 REFERENCE DOCUMENTS.

I.5.1 Document A. TBD

I.5.2 Document B. TBD

I.6 DEFINITIONS.

TBD

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APPENDIX J

CERTIFICATION LESSONS LEARNED

Reserved

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APPENDIX K

ENVIRONMENTAL, SAFETY AND OCCUPATIONAL HEALTH

K.1 SCOPE.

K.1.1 General. The detailed guidelines for a program environmental assessment are provided in the "Development Guide for Program Managers", December 2003. The following paragraphs have been abstracted from this document in order to provide the reader with a better understanding of the process and steps required to accomplish this assessment.

K.1.2 Background.

K.1.2.1 Requirements. DoD Instruction (DODI) 5000.2 requires that all defense technology projects and acquisition programs, regardless of acquisition category, and throughout their life cycle, perform and maintain an Environmental, Safety and Occupational Health (ESOH) evaluation. This evaluation, the Programmatic Environmental, Safety and occupation Health Evaluation (PESHE), is a statutory requirement as stated in DODI 5000.2, Table E3.T1. It is a living document that is required at Program Initiation for Ships and at Milestone B, Milestone C, and Full Rate Production Decision Review for all systems, and have to be summarized in the Acquisition Strategy. The evaluation will include, as a minimum, the following:

ESOH risks

For acceptance of identified ESOH mishap risks, the Component Acquisition Executive (CAE) is the acceptance authority for high risks, the Program Executive Office (PEO) for serious risks, and the Program Manager (PM) for medium and low risks.

Strategy for integrating ESOH considerations into the systems engineering process

Identification of ESOH responsibilities

Method for tracking progress

Completion schedule for National Environmental Policy Act compliance

Identification of hazardous materials (HAZMATs) used in the system and plan for the system's demilitarization and disposal

K.1.2.2 ESOH Evaluation. The ESOH evaluation is the program's way of assessing the environmental, occupational health and safety impacts of the weapon system on people and the environment and evaluating the impact of ESOH requirements on the mission readiness. The ESOH evaluation ensures potential "show stoppers" are identified and resolved as early in the acquisition process as possible.

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In addition, ESOH considerations should be part of an overall Operational Safety, Suitability, and Effectiveness (OSS&E) plan. This plan should ensure OSS&E throughout the life cycle of the system and should include follow-on evaluations, systems engineering, and requirements review.

The ESOH analysis process shown on Figure K-1 is initiated by completion of the Environmental Impact Analysis Request Form (AF IMT 813) (See Figure K-2) and its submission to Air Force Materiel Command.

K.1.2.3 PESHE Content. The PESHE addresses a program's status concerning the National Environmental Policy Act (NEPA) of 1969, ESOH compliance, safety, health, hazardous materials (HAZMATs) management, pollution prevention, and explosives safety.

K.1.2.4 PM Responsibilities. DoD systems acquisition policy clearly directs the PM to optimize total system performance, operational effectiveness and suitability, and affordability over the total life cycle. DoD acquisition policy states that the PM has to plan for human systems integration (HSI), which includes considerations, early in the acquisition process. As part of systems engineering, the PM will prevent ESOH risks, where possible, and will manage ESOH risks where they cannot be avoided. The process for integrating ESOH into systems engineering is documented in the PESHE.

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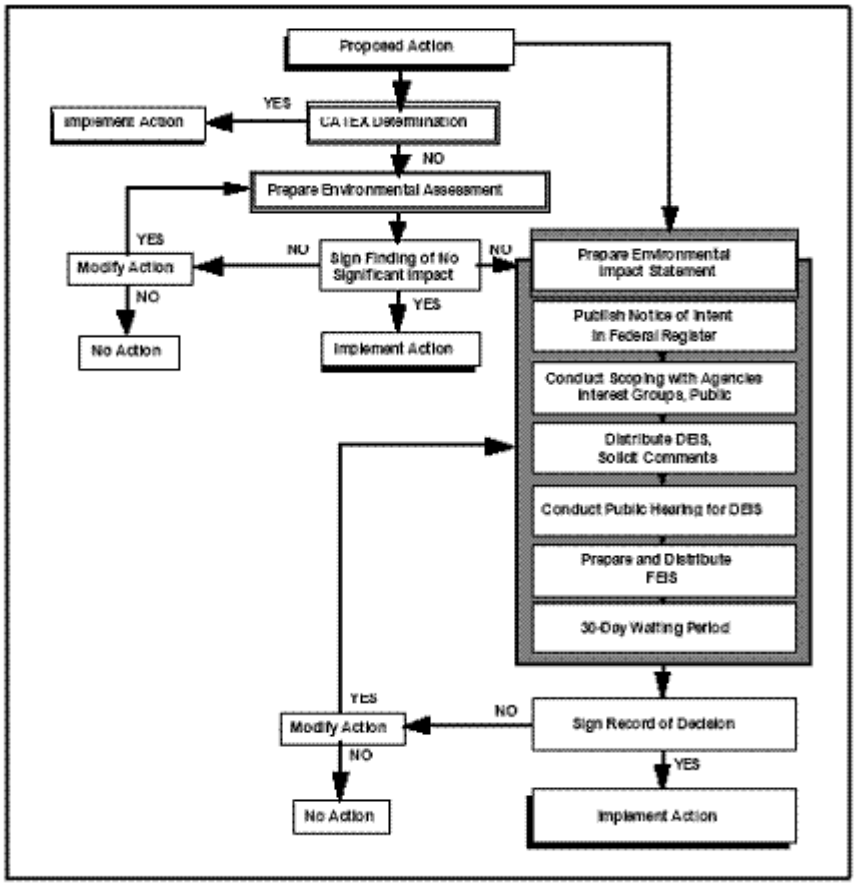


FIGURE K-1. Environmental impact analysis process.

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REQUEST FOR ENVIRONMENTAL IMPACT ANALYSIS

Report Control Symbol RCS:

INSTRUCTIONS: Section I to be completed by Proponent; Sections II and III to be completed by Environmental Planning Function. Continue on separate sheets as necessary. (Reference appropriate item number(s).

SECTION I - PROPONENT INFORMATION

1. TO (Environmental Planning Function)	2. FROM (Proponent organization and functional address symbol)	2a. TELEPHONE NO.
(Base CE Environmental POC)	(Single Manager of System demonstrating JP-8/SPK)	

3. TITLE OF PROPOSED ACTION
Demonstrate Fischer-Tropsch (F-T) Synthetic Paraffinic Kerosene (SPK) blended with JP-8, to fuel aircraft and equipment

4. PURPOSE AND NEED FOR ACTION (Identify decision to be made and need date)
Demonstrate (specify) aircraft and equipment using JP-8/SPK 50/50 blended fuel. Base approval to use JP-8/SPK (from date..... to date ....).

5. DESCRIPTION OF PROPOSED ACTION AND ALTERNATIVES (DOPAA) (Provide sufficient details for evaluation of the total action.)
Perform demonstration with JP-8/SPK to certify performance in aircraft and equipment. MIL-DTL-83133F specifies the use of JP-8/SPK for operational flight demonstration.

6. PROPONENT APPROVAL (Name and Grade)	6a. SIGNATURE	6b. DATE
(Single Manager of System)		

SECTION II - PRELIMINARY ENVIRONMENTAL SURVEY. (Check appropriate box and describe potential environmental effects including cumulative effects.) (+ = positive effect; 0 = no effect; " = adverse effect, U= unknown effect)	+	0	-	U
7. AIR INSTALLATION COMPATIBLE USE ZONE/LAND USE (Noise, accident potential, encroachment, etc.)	☐	☒	☐	☐
8. AIR QUALITY (Emissions, attainment status, state implementation plan, etc.)	☒	☐	☐	☐
9. WATER RESOURCES (Quality, quantity, source, etc.)	☐	☒	☐	☐
10. SAFETY AND OCCUPATIONAL HEALTH (Asbestos/radiation/environmental exposure, explosives safety quantity distance, bird/wildlife aircraft hazard, etc.)	☒	☐	☐	☐
11. HAZARDOUS MATERIALS/WASTE (Use/storage/generation, solid waste, etc.)	☐	☒	☐	☐
12. BIOLOGICAL RESOURCES (Wetlands/floodplains, threatened or endangered species, etc.)	☐	☒	☐	☐
13. CULTURAL RESOURCES (Native American burial sites, archeological, historical, etc.)	☐	☒	☐	☐
14. GEOLOGY AND SOILS (Topography, minerals, geothermal, Installation Restoration Program, seismicity, etc.)	☐	☒	☐	☐
15. SOCIOECONOMIC (Employment/population projections, school and local fiscal impacts, etc.)	☐	☒	☐	☐
16. OTHER (Potential impacts not addressed above.)	☐	☒	☐	☐

SECTION III - ENVIRONMENTAL ANALYSIS DETERMINATION

17. ☒ PROPOSED ACTION QUALIFIES FOR CATEGORICAL EXCLUSION (CATEX) A2,3,7 ; OR
☐ PROPOSED ACTION DOES NOT QUALIFY FOR A CATEX; FURTHER ENVIRONMENTAL ANALYSIS IS REQUIRED.

18. REMARKS
Aircraft and equipment are being demonstrated with a 50/50 blend of JP-8/SPK aviation jet fuel (authorized by MIL-DTL-83133F) to acquire functional certification (outlined in MIL-HDBK-510, Aerospace Fuels Certification). Each systems' and equipments' capability to use JP-8/SPK will be recorded in their governing technical airworthiness/operational documents, following demonstration, so that the fuel blend can be used to carry out assigned missions. Aircraft certification must be completed in preparation for the time when SPK fuel is readily available. Demonstrations of JP-8/SPK (2006-2008) have occured at Edwards, McChord, Dyess, Minot, Selfridge AEDC, and OC-ALC. The 50/50 blend presents no greater impact than JP-8.

19. ENVIRONMENTAL PLANNING FUNCTION CERTIFICATION (Name and Grade)	19a. SIGNATURE	19b. DATE

AF IMT 813, 19990901, V1 — THIS FORM CONSOLIDATES AF FORMS 813 AND 814. PREVIOUS EDITIONS OF BOTH FORMS ARE OBSOLETE. — PAGE 1 OF PAGE(S)

FIGURE K-2. Request for environmental impact analysis.

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K.2 APPLICABLE DOCUMENTS.

K.2.1 General.

The documents listed below are not necessarily all of the documents referenced herein but are those needed to understand the information provided by this Appendix.

K.2.2 Government documents.

K.2.2.1 Specifications, standards, and handbooks.

The following specifications, standards, and handbooks form a part of this document to the extent specified herein.

K.2.2.2 Other Government documents, drawings, and publications.

The following other Government documents, drawings, and publications form a part of this document to the extent specified herein.

DEFENSE ACQUISITION GUIDEBOOK SECTIONS C2.8.6, C2.8.7

PESHE Development Guide for Program Managers, December 2003, AFMC

(This information may be accessed at https://akss.dau.mil/DAG/.)

UNITED STATES AIR FORCE INSTRUCTIONS

AFI 32-7061 Environmental Impact Analysis Process (EIAP)

(Copies of this Instruction are available at http://www.e-publishing.af.mil.)

Executive Order 13423 of January 24, 2007, Strengthening Federal Environmental, Energy, and Transportation Management, and National Defense Authorization Act for Fiscal Year 2007, CHAPTER 173—ENERGY SECURITY, 2911. Energy performance goals and plan for Department of Defense, (c) SPECIAL CONSIDERATIONS.

Energy Independence and Security Act of 2007, SEC. 526. PROCUREMENT AND ACQUISITION OF ALTERNATIVE FUELS

The ENERGY INDEPENDENCE AND SECURITY ACT of 2007 (PUBLIC LAW 110–140—DEC. 19, 2007) has been enacted to ensure that Federal Agencies only procure alternative or synthetic fuels that don't create more GHG. This law may constrain the procurement of alternative fuel to meet the Air Force's second goal. The pertinent section is excerpted for reference:

Energy Independence and Security Act of 2007, SEC. 526. PROCUREMENT AND ACQUISITION OF ALTERNATIVE FUELS: No Federal agency will enter into a contract for procurement of an alternative or synthetic fuel, including

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a fuel produced from nonconventional petroleum sources, for any mobility-related use, other than for research or testing, unless the contract specifies that the lifecycle greenhouse gas emissions associated with the production and combustion of the fuel supplied under the contract will, on an ongoing basis, be less than or equal to such emissions from the equivalent conventional fuel produced from conventional petroleum sources. Compliance with this law would rest with the procuring agency, Defense Energy Support Center (DESC), through contracting action with suppliers.

(Copies of these documents are available from www.nist.gov.)

K.2.3 Non-Government publications.

The following documents form a part of this document to the extent specified herein.

"Class- and Structure-Specific Separation, Analysis, and Identification Techniques for the Characterization of the Sulfur Components of JP-8 Aviation Fuel Energy & Fuels," Volume 17, pp. 1292-1302, 2003, Link, D. D.; Baltrus, J. P.; Rothenberger, K. S.; Zandhuis, P.; Minus, D. K.; Striebich, R. C. .

"Qualification of a Semi-Synthetic Jet A-1 as Commercial Jet Fuel," SwRI-8531, November 1997, Moses, C. A., Stavinoha, L. L., Roets, P. .

"Fischer-Tropsch Jet Fuels – Characterization for Advanced Aerospace Applications", AIAA 2004-3885, July 2004, Edwards, T., Don Minus, William Harrison, Edwin Corporan, Matt DeWitt, Steve Zabarnick, Lori Balster, .

"World Fuel Sampling Program", CRC Report No. 647, June 2006, O J. Hadaller, J.M. Johnson,

"Chemical Class Composition of Commercial Jet Fuels and Other Specialty Kerosene Fuels," AIAA Paper 2006-7972, November 2006, Shafer, L, Striebich, R., Gomach, J., Edwards, T..

"Petroleum Quality Information System," Defense Energy Support Center, 2005 report (http://www.desc.dla.mil/DCM/DCMPage.asp?pageid=99)

"Reduction of Turbine Engine Particulate Emissions Using Synthetic Jet Fuel," ACS Fuel Chemistry Division Preprints, Vol. 50(1), p. 338-341, 2005, Corporan, E., DeWitt, M. J., Monroig, O., Ostdiek, D., Mortimer, B., Wagner, M.

"Emissions Characteristics of a Turbine Engine and Research Combustor Burning a Fischer-Tropsch Jet Fuel," Energy & Fuels, 2007, Edwin Corporan, Matthew J. DeWitt, Vincent Belovich, Robert Pawlik, Amy C. Lynch, James R. Gord, and Terrence R. Meyer

"DoD Assured Fuels Initiative: B-52 Aircraft Emissions Burning a Fischer-Tropsch/JP-8 Fuel Blend", Edwin Corporan, Paper Presented at: IASH 2007,

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Tenth International Conference on Stability, Handling, And Use Of Liquid Fuels, Oct 2007. Approved for public release. Case number: AFRL-WS 07-2187.

(Copies of these documents are available online at The Defense Technical Information Center Website and are available from DTIC Headquarters, 8725 John J. Kingman Rd., Ft. Belvoir VA 22060-6218; telephone (800) 225-3842.)

K.3 Environmental Safety and Occupational Health. Appendix E provides a detailed discussion of the toxicity testing and a baseline of known information regarding JP-8 jet fuel. When sufficient toxicity data are available, environmental and occupational exposure standards can be developed that are protective of ecosystems (the environment) and workers (occupational medicine and health).

K.3.1 Toxicity and Safety Considerations. The fuel blend will be handled as JP-8, using the same infrastructure and procedures. Toxicity tests are ongoing, and the SPK fuel blend is expected to be less toxic than current fuels (which are relatively non-toxic kerosene fuels) due to the reduced level of aromatics. This testing covers only the impact of the use of the SPK fuel blend as a dispersed environmental impact from mobile sources, and does not cover the production process.

SPK fuels are refined to the same flash point and distillation specification limits as their petroleum-derived counterparts, so their vapor pressures are similar. Thus, personnel exposure to SPK fuel vapors will be similar to that of conventional jet fuels. Aromatics are widely accepted as the most toxic of the hydrocarbon components in jet fuel, thus the addition of SPK fuels, which dilutes the aromatic concentration, can be expected to decrease the toxicity of jet fuels. Testing is underway to assess this expectation, and is expected to be complete in 2008.

The use of a 50/50 SPK/JP-8 fuel blend in aircraft in place of current petroleum-derived JP-8 will create several minor environmental impacts – in this case, essentially all are positive. Fischer-Tropsch SPK fuels will be procured to a specification very similar to that for current jet fuels, thus the boiling range, molecular weight range, flash point, and freeze point, for example, will be similar to current fuels. Jet fuels contain four general classes of hydrocarbons: n-paraffins, iso-paraffins, naphthenes (cycloparaffins), and aromatics. SPK fuels do not contain the aromatic and naphthenic components present in JP-8 and commercial Jet A fuels, thus SPK fuels have a higher hydrogen/carbon ratio (note that the n-paraffins and iso-paraffins in F-T and JP-8 are similar). SPK fuels also do not contain the ~500 ppm of sulfur compounds present in current jet fuels (the specification allows up to 0.3 wt % (~3000 ppm)).

K.3.2 HAZMAT Summary. The 50/50 SPK/JP-8 fuel blend is a hazardous material. This material replaces JP-8 jet fuel. Although expected to be less hazardous than JP-8, the 50/50 fuel blend is still hazardous and will be treated as such. There are no new hazards expected with the implementation of use of this alternative fuels.

K.3.2.1 Locations and Quantities of HAZMAT in the System. It is expected that this fuel will replace all JP-8 currently stored, maintained, used and disposed of on all AF installations. Quantities and locations are established by the installation.

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K.3.2.2 Energetic Qualification Information. The 50/50 SPK/JP-8 fuel blend is thoroughly tested on each system where it is used. The contractor is responsible for this information.

K.3.2.3 Hazardous Byproducts and Discharges. Pollutant emissions contain CO, NOx and unburned hydrocarbons, which are very similar to current fuels.

K.3.2.4 Special HAZMAT Training, Handling, and Storage Needs. SPK fuels are refined to the same flash point and distillation specification limits as their petroleum-derived counterparts, so their vapor pressures are similar. Thus, personnel exposure to SPK fuel vapors will be similar to that of conventional jet fuels. The 50/50 SPK/JP-8 fuel blend utilizes existing infrastructure of JP-8 fuel. No new training, handling or storage requirements are expected.

K.3.2.5 Pollution Prevention (P2). The 50/50 blend of SPK and JP-8 fuel is a P2 initiative. This fuel provides lower emissions and less hazards that a regularly associated with the use of JP-8 fuels.

K.3.2.6 Emissions Measurements. In support of the DoD Assured Fuels Initiative, the emissions of two TF33 P-103 engines burning JP-8 and a 50/50 blend by volume of JP-8 and Fischer Tropsch (FT) synthetic fuel (referred herein as FT blend) were characterized to determine the impacts of the synthetic fuel on the engine emissions. This effort represents the first emissions evaluation of an actual United States (US) military aircraft operating with a semi-synthetic fuel.

Gaseous and particulate matter (PM) emissions were quantified. Measurements of the mostly non-volatile PM emissions were performed using conventional instrumentation to determine particle number (concentration), size, mass and smoke number. Soot samples were collected on quartz filters for subsequent chemical analysis. The engines were operated with each fuel at four power settings, idle-to-maximum thrust, and a 50-hour engine endurance test with the FT blend was conducted in the test cell at normal rated thrust.

The particle numbers (PN) were normalized for the fuel consumed to obtain particle number emission indices (PN-EI) (number of particles per kilogram of fuel). This accounts for any variation in fuel-to-air (F/A) ratio between test runs of the same engine condition and provides a normalized basis for comparison of engine settings. The engine F/A ratios were determined based on CO_2 and CO emissions. Corrected PN-EI for the TF33 engine tested at Tinker and those at Edwards AFB with JP-8 were very similar and ranged from 4.6E+15 to 8.2E+15 particles per cubic centimeter (#/cm3). For the engines operating with the FT blend the PN-EI ranged from 3.4E+15 to 6.2E+15 #/cm3 for the same conditions.

Changes in TF33 engine PN-EI burning the FT blend relative to operation with JP-8 are shown on Figure K-3. These data are based on an average of multiple test runs sampled with two probes with each point including over 1000 data scans. The data uncertainty (random error) for each PN measurement was <7% for most conditions. As shown, significant reductions in PN-EI were measured when the engine was operated with the FT blend.

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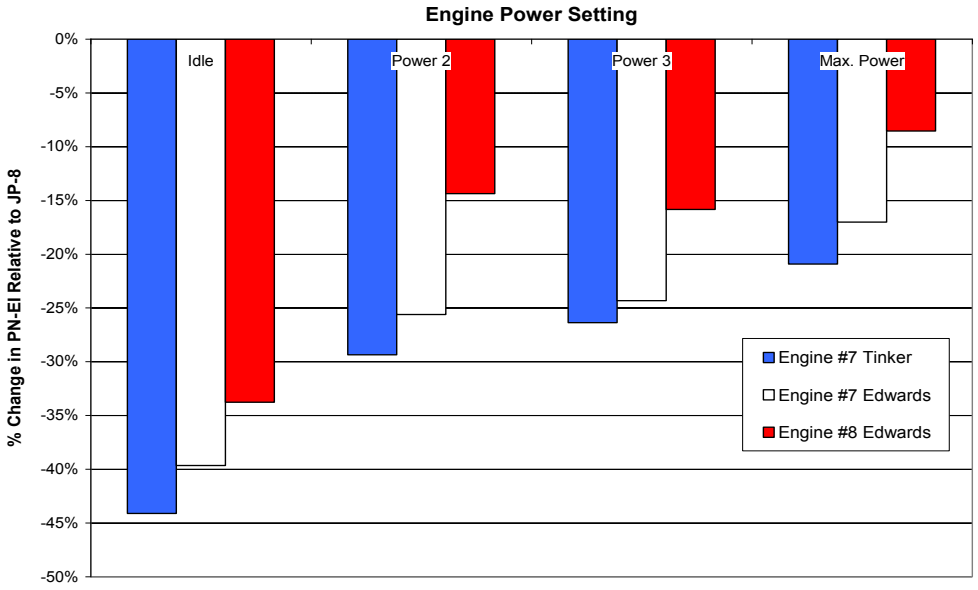


FIGURE K-3. Change in particle number emissions of TF33 engines burning JP-8 and FT blend.

For both engines, the results clearly show a trend of higher fuel impact (larger reductions) at the lower engine power settings. At low power, the engine produces small soot particles and relatively high concentrations of volatile organics, some of which condense in the sample lines to become particles. Due to the aliphatic-only nature of the FT fuel it produces even smaller particles than JP-8, which readily oxidize in the engine combustor. A relatively low but still significant effect on PN-EI at maximum power was observed. These results are in agreement with those observed in a previous study on a T63 engine operated on FT fuel blends. In general, the reduction in PM emissions is primarily the result of the reduced aromatics and higher H/C ratio in the FT blend. Aromatic species in the fuel act as seeds for the formation of PAH molecules, which coagulate to subsequently produce soot nuclei. The propensity of aromatics to produce soot has been well established and observed in large scale combustors and laboratory flames.

Significant reductions (50%) in sulfur-based particulate emissions are also anticipated as result of the sulfur-free FT fuel. These sulfur-based species exist in the gaseous phase at the engine exhaust and nucleate into particles (aerosols) as they react in the atmosphere.

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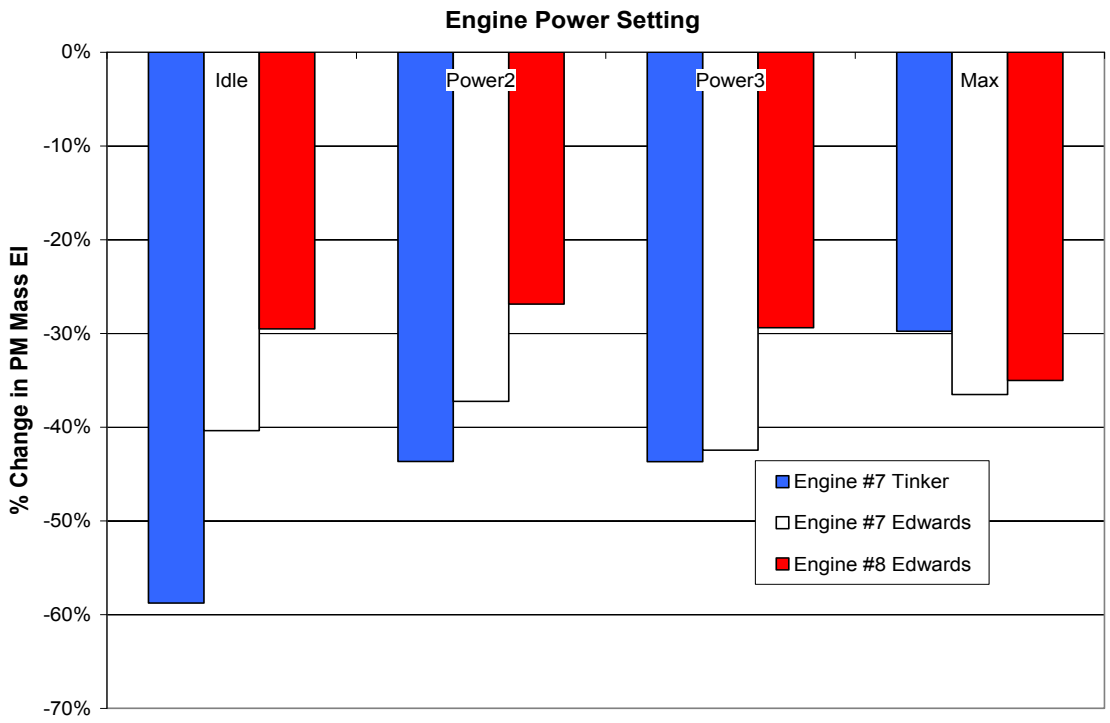


FIGURE K-4. Change in particle mass EI for TF33 engines using FT blend relative to JP-8.

Direct PM mass measurements were made with a TEOM. The average calculated PM mass EI for the two TF33 engines on JP-8 ranged between 1.4 to 3.0 g/kg-fuel with the lowest PM mass EI at idle and highest at Power 2. The PM mass EI using the FT blend were reduced to 0.81-1.9 g/kg-fuel. Changes in PM mass EI as a function of power setting and engine are shown on Figure K-4. Consistent with all other particle emissions, excellent agreement was observed for the measurements and trends of Engine #7 at Tinker Air Force Base and Edwards Air Force Base. Average reductions in PM mass EI of 40% for engine #7 and 30% for engine #8 were observed with the FT blend relative to JP-8. The larger effect of the FT fuel on particle mass compared to PN-EI for the higher power settings is due to the reduction in particle size and the relation of particle diameter with mass.

Negligible differences in most measured gaseous emissions were observed for the engine burning JP-8 and FT blend. Approximately a 10% reduction in the total unburned hydrocarbons (UHC) was observed at idle with the FT blend. As anticipated, reductions in sulfur oxide emissions with the FT blend were observed due to the sulfur-free nature of the FT fuel. These results are in good agreement with concurrent measurements made at Tinker AFB by AEDC. Although slight reductions in CO_2 emissions (~1.8%) and increases in water vapor were anticipated as result of the higher H/C ratio in the FT fuel blend, no statistically significant differences in these emissions were observed.

The results from this study further demonstrate the potential environmental benefits of using an FT fuel in turbine engine systems. In addition to the lower PM emissions, which help reduce the environmental burden of aircraft, use of FT fuels may potentially increase engine life

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and reduce engine maintenance and aircraft exhaust signature. Average reductions of 30-45% in PM mass emissions relative to JP-8 were observed in both engines burning the FT blend. In addition, significant reductions in particle number and engine smoke numbers were also observed. Chemical analysis of the soot samples showed reductions of 50-70% in three and four-ring polycyclic aromatic hydrocarbons (PAH) concentration with the FT blend. Effects of the synthetic fuel on gaseous emissions were negligible except for slight reductions in unburned hydrocarbons and the anticipated reductions in sulfur oxides. In summary, only beneficial impacts on TF33 engine emissions were observed with the use of the FT blend.

K.4 ACRONYMS.

AFMC	- Air Force Material Command
CAE	- Component Acquisition Executive
CATEX	- categorical exclusion
CFR	- Code of Federal Regulation
DESC	- Defense Energy Support Center
DoD	- Department of Defense
EA	- Environmental Assessment
EIAP	- Environmental Impact Analysis Process
EIS	- Environmental Impact Statement
EOs	- Executive Orders
EPA	- Environmental Protection Agency
ESC	- Electronic Systems Center
ESOH	- Environment, Safety and Occupational Health
FAR	- Federal Acquisition Regulation
GHG	- Green House Gases
HAPs	- Hazardous Air Pollutants
HAZMAT	- hazardous material
IMP	- Integrated Master Plan
NAS	- National Aerospace Standard
NEPA	- National Environmental Policy Act
OSS&E	- Operational Safety, Suitability, and Effectiveness
PEA	- Programmatic Environmental Analysis
PEO	- Program Executive Officer
PESHE	- Programmatic Environmental, Safety and Occupational Health Evaluation

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PM - Program Manager

Pub. L. - Public Law

PQIS - The Petroleum Quality Information System

SPK - Synthetic Paraffininc Kerosene

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APPENDIX L

FUEL PROPERTY TRACEABILITY INDEX

L.1 SCOPE.

This Appendix provides a cross reference in Table L-I where each fuel property/characteristic appearing in Appendix C is listed along with the required test method and nominal certification criteria (Entrance, Subset 1,2, or 3). The table further shows where in the appropriate appendix or appendices (B-H) the property and test method are applied. The Air Force SMEs associated with each Appendix have evaluated all fuel properties and characteristics based on the requirements decomposition process that correlated requirements to safety, performance, durability and supportability, as described in Appendix A. A spreadsheet containing this information in a sortable form is embedded on Figure L-1. The Appendix C Criteria column shows the highest priority of each property (Entrance, or Subset 1, 2, or 3) in the entire certification process. This priority may be higher than the priorities shown in the other appendices (listed in the last column of Table L-I) because the holistic view in the systems engineering process elevates the importance of the property above that of any of the more specialized subject areas addressed by each of the Appendices.

TABLE L-I. Fuel property/test method application.

C-Para.	Appendix C			Appendix Where Applied	
	Property	Test Method	Criteria	Paragraph	Criteria
C.4	Acid Number, Total	ASTM D3242	Entrance	B.4.1.1 Entrance Criteria MIL DTL-83133F	Entrance
				D.6.3 Alternative Fuels	1
C.5	Additive Compatibility	ASTM D4054	1	B.4.2.1 Fuel Property Tests	1
				D.6.5 Baseline Test Fluids D 6.6 Additive Testing	1
				H 4.1 Electrostatic Buildup	1
C.6	Aromatics	ASTM D1319	Entrance	B.4.1.1 Entrance Criteria MIL DTL-83133F	Entrance
				D.6.3 Alternative Fuels	1
C.7	Autoignition Temperature	ASTM E659	1	B.4.2.1 Fuel Property / Characteristic Tests	1
				F.4.6 Autoignition Temperatures in Aviation Fuels	1
C.8	Bulk Modulus	ASTM D6793	1	B.4.2.1 Fuel Property / Characteristic Tests	1
				G.4.5 Table G-III. Bulk Modulus versus Temperature	2

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TABLE L-I. Fuel property/test method application - Continued.

C-Para.	Appendix C			Appendix Where Applied	
	Property	Test Method	Criteria	Paragraph	Criteria
C.9	Calculated Cetane Index	ASTM D976	Entrance	B.4.1.1 Entrance Criteria MIL DTL-83133F	Entrance
				B.4.2.2.2.2 Support Equipment & Vehicles (SE&V) (Cetane Index) Diesel ignition quality	1
C.10	Cetane Number	ASTM D613	2	B.4.3.2.2 SE&V A measure of the ignition quality of a fuel in a diesel engine	2
C.11	Copper Strip Corrosion	ASTM D130	Entrance	B.4.1.1 Entrance Criteria MIL DTL-83133F	Entrance
C.12	Density (versus Temperature – Subset 1, Thermal Expansion – Subset 2)	ASTM D1298 ASTM D4052 ASTM D1903 (App H)	Entrance	B.4.1.1 Entrance Criteria MIL DTL-83133F	Entrance
				B.4.2.1 versus Temperature, Fuel Property/Characteristic Tests	1
				B.4.3 Thermal Expansion, Fuel Properties/Characteristics	2
				G.4.4 Table G-II.	1
				H.5.4 Fuel Distribution Utilities (Thermal Expansion)	2
				H.5.5 Subset 2 Criteria Evaluation (Thermal Expansion)	2
C.13	Dielectric Constant versus Density versus Temperature	ASTM D924	1	B.4.2.1 Fuel Property / Characteristic Tests	1
C.14	Distillation Curve	ASTM D86 ASTM D2887	Entrance	B.4.1.1 Entrance Criteria MIL DTL-83133F	Entrance
C.15	Electrical Conductivity at Standard Temp.	ASTM D2624	Entrance	B.4.1.1 Entrance Criteria MIL DTL-83133F	Entrance
				D.6.3 Alternative Fuels	1
C.15	Electrical Conductivity versus Temperature	ASTM D2624	2	B.4.2.1 Fuel Property / Characteristic Tests	2
				H.5.1 Electrostatic Charge Dissipation	2
C.16	Enthalpy versus Temperature	TBD	3	B.4.4.1 Fuel Property / Characteristic Tests	3
				G.4.6 Table G-IV.	

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TABLE L-I. Fuel property/test method application – Continued.

C-Para.	Appendix C			Appendix Where Applied	
	Property	Test Method	Criteria	Paragraph	Criteria
C.17	Existent Gum	ASTM D381	Entrance	B.4.1.1 Entrance Criteria MIL DTL-83133F	Entrance
C.18	Filtration Time	MIL-DTL-83133	Entrance	B.4.1.1 Entrance Criteria MIL DTL-83133F	Entrance
C.19	Flame Speed	TBD	1	B.4.2.1 Fuel Property / Characteristic Tests	1
				Table F-I. Baseline fire detection sensors and flame speed criteria	
C.20	Flammability Limits	ASTM E681	1	B.4.2.1 Fuel Property / Characteristic Tests	1
				F.4.1 Flammability Limits	1
				G.4.4 Table G-II.	1
C.21	Flash Point	ASTM D56 or ASTM D93	Entrance	B.4.1.1 The Entrance Criteria Definition MIL-DTL-83133F	Entrance
				B.4.2.2.1 Auxiliary and Emergency Power Units (APU/EPU) Evaluation	1
				D.6.3 Alternative Fuels	1
				F.4.5 Flash Point for Aviation Fuels	1
				G.4.3 Entrance Criteria Evaluation	Entrance
C.22	Freezing Point	ASTM D2386 ASTM D5972	Entrance	B.4.1.1 The Entrance Criteria Definition MIL-DTL-83133F	Entrance
				G.4.3 Entrance Criteria Evaluation	
				B.4.2.2.1 Auxiliary and Emergency Power Units (APU/EPU) Evaluation	1
C.23	Fuel System Icing Inhibitor	ASTM D5006	Entrance	B.4.1.1 The Entrance Criteria Definition MIL-DTL-83133F	Entrance
				B.4.4.2.1 Fuel System Icing Inhibitor (FSII) Rig Test	3
C.24	Gas Chromatograph	ASTM D2887	Entrance	B.4.1.1 The Entrance Criteria Definition MIL-DTL-83133F	Entrance

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TABLE L-I. Fuel property/test method application - Continued.

C-Para.	Appendix C			Appendix Where Applied	
	Property	Test Method	Criteria	Paragraph	Criteria
C.25	Heat of Combustion, Net	ASTM D4809 ASTM D4529	Entrance	B.4.1.1 The Entrance Criteria Definition MIL-DTL-83133F	Entrance
				G.4.4 Table G-II.	1
				F.4.7 Heat of Combustion	1
C.26	Heat of Vaporization, Latent – see C.46 Vapor Pressure				
C.27	Hot Surface Ignition	Fed Test Std 791C ISO 20823	1	B.4.2.1 Fuel Property / Characteristic Tests	1
				B.4.3.1 Fuel Property / Characteristic Tests Under Turbulent Airflow	2
				Table F-II. Safety Related Fuel Characteristics	
				F.5.2 Under Turbulent Airflow	
C.28	Hydrogen Content	ASTM D3701 ASTM D3343	Entrance	B.4.1.1 The Entrance Criteria Definition MIL-DTL-83133F	Entrance
C.29	Lubricity	ASTM D5001	1	B.4.2.1 Fuel Property / Characteristic Tests	1
				B.4.2.2.2.2 SE&V Properties	1
				B.4.4.2.2.1 SE&V Evaluation Injection Pump Durability	3
				G.4.6 Table G-IV.	3
C.30	Minimum Spark Ignition Energy	TBD The criterion is defined as "no easier to ignite than Jet A/JP-8."	2	B.4.3.1 Minimum Energy, Ignition	2
				F.5.3 Safety Related Fuel Characteristics Minimum Ignition Energy (MIE)	2

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TABLE L-I. Fuel property/test method application - Continued.

C-Para.	Appendix C			Appendix Where Applied	
	Property	Test Method	Criteria	Paragraph	Criteria
C.31	Napthalenes	ASTM D1840	Entrance	B.4.1.1 The Entrance Criteria Definition MIL-DTL-83133F	Entrance
C.32	Ostwald Coefficient/ Gas Solubility	ASTM D2779	2	B.4.3.1 Fuel Property / Characteristic Tests	2
C.33	Particulate Matter	ASTM D2276 ASTM D5452	Entrance	B.4.1.1 The Entrance Criteria Definition MIL-DTL-83133F	Entrance
C.34	Pour Point	ASTM D.97	2	B.4.2.1 Fuel Property / Characteristic Tests SE&V design critical	2
C.35	Saybolt Color	ASTM D156	Entrance	B.4.1.1 The Entrance Criteria Definition MIL-DTL-83133F	Entrance
C.36	Smoke Point	ASTM D1322	Entrance	B.4.1.1 The Entrance Criteria Definition MIL-DTL-83133F	Entrance
C.37	Specific Heat (as a Function of Temperature)	currently calculated	1	B.4.2.1 Fuel Property / Characteristic Tests	1
				B.4.4.2.2 SE&V Evaluation	3
				G.4.6 Table G-IV.	3
C.38	Storage Stability	MIL-STD-3004	1	B.4.2.1 Fuel Property/Characteristic Tests	1
C.39	Sulfur, Mercaptan	ASTM D3227 ASTM D4952	Entrance	B.4.1.1 The Entrance Criteria Definition MIL-DTL-83133F	Entrance
				D.7.5 Related Materials Testing	2
C.40	Sulfur, Total	ASTM D129 ASTM D1266 ASTM D2622 ASTM D3120 ASTM D4294 ASTM D5453	Entrance	B.4.1.1 The Entrance Criteria Definition MIL-DTL-83133F	Entrance
				D.6.3 Alternative Fuels	1
				D.7.5 Related Materials Testing	2
C.41	Surface Tension versus Temperature	ASTM D971	1	B.4.2.1 Fuel Property / Characteristic Tests	1
				G.4.4 Table G-II.	1

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TABLE L-I. Fuel property/test method application - Continued.

C-Para.	Appendix C			Appendix Where Applied	
	Property	Test Method	Criteria	Paragraph	Criteria
C.42	Thermal Conductivity versus Temperature	CRC Handbook	1	B.4.2.1 Fuel Property / Characteristic Tests	1
				B.4.4.2.2 SE&V Evaluation	3
				G.4.6 Table G-IV.	3
				H.5.4 Fuel Distribution Utilities. ASTM D2717	2
				H.5.5 Subset 2 Criteria Evaluation	2
C.43	Thermal Expansion see C.12 Density (Thermal Expansion)				
C.44	Thermal Stability	ASTM D3241	Entrance	B.4.1.1 The Entrance Criteria Definition MIL-DTL-83133F	Entrance
				B.4.2.1 Fuel Property / Characteristic—as affected by Trace Elements	1
				B.4.2.2.2.1.4 SE&V Heat Removal	1
				B.4.3.2.1 APU and Emergency Power Unit (EPU) Evaluation	2
				G.4.6 Table G-IV.	3
C.45	Trace Elements	ASTM D7111	1	B.4.2.1 Fuel Property / Characteristic Tests	1
				G.4.4 Table G-II.	1
				G.6.2 Risk Assessment Process - Flame Tube Test	3
C.46	Vapor Pressure, True versus Temperature	ASTM D323 or ASTM D5191	1	B.4.2.1 True, Fuel Property / Characteristic Tests	1
				B.4.2.2.1 Auxiliary and Emergency Power Units (APU/EPU) Evaluation	1
				B.4.2.2.2.2 SE&V Properties	1
				G.4.4 Table G-II.	1
C.46A	Velocity of Sound		2	B.4.3.1 Subset 2 Fuel Property/Characteristic Tests	2

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TABLE L-I. Fuel property/test method application - Continued.

C-Para.	Appendix C			Appendix Where Applied	
	Property	Test Method	Criteria	Paragraph	Criteria
C.47	Viscosity at -20° C	ASTM D445	Entrance	B.4.1.1 The Entrance Criteria Definition MIL-DTL-83133F	Entrance
				G.4.3 Entrance Criteria Evaluation MIL-DTL-83133F	Entrance
C.47	Viscosity versus Temperature	ASTM D445	1	B.4.2.1 Fuel Property / Characteristic Tests	1
				B.4.2.2.1 Auxiliary and Emergency Power Units (APU/EPU) Evaluation	1
				B.4.2.2.2.2 SE&V Properties at -20° C and +40° C	1
				B.4.4.2.2.1 Impact of Fuel on Injection Pump Durability	3
				G.4.4 Table G-II.	1
C.48	Water Reaction Interface Rating	ASTM D1094 ASTM D3242	Entrance	B.4.1.1 The Entrance Criteria Definition MIL-DTL-83133F	Entrance
C.49	Water Separation Index	ASTM D3948	Entrance	B.4.1.1 The Entrance Criteria Definition MIL-DTL-83133F	Entrance
C.50	Water Solubility	ASTM D6304	1	B.4.2.1 Fuel Property / Characteristic Tests	1



FIGURE L-1. Index spreadsheet.

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Air Force – 11
(Project 91GP-2009-004)

Review activity:

Air Force – 68

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